

COBORDISM MAPS AND ATIYAH–FLOER ISOMORPHISMS FOR FRAMED INSTANTON HOMOLOGY

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ABSTRACT. Daemi, Fukaya and Lipyanskiy proved that the framed instanton homology of a closed oriented three-manifold is isomorphic to the Lagrangian Floer homology of the two handlebody Lagrangians of a Heegaard splitting, in the moduli space of flat $\mathrm{SO}(3)$ connections with $w_2 \neq 0$ on the surface obtained from the Heegaard surface by adding a torus summand. Their isomorphism counts solutions of a mixed equation, which couples the anti-self-duality equation to the holomorphic curve equation. We construct isomorphisms of the same type, counting solutions of the mixed equation whose holonomy perturbations respect the $\mathbb{Z}/2$ symmetry that defines the framed theory, and prove that, with coefficients in $\mathbb{F}_2$, they intertwine the cobordism maps of the two theories. On the gauge side the maps are those of Kronheimer and Mrowka. On the symplectic side they are defined by holomorphic curves alone: counts of holomorphic triangles in the moduli space of the Heegaard surface and, for one- and three-handles, in a product of two such spaces with the correspondence of the compression body as a boundary condition. The statement holds for every connected cobordism between connected three-manifolds, equipped with an embedded arc joining the ends, with a framing of its normal bundle, and an $\mathrm{SO}(3)$ bundle trivialized over the ends and along the arc. As a consequence the symplectic cobordism maps are independent of the handle decomposition and of the Heegaard data, and form a functor isomorphic to framed instanton homology. At the data where both constructions are defined, with secondary perturbations invariant under that symmetry, our isomorphisms coincide with those of Daemi, Fukaya and Lipyanskiy reduced modulo two. The analytic input consists of two new results for the mixed equation. The first is an adiabatic limit in which the metric on a compression body collapses next to a holomorphic strip: the compression body becomes the Lagrangian boundary condition of its flat connections, two ends appear along which the matching surface runs to infinity, with a Morse–Bott family of stationary solutions, the index-zero counts are unchanged, and a single affine relation between energy and index holds at every scale. The second is an excision theorem for the mixed equation along a torus, in a situation where no closed three-manifold separates the two sides.

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1. INTRODUCTION

1.1. The Atiyah–Floer isomorphism. Let M be a closed, connected, oriented three-manifold. Its framed instanton homology $I^\#(M)$ is the instanton Floer homology of the connected sum $M \# T^3$ with the $\mathrm{SO}(3)$ bundle that is trivial over M and nontrivial over T^3 [21, 38]. A Heegaard splitting $M = H_0 \cup_\Sigma H_1$ of genus g gives a second invariant. Take the boundary sum of each handlebody with $[0, 1] \times T^2$. The two resulting compression bodies C_0, C_1 have boundary $\Sigma_h \sqcup T^2$ with $h = g + 1$. Let $\mathbf{M}_h$ denote the moduli space of flat connections on the nontrivial $\mathrm{SO}(3)$ bundle over Σ_h . The flat connections that extend over C_i form an embedded Lagrangian $L(C_i) \subset \mathbf{M}_h$. The Lagrangian Floer homology $HF(L(C_0), L(C_1))$ is the symplectic framed Floer homology of [11], introduced by Wehrheim and Woodward as torus-summed Floer homology [43, Definition 4.4.1]; they show by Cerf theory and geometric composition that it is an invariant of M . Atiyah conjectured that, for an integral homology sphere, instanton Floer homology [16] is recovered in this way from the $\mathrm{SU}(2)$ character varieties of a Heegaard splitting [2]; for the version for admissible $\mathrm{SO}(3)$ bundles adopted here see [10, Section 10]. Daemi, Fukaya and Lipyanskiy proved that the two groups are isomorphic [11, Theorem 3]. Their isomorphism counts solutions of a *mixed equation*, introduced by Lipyanskiy [24]: the anti-self-duality equation on part of a four-dimensional domain, coupled along a matching surface to the holomorphic curve equation in $\mathbf{M}_h$ on the rest [11, 12].

Both groups carry cobordism maps. On the gauge side a cobordism W with a path between its ends induces a map $I^\#(W): I^\#(M_0) \rightarrow I^\#(M_1)$, and these maps form a functor [21, Section 5], [38, Section 7]. On the symplectic side a handle decomposition of W can be assigned a composite of counts of holomorphic polygons and quilts, using the quilted Floer theory of Wehrheim and Woodward [45, 42]. For the symplectic instanton homology of Manolescu and Woodward [26], such handle maps were defined and shown to be independent of the decomposition by Cazassus [8], using Cerf theory, and Horton proved the corresponding functoriality for his theory built from traceless character varieties [18], [19, Theorems 1.1–1.2]. For the symplectic framed theory used here no maps for four-dimensional cobordisms were available, and in none of these settings had the symplectic maps been compared with the gauge cobordism maps. The comparison closest in spirit is the open–closed isomorphism of Salamon and Wehrheim [36, Theorem 5.8], which intertwines relative invariants of four-manifolds, with gauge theory on both sides. This paper compares the two families of maps.

1.2. Main result. We work over $\mathbb{F}_2$ throughout. Objects of the connected framed cobordism category $\mathbf{B}$ are closed, connected, oriented three-manifolds M with a parametrized ball, which fixes where the torus is attached (Definition 2.1). A morphism $(W, E, \nu): M_0 \rightarrow M_1$ is an isomorphism class of connected oriented cobordisms W with a framed path tube ν joining the two balls, and an $\mathrm{SO}(3)$ bundle E on W trivialized over M_0 , M_1 and along the tube.

The bundle is arbitrary in the interior; its class relative to the ends lies in $H^2(W, \partial W; \mathbb{F}_2)$. Composition is gluing.

Framed instanton homology is a functor $I^\# : \mathbf{B} \rightarrow \text{Vect}_{\mathbb{F}_2}$ (Proposition 2.4). For each object M and each admissible presentation p (a Heegaard splitting adapted to the ball, together with the analytic data of Section 3) there are the gauge complex $C_G(M, p)$, the symplectic complex $C_S(M, p) = CF(L(C_0), L(C_1))$ and a mixed chain map $N_p : C_G(M, p) \rightarrow C_S(M, p)$. It is a quasi-isomorphism by the triangularity argument of [11, Theorem 3.29] (Theorem 3.6).

For a morphism we define symplectic maps by counting holomorphic curves in the Heegaard moduli spaces, without reference to gauge theory:

- the trace of surgery on a framed link is sent to the triangle map $\mu_S(\cdot \otimes \theta_S)$ of a subordinate Heegaard triple $(\Sigma; H_0, H_1, H_2)$, where θ_S is the generator of maximal relative grading (the maximum of a height function) of the Floer complex of the Lagrangians of H_1 and H_2 (Section 9);
- a k -fold one-handle is sent to a compression cup: the triangle map in $M_h^- \times M_{h+k}$ of the pairs formed by the handlebody Lagrangians and the correspondence L_C of the compression body, evaluated on the top classes of the two auxiliary pairs; a three-handle is sent to the transposed compression cap (Section 11);
- a change of trivialization $u : M \rightarrow \text{SO}(3)$ of the bundle (a *frame change*) is sent to the pushforward by the induced symplectomorphism of M_h ;
- a change of presentation is sent to the corresponding word of continuations, cups, caps and triangles (Section 12).

An *admissible decomposition* d of (W, E, ν) is a factorization into presentation changes, frame changes, single one- and three-handles and one two-handle cobordism, from (M_0, p_0) to (M_1, p_1) (Section 13.2); it gives a composite map $S_d(W, E, \nu)$ from $HF(M_0, p_0) = H(C_S(M_0, p_0))$ to $HF(M_1, p_1)$. A presentation is *admissible* if its primary holonomy perturbations are small in the sense of Definition 3.3: they must lie in a neighbourhood of zero that depends only on the splitting, the metric data and a finite family of holonomy perturbations, on which the mixed maps are natural under changes of primary data and the transpose of the reflected mixed map inverts N_p (Propositions 11.9 and 11.10). This smallness is used in the one-handle comparison.

Theorem A. *For every morphism $(W, E, \nu) : M_0 \rightarrow M_1$ of $\mathbf{B}$, all admissible presentations p_0 of M_0 and p_1 of M_1 , and every admissible decomposition d ,*

$$H(N_{p_1}) \circ I^\#(W, E, \nu) = S_d(W, E, \nu) \circ H(N_{p_0}).$$

Consequently $S_d(W, E, \nu)$ depends only on the morphism and the end presentations. The groups $HF(M, p)$ form a transitive system over the presentations of M , and with its colimit $HF^\#(M)$ the maps $S(W, E, \nu)$ form a functor $HF^\# : \mathbf{B} \rightarrow \text{Vect}_{\mathbb{F}_2}$. The maps $H(N_p)$ descend to a natural isomorphism $I^\# \Rightarrow HF^\#$. The same statements hold in the homotopy category of finite chain complexes over $\mathbb{F}_2$, with the colimit replaced by a choice of representative of the transitive system of homotopy equivalences.

The symplectic side of Theorem A contains no gauge theory. In particular, independence of the decomposition and functoriality of the symplectic maps are consequences, not hypotheses. The argument uses no Cerf theory on the symplectic side, in contrast with the direct constructions of [8, 19] for other symplectic models, and no geometric composition isomorphism: the embedded compositions of the compression correspondence with handlebody Lagrangians enter only as sets, to identify generators and gradings. The quilted description

of the compression cup, which rests on the composition theorems of Wehrheim–Woodward and Lekili–Lipyanskiy, is recorded in Remark 11.5 and is not used.

The maps N_p are constructed by the method of [11]. Where the primary perturbations already regularize the gauge complex, as in [11, Lemma 2.56], and the secondary perturbations are invariant under the framed symmetry, N_p , computed with $f = 0$ and with the almost complex structures and secondary perturbations of [11], is the isomorphism of [11, Theorem 3.29] reduced modulo two, as a chain map (Proposition 3.9). Consequently, at admissible presentations with such primary data, Theorem A holds for the isomorphism of Daemi, Fukaya and Lipyanskiy itself, reduced modulo two, and this isomorphism on homology is independent of the choices within that class, a point not addressed in [11] (Corollary 13.4). DFL state that such primary data exist after a constant rescaling of the metric [11, Proposition 6.8]; whether they exist at the fixed metric data of our presentations is a transversality question that we leave open (Remark 3.5).

1.3. The collapse theorem. Every elementary comparison in Theorem A reduces to one degeneration, stated here informally. Let X be a domain for the mixed equation containing a *compression rectangle*: a product $I \times C$ of an interval with a compression body, attached to a holomorphic strip along the matching surface $I \times \partial_+ C$. Scale the metric of C by ε^2 . After a conformal change each endpoint of I becomes a cylinder over (C, g_C) of length $\log(1/\varepsilon)$, and ε enters only through explicit weights. As $\varepsilon \rightarrow 0$ the rectangle disappears. In the limit the strip meets I along the Lagrangian boundary condition $L(C)$, and two mixed ends appear at the endpoints.

Let X_0 denote the limiting domain, in which the compression rectangle is replaced by the boundary condition $L(C)$, and X_ε the domain at scale ε .

Theorem B. *Assume that all data on X_ε outside a fixed neighbourhood of the rectangle are independent of ε , and that the moduli spaces on X_0 of index at most zero are regular. Then the following hold.*

- (i) *If on X_0 the Fredholm index and the topological energy κ , normalized as in [11, Proposition 3.20], satisfy $\text{ind} = c + 8\kappa$ for all configurations with fixed asymptotic labels, then the same relation, with the same constant c , holds on X_ε for every $\varepsilon > 0$.*
- (ii) *For every sequence of solutions of index d on X_{ε_ν} , with $\varepsilon_\nu \rightarrow 0$, a subsequence converges to a solution on X_0 of index d_0 and broken trajectories of indices d_j , and $d - d_0 - \sum_j d_j = 4n$ for an integer $n \geq 0$.*
- (iii) *There is $\varepsilon_0 > 0$ such that for $0 < \varepsilon < \varepsilon_0$ gluing the constant flat solution on the rectangle defines a bijection $\mathcal{G}_\varepsilon: \mathcal{M}^0(X_0) \rightarrow \mathcal{M}^0(X_\varepsilon)$ between the index-zero moduli spaces, and the solutions at scale ε are regular.*

The precise statement is Theorem 6.6. The limit is of the type proposed in the programme of Salamon and Wehrheim [35, Conjecture 3.4], [41, Section 4], in which the metric on a handlebody degenerates onto its spine and the handlebody is replaced by the Lagrangian boundary condition of its flat connections. (The second degeneration of that programme, shrinking Σ in $[0, 1] \times \Sigma$, is the adiabatic limit of Dostoglou and Salamon [15]; Fukaya [14] coupled the anti-self-duality equation to holomorphic curves through a degenerate metric.) Here the collapsing body is a compression body whose boundary is matched, along part of it, to a holomorphic strip that does not collapse, and two mixed ends appear. To our knowledge the handlebody degeneration of [35, 41] has not been carried out; Salamon and Wehrheim prove their open–closed isomorphism by a different route [36]. After a conformal change the collapse becomes neck stretching along cylinders over C . In the compactness theorem of Xu

[46, Theorem 1.4], for instantons on $\mathbb{R} \times M$ with M a three-manifold with a cylindrical end over Σ , energy escaping along the end produces holomorphic curves in the moduli space of Σ with the Lagrangian boundary condition of M . Theorem B adds to compactness the index–energy law at every scale and the bijection of index-zero counts. The symplectic analogue is strip shrinking in quilted Floer theory [42, 7]. Three features of the proof are specific to this situation. First, the model linear estimates (Section 7) lose no power of $\log(1/\varepsilon)$; the patched inverse on the whole domain loses exactly one factor $1 + \log(1/\varepsilon)$, which is sharp, and the surjectivity argument is arranged so that this factor is always multiplied by a positive power of ε . The reason for the first point is that the slow harmonic directions of C are tied, pointwise in time, to the boundary value of the strip and are controlled by the strip’s unweighted norm. This depends on the orientation convention 3.1; with the other sign the estimate is false. Second, compactness requires no classification of bubbles. Energy loss is excluded by the affine law (i), which holds at every scale, together with nonnegativity of the limiting indices. Third, surjectivity of the gluing map is proved without entering a logarithmically shrinking uniqueness ball directly. A solution is first brought within a power of ε of the glued solution, and only then compared with it.

1.4. Excision along a torus. The one-handle comparison needs a second new result. Let M have a Heegaard splitting and let $M^+ = M \#^k (S^1 \times S^2)$ have the splitting obtained by adding k trivial handles. Cutting $(-M)^\#$ and $(M^+)^\#$ along torus fibres and regluing gives the closed three-manifold $Z \cong (-M) \# M^+ \# T^3$. The pair of compression bodies (D_0, D_1) splitting Z defines a Lagrangian pair (A_0, A_1) in $\mathbf{M}_h^- \times \mathbf{M}_{h+k}$, the *folded pair* (Section 10.1). Let N_Z be its mixed map, N_- the mixed map of the reflected presentation of $-M$ (valued in the dual Floer complex), N_+ that of M^+ , E Floer’s excision map [5, 22], [21, Theorem 5.6] from the instanton complex of Z to the tensor product of those of $(-M)^\#$ and $(M^+)^\#$, and $\mathcal{K}$ the folding isomorphism of Lagrangian Floer complexes for a split almost complex structure.

Theorem C. *With N_Z , $N_\pm$, E and $\mathcal{K}$ as above, $[\mathcal{K}][N_Z] = ([N_-] \otimes [N_+])[E]$ on homology over $\mathbb{F}_2$.*

This is Theorem 10.2. It is not an instance of neck stretching along a closed hypersurface: the torus region runs into the mixed ends, so no closed three-manifold separates the two sides. Near the torus the domain is a product of T^2 with a planar region. We separate the two sides by a flat product region of width 2λ (the *bridge*), on whose boundary only flatness of the torus restriction is imposed (the *flat-fibre condition*); the boundary on which it is imposed is the *flat face*, and letting the bridge become infinite gives the *completed tail*. The proof proceeds by a closed cut at a fixed bridge width, a homotopy of the width and a completion of the torus region (Section 10, plan of the proof). Three features make the completion work. The linearized operator at the odd flat torus is invertible on any base of bounded geometry, with the flat-fibre boundary condition and with bound given by the spectral gap μ_T of the fibre (Proposition 10.3), so all estimates are uniform in the collar length. Small-energy regularity holds up to the flat face, because the boundary flux is cubic in the curvature (Theorem 10.5). Solutions decay exponentially along the completed tail (Proposition 10.10).

1.5. Strategy. Theorem A is deduced in Section 13 from four inputs by a formal argument (Lemma 13.1):

- (1) $I^\#$ is a functor on $\mathbf{B}$ (Section 2);
- (2) N_p is an isomorphism on homology, and its homology class is independent of auxiliary choices (Section 3);

- (3) every morphism is a composite of elementary arrows: presentation moves, frame changes, one-handles, a single two-handle cobordism and three-handles (Section 4);
- (4) every elementary arrow e satisfies the square $H(N)I^\#([e]) = S_e H(N)$, where N denotes the mixed isomorphism N_p of the chosen presentation at each end (the *vertex maps*) (Sections 9–12).

The maps $H(N_p)$ are invertible. Hence (3) and (4) determine every symplectic word as $H(N)I^\#(W)H(N)^{-1}$. Independence of the word and functoriality are then inherited from the gauge side.

The squares for two-handles are proved as follows. The gauge two-handle map is a triangle map with an auxiliary input (Section 9). A mixed triangle degenerates in two ways. Stretching one family of necks produces the gauge triangle followed by N . Collapsing a compression rectangle by Theorem B and then stretching ordinary necks produces $N \otimes N$ followed by the symplectic triangle. A compact family of mixed triangles joins the two (Section 5).

For a k -fold one-handle the symplectic cup is a triangle count in the product $M_h^- \times M_{h+k}$ of two moduli spaces, with the correspondence of the compression body as middle Lagrangian and top classes of two auxiliary pairs as inputs (Section 11). Folding the triangle along the diagonal and applying the same two degenerations identifies the cup with the mixed map of a folded pair composed with a gauge triangle, and Theorem C splits off the extra three-torus summand. The result is the *cup formula*

$$U_C = N_{C \circ p} \circ I^\#(W_k) \circ N_p^\dagger,$$

where W_k is the one-handle cobordism and $N_p^\dagger$ is the transpose of the mixed map $N_{\bar{p}}$ of the reflected presentation. It remains to show that $N_p^\dagger = N_p^{-1}$ on homology, and this is done without any composition theorem for Lagrangian correspondences. For $k = 0$ the cup is a continuation map. At a common auxiliary presentation A , the cup formula for the reflections of two presentations with primary data P and Q reads (indexing mixed maps by primary data, and using $N_{\bar{P}}^\dagger = N_P^\vee$)

$$\text{cont}_S(\bar{P}, A) = N_A \circ \text{cont}_G(\bar{P}, A) \circ N_P^\vee, \quad \text{cont}_S(\bar{Q}, A) = N_A \circ \text{cont}_G(\bar{Q}, A) \circ N_Q^\vee.$$

Eliminating the invertible N_A gives the naturality of N under small changes of primary data (Proposition 11.9). Applying the $k = 0$ cup formula once more, with P itself as the lower presentation, and comparing with that naturality gives $N_P \circ N_P^\dagger = 1$ (Proposition 11.10). The three-handle squares follow by reflection.

1.6. Scope. The theorem is stated over $\mathbb{F}_2$. Over $\mathbb{Z}$ the symplectic counts require coherent orientations of the polygon and quilt moduli spaces, together with a comparison of those orientations with the homology orientations of the gauge side. Neither is addressed here. The objects carry the trivial bundle on the original three-manifold. Disconnected objects, the units and counits of a field theory, and closed pairings are not treated. The mixed maps N_p are constructed by the method of [11], with the invariant perturbations of Section 3; they agree with the isomorphism of [11], reduced modulo two, at the data where both are defined (Proposition 3.9); the existence of such data at the metric data of a presentation is left open (Remark 3.5). The maps of the transitive system of Theorem A are symplectic words; their independence of the word is proved through the gauge side, and we do not compare them with the Cerf-theoretic identifications of [43].

Organization. Section 2 defines $\mathbf{B}$ and the gauge functor. Section 3 recalls the Heegaard moduli spaces and the mixed isomorphism and fixes conventions. Section 4 contains the topology of handle decompositions and bundles. Section 5 treats neck-stretching, families and regularity in the invariant perturbation class. Sections 6–8 prove Theorem B. Section 9 proves the two-handle square, Section 10 proves the excision theorem along a torus (Theorem C), which the one-handle comparison requires, Section 11 proves the one- and three-handle squares, Section 12 treats changes of presentation and frame, and Section 13 proves Theorem A. A table of notation follows.

Notation.

$M^\#, E_M^\#$	torus-summed object and its bundle (Section 2)
ϕ_M	lifting group, the subgroup ϕ of [21, Section 5.1]
$\Sigma_h, \mathbf{M}_h$	Heegaard surface of genus h and its moduli space of odd flat connections
$C_0, C_1, L(C)$	compression bodies of a presentation and their Lagrangians
p, N_p	presentation and its mixed map (the vertex map)
$P, N_P, N_P^\dagger$	primary data of p , $N_P = N_p$, and $N_P^\dagger = N_P^\vee$
$C_G(M, p), C_S(M, p)$	gauge and symplectic complexes
$\mathfrak{h}_i, f, \eta$	primary perturbations, end perturbation, secondary perturbation
$\kappa, \mathfrak{c}$	topological energy and the index constants of [11, Proposition 2.53]
$\text{cont}_S, \text{cont}_G$	symplectic and gauge continuation maps
X_0, X_ε, I	collapsing family and the collapsing arc (Section 6.2)
$\mathfrak{s}, \mathbf{A}$	compression scale, and $\mathbf{A} = 1 + \log(1/\varepsilon)$
$W_R, W_\infty, W_\lambda^J$	collared, completed and joined domains (Definition 10.1)
μ_T	spectral gap of the odd flat connection on the torus
$R_i, \tilde{P}_i, L_C$	height Lagrangians, lower Lagrangians, correspondence of $C_{h,g}$
$A_0, A_1, \mathcal{K}, \mathbf{E}$	folded pair, folding isomorphism, Floer’s excision map
U_C, V_C	compression cup and cap (Definition 11.3)

2. THE FRAMED COBORDISM CATEGORY AND THE GAUGE FUNCTOR

2.1. The torus attachment. Write $T^3 = S^1 \times T^2$ and let $P_T \rightarrow T^3$ be the pullback of the $\text{SO}(3)$ bundle on T^2 with $w_2 \neq 0$. Fix once and for all a ball $B_T \subset T^3$ disjoint from a fibre $\{\theta_0\} \times T^2$, a parametrization of B_T by the unit ball B^3 , a trivialization of $P_T|_{B_T}$ and a metric on T^3 that is standard near B_T and flat on a neighbourhood of the fibre $\{\theta_0\} \times T^2$; the flat metric it induces on T^2 is used for every torus collar below, in all presentations. Put $T^\circ = T^3 \setminus \text{int } B_T$. Let

$$(2.1) \quad \alpha_T = \text{pr}_{S^1}^*(\text{generator of } H^1(S^1; \mathbb{F}_2)) \in H^1(T^3; \mathbb{F}_2).$$

Then $\langle w_2(P_T), [T^2] \rangle = 1$, $\alpha_T|_{T^2} = 0$ and $\langle \alpha_T \smile w_2(P_T), [T^3] \rangle = 1$.

Definition 2.1. An *object* of $\mathbf{B}$ is a closed, connected, oriented three-manifold M together with an orientation-preserving embedding $b: B^3 \rightarrow M$. The *torus-summed object* is

$$M^\# = (M \setminus b(\text{int } B^3)) \cup_{S^2} T^\circ, \quad E_M^\# = \mathbb{R}^3 \cup P_T|_{T^\circ},$$

glued along the boundary spheres by the fixed parametrizations and trivializations.

A *decorated cobordism* $(W, E, \nu): (M_0, b_0) \rightarrow (M_1, b_1)$ consists of:

- a compact, connected, oriented four-manifold W with $\partial W = -M_0 \sqcup M_1$ and product collars at both ends;
- an embedding $\nu: B^3 \times [0, 1] \rightarrow W$ with $\nu(\cdot, i) = b_i$, a product in the collars (the *tube*);

- an $\mathrm{SO}(3)$ bundle $E \rightarrow W$ with trivializations over M_0 , M_1 and over the image of ν , which agree on the overlaps.

An *isomorphism* of decorated cobordisms is an orientation-preserving diffeomorphism covered by a bundle isomorphism, both respecting the boundary identifications, the tubes and the trivializations. The morphisms of $\mathbf{B}$ are isomorphism classes of decorated cobordisms. Composition glues along the ends using the collars, tubes and trivializations; the identity of (M, b) is the product $[0, 1] \times M$ with the vertical tube and the product bundle.

Associativity and the unit laws hold on isomorphism classes, by the usual collar-reparametrization isomorphisms. A product $[0, 1] \times M$ whose bundle identifications at the two ends differ by a nontrivial automorphism is not the identity. Its image under the functor below is the corresponding bundle automorphism map.

For a decorated cobordism put $W^\circ = W \setminus \nu(\mathrm{int} B^3 \times [0, 1])$. Then

$$(2.2) \quad W^\# = W^\circ \cup_{S^2 \times [0, 1]} (T^\circ \times [0, 1]), \quad E^\# = E|_{W^\circ} \cup (P_T|_{T^\circ} \times [0, 1]),$$

glued by the tube parametrization and the trivializations. Its ends are $-M_0^\# \sqcup M_1^\#$ with the bundles $E_{M_i}^\#$.

Lemma 2.2. *Restriction gives an isomorphism $H^1(W^\#; \mathbb{F}_2) \rightarrow H^1(W; \mathbb{F}_2) \oplus H^1(T^3; \mathbb{F}_2)$.*

Proof. W° is connected, since a path in the four-manifold W can be displaced off the core arc of the tube and then off its tubular neighbourhood. The pieces W° and $T^\circ \times [0, 1]$ are connected, their intersection $S^2 \times [0, 1]$ is connected and has vanishing H^1 , so Mayer–Vietoris gives $H^1(W^\#) = H^1(W^\circ) \oplus H^1(T^\circ)$. The same argument applied to $W = W^\circ \cup (B^3 \times [0, 1])$ and $T^3 = T^\circ \cup B_T$ identifies $H^1(W^\circ) = H^1(W)$ and $H^1(T^\circ) = H^1(T^3)$. $\square$

Let $\alpha_W \in H^1(W^\#; \mathbb{F}_2)$ correspond to $(0, \alpha_T)$ and put $\phi_W = \{0, \alpha_W\}$. On each end $M_i^\#$ the class α_W restricts to the class α_{M_i} corresponding to $(0, \alpha_T)$, and $\phi_{M_i} = \{0, \alpha_{M_i}\}$. The class α_W is realized by a bundle automorphism of $E^\#$: take the standard automorphism of P_T with lifting class α_T (the gauge transformation g of [38, Section 7.1]), deform it to be the identity over B_T , use it constantly along $T^\circ \times [0, 1]$, and extend by the identity over W° .

2.2. The gauge functor. Recall from [21, Section 5.1] that for an $\mathrm{SO}(3)$ bundle on a three-manifold the quotient of the automorphism group by the image of the determinant-one gauge group is $H^1(\cdot; \mathbb{F}_2)$, so a subgroup $\phi \subset H^1$ acts on the configuration space of connections modulo determinant-one gauge transformations. The class in $H^1(\cdot; \mathbb{F}_2)$ of an automorphism g is its *lifting class* $c(g)$, the obstruction to lifting g to a determinant-one gauge transformation; g is *even* if its lifting class vanishes. A subgroup ϕ of lifting classes used in this way is the *lifting group* of the problem; it is the subgroup ϕ of [21, Section 5.1]. A triple satisfies the *ϕ -non-integral condition* if it contains a surface on which w_2 is odd and ϕ vanishes [21, Definition 5.1]. For $M^\#$ the torus $\{\theta_0\} \times T^2$ is such a surface for ϕ_M . Under this condition ϕ acts freely on the critical set, the instanton homology of the quotient $I(M^\#, E_M^\#)^{\phi_M}$ is defined, and a cobordism with a subgroup $\phi \subset H^1(W; \mathbb{F}_2)$ restricting to the given subgroups at the ends induces a map $I(W^\#, E^\#)^\phi$ by counting points in the quotient moduli spaces [21, pp. 154–155]. Over $\mathbb{F}_2$ no orientation data is needed.

Definition 2.3. For an object (M, b) put $I^\#(M) = I(M^\#, E_M^\#; \mathbb{F}_2)^{\phi_M}$. For a decorated cobordism put $I^\#(W, E, \nu) = I(W^\#, E^\#; \mathbb{F}_2)^{\phi_W} : I^\#(M_0) \rightarrow I^\#(M_1)$.

Here $I(M^\#, E_M^\#; \mathbb{F}_2)^{\phi_M}$ denotes a single vector space: the Floer groups for different metrics and perturbations are identified by continuation maps, which form a transitive system, and $I^\#(M)$ is its limit, that is, any one of the groups together with these identifications. The group $I^\#(M)$ is the framed instanton homology of [21, 38] with $\mathbb{F}_2$ coefficients. Since $\langle \alpha_T \smile w_2(P_T), [T^3] \rangle = 1$, the nonzero element of ϕ_M shifts the grading of the determinant-one (unquotiented) theory by four (the proof of [21, Proposition 5.2(a)]), and the determinant-one Floer homology of $(M^\#, E_M^\#)$ is two copies of $I^\#(M)$.

Proposition 2.4. $I^\#$ is a functor $\mathbf{B} \rightarrow \text{Vect}_{\mathbb{F}_2}$.

Proof. Well defined. Isomorphic decorated cobordisms give isomorphic data $(W^\#, E^\#, \phi_W)$, hence equal maps by the usual continuation argument for metrics and perturbations [21, Section 3], [13, Section 5.3].

Composition. Let $W_{01}: M_0 \rightarrow M_1$ and $W_{12}: M_1 \rightarrow M_2$ be composable, with composite W_{02} . Then $W_{02}^\# = W_{01}^\# \cup_{M_1^\#} W_{12}^\#$ as bundles. By [21, Proposition 5.4], $I(W_{02}^\#)^\phi = I(W_{12}^\#)^{\phi_{12}} \circ I(W_{01}^\#)^{\phi_{01}}$ provided that ϕ contains the image ψ of the Mayer–Vietoris connecting map $H^0(M_1^\#; \mathbb{F}_2) \rightarrow H^1(W_{02}^\#; \mathbb{F}_2)$, and ϕ is the largest subgroup restricting to ϕ_{01} and ϕ_{12} on the pieces. Since $M_1^\#$ is connected, the restriction $H^0(W_{01}^\#) \oplus H^0(W_{12}^\#) \rightarrow H^0(M_1^\#)$ is onto, so $\psi = 0$. The class $\alpha_{W_{02}}$ restricts to $\alpha_{W_{01}}$ and $\alpha_{W_{12}}$. A class restricting to 0 or to α on both pieces is determined by its restrictions, because $H^1(W_{02}^\#) \rightarrow H^1(W_{01}^\#) \oplus H^1(W_{12}^\#)$ is injective when $\psi = 0$. Hence the largest such subgroup is $\phi_{W_{02}}$.

Unit. For the product $[0, 1] \times M$ with translation-invariant regular data the index-zero solutions are the constant trajectories. The group ϕ_M acts freely on the critical set. A trajectory from a critical point β to $\alpha_M \cdot \beta$ has index in the determinant-one theory congruent to 4 modulo 8 by the proof of [21, Proposition 5.2(a)], so it does not contribute in index zero. The map is therefore the identity on the chain level for such data, and on homology for all data. $\square$

Remark 2.5. The functor uses the enlarged gauge group of [21, Section 5] with the subgroup ϕ_W . It is not the projective functor of [21, Section 4.1], which uses determinant-one gauge transformations and yields two copies of $I^\#$. Scaduto enlarges the gauge group on $W^\#$ by the transformations that restrict to g on the torus [38, Section 7.2]; restricted to classes compatible with the ends, this is ϕ_W . The subgroup ϕ_W vanishes on the original piece W° . Classes of $H^1(W)$ that vanish on ∂W are not added.

Remark 2.6. With the $\mathbb{Z}/4$ grading of [38, Proposition 7.1], $I^\#(W, E, \nu)$ is homogeneous of degree $-\frac{3}{2}(\chi(W) + \sigma(W)) + \frac{1}{2}(b_1(M_1) - b_1(M_0)) + 2P(E)$ modulo 4, where $P(E)$ is the mod-two self-intersection of a surface in the interior of W dual to $w_2(E)$ relative to the trivialized ends.

3. HEEGAARD DATA, THE TWO COMPLEXES AND THE MIXED ISOMORPHISM

3.1. The moduli space of a surface. For $h \geq 1$ let Σ_h be a closed oriented surface of genus h and $F_h \rightarrow \Sigma_h$ the $\text{SO}(3)$ bundle with $w_2 \neq 0$. Let $\mathbf{M}_h$ be the moduli space of flat connections on F_h modulo determinant-one automorphisms. It is a closed, simply connected, monotone symplectic manifold of dimension $6h - 6$, with $\pi_2(\mathbf{M}_h) = \mathbb{Z}$ and minimal Chern number 2 [3, 29]. Its tangent space at $[\alpha]$ is $H^1(\Sigma_h; \text{ad } \alpha)$.

Convention 3.1 (orientation). Let C be an oriented three-manifold and $\Sigma \subset \partial C$ a boundary component with the outward-normal-first orientation. On the moduli space of Σ viewed as a

boundary component of C we use

$$\omega(v, w) = \int_{\Sigma} \langle v \wedge w \rangle, \quad J = *_{\Sigma},$$

where $\langle \cdot, \cdot \rangle$ is the inner product $\langle a, b \rangle = -\operatorname{tr}(ab)$ on $\operatorname{ad} F \cong \mathfrak{su}(2)$, used for all L^2 norms. In the notation of [11], whose bilinear form tr is $-\frac{1}{2}$ of the Euclidean inner product on $\mathbb{R}^3 \cong \mathfrak{so}(3)$, the matching surface is $\Sigma_D = -\partial C$, with $\Omega_D = -\omega$ and $J_D = -J$; the two descriptions define the same mixed equation.

Convention 3.1 is not a matter of taste. It is the sign for which the Green identity of the compression body cancels the Green identity of the adjacent strip (Lemma 6.8). With the opposite sign the uniform estimate of Proposition 7.4 fails, already for constant scale (Remark 7.5).

3.2. Compression bodies and their Lagrangians. A *torus compression body* of genus h is a compact oriented three-manifold C with $\partial C = \Sigma_h \sqcup T^2$ obtained from a handlebody H of genus $h - 1$ by boundary sum with $[0, 1] \times T^2$, together with the $\operatorname{SO}(3)$ bundle that is pulled back from the odd bundle on T^2 . More generally we consider compression bodies C from Σ_h to $\Sigma_{h'}$ with the odd bundle. Every flat connection on C is irreducible with $H^0(C; \operatorname{ad}) = H^2(C; \operatorname{ad}) = 0$, the flat moduli space $\mathcal{R}(C)$ is a compact manifold, and restriction to the boundary is a Lagrangian embedding into the product of the moduli spaces of the boundary components with the orientation of ∂C [11, Proposition 2.21]. For a torus compression body the moduli space of the torus boundary is a point, so restriction to Σ_h is a Lagrangian embedding

$$L(C) \subset \mathbf{M}_h,$$

with $L(C) \cong (S^3)^{h-1}$, simply connected with $\pi_2 = 0$. The pair $(L(C_0), L(C_1))$ of two torus compression bodies is monotone with minimal Maslov number 4 [11, Theorem 1]. For a compression body from Σ_h to $\Sigma_{h'}$ with $h' \geq 2$ the Lagrangian lies in $\mathbf{M}_h^- \times \mathbf{M}_{h'}$ (Definition 3.2 and Section 10.1).

The properties used in the collapse theorem are the following.

Definition 3.2. An *odd compression body* is a compact oriented three-manifold C with a metric g_C that is a product near ∂C , and an $\operatorname{SO}(3)$ bundle $\mathbb{E} \rightarrow C$, such that:

- (C1) $w_2(\mathbb{E})$ is nonzero on every component of ∂C ;
- (C2) every flat connection B on $\mathbb{E}$ has $H^0(C; \operatorname{ad} B) = H^2(C; \operatorname{ad} B) = 0$;
- (C3) the restriction $H^1(C; \operatorname{ad} B) \rightarrow H^1(\partial C; \operatorname{ad} B)$ is injective, and the flat moduli space of C restricts to an embedded Lagrangian $L(C)$ in the product of the moduli spaces of the boundary components with the signs of Convention 3.1.

The name refers to the examples of Sections 3.2 and 6.1. Beyond (C1)–(C3), Sections 6–8 use only that $L(C)$ is compact and connected, not the structure of a compression body.

For $v \in T_q L(C)$ let h_v denote the harmonic one-form on C with values in $\operatorname{ad} B_q$, with vanishing normal component and boundary cohomology class v . The map $v \mapsto h_v$ is well defined because $H^1(C; \operatorname{ad}) \rightarrow H^1(\Sigma_h; \operatorname{ad})$ is injective.

Primary perturbations. Following [11, Section 2.3], a *primary perturbation* of C is a holonomy perturbation $\mathfrak{h}$ supported in the interior of C° , the part of C outside the unit collar on which [11, (2.1) and p. 9] require the perturbations to vanish (the special quintuple and the torus analysis of Section 10 use this collar unperturbed), small enough that the perturbed flat connections still form a compact manifold whose restriction $L_{\mathfrak{h}}(C) \subset \mathbf{M}_h$ is an embedded Lagrangian. It is Hamiltonian isotopic to $L(C)$: for $\mathfrak{h}$ small the implicit function theorem makes

$L_{\mathfrak{h}}(C)$ C^1 -close to $L(C)$, hence the graph of a closed one-form in a Weinstein neighbourhood, and the one-form is exact because $L(C)$ is simply connected. ([11] call these perturbations cylinder functions. In their notation C_0 and C_1 are the halves $Y_1 = Y_0 \cup [0, 1] \times \Sigma$ and Y'_1 of $Y_{\#}$ cut along $\{0\} \times \Sigma$ [11, p. 79], and C_i° is Y_0, Y'_0 ; Section 10 writes $(Y_{\pm})_0$.) The universal restriction map from pairs (perturbation, flat connection) to M_h is a submersion [11, Section 2.3], so finitely many primary perturbations make any finite collection of such Lagrangians pairwise transverse with empty triple intersections.

3.3. Admissible presentations.

Definition 3.3. Let (M, b) be an object of **B**. *Presentation data* p for M consist of:

- (i) a Heegaard splitting $M = H_0 \cup_{\Sigma} H_1$ of genus $g \geq 1$ adapted to b ;
- (ii) the torus compression bodies $C_i = H_i \natural ([0, 1] \times T^2)$, the boundary sums taken inside $b(B^3)$ through the equatorial disk, so that $C_0 \cup C_1$ is the torus-summed object $M^{\#}$ of Definition 2.1, with $h = g + 1 \geq 2$;
- (iii) identifications of collars of $\partial C_0, \partial C_1$ as in [11, Section 2.1], and metrics on C_0, C_1 that are products on these collars and agree on Σ_h and on the collar of T^2 , which carries the flat metric fixed in Section 2 (the same for all presentations) (*metric data*);
- (iv) primary perturbations $\mathfrak{h}_0, \mathfrak{h}_1$ with $L_{\mathfrak{h}_0}(C_0) \natural L_{\mathfrak{h}_1}(C_1)$ (*primary data*);
- (v) the regularizing data of Section 3.4, with the end perturbation f and the secondary perturbation below the thresholds of Theorem 3.6.

They form an *admissible presentation* if, for some finite family F of primary perturbations containing the height functions of Sections 9 and 10 and enough further perturbations that the universal restriction map of the span of F is a submersion at every flat connection (possible by compactness and [11, Section 2.3]), the primary data lie in the neighbourhood $\mathcal{U}(F)$ of zero in the span of F constructed after Proposition 11.10; it depends only on the splitting, the metric data and F .

Theorem 3.6 and Sections 5–10 apply to all presentation data, and more generally to the pairs of Remark 3.7; admissibility is used only in Sections 11–13.

3.4. The two complexes. Let p be presentation data and put $P_i = L_{\mathfrak{h}_i}(C_i)$.

The symplectic complex. $C_S(M, p) = CF(P_0, P_1)$ is the Floer complex over $\mathbb{F}_2$ generated by $P_0 \cap P_1$, with the differential counting index-one holomorphic strips for a generic t -dependent compatible almost complex structure equal to J near the boundary [30, 11]. Monotonicity and minimal Maslov number 4 exclude disk bubbling in dimensions at most one.

The gauge complex. Glue C_0 and $-C_1$ along $\Sigma_h \sqcup T^2$ to obtain $M^{\#}$ with its bundle. The primary perturbations define a perturbed Chern–Simons functional $\text{CS}_{\mathfrak{h}}$ on $M^{\#}$, whose critical orbits correspond bijectively to $P_0 \cap P_1$ and are nondegenerate [11, Lemmas 2.45, 2.47]. The nonzero element of ϕ_M acts freely on the critical set. Choose an *end perturbation* f : a finite holonomy perturbation of the Chern–Simons functional on $M^{\#}$, invariant under ϕ_M , vanishing on a neighbourhood of every critical connection, and such that all nonconstant trajectories of $\text{CS}_{\mathfrak{h}} + f$ of index at most seven in the determinant-one theory are regular. Such f exist, may be taken arbitrarily small, and do not change the critical set or the Hessians (Proposition 3.4). Regularity is achieved by f rather than by the primary perturbations because in the two-handle comparison the primary perturbation of the middle compression body must vanish (Remark 3.8). If the trajectories of $\text{CS}_{\mathfrak{h}}$ of index at most seven are already regular, $f = 0$ is allowed; we then call the primary data *DFL-regular*. These are the primary data required in [11, Lemma 2.56] and Theorem 3.29 there; see Remark 3.5 on their existence

at the metric data of a presentation. The gauge complex $C_G(M, p)$ is the Floer complex of $CS_{\mathfrak{h}} + f$ modulo ϕ_M . Its homology is canonically $I^\#(M)$.

Both complexes have the same generator set $P_0 \cap P_1$. This is an identification of generators, not of differentials.

Proposition 3.4. *For every choice of primary data there are end perturbations f as above, of arbitrarily small C^0 norm, with $f = 0$ near the critical set.*

Proof. This is the standard transversality argument for holonomy perturbations [13, Section 5.5], carried out with perturbations that vanish near the critical set; we indicate the three points to check.

The perturbations reach every nonconstant trajectory. The critical set is finite and nondegenerate. Let γ be a nonconstant trajectory of index at most seven in the determinant-one theory. Since γ is nonconstant, there is a time s_0 at which $\gamma(s_0)$ lies outside a fixed neighbourhood of the critical set, and there the perturbation f may be nonzero.

The perturbations detect cokernel elements. Let ψ be a nonzero element of the cokernel of the linearization at γ . By unique continuation for the trajectory equation, forward and backward in time [37, Proposition 8.6 and Theorem C.1], both ψ and $\partial_s \gamma$ are nonzero at s_0 , which we may choose generic. Holonomy functions separate $\gamma(s_0)$ from the rest of γ , from its translate $\alpha_M \cdot \gamma$ and from the critical set, and their gradients at $\gamma(s_0)$ span a dense subspace of the tangent space of the configuration space (compare the finite embedding argument of [13, Section 5.5, Lemma 5.16]). So some holonomy direction pairs nontrivially with ψ , and the Sard–Smale theorem makes all such trajectories regular for generic f .

Invariance and finiteness. Averaging over ϕ_M preserves the construction: a cylinder function whose gradient is supported, in the configuration space, near $\gamma(s_0)$ has its α_M -translate supported near $\alpha_M \cdot \gamma(s_0)$, and these neighbourhoods are disjoint because ϕ_M acts freely. There are finitely many energy levels below the index-seven bound, and the choice is made by induction on the energy. $\square$

Remark 3.5. DFL state that DFL-regular data exist after a constant rescaling of the metric of $M^\#$ [11, Lemma 2.56 and Proposition 6.8]. Their inductive step adds cylinder functions with loops in $\text{int } C_0$ and in $\text{int } C_1$, and uses, for each cokernel element ψ of the linearization at a trajectory γ , the independence of ψ and $\partial_s \gamma$ on one of the two pieces, which is inferred there from their linear independence on $M^\# \setminus \Sigma$; the rescaling then moves a unit collar off the supports. It preserves the regularity of the trajectories, which it only reparametrizes in time, and the Lagrangians, since the two-form $*_3 \nabla h$ does not change under a constant rescaling of the metric, but it changes the flat metric of the torus collar. At the metric data of a presentation the same step is needed with C_i° in place of $\text{int } C_i$. Neither version follows from the L^2 -orthogonality of ψ and $\partial_s \gamma$ on $M^\#$ alone: $\psi = c_i \partial_s \gamma$ modulo gauge on C_i with constants $c_0 \neq c_1$ is compatible with it, and for such ψ every cylinder function with loops in one piece pairs to zero. The argument of [13, Section 5.5, Proposition 5.17] goes through if, at each stage of the induction, every such pair (γ, ψ) admits a time near which ψ is not a constant multiple of $\partial_s \gamma$ modulo gauge on C_0° or on C_1° and the restricted path is injective and irreducible. We do not settle this here, and the results below that concern DFL-regular data are stated for data that have them.

3.5. The mixed isomorphism. Let q_s be the special quintuple of [11, Section 3.1] associated to the compression bodies C_0, C_1 : a four-manifold X containing $\mathbb{R} \times C_0^\circ, \mathbb{R} \times (-C_1^\circ)$ and $\mathbb{U}_- \times \Sigma_h$, with a gauge-theoretic end $(-\infty, -3] \times M^\#$, and a surface $S = \mathbb{U}_+$ with a symplectic end $[3, \infty) \times [-1, 1]$ and two boundary components labelled by P_0 and P_1 . Here

$\mathbb{C}$ is cut by the matching line $\mathbb{U}_\partial = i\mathbb{R}$ into $\mathbb{U}_-$ and $\mathbb{U}_+$; the planar region $\mathbb{U}_-$ contains the input arm $(-\infty, -3] \times [-1, 1]$, two arms $[-2, 0] \times [2, \infty)$ and $[-2, 0] \times (-\infty, -2]$ and the square $[-1, 0] \times [-2, 2]$, and $\mathbb{R} \times C_0^\circ$ and $\mathbb{R} \times (-C_1^\circ)$ are glued to $\mathbb{U}_- \times \Sigma_h$ along the two boundary curves of $\mathbb{U}_-$ other than $\mathbb{U}_\partial$ [11, Figure 1]. The ends of the two arms are the mixed ends of the quintuple. We call a domain of this type, for any pair covered by the results below, a *DFL domain*. A *DFL disk* is a DFL domain obtained by cutting a larger domain along ordinary strip ends.

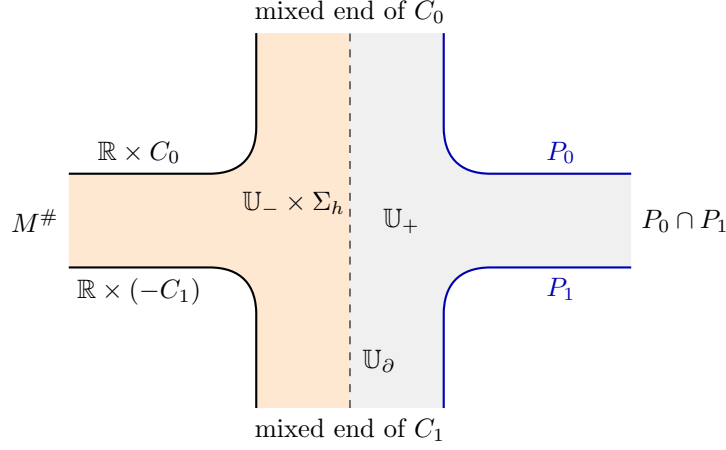


FIGURE 1. The planar base of the special quintuple of [11, Section 3.1], after [11, Figure 1]. Over the region $\mathbb{U}_-$ (shaded) the gauge domain is $\mathbb{U}_- \times \Sigma_h$, with $\mathbb{R} \times C_0$ and $\mathbb{R} \times (-C_1)$ glued along the two boundary curves; the section is defined over $\mathbb{U}_+$, with boundary conditions P_0 and P_1 ; the two are matched along the line $\mathbb{U}_\partial$. The ends are the gauge input $M^\#$, the strip output at $P_0 \cap P_1$, and two mixed ends along the matching line.

The torus T^2 and its point moduli space are carried along; the matching condition on $\mathbb{U}_\partial \times T^2$ imposes flatness of the torus restriction. On the gauge-theoretic end we add χf , where χ is a cutoff equal to 1 near $-\infty$ and supported before the matching region, and we choose a secondary perturbation η : a finite holonomy perturbation supported in a compact subset of X away from the matching line and invariant under ϕ_M , together with a compact deformation of the almost complex structure away from the matching line. The mixed equation is

$$(3.1) \quad F_A^+ + (*_3 \nabla \mathfrak{h})^+ + \chi(*_3 \nabla f)^+ + \eta(A) = 0, \quad \partial_s u + J_{s,\theta}(u) \partial_\theta u = 0,$$

with the matching condition along $\mathbb{U}_\partial \times \Sigma_h$ and the Lagrangian boundary conditions on S . For generators a, x let $\mathcal{M}^d(a, x)$ be the space of finite-energy solutions of index d asymptotic to a at the gauge end and to x at the symplectic end, modulo the gauge group enlarged by ϕ_M .

Theorem 3.6. (i) For f sufficiently small and generic invariant secondary perturbations η , every $\mathcal{M}^d(a, x)$ with $d \leq 1$ is regular, the zero-dimensional spaces are finite, and

$$N_p(a) = \sum_x \# \mathcal{M}^0(a, x) x$$

is a chain map $C_G(M, p) \rightarrow C_S(M, p)$. With respect to the filtration by the constants $\mathfrak{c}$ of [11, Proposition 2.53], which normalize the index formula, $N_p(a) = a +$ (terms of lower filtration); in particular N_p is a chain isomorphism.

- (ii) *The chain homotopy class of N_p , composed with the gauge continuation map between two choices of f , is independent of f , χ and η .*

We call N_p the *vertex map* at p ; in Section 12 the presentations are the vertices of a graph whose edges are changes of presentation.

Proof. Energy and index. For a solution of (3.1) in the homotopy class of (A, u) , the analytic energy equals the topological energy plus $2\|\eta(A)\|^2$ plus the term $\int \chi'(s)f(A_s)ds$, whose absolute value is at most $\sup |f|$, since $\int |\chi'| = 1$ and f vanishes at the critical point at the gauge end. The index is $\text{ind } D_{(A,u)} = 8\kappa(A, u) + \mathfrak{c}(x) - \mathfrak{c}(a)$, where κ is the topological energy normalized as in [11, Proposition 3.20]; the *asymptotic labels* of a configuration are the critical points, intersection points or components of critical manifolds to which it converges at its ends. For two configurations with the same labels, $2(\kappa - \kappa') \in \mathbb{Z}$ [11, Lemma 3.15], and $\mathfrak{c}$ is the function on generators of [11, Proposition 2.53], determined up to an additive constant by this formula.

Regularity. Transversality for secondary perturbations is obtained by DFL's induction over finite-dimensional families of perturbations and the Sard–Smale theorem [11, Section 6.2]. The perturbations can be chosen invariant under ϕ_M by Lemma 5.3: holonomies along loops with vanishing lifting class span the relevant test fields. Regularity in index at most one is achieved by a finite choice, proceeding upward from the lowest index allowed by the energy bound.

Chain map. The one-dimensional spaces compactify with boundary the broken configurations of [11, Proposition 3.26]: a gauge trajectory followed by an index-zero mixed solution, or an index-zero mixed solution followed by a holomorphic strip. Sphere and disk bubbles and energy concentration are excluded in dimension at most one by the index-energy relation and monotonicity [11, Section 5.4]. Hence $\partial_S N_p = N_p \partial_G$.

Triangularity. If $\sup |f| + \sup \|\eta\|^2$ is less than the smallest positive topological energy that can occur, an index-zero solution has $\kappa \geq 0$, so $\mathfrak{c}(a) \geq \mathfrak{c}(x)$, and equality forces $\kappa = 0$. The index-zero solutions with $\kappa = 0$ converge, as $(f, \eta) \rightarrow 0$, to the constant pairs of [11, Section 3.2], which are regular by [11, Proposition 3.19]; by the implicit function theorem there is exactly one near each constant pair. This is the argument of [11, Theorem 3.29].

Independence. At fixed f , a path of choices (χ_t, η_t) has parameter derivative supported in a compact core, and defines a parametrized moduli space whose index -1 elements occur at isolated parameters and are glued using the parametrized operator $(\xi, t) \mapsto D\xi + t \partial_\lambda F$ (Theorem 5.2). Its one-dimensional part has boundary $N^1 - N^0$ and the broken configurations, giving a chain homotopy $N^1 \simeq N^0$. A change of f changes the equation on the whole gauge end and is not a compactly supported variation, so it is treated separately: insert a long gauge cylinder carrying the continuation from f_0 to f_1 at the gauge end. By Theorem 5.1 applied at the closed cut $M^\#$, for long necks the count on the glued domain is $N_{f_1} \circ \text{cont}_G(f_0, f_1)$, while the glued domain is a compactly supported deformation of the domain for f_0 , so the previous argument identifies the count with N_{f_0} up to chain homotopy. $\square$

Remark 3.7. Theorem 3.6 and its proof hold verbatim for any pair of odd compression bodies (Definition 3.2) with common boundary, possibly disconnected, product target and transverse primary Lagrangians, including boundary markings twisted by a frame transition: in particular for the pairs (C_0, C_j) with the ρ_ζ -marking and (C_1, C_2) of Section 9 (whose critical sets are described by Lemma 9.8), and for the pairs (D_i, C) and (D_0, D_1) of Section 10.1, whose matching surface is $\Sigma_h \sqcup \Sigma_g$ and target $M_h^- \times M_g$. The inputs [11, Lemma 3.15, Proposition 3.20] and the triangularity argument use only embeddedness and monotonicity of the

Lagrangians, and [12] allows a disconnected odd matching surface. The two further inputs of the triangularity step also hold: regularity of the constant pairs [11, Proposition 3.19] reduces to transversality of the pair at the corresponding intersection point, and the constants $\mathfrak{c}$ of [11, Proposition 2.53] are computed from the harmonic forms on each boundary component separately, the reversed factor contributing with the opposite sign.

Remark 3.8. The map N_p is defined by the construction of [11, Section 3.2], with two changes: the gauge complex may be regularized by an end perturbation instead of by the primary perturbations, and the secondary perturbations are invariant under ϕ_M . The first is needed in the two-handle comparison, where the primary perturbation of the middle compression body must vanish, so that [11, Lemma 2.56] is not available there. The second makes the counts independent of the lift of a generator to the determinant-one configuration space: DFL count solutions asymptotic to a fixed representative of each generator [11, p. 27], and pass to the quotient by the involution ι of [11, p. 18], which is α_M . Where both constructions apply they agree (Proposition 3.9).

Proposition 3.9 (comparison with the DFL isomorphism). *Let p be presentation data, or more generally a pair of torus compression bodies as in Remark 3.7 with collars and metrics as in Definition 3.3(iii) but an arbitrary flat metric on the torus collar, with DFL-regular primary data. Take $f = 0$; a family of almost complex structures as in [11, Section 3.1 and Lemma 2.35], equal near $\theta = \pm 1$ to DFL’s standard structure (which is J under Convention 3.1); and a ϕ_M -invariant secondary perturbation η , sufficiently small and supported, with the deformation of the almost complex structure, in a compact set away from the matching line as in [11, Proposition 3.23(i)], for which the mixed moduli spaces of expected dimension at most three are regular. Such η exist, the isomorphism N^{DFL} of [11, Theorem 3.29] is defined at these data, and*

$$N_p = N^{\text{DFL}} \otimes \mathbb{F}_2 : C_G(M, p) \rightarrow C_S(M, p)$$

as chain maps.

Proof. The complexes. DFL’s gauge complex is generated by the critical points of the perturbed functional modulo determinant-one gauge transformations and the involution ι given by the class Poincaré dual to a component of the matching surface [11, pp. 18–19, Lemmas 2.43 and 2.45]. In the framed splitting this component may be taken to be the torus, so $\iota = \alpha_M$ and DFL’s generators are ours. With $f = 0$ the gauge equations agree, so the differentials agree mod two; the strip complexes agree because the Lagrangians and almost complex structures do. Regular families equal to the standard structure near $\theta = \pm 1$ exist, since the transversality perturbations for nonconstant strips can be supported in the interior of the strip.

The equations. With $f = 0$, equation (3.1) is [11, (3.22)] on the same special quintuple; Convention 3.1 records that the two sign conventions define the same equation.

The counts. DFL’s zero-dimensional space $\mathcal{M}_\eta(\tilde{a}, x)_0$ consists of solutions asymptotic to a fixed determinant-one representative $\tilde{a}$ of a , modulo determinant-one gauge transformations. The automorphism α_M extends over the special quintuple [11, Remark 3.7], it maps solutions for η to solutions for $\alpha_M^* \eta$, and $\alpha_M^* \eta = \eta$ by invariance. So it identifies the spaces for the two representatives $\tilde{a}$, $\alpha_M \tilde{a}$, and $\mathcal{M}_\eta(\tilde{a}, x)_0$ maps bijectively onto our $\mathcal{M}^0(a, x)$, the space of solutions modulo the gauge group enlarged by ϕ_M (which acts freely by Theorem 5.4(i)). Hence the counts agree mod two.

Existence of η and DFL’s hypotheses. By Theorem 5.4(ii) the linearized operator together with the ϕ_M -invariant holonomy directions supported in a compact set away from the

matching line is onto at every solution, and the Sard–Smale theorem, applied finitely many dimensions at a time, gives η , small and compactly supported away from the matching line, with all moduli spaces of expected dimension at most three regular. These are the requirements of [11, Propositions 3.23 and 3.26], whose remaining conclusions (nonnegative energy, the unique solution in $\mathcal{M}_\eta(a, a)_0$, compactness and the boundary description) follow from smallness and compact support as in [11, Section 5]. $\square$

If for an admissible presentation p with primary data $\mathfrak{h}$ and some finite family G there are DFL-regular primary data $\mathfrak{h} + g$ with $g \in \text{span } G$ arbitrarily small (Remark 3.5), then those with g small enough are admissible, by the monotonicity of the admissible neighbourhoods (the paragraph after Proposition 11.10), and Proposition 3.9 applies there.

3.6. Energy, gradings and filtrations. We use DFL’s normalization throughout: the topological energy κ of a mixed pair is the Chern–Weil integral of the gauge part plus the symplectic area of the section, divided so that $\text{ind} = 8\kappa + \text{const}$ [11, Proposition 3.20], and the analytic energy of a solution is $8\pi^2\kappa$ plus the secondary terms [11, (3.24)]. For two configurations with the same asymptotic labels $2(\kappa - \kappa') \in \mathbb{Z}$ [11, Lemma 3.15]; we write κ_0 for the resulting coset representative. Thus the smallest change of index between homotopy classes with the same labels is 4. In the gauge-theoretic normalization used by some authors, $\kappa = \frac{1}{8\pi^2} \int \text{tr}(F_A \wedge F_A)$ plus the area term; a unit of 2κ corresponds to an $\text{SO}(3)$ instanton of energy $\int |F_A|^2 = 2\pi^2$ with the inner product $-\frac{1}{2} \text{tr}$ on $\mathfrak{su}(2)$, or $4\pi^2$ with the inner product $-\text{tr}$. We fix the inner product $-\text{tr}$ in Convention 3.1. The half-unit occurs because the framed quotient uses the full $\text{SO}(3)$ automorphism group through ϕ ; for the determinant-one group alone the minimal step would be twice as large, which is harmless for every argument below.

4. HANDLE DECOMPOSITIONS AND BUNDLES

This section contains the topology needed to write every morphism of $\mathbf{B}$ as a composite of elementary pieces. All handle decompositions are relative to the tube: handles are attached away from ν , and the tube is a product throughout.

4.1. Decompositions relative to the tube.

Proposition 4.1. *Let $(W, E, \nu): M_0 \rightarrow M_1$ be a morphism of $\mathbf{B}$. Then W has a handle decomposition relative to $M_0 \times [0, 1] \cup \nu(B^3 \times [0, 1])$ with handles of index 1, 2 and 3 only, attached in this order and away from the tube, such that every regular level is connected. All two-handles can be attached along one framed link in a single level. Consequently*

$$W = W_3 \circ W_2 \circ W_1,$$

where $W_1: M_0 \rightarrow Z_0$ consists of one-handles, $W_2: Z_0 \rightarrow Z_1$ is the trace of surgery on a framed link, and $W_3: Z_1 \rightarrow M_1$ consists of three-handles, with Z_0, Z_1 connected.

Proof. Let W° be the complement of the open tube; it is connected (proof of Lemma 2.2). Choose a Morse function $f: W^\circ \rightarrow [0, 1]$ that equals the $[0, 1]$ coordinate on a collar $S^2 \times [0, \delta] \times [0, 1]$ of the vertical boundary and near the ends, and a gradient-like vector field equal to ∂_t on the collar. The collar is a union of flow lines, so every stable and unstable manifold of an interior critical point, and every modification below, lies outside it.

Order of indices. If a k -handle is attached after a j -handle with $k \leq j$, the attaching sphere of dimension $k - 1$ and the belt sphere of dimension $3 - j$ in a three-dimensional level

satisfy $(k - 1) + (3 - j) < 3$. An isotopy supported off the collar separates them, and the critical values can be interchanged [28, Section 4], [17, Propositions 4.2.7 and 4.2.13].

No zero- or four-handles. After ordering, the union of the incoming collar with the zero- and one-handles is connected, because later handles have connected attaching regions and cannot join components, while W° is connected. Choose a spanning tree of the graph whose vertices are the zero-handles and the incoming collar and whose edges are the one-handles. A leaf zero-handle meets its tree one-handle in one point of the attaching sphere, and the pair cancels [28, Theorem 5.4]. Proceeding from the leaves removes all zero-handles. Four-handles are removed by the same argument applied to $1 - f$.

Connected levels. Attaching a one-handle to a connected level gives a connected level. Surgery on a circle in a connected three-manifold gives a connected manifold, since the complement of a circle is connected. Surgery on a two-sphere changes the number of components by 0 or +1. Starting from the connected M_0 , all levels before the three-handles are connected. Since there are no four-handles, the number of components does not decrease from the first three-handle on, and it equals one at M_1 . Hence all levels are connected.

One level of two-handles. A two-handle can be attached below the belt circle of an earlier two-handle, since $1 + 1 < 3$. Pushing all attaching circles into the level just above the one-handles gives a single framed link. $\square$

Proposition 4.2. *In the decomposition of Proposition 4.1, the trivialization of E over M_0 and the tube extends over W_1 , and the trivialization over M_1 and the tube extends over W_3 . In particular $E|_{Z_0}$ and $E|_{Z_1}$ are trivial, and (W, E, ν) is the composite of $(W_1, \mathbb{R}^3)$, $(W_2, E|_{W_2})$ with the induced frames at Z_0 , Z_1 , and $(W_3, \mathbb{R}^3)$.*

Proof. Up to homotopy equivalence relative to $M_0 \cup \nu$, the pair $(W_1, M_0 \cup \nu)$ is $(M_0 \times [0, 1], M_0 \times 0 \cup b_0 \times [0, 1])$ with relative one-cells attached along pairs of points. The trivialization extends over the product. Over a one-handle $D^1 \times D^3$ the two transition functions on the attaching balls are null-homotopic maps $D^3 \rightarrow \mathrm{SO}(3)$, and a path in the connected group $\mathrm{SO}(3)$ joining their values extends the frame across the handle. The statement for W_3 is the same with f replaced by $1 - f$. The decorated pieces glue to (W, E, ν) by construction. $\square$

The extension in Proposition 4.2 is not unique: frames on Z_0 may differ by an element of $\pi_1(\mathrm{SO}(3))$ for each one-handle. This ambiguity is absorbed by the frame-change arrows of Section 4.5, and it is invisible in the final statement, because the composite morphism does not depend on it.

4.2. One- and three-handles and adapted presentations. A Heegaard splitting $M = H_0 \cup_\Sigma H_1$ is *adapted to b* if $\Sigma \cap b(B^3)$ is the equatorial disk D and Σ is standard on a collar of $b(B^3)$. The attaching balls of a one-handle can be moved, by an isotopy of the level fixed near $b(B^3)$, into a neighbourhood of a Heegaard surface meeting $b(B^3)$ in D , because the complement of a ball in a connected three-manifold is connected. The isotopy extends to the cobordism by a diffeomorphism fixed on the tube, so the decorated morphism is unchanged. Reading W_3 from M_1 , each three-handle becomes a one-handle attached to $-M_1$, and its attaching sphere is the belt sphere of that one-handle; by Proposition 4.1 it is nonseparating in its level.

4.3. Subordinate Heegaard triples and the tube. Let $L \subset Z$ be a framed link of n components in a connected three-manifold with a ball b disjoint from L , and let $W(Z, L)$ be the trace of surgery. By [32, Lemma 4.5] there is a Heegaard triple $(\Sigma; H_0, H_1, H_2)$ of genus g subordinate to a bouquet for L , in the sense of [32, Definition 4.2]. After an isotopy in the

complement of the bouquet, Σ meets $b(B^3)$ in the equatorial disk. Then $Y_{01} = H_0 \cup H_1 = Z$, $Y_{02} = H_0 \cup H_2 = Z(L)$ and $Y_{12} = H_1 \cup H_2 \cong \#^r(S^1 \times S^2)$ with $r = g - n$. Let X be the four-manifold

$$X = (\Delta \times \Sigma) \cup (e_0 \times H_0) \cup (e_1 \times H_1) \cup (e_2 \times H_2)$$

of [32, Section 2.2], where Δ is a triangle with edges e_0, e_1, e_2 , and let $H_r = \natural^r(S^1 \times D^3)$.

Proposition 4.3. *There is a diffeomorphism $\Phi: X \cup_{Y_{12}} H_r \rightarrow W(Z, L)$ that is the identity on Z and on $Z(L)$, and that restricts to the identity from a tube $D_z \times [-1, 1] \times [-1, 1]$ onto $b(B^3) \times [-1, 1] \subset Z \times [-1, 1]$, where $D_z \subset \Sigma$ is a disk around the basepoint missing all attaching curves. The core of the tube is carried to the product arc with its product normal framing. The same holds for any other identification of the filling with H_r , after an isotopy relative to the boundary.*

Proof. Let V be a regular neighbourhood of the bouquet, a handlebody of genus n containing L . By [32, Definition 4.2], $Z \setminus \text{int } V = H_0 \cup_\Sigma \mathcal{K}$, where $\mathcal{K}$ is the compression body obtained from $\Sigma \times [0, 1]$ by attaching two-handles along the curves $\beta_{n+1}, \dots, \beta_g$; moreover $H_1 = \mathcal{K} \cup V$ and $H_2 = \mathcal{K} \cup V(L)$, where $V(L)$ is V surgered along L . Fix a bicollar $\Sigma \times [-1, 1]_s \subset H_0 \cup \mathcal{K}$ and put $B = D_z \times [-1, 1]_s$, a ball with $B \cap \Sigma = D_z$, disjoint from L .

A product model of X . For a planar hexagon Q whose sides alternate between edge sides $\partial_0, \partial_1, \partial_2$ and vertex sides, set $X(Q) = (\Sigma \times Q) \cup \bigcup_i H_i^0 \times \partial_i Q$, where H_i^0 is H_i minus a collar, with corners smoothed. A diffeomorphism of hexagons matching labelled sides, and a product near the edge sides, induces a diffeomorphism of the corresponding spaces. Take $Q' = [-1, 1]_s \times [-1, 1]_t \setminus ((0, 1] \times (-\epsilon, \epsilon))$ with $\partial_0 = \{s = -1\}$, $\partial_1 = \{s = 1, t \leq -\epsilon\}$ and $\partial_2 = \{s = 1, t \geq \epsilon\}$. Then

$$X \cong Z \times [-1, -\epsilon] \cup H_0 \times [-\epsilon, \epsilon] \cup Z(L) \times [\epsilon, 1],$$

and near the basepoint X contains the product $D_z \times Q'$.

The filling. The complement of X in $(Z \setminus V) \times [-1, 1]$ is $\mathcal{K}^{s \geq 0} \times (-\epsilon, \epsilon)$. Let Q_V be a four-ball bounding the three-sphere $V \times \{-\epsilon\} \cup \partial V \times [-\epsilon, \epsilon] \cup V(L) \times \{\epsilon\}$, a Heegaard splitting of S^3 of genus n . Put $H = (\mathcal{K}^{s \geq 0} \times [-\epsilon, \epsilon]) \cup Q_V$. Since $\mathcal{K} \times I$ is $\partial V \times I \times I$ with $g - n$ four-dimensional one-handles, $H \cong \natural^r(S^1 \times D^3)$ with $\partial H = Y_{12}$. Near the basepoint H contains the half-ball $Q_z = D_z \times [0, 1]_s \times [-\epsilon, \epsilon]$.

The diffeomorphism. Now $X \cup H = (Z \setminus V) \times [-1, 1] \cup W'_V$ with $W'_V = V \times [-1, -\epsilon] \cup Q_V \cup V(L) \times [\epsilon, 1]$, while $W(Z, L) = (Z \setminus V) \times [-1, 1] \cup W_V$ with W_V the union of $V \times [-1, 1]$ and the two-handles along $L \times \{1\}$. Both W_V and W'_V are four-balls: in W_V each two-handle cancels a distinct one-handle of $V \times I$. Their boundaries are identified by the identity on $V \times \{-1\} \cup \partial V \times [-1, 1]$ and the tautological identification of the surgered handlebody at the top. By Cerf's theorem [9] this boundary diffeomorphism extends over the balls. Define Φ to be the identity on $(Z \setminus V) \times [-1, 1]$ and this extension on W'_V .

The tube. The tube $(D_z \times Q') \cup Q_z = B \times [-1, 1]$ lies in $(Z \setminus V) \times [-1, 1]$, where Φ is the identity. Its core and its normal frame $(\partial_x, \partial_y, \partial_s)$ are carried to themselves. For another filling H'_r , a diffeomorphism $H'_r \rightarrow H$ can be chosen to be the identity on the boundary by Laudenbach–Poénaru [23]. Since Q_z is a regular neighbourhood of a boundary ball inside a collar, and collars are unique up to isotopy, the diffeomorphism may be isotoped relative to the boundary to preserve Q_z . $\square$

4.4. Bundles on a two-handle cobordism. Let $V = H^1(\Sigma; \mathbb{F}_2)$, and let $A_i \subset V$ be the image of $H^1(H_i; \mathbb{F}_2)$, a Lagrangian subspace of dimension g . Put $W_2 = X \cup_{Y_{12}} H_r$.

Proposition 4.4. *Let $\delta: V^3 \rightarrow H^2(X; \mathbb{F}_2)$ be the Mayer–Vietoris coboundary for the cover of X by $\Delta \times \Sigma$ and $e_i \times H_i$. Then $x \mapsto \delta(x, 0, 0)$ induces an isomorphism*

$$V / (A_0 + (A_1 \cap A_2)) \xrightarrow{\cong} H^2(W_2; \mathbb{F}_2).$$

The class $\delta(x, 0, 0)$ restricts to x modulo $A_0 + A_1$ on Y_{01} and to x modulo $A_0 + A_2$ on Y_{02} . In particular every $\mathrm{SO}(3)$ bundle on W_2 that is trivial over both ends has $w_2 = \delta(x, 0, 0)$ with $x \in (A_0 + A_1) \cap (A_0 + A_2)$.

Proof. The pieces $e_i \times H_i$ are pairwise disjoint and meet $\Delta \times \Sigma$ in copies of $e_i \times \Sigma$. Mayer–Vietoris reads

$$V \oplus \bigoplus_i H^1(H_i) \xrightarrow{\rho} V^3 \xrightarrow{\delta} H^2(X) \rightarrow H^2(\Delta \times \Sigma) \rightarrow H^2(\Sigma)^3,$$

with $\rho(v, u_0, u_1, u_2) = (v - u_0, v - u_1, v - u_2)$. The last map is the diagonal $\mathbb{F}_2 \rightarrow \mathbb{F}_2^3$ and is injective, so $H^2(X) = (V/A_0 \oplus V/A_1 \oplus V/A_2)/\Delta(V)$. The same cover of Y_{ij} gives $H^2(Y_{ij}) = V/(A_i + A_j)$, with restriction $[x_0, x_1, x_2] \mapsto x_i - x_j$. Since $H^1(H_r) \rightarrow H^1(Y_{12})$ is onto and $H^2(H_r) = 0$, Mayer–Vietoris for $W_2 = X \cup H_r$ identifies $H^2(W_2)$ with the kernel of $H^2(X) \rightarrow H^2(Y_{12})$. For a class in this kernel, $x_1 - x_2 \in A_1 + A_2$. Subtracting $\Delta(x_2)$ and then $\Delta(a_2)$ for suitable $a_2 \in A_2$ brings it to the form $[x, 0, 0]$. Two such classes agree exactly when $x - x' \in A_0 + (A_1 \cap A_2)$. $\square$

Example 4.5. For the (-1) -framed unknot in S^3 one may take $g = 1$, $\alpha = \lambda$, $\beta = \mu$, $\gamma = \lambda - \mu$. Then $r = 0$, $A_1 \cap A_2 = 0$ and $H^2(W_2; \mathbb{F}_2) = \mathbb{F}_2^2 / \langle \lambda \rangle \cong \mathbb{F}_2$, the class of the nontrivial bundle on $\overline{\mathbb{C}P}^2$ minus two balls. Its restrictions to both ends vanish.

Definition 4.6. Let E be a bundle on W_2 trivial over both ends, with $w_2(E) = \delta(\zeta, 0, 0)$. A *filling frame* for E consists of a trivialization of E over the central product $\Delta \times \Sigma$, over $e_1 \times H_1$, $e_2 \times H_2$ and H_r with identity transitions, and over $e_0 \times H_0$ with transition a map $\rho_\zeta: \Sigma \rightarrow \mathrm{SO}(3)$ of lifting class ζ , equal to the identity near D_z .

Filling frames exist, since $[\Sigma/D_z, \mathrm{SO}(3)] = H^1(\Sigma; \mathbb{F}_2)$ (the map $\mathrm{SO}(3) \rightarrow \mathbb{R}P^\infty$ is 3-connected). A filling frame is not a trivialization of E over $Z_0 = H_0 \cup (-H_1)$: its two handlebody frames differ along Σ by ρ_ζ . In the surface coordinates of the frame on $\Delta \times \Sigma$, the boundary conditions of the triangle are the three Lagrangians $(\tau_\zeta L(C_0), L(C_1), L(C_2))$ of Section 9.1, where τ_ζ is the symplectomorphism of the surface moduli space induced by ρ_ζ .

Global frames from the filling. Since $E|_{Z_0}$ is trivial, $\zeta \in A_0 + A_1$ (Proposition 4.4); write $\zeta = a_0 + a_1$ with $a_i \in A_i$. Choose $w_1: H_1 \rightarrow \mathrm{SO}(3)$, equal to 1 near the ball, with lifting class a_1 . The map $w_1 \rho_\zeta^{-1}|_\Sigma$ has lifting class $a_0 \in A_0$, hence extends, after a homotopy absorbed in a collar, to $w_0: H_0 \rightarrow \mathrm{SO}(3)$. Then $\phi_0^{\mathrm{gl}} = (w_0 \cdot \text{frame on } H_0) \cup (w_1 \cdot \text{frame on } H_1)$ is a trivialization of $E|_{Z_0}$; in the surface coordinates it acts by τ_{a_1} , which carries $(\tau_\zeta L(C_0), L(C_1))$ to $(L(C_0), L(C_1))$, because $\tau_{a_1} \tau_\zeta = \tau_{a_0}$ and τ_{a_i} preserves $L(C_i)$. The same construction at Z_1 , with $\zeta = a'_0 + a'_2$, gives ϕ_1^{gl} .

4.5. Frame changes. A *frame change* of an object M is a map $u: M \rightarrow \mathrm{SO}(3)$ equal to 1 near $b(B^3)$. It defines a morphism $[u]: M \rightarrow M$ of $\mathbf{B}$: the product $[0, 1] \times M$ with the vertical tube and the product bundle, whose trivialization at the outgoing end is changed by u . Frame changes compose by pointwise multiplication, $[v] \circ [u] = [vu]$, and $[u]$ depends only on the homotopy class of u relative to the ball.

Lemma 4.7. *Let $(W_2, E; \sigma_0, \sigma_1): Z_0 \rightarrow Z_1$ be the two-handle cobordism with the trivializations σ_i induced by Proposition 4.2, and put $u_i = \sigma_i(\phi_i^{\text{gl}})^{-1}$. Then in $\mathbf{B}$*

$$(4.1) \quad (W_2, E; \sigma_0, \sigma_1) = [u_1] \circ (W_2, E; \phi_0^{\text{gl}}, \phi_1^{\text{gl}}) \circ [u_0]^{-1}.$$

If W_2 has no two-handles, $E|_{W_2}$ is trivial and the morphism $(W_2, E; \sigma_0, \sigma_1)$ is the single frame change $[\sigma_1 \tilde{\sigma}_1^{-1}]$, where $\tilde{\sigma}$ extends σ_0 as a product.

Proof. Both sides have the same underlying cobordism and bundle; the trivializations at the ends agree after composing with the product cobordisms $[u_i]$, since $u_i = 1$ on the ball. $\square$

5. NECKS, FAMILIES AND INVARIANT REGULARITY

This section collects three statements used repeatedly: gluing along necks with invertible cross-sections, one-parameter families, and regularity within the class of perturbations invariant under the lifting group. They are standard in structure; we state them in the form needed and indicate the points specific to the present setting.

5.1. Pieces, ends and hypotheses. A *piece* is a pure gauge cobordism with cylindrical ends, a DFL mixed domain [11, Section 3.1], possibly with several gauge inputs, or a holomorphic polygon domain with strip-like ends on pairwise transverse Lagrangians. Ends are *closed* ($\mathbb{R}_+ \times Y$ with a nondegenerate irreducible critical set), *ordinary* (strip ends at transverse intersections) or *mixed* (DFL mixed ends, whose stationary solutions form a Morse–Bott family, treated in extended weighted spaces). Only closed and ordinary ends are joined. We assume:

- (H1) *Lattice.* For fixed labels every finite-energy configuration on a piece, a joined problem or a family fibre satisfies $\text{ind} = c + 8\kappa$ with $2\kappa \in \mathbb{Z} + 2\kappa_0$.
- (H2) *Component regularity.* Each piece is regular in index at most zero (at most one when chain identities are needed), and every nonconstant trajectory on a joined end has index at least one.
- (H3) *Supports.* All holonomy dependence regions (the unions of the loops of the holonomy perturbations), the regions where the cutoffs χ multiplying the end perturbations are nonconstant, and the parameter derivatives lie in compact regions away from the necks.
- (H4) *Energy.* On solutions, analytic energy equals topological energy plus terms bounded independently of neck lengths and family parameters.
- (H5) *Lifting group.* A finite group Φ of lifting classes vanishing on every matching surface, satisfying the ϕ -non-integral condition of [21, Definition 5.1] on every component of every cut (a surface with $w_2 \neq 0$ on which Φ vanishes), acts freely on the solution spaces; cuts are connected, so the group of the joined problem is the fibre product of the groups of the pieces [21, Proposition 5.4].

5.2. Neck gluing.

Theorem 5.1 (neck gluing). *Let pieces $X_1, \dots, X_k$ be joined along a tree of closed or ordinary ends with lengths $R = (R_e)$, under (H1)–(H5). There is R_0 such that for $\min_e R_e \geq R_0$:*

- (i) *the joined problem has no solutions of negative index;*
- (ii) *gluing is a Φ -equivariant bijection from the union, over matched intermediate labels, of products of index-zero component spaces onto the index-zero space of the joined problem, whose elements are regular;*
- (iii) *the joined count is the composite of the component counts over $\mathbb{F}_2$;*

- (iv) *the same holds with one component an index-one trajectory, modulo translation, and with one component an index -1 piece of a family together with the parametrized operator.*

Proof. The proof is the standard neck-stretching argument of [13, Chapters 3–5]; we describe its four steps and the one point that differs for the mixed equation.

Compactness. By (H4) and (H1) the energy is bounded at fixed index. By the compactness theorems for the pieces [11, Theorems 5.1–5.2], [12, Theorem 3], [13, Theorem 5.4] and Gromov compactness, a sequence with $\min_e R_e \rightarrow \infty$ converges to broken configurations on the pieces. The energy lost in the limit is nonnegative, and the lattice turns it into an index loss $4N$ with $N \geq 0$; with (H2) this excludes negative indices and, at index zero, all breaking except at the joined necks.

Uniform inverse. Each neck has an invertible cross-section: on closed necks the extended Hessian, including the gauge-fixing component, has a spectral gap because the critical points are nondegenerate and irreducible; on ordinary necks the intersections are transverse. Hence the linearized operator of the joined problem at the preglued configuration has a right inverse bounded independently of the neck lengths, by the patching argument of the proof of [13, Proposition 3.9].

Existence and local uniqueness. The preglued configuration has error decaying exponentially in $\min_e R_e$, because the component solutions converge exponentially to nondegenerate limits. The quadratic estimate holds with uniform constants: on closed necks this is [13, Theorem 4.17], and on ordinary necks, whose geometry is translation invariant, it follows from the local estimates of [12, Section 4] on unit patches. The contraction mapping principle gives a unique solution near the preglued configuration. This is the only point where the mixed equation enters, and monotonicity is used only through compactness.

Surjectivity. By compactness and energy exhaustion, every sequence of solutions with $\min_e R_e \rightarrow \infty$ is eventually exponentially close to a preglued configuration on the necks, uniformly because the cross-sections are invertible, and hence lies in the neighbourhood where the contraction gives uniqueness [13, Section 5.3]. Equivariance and the fibre-product description of the framed quotient follow from (H5). $\square$

5.3. Families.

Theorem 5.2 (families). *Let F_λ , $\lambda \in [0, 1]$, be a smooth family of data on one domain, constant near $\lambda = 0, 1$, with ends and critical data independent of λ , $\partial_\lambda F$ supported in a compact region, all scales in a compact positive range, and Φ -invariant. Assume both endpoint fibres are regular in index at most zero. After a finite relative invariant regularization vanishing near the endpoints:*

- (i) *there are no solutions of fibre index less than -1 ; the index -1 solutions occur at finitely many parameters, and at each the augmented operator $(\xi, t) \mapsto D\xi + t \partial_\lambda F$ is invertible;*
- (ii) *for each such solution and each index-one trajectory at a joinable end there is exactly one boundary branch of the parametrized index-zero space;*
- (iii) *the compactified parametrized index-zero space is a compact one-manifold whose boundary consists of the two endpoint index-zero spaces and the branches of (ii);*
- (iv) *consequently the two endpoint counts P_0, P_1 satisfy $P_0 + P_1 = dK + Kd$ over $\mathbb{F}_2$, where K counts the index -1 solutions.*

Proof. *Compactness in the family.* The compactness theorem [12, Theorem 3] is stated for the fixed collar geometry $ds^2 + d\theta^2 + g_\Sigma$ of [12, (1.1)], and only its interior estimate [12,

Lemma 4.50] is formulated for a C^k -compact family of metrics. We explain why it holds uniformly over the family. Near the seams, the metrics of the family are of the form $e^{2f_\lambda}(ds^2 + d\theta^2) + c_\lambda^2 g_\Sigma$ with f_λ in a C^k -compact set and scales c_λ in a compact positive range. The four-dimensional conformal change by e^{-2f_λ} turns this into the warped product $ds^2 + d\theta^2 + (e^{-f_\lambda} c_\lambda)^2 g_\Sigma$, whose fibre scale has logarithmic derivatives bounded uniformly in λ . On a unit patch, after rescaling (s, θ) by the fibre scale at its centre, the metric is a C^k -small perturbation of the fixed product $ds^2 + d\theta^2 + g_\Sigma$, with smallness controlled by the patch size. The anti-self-duality equation and the curvature energy are invariant under four-dimensional conformal changes, the Cauchy–Riemann equation and the energy of the section depend only on the conformal structure of the two-dimensional factor, the matching condition does not involve the metric, and the holonomy perturbations vanish on the collars. The mean-value inequality [12, Theorem 4.2] and the seam derivative bounds [12, Lemma 4.25] are proved by elliptic estimates whose constants depend continuously on the metric in C^k , so they hold on these patches with uniform constants, and so does seam regularity. Away from the seams, Uhlenbeck compactness and [12, Lemma 4.50] apply to compact families of metrics, and the perturbations depend smoothly on λ .

Consequently a sequence of solutions U_ν of fibre index at most one and bounded energy at parameters $\lambda_\nu \rightarrow \lambda$ has a subsequence converging, after gauge transformations, in C^∞ on compact subsets of the complement of a finite set Z and modulo breaking at the ends, to a solution at λ , and exactly one of the following holds: $Z = \emptyset$ and no energy is lost, in which case the convergence is C^∞ on compact sets up to breaking; or $Z \neq \emptyset$ and at least one quantum of energy is lost at Z , in which case convergence fails in C^∞ near Z and holds only in L_1^p on compact subsets of the complement. In the second case (H1) converts the energy loss into an index loss of at least four, so the limit has fibre index at most -3 , which the regularity of the family ((H2) and (i)) excludes. Hence in index at most one only the first alternative occurs. Regularity of the parametrized problem is Theorem 5.4 applied with the parameter. Gluing at an index -1 piece uses the augmented operator [13, Section 5.3, (5.6)]. The remaining statements are the standard boundary count. $\square$

5.4. Invariant regularity. We use holonomy perturbations in the sense of [11, Section 6.2, (6.13)–(6.14)]: finite sums of terms $\pi_{\text{su}(2)} \text{Hol}_\gamma(\tilde{A}) \otimes \omega$, where γ is a family of loops, $\tilde{A}$ a local unitary lift and ω a compactly supported form, with every loop having zero value under every element of Φ .

Lemma 5.3. *Let Σ be a closed oriented surface of genus at least one, $U \rightarrow \Sigma$ a rank-two Hermitian bundle of odd degree with fixed determinant connection, and $\tilde{B}$ a unitary connection inducing it whose associated $\text{PU}(2)$ connection is flat. Let $\mathcal{H} \subset \text{U}(2)$ be its based holonomy group. Then $\mathcal{H}$ acts irreducibly on $\mathbb{C}^2$, $\mathcal{H}$ contains $\text{U}(1) \cdot \text{Id}$, and the real span of $\mathcal{H}$ is $M_2(\mathbb{C})$. The same holds for holonomies defined only up to sign.*

Proof. An invariant line would carry half the curvature of the determinant, hence have half-integer degree. Small discs where the central part of the curvature is nonzero give a continuous arc of scalar holonomies, which generates $\text{U}(1)$. By Burnside’s theorem the complex span of $\mathcal{H}$ is $M_2(\mathbb{C})$, and since $i\mathcal{H} \subset \mathcal{H}$ its real span is the same. A sign change does not change a real linear span. $\square$

Theorem 5.4 (invariant regularity). *(i) Let Y be a closed connected three-manifold containing a surface Σ with $w_2(E)|_\Sigma \neq 0$, and g an automorphism of the $\text{SO}(3)$ bundle E whose lifting class vanishes on Σ . If g fixes a connection that is flat on a collar*

- of Σ , then $g = 1$. In particular Φ acts freely on critical points and on finite-energy solutions.
- (ii) At every solution of a piece, joined or family problem, the linearized operator together with the invariant holonomy directions supported in a prescribed open set is onto.
 - (iii) Generic small invariant perturbations, chosen finitely many at a time after the compactness step of each index, make all spaces used in Theorems 5.1 and 5.2 regular.

Proof. (i) A fixed automorphism commutes with the holonomy of the connection on the collar, which is irreducible by Lemma 5.3; so g is the identity on the collar, and hence everywhere by unique continuation of parallel sections along paths, Y being connected. The hypotheses cannot be dropped, as the example of T^3 with holonomy the Klein four-group shows.

(ii) Let ψ be an element of the cokernel of the linearized operator enlarged by the invariant holonomy directions. The holonomy perturbations are nonlocal, so we first project ψ off the gauge directions, using the Neumann inverse of $d_A^* d_A$; after this the adjoint equation for ψ is a local elliptic system. Pairing with the holonomy directions shows that ψ is pointwise orthogonal, on the support of their forms, to every $\pi_{\mathfrak{su}(2)} \text{Hol}_\gamma$ with γ a loop of zero lifting class based at an arbitrary point. These span $\mathfrak{su}(2)$ pointwise by Lemma 5.3, since loops in the odd surface of a collar have zero lifting class. Hence ψ vanishes on an open set, and by unique continuation [1] for the gauge part and for the section, together with the matching conditions, it vanishes everywhere.

(iii) Standard: after compactness at a given index, finitely many perturbation directions span the cokernels on a neighbourhood of the compact solution set; the Sard–Smale theorem applies to the resulting finite-dimensional family, and the smallness is preserved by induction over the finitely many indices involved. Each term is individually Φ -invariant, so no averaging is needed. $\square$

6. THE COLLAPSE THEOREM: GEOMETRY, INDEX AND COMPACTNESS

This section and the next two prove Theorem B. The present section fixes the geometry, states the theorem, proves the index-energy law at every scale and proves compactness with the loss formula. Section 7 contains the uniform linear estimates, and Section 8 the nonlinear gluing and the surjectivity of the gluing map.

6.1. Odd compression bodies. Let $(C, g_C, \mathbb{E})$ be an odd compression body (Definition 3.2), and write $n = \dim L(C)$ and $P = L(C)$; Σ denotes the union of the boundary components matched to the section and $\mathbf{M}$ their moduli space.

The two instances used in this paper are the torus compression body $C = C_1$ with $\partial C = \Sigma_h \sqcup T^2$, where the torus has a one-point moduli space, $L(C) \subset \mathbf{M}_h$ and $n = 3h - 3$; and the compression body $C_{h,g}$ from Σ_h to Σ_g used in the folded geometry of Section 10, with $L(C_{h,g}) \subset \mathbf{M}_h^- \times \mathbf{M}_g$ and $n = 3(h + g - 2)$. Conditions (C1)–(C3) hold for both: irreducibility, forced by the odd bundle on a boundary component, gives $H^0 = 0$; since $H^0(\partial C; \text{ad}) = 0$, duality gives $H^2(C; \text{ad}) \cong H^1(C, \partial C; \text{ad})^*$, which vanishes exactly when $H^1(C; \text{ad}) \rightarrow H^1(\partial C; \text{ad})$ is injective. Injectivity holds because the inclusion of the large boundary surface $\partial_+ C$ is π_1 -surjective: first cohomology with local coefficients is group cohomology of π_1 , and inflation $H^1(\pi_1 C; \text{ad}) \rightarrow H^1(\pi_1 \partial_+ C; \text{ad})$ is injective for a surjective homomorphism. The same surjectivity makes the image an embedded Lagrangian [11, Proposition 2.21].

For $v \in T_q L(C)$ let h_v be the harmonic $\text{ad } B_q$ -valued one-form on C with vanishing normal component and boundary class v ; it is unique by (C3).

Lemma 6.1 (distance to the Lagrangian). *There is c_C such that every smooth connection B on $\mathbb{E}$ with flat boundary restriction satisfies $\text{dist}([B|_{\partial C}], L(C)) \leq c_C \|F_B\|_{L^2(C)}$.*

Proof. Since the target is compact, it suffices to treat small curvature. If $\|F_{B_i}\|_{L^2} \rightarrow 0$, weak Uhlenbeck compactness on the compact three-manifold with boundary gives gauges in which $B_i \rightharpoonup B_\infty$ in $W^{1,2}$ and $B_i \rightarrow B_\infty$ in L^4 , with B_∞ flat; the relative Coulomb slice, with vanishing normal component, and the normal Hodge estimate $\|a\|_{W^{1,2}} \leq C(\|d_{B_\infty} a\| + \|d_{B_\infty}^* a\| + \|a\|)$ upgrade this to strong $W^{1,2}$ convergence. In a flat chart about B_q , fix the gauge so that $a = B - B_q$ is in Coulomb gauge with vanishing normal component and L^2 -orthogonal to the harmonic fields, adjusting q ; this is possible by the implicit function theorem because the gauge and flat directions are transverse. Then $\|a\|_{W^{1,2}} \leq C\|d_{B_q} a\| \leq C(\|F_B\| + \|a\|_{L^4}^2)$, so $\|a\|_{W^{1,2}} \leq C\|F_B\|$, and the trace inequality bounds the distance of the boundary class from $q \in L(C)$. $\square$

6.2. The collapsing family X_ε .

Definition 6.2. Let X_0 be a domain for the mixed equation of [11]: a four-manifold with a bundle, a surface S_0 with boundary arcs carrying Lagrangians (the surface on which the holomorphic part u of a configuration, called the *section* as in [11], is defined), matching seams, and ends of three kinds: gauge-theoretic cylindrical ends, strip ends, and mixed ends of Morse–Bott type. Assume that S_0 has a boundary arc I carrying $L(C)$ between two boundary punctures $z_\pm$, and that at $z_\pm$ the domain has the standard mixed end of C : in polar coordinates $z = z_+ + xe^{i\theta}$ at z_+ and $z = z_- - xe^{-i\theta}$ at z_- , $0 < x < x_0$, $0 \leq \theta \leq \pi$, with $x_0 < |I|/2$ so that the two ends do not overlap, the gauge piece $\{0 < x < x_0\} \times C$ is matched to the section along the ray $\theta = 0$, and the ray $\theta = \pi$ lies in I . The ends of X_0 other than the two mixed ends at $z_\pm$ are its *original ends*. For $\varepsilon \in (0, x_0/4)$ the domain X_ε is obtained as follows.

- (i) The section surface of X_ε is S_0 with I and $z_\pm$ restored: near I it is the upper half-plane chart with coordinate z , and I becomes part of a single matching seam $(-2, 2) \subset \mathbb{R}$. Its complex structure j_ε agrees with that of S_0 outside two half-disks of radius $C\varepsilon$ about $z_\pm$.
- (ii) The gauge piece over the seam is $(-2, 2) \times C$ with the metric $g_\varepsilon = dt^2 + \lambda_\varepsilon(t)^2 g_C$, where $\lambda_\varepsilon = \varepsilon \ell(x/\varepsilon)$ near the endpoints (x the distance to the nearer endpoint, positive outside I), $\lambda_\varepsilon = \varepsilon$ on I , and $\lambda_\varepsilon = 1$ near $t = \pm 2$; here $\ell: \mathbb{R} \rightarrow [1, \infty)$ is smooth with $\ell(s) = 1$ for $s \leq 1$ and $\ell(s) = s$ for $s \geq 2$. For $x \geq 2\varepsilon$, where $\lambda_\varepsilon = x$, this piece is the exterior gauge piece of X_0 at $z_\pm$.
- (iii) The matching seam is $t \mapsto t$ along $(-2, 2)$, the boundary of the section collar embedded by $F_\varepsilon(t, \theta) = t + \lambda_\varepsilon(t) [\sigma b_\varepsilon(t)(\cos \theta - 1) + i((1 - b_\varepsilon(t))\theta + b_\varepsilon(t) \sin \theta)]$ for $0 \leq \theta < \theta_0$, a fixed small angle, with $\sigma = \pm 1$ at the right and left endpoints and b_ε a cutoff equal to 0 for $x \leq \varepsilon$ and 1 for $x \geq 2\varepsilon$. On I it is $t + i\varepsilon\theta$; for $x \geq 2\varepsilon$ it is the polar collar $z_\pm \pm xe^{\pm i\theta}$. For θ in this range F_ε is an embedding.
- (iv) Outside a fixed neighbourhood N of I and its endpoints, X_ε coincides with X_0 .

Lemma 6.3. (i) *On the collar, $\lambda_\varepsilon^{-2} g_\varepsilon = ds^2 + g_C$ with $ds = dt/\lambda_\varepsilon$, and the section metric is conformal to $ds^2 + d\theta^2$. The matching conditions on ∂C therefore have the product form of [11, Section 3.1] at every scale, with the fixed surface metric.*
(ii) *Near each endpoint, for $2\varepsilon < x < x_0$, $s = -\log x$ and polar coordinates on S_0 , the domain is conformal to a cylinder of length $\log(x_0/2\varepsilon)$ with slice*

$$(C, g_C) \sqcup [0, \pi] \sqcup (C, \mathfrak{s}^2 g_C), \quad \mathfrak{s} = \varepsilon e^s = \varepsilon/x,$$

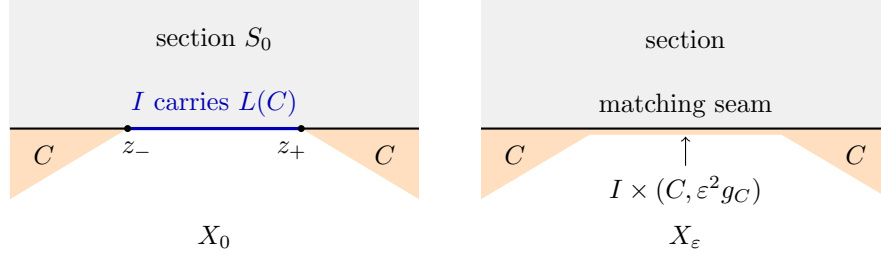


FIGURE 2. The collapsing family of Definition 6.2 near the arc I . The shaded region below the seam is the gauge piece, drawn with thickness equal to the scale of the metric on C : it is proportional to the distance to $z_\pm$ outside I , and equal to ε over I on X_ε . On X_0 (left) the arc I carries the Lagrangian boundary condition $L(C)$, and after the conformal change of Lemma 6.3 the two points $z_\pm$ become mixed ends. On X_ε (right) the whole segment is a matching seam, and the thin part is the compression rectangle.

in which the first copy is the exterior gauge piece at the point $z_\pm \pm x$, matched at $\theta = 0$, and the last is the rectangle at the point $z_\pm \mp x \in I$, matched at $\theta = \pi$.

- (iii) Along I , at distance $r \geq 2\varepsilon$ from the nearer endpoint, the conformal change by r^{-2} gives the compression the metric $dT^2 + (\varepsilon/r)^2 g_C$ with $dT = dt/r$; so $\mathfrak{s} = \varepsilon/r$ along I .

The anti-self-duality equation, the Cauchy–Riemann equation and their energies are conformally invariant.

Proof. (i) and (iii) are direct computations. For (ii), $dx^2 + x^2 g_C = x^2(ds^2 + g_C)$, and the collar embedding is the polar map. Conformal invariance holds in dimensions four and two respectively. $\square$

The gauge-fixing (Coulomb) condition is not conformally invariant. In the conformal coordinates of Lemma 6.3 it acquires a zeroth-order term $2h P_p$ on the collapsing component, where $h = \partial_s \log \mathfrak{s}$ and P_p is the projection onto the scalar component.

Definition 6.4. A configuration on X_ε ($\varepsilon \geq 0$) with given asymptotic labels is a pair (A, u) of a connection on the bundle and a section map, of class $L^2_{k, \text{loc}}$ for a fixed large k , such that: the restriction of A to each matching surface is flat and represents the value of u on the seam; u takes values in the given Lagrangians on the Lagrangian arcs (in $L(C)$ on I when $\varepsilon = 0$); and at every end (A, u) converges exponentially, at the fixed weights below the gaps, to the labelled critical data, with free limits in the critical manifolds at mixed ends. Configurations are considered up to the determinant-one gauge group enlarged by the lifting group, and carry their relative homotopy class and bundle marking.

6.3. Data and hypotheses.

Hypothesis 6.5. The data on X_ε , $\varepsilon \geq 0$, satisfy:

- (X1) *Structures near the collapse.* On N the primary perturbation of C vanishes, the section almost complex structure is the complex structure $*_\Sigma$ of Convention 3.1 with no inhomogeneous term, and all metrics are products near the seams.
- (X2) *Perturbations.* All secondary perturbations and end perturbations have outputs (the supports of their forms) and dependence regions (the unions of their loops) in a compact set disjoint from N , and $\sup \|\eta\|_{L^2} \leq b$.

- (X3) *Ends.* The original ends are nondegenerate gauge ends, transverse strip ends or mixed ends with fixed weights below their gaps. Every nonconstant autonomous trajectory on an original end that can occur has index at least one, where at mixed ends the index is taken with fixed incoming and free outgoing limit.
- (X4) *Regularity at scale zero.* For every choice of asymptotic labels, there are no solutions on X_0 of index less than zero, and the index-zero solutions form a finite set of regular points, the index being computed in the extended weighted spaces at the two new ends.
- (X5) *Index-energy law at scale zero.* For fixed labels, $\text{ind } D_V - 8\kappa(V)$ is the same constant c for all configurations V on X_0 , and $2\kappa(V) - 2\kappa_0 \in \mathbb{Z}$ for a fixed κ_0 .

The data outside N , including the neck lengths of any ordinary necks, are chosen before ε .

Hypothesis (X5) is verified for the domains used in this paper in Lemma 6.7.

6.4. Statement.

Theorem 6.6 (collapse theorem). *Let $(C, g_C, \mathbb{E})$ be an odd compression body and $\{X_\varepsilon\}_{\varepsilon \geq 0}$ the domains of Definition 6.2, with data satisfying Hypothesis 6.5. Fix asymptotic labels.*

- (i) *Index-energy law.* Every configuration U with these labels on X_ε , for any $\varepsilon > 0$, has $\text{ind } D_U = c + 8\kappa(U)$ with $2\kappa(U) - 2\kappa_0 \in \mathbb{Z}$, with the constants of (X5). Every solution of index d has analytic energy at most $\pi^2(d - c) + B$ in the normalization of Section 3.6, with $B = B(b)$ independent of ε .
- (ii) *Loss formula.* For every sequence $\varepsilon_\nu \rightarrow 0$ and solutions U_ν on X_{ε_ν} of index d , a subsequence converges in the sense of Proposition 6.15 to a solution V on X_0 and broken trajectories Γ_j on the original ends, with

$$d - \text{ind}(V) - \sum_j \text{ind}(\Gamma_j) = 4n, \quad n \in \mathbb{Z}_{\geq 0},$$

the indices of Γ_j taken as in (X3). If $d \leq 3$, $\text{ind}(V) \geq 0$ and every $\text{ind}(\Gamma_j) \geq 1$, then $n = 0$ and no energy is lost.

- (iii) *Counts.* There is $\varepsilon_0 > 0$ such that for $0 < \varepsilon < \varepsilon_0$ there are no solutions on X_ε of negative index, and gluing (the construction of Section 8) defines a bijection

$$\mathcal{G}_\varepsilon: \mathcal{M}^0(X_0) \longrightarrow \mathcal{M}^0(X_\varepsilon)$$

onto a set of regular points, preserving asymptotic labels and bundle markings.

Lemma 6.7. *Hypothesis (X5) holds when X_0 is obtained from DFL domains of compression pairs and a polygon domain with pairwise transverse Lagrangians by joining along ordinary strip ends, as in Sections 9 and 11, in the target $\mathbf{M}_h$ or in the product target $\mathbf{M}_h^- \times \mathbf{M}_g$ with labels among products of handlebody Lagrangians and the correspondence L_C .*

Proof. For a DFL domain, $\text{ind} = 8\kappa + \mathfrak{c}(x) - \mathfrak{c}(a)$ [11, Proposition 3.20], and two configurations with the same labels have $2(\kappa - \kappa') \in \mathbb{Z}$ [11, Lemma 3.15]. For a polygon domain with simply connected Lagrangians with $\pi_2 = 0$ in the simply connected $\mathbf{M}_h$, relative classes with the same corners form a torsor over $\pi_2(\mathbf{M}_h) = \mathbb{Z}$, on which the area changes by the positive generator, which is $8\pi^2 \cdot \frac{1}{2}$ in the normalization of Section 3.6, and the index by $2c_1 = 4$; this gives the law with its own constant. In the product target $\mathbf{M}_h^- \times \mathbf{M}_g$ of Section 11, with Lagrangians among products of simply connected Lagrangians with $\pi_2 = 0$ and the correspondence L_C , relative classes with the same corners form a torsor over $\pi_2(\mathbf{M}_h^- \times \mathbf{M}_g) = \mathbb{Z}^2$ modulo the image of $\pi_2(L_C) = \mathbb{Z}$, which is generated by spheres of zero area and zero Chern number and maps

to $\mathbb{Z}(1, 1)$. The quotient is $\mathbb{Z}$, and its generator changes the area by one half and the index by 4, so the law holds with its own constant. Along an ordinary strip end with transverse intersection the cross-section is invertible, so the index is additive [13, Proposition 3.9] and the topological energy is additive by locality. The constants add. $\square$

6.5. Fredholm theory and excision at a fixed scale.

Lemma 6.8 (Green identity). *Let B be a flat connection on C , and let $x = (p, b)$, $y = (\pi, c)$ be pairs of an $\text{ad } B$ -valued function and one-form on C with $b(n) = c(n) = 0$ on ∂C . Put $\mathcal{B}(p, b) = (d_B^* b, - * d_B b + d_B p)$. Then*

$$\langle \mathcal{B}x, y \rangle_{L^2(C)} - \langle x, \mathcal{B}y \rangle_{L^2(C)} = - \int_{\partial C} \langle b \wedge c \rangle.$$

If the traces $b|_{\partial C}$, $c|_{\partial C}$ are closed, the right side equals $-\omega(\tau b, \tau c)$, where τ denotes the harmonic part of the trace and ω is as in Convention 3.1. For a section variation ν on a strip $[0, 1]_r$ with J -compatible metric, $\langle -J\partial_r \nu, \mu \rangle - \langle \nu, -J\partial_r \mu \rangle = -[\omega(\nu, \mu)]_{r=0}^{r=1}$.

Proof. The scalar terms cancel because the normal components vanish. For the one-form terms, $\langle - * db, c \rangle - \langle b, - * dc \rangle = - \int_C (db \wedge c - b \wedge dc) = - \int_{\partial C} b \wedge c$ by Stokes. Exact parts of closed traces pair to zero with closed forms, and on a torus boundary component every closed trace is exact. The strip identity is integration by parts in r . $\square$

With the matching condition and Convention 3.1, the seam contributions of Lemma 6.8 from the compression and from the section cancel. This makes the mixed tangential operators symmetric, and it is the mechanism behind every uniform estimate below.

At a mixed end the stationary solutions form a Morse–Bott family, and we use *extended weighted spaces*, as in [11, Definitions 3.8–3.9]: the exponentially weighted Sobolev space of decaying fields plus the span of the stationary fields $c_e(v)$, v in the tangent space of the critical family, cut off to the end. The coefficient v is the *asymptotic value* of the field. This is the extended L^2 space of Atiyah, Patodi and Singer [4] in the weighted form.

Proposition 6.9. *For every $\varepsilon > 0$ and every configuration U on X_ε , the linearized mixed operator D_U , with its gauge-fixing component and on the extended weighted spaces at the mixed ends, is Fredholm [11, Proposition 3.18]. The same holds on X_0 , where the two new ends are mixed ends with asymptotic values in $T_{q_\pm} L(C)$. The index is locally constant along continuous paths of configurations with fixed asymptotic data.*

Proof. At fixed $\varepsilon > 0$ the region N is compact with a smooth metric, and by Lemma 6.3(i) its seam is a product mixed seam after a conformal change that does not affect the Fredholm property. The local estimates on the compact part are [11, Theorem 4.17 and Remark 4.19]; the end parametrices are those of [11, Section 4.2.1]. On X_0 the new ends are mixed ends of the same type. Paths of configurations are identified with paths of operators on fixed spaces by local matching charts, and the index is continuous. $\square$

Lemma 6.10 (excision). *Let D_1, D_2, D_3, D_4 be mixed operators, Fredholm on weighted spaces, on domains $X_1 = P \cup F$, $X_2 = P' \cup F'$, $X_3 = P \cup F'$, $X_4 = P' \cup F$, glued along a common collar $K = [-1, 1] \times \Gamma$, where Γ is a slice of the form $(\pm C) \sqcup [0, \pi] \sqcup (\pm C)$ with the boundary conditions of Definition 6.4. Suppose the operators agree on the pieces P , P' , F , F' as indicated and on K . Then $\text{ind } D_1 + \text{ind } D_2 = \text{ind } D_3 + \text{ind } D_4$.*

Proof. Let ϕ, ψ be functions of the collar coordinate with $\phi^2 + \psi^2 = 1$, $\phi = 1$ on the P and P' sides and $\psi = 1$ on the F and F' sides. The map $R(x_1, x_2) = (\phi x_1 + \psi x_2, -\psi x_1 + \phi x_2)$,

where x_1 is read on X_1 and x_2 on X_2 and the products are transplanted to X_3 and X_4 through the common collar, is an isomorphism between the domains of $D_1 \oplus D_2$ and $D_3 \oplus D_4$: multiplication by a function of the collar coordinate alone preserves the vanishing of the normal component, the closedness of the traces and the harmonic matching, because these conditions are linear and pointwise in the collar coordinate. Then $(D_3 \oplus D_4)R - R(D_1 \oplus D_2)$ is multiplication by $(\partial\phi, \partial\psi)$ on the compact collar, which is compact from the first-order domain to L^2 by the Rellich lemma on the compact domain K , whose corners, where the seams meet the cut, are those of a product of an interval with a manifold with boundary. Hence the two Fredholm operators have the same index (compare the addition property of [13, Section 3.3]). $\square$

6.6. The index–energy law at positive scale.

Lemma 6.11 (flattening). *Let U be a configuration on X_ε , $\varepsilon > 0$, and $q_* \in L(C)$. There is a continuous path of configurations, with fixed ends and fixed data outside a compact neighbourhood K_2 of the rectangle, from U to a configuration $U^\flat$ equal to the constant flat pair (B_{q_*}, q_*) on a smaller neighbourhood K_0 of the rectangle. The topological energy is constant along the path, and $\kappa(U) = \kappa_0(V)$, where V is the configuration on X_0 obtained from $U^\flat$ by removing the rectangle and completing the two new ends by (B_{q_*}, q_*) .*

Proof. The section. In a star-shaped half-disk chart about the rectangle, contract the section to its value $u(0)$ near the rectangle by $u_s = u \circ ((1 - s\chi)z)$, with χ a cutoff; then move $u(0)$ to q_* along a path c in the connected moduli space, $u_{1+s} = c(s\rho)$, with ρ a smaller cutoff. The seam carries no Lagrangian condition at positive scale, so the path may leave $L(C)$.

Flat surface traces. Lift the seam values to actual flat connections on the matching surfaces by path lifting in the principal bundle of flat connections over the moduli space, with structure group the determinant-one gauge group modulo its centre, keeping the lift constant where the base path is constant. At the end the trace over a central time interval differs from B_{q_*} by a path h in the gauge group. The correction needed is a path of surface gauge transformations over the interval, equal to h in the middle and to the identity at both ends; it exists because the determinant-one gauge group of the surface is connected. No loop of gauge transformations arises, so the correction introduces no lifting class. Do the same on the other boundary components.

Connections. Let β_s be the resulting boundary connections. Choose an auxiliary connection equal to B_{q_*} near the rectangle and to A outside K_2 , interpolate affinely, and add a collar extension of the difference between β_s and the trace of the interpolant. The result has the prescribed flat traces at every parameter and equals the flat product near the rectangle at the end.

Energy. The variation of the topological energy along a compactly supported path is a boundary integral; on the matching surfaces the gauge term cancels the section term by the flat matching condition and Convention 3.1, exact parts integrate to zero, and on a torus face the variations are exact because $H^1(T^2; \text{ad}) = 0$. So the energy is constant. The removed region is flat with constant section, and the added tails have zero energy density, so $\kappa(U^\flat) = \kappa_0(V)$. $\square$

Let $F_\varepsilon(q)$ denote the linearized operator of the *flat cap at scale ε* : the section is a half-disk bounded by the seam over $(-2, 2)$, the gauge piece is $(-2, 2) \times C$ with the metric g_ε , the two lateral interfaces are completed by the cylinder with slice $(-C) \sqcup [0, \pi] \sqcup C$, and the configuration is the constant flat pair (B_q, q) . Let $F_0(q)$ be the zero-scale operator: the section is $S_0 = \mathbb{H} \setminus \{\pm 1\}$ with boundary arcs $\ell_- = (-\infty, -1)$, $I = (-1, 1)$, $\ell_+ = (1, \infty)$, the

gauge pieces are $Z_{\pm} = \ell_{\pm} \times C$ matched along $\ell_{\pm}$, the condition $L(C)$ is imposed on I , and there are three ends: the mixed ends $E_{\pm 1}$ with slice $C \sqcup [0, \pi]$ and the end E_{∞} with slice $C \sqcup [0, \pi] \sqcup C$. All operators act on extended weighted spaces, with one asymptotic value at each end, represented by the cut-off stationary fields $c_e(v) = (0, h_v, v)$ at $E_{\pm 1}$ and (h_v, v, h_v) at E_{∞} , for $v \in T_q L(C)$.

Proposition 6.12. *For every $q \in L(C)$, $\ker F_{\varepsilon}(q) \cong \ker F_0(q) \cong T_q L(C)$ and $\operatorname{coker} F_{\varepsilon}(q) = \operatorname{coker} F_0(q) = 0$. In particular $\operatorname{ind} F_{\varepsilon}(q) = \operatorname{ind} F_0(q) = n$.*

Proof. We give the proof for $F_0(q)$. The positive-scale case is the same with one end and no Lagrangian arc; there the kernel lies in $T_q L(C)$ because the constant section value must be the boundary class of a closed form on all of C , namely the harmonic field h_v . Use the weight δ below all end gaps and below the Coulomb gap of the Neumann Laplacian on C .

Energy identity. For (ζ, ν) , (ζ', ν') in the domain,

$$\int_{Z_+ \sqcup Z_-} \operatorname{tr}(d\zeta \wedge d\zeta') + \int_{S_0} \omega(d\nu \wedge d\nu') = 0.$$

Indeed both integrands are exact because B_q is flat. On $\ell_{\pm} \times T^2$ the boundary terms vanish since closed traces are exact; on $\ell_{\pm} \times \Sigma$ the gauge term is $\pm \int \omega(\nu, \partial_t \nu') dt$ and cancels the seam term of the section by Convention 3.1; on I the term vanishes because $T_q L(C)$ is Lagrangian; at the ends the cross terms between a stationary field and a decaying remainder form the Green pairing of the self-adjoint end operator and vanish, and the remaining terms tend to zero.

Kernel. For (ζ, ν) in the kernel the identity gives $\|d\zeta\|^2 + \|d\nu\|^2 = 0$, so $\nu \equiv v$ with $v \in T_q L(C)$ by the condition on I . On $Z_+ = \mathbb{R} \times C$ write $\zeta = b_t + \varphi_t dt$; $d\zeta = 0$ gives $d_C b_t = 0$ and $\partial_t b_t = d_C \varphi_t$, so the class of b_t is constant with boundary value v , and $b_t = h_v + d_C \psi_t$ with $\partial_t \psi_t = \varphi_t$. Hence $\zeta = \pi^* h_v + d\psi$. The pullback $\pi^* h_v$ is closed, coclosed and has vanishing normal component for any metric conformal to $dt^2 + g_C$ by a factor depending on t , so the gauge-fixing equation gives $\Delta \psi = 0$ with Neumann condition and decay, and $\psi = 0$ by the maximum principle. So the kernel is $\{(\pi^* h_v, v)\} \cong T_q L(C)$.

Adjoint domain. Let $w = (\xi, \mu, \nu)$ in the dual weighted space annihilate the range. Testing with $(d\chi, 0)$, χ with Neumann condition, and using that the Neumann Laplacian is onto on the weighted spaces, gives $\xi = 0$. Interior tests give $d\mu = 0$ with μ self-dual and $d\nu = 0$ for the section $(0, 1)$ -form. Near $\ell \times \partial C$ write $\mu = dt \wedge m + *_C m$. Boundary tests give $m(n) = 0$, closed restrictions of m to Σ and to T^2 , and, by Convention 3.1, $[m]_{\Sigma} = v(\partial_t)$ on the seams; on I , $v(\tau) \in T_q L(C)$, since a Lagrangian subspace is its own symplectic complement. These are the conditions of the original domain under $\mu \mapsto m$.

Decay. On each end w solves the translation-invariant adjoint equation, so $w = w_0 + O(e^{-\gamma \rho})$ with w_0 in the kernel of the end operator. Pair with $F_0(q)(\beta_e(\cdot - R)c_e(v)) = \beta'_e(\cdot - R)c_e(v)$, where β_e is a cutoff; letting $R \rightarrow \infty$ removes the contribution of the decaying part and gives $\langle c_e(v), w_0 \rangle = 0$ for all v , so $w_0 = 0$ and w decays at a rate exceeding δ .

Vanishing. The section surface S_0 is contractible and ν is closed and decaying, so $\nu = d\nu'$ with ν' converging at each puncture; normalize $\nu'(1) = 0$; on I , $\nu' \in T_q L(C)$. On Z_+ put $\zeta'_+(t) = \pi^* h_{\nu'(1)} + \int_{-\infty}^t m dt'$; then $d\zeta'_+ = \mu$, $\zeta'_+(n) = 0$, its traces are closed, and by the seam condition its boundary class is $\nu'(t)$. At the other end ζ'_+ converges to $h_{\nu'(1)} + M$ with $M = \int m$ closed, coclosed and normal-free, hence harmonic with boundary class $\nu'(\infty) - \nu'(1)$; so $\nu'(\infty) \in T_q L(C)$ and $\zeta'_+ \rightarrow h_{\nu'(\infty)}$. The same on Z_- . Thus (ζ', ν') lies in the domain, with asymptotic values at the three ends, and $0 = \langle F_0(q)(\zeta', \nu'), w \rangle = \|\mu\|^2 + \|\nu\|^2$.

The primitive $\zeta'_\pm$ need not decay, since M may be nonzero; it is asymptotically constant, and this is exactly what the extended spaces allow. The index is therefore $\dim \ker = n$. $\square$

Proof of Theorem 6.6(i). Let U be a configuration on X_ε . By Lemma 6.11 and Proposition 6.9, $\text{ind } D_U = \text{ind } D_{U^b}$ and $\kappa(U) = \kappa_0(V)$. The four operators D_{U^b} , $F_0(q_*)$, D_V and $F_\varepsilon(q_*)$ agree on a fixed collar around the cut with slice $(-C) \sqcup [0, \pi] \sqcup C$: on the U^b and F_ε side it is the rectangle at positive scale near its lateral interfaces, and on the V and F_0 side the corresponding part of the new mixed ends, and all four configurations are the flat pair there. Lemma 6.10 gives $\text{ind } D_{U^b} + \text{ind } F_0(q_*) = \text{ind } D_V + \text{ind } F_\varepsilon(q_*)$. By Proposition 6.12 the cap indices are equal, so $\text{ind } D_U = \text{ind } D_V = c + 8\kappa_0(V) = c + 8\kappa(U)$ by (X5).

For a solution, the analytic energy equals $8\pi^2\kappa$ plus $2\|\eta\|^2 \leq 2b^2$ plus the term $\int \chi' f$ of the end perturbations, bounded by $\sup |f|$; the first two terms are [11, (3.24)], and the term $\int \chi' f$ is estimated as in the proof of Theorem 3.6. The bound follows. $\square$

6.7. Regularity along the collapsing seam. We use the small-energy estimates for the mixed equation in the fixed product geometry of [12, (1.1)]: there are $e_* > 0$ and constants C_k such that a solution on a unit half-ball centred on the seam, with energy $e \leq e_*$ on the ball of twice the radius, satisfies $|\nabla^k F_A| + |\nabla^k du| \leq C_k \sqrt{e}$ on the unit half-ball, up to the seam [12, Theorem 4.2 and Lemma 4.25]; away from the seam these are the interior estimates of [12, Lemma 4.50].

The following lemma is the source of all uniform local control near the collapse. For a seam point b (on I , or on an endpoint neck) let $\mathfrak{s}_b$ be the local compression scale ($\mathfrak{s}_b = \varepsilon$ on I , and the value of Lemma 6.3(ii) on the necks), and let $\mathcal{H}_r(b)$ be the half-disk of radius r in the section about b together with the gauge region over $[b-r, b+r]$. Let $E(b, r)$ be the energy of a solution on $\mathcal{H}_r(b)$, and $r_0 > 0$ fixed.

In the following lemma $\Sigma \subset \partial C$ is the union of the boundary components matched to the section and $\mathbf{M}$ their moduli space, the remaining components carrying the flatness condition, as in Section 7.1. The topological energy $\mathcal{Q}$ of a configuration is that of Section 3.6: the Chern–Weil integral of the gauge part plus the symplectic area of the section, normalized so that it equals the analytic energy of a solution of the unperturbed equations.

Lemma 6.13 (quantization of capped configurations). *There is $\hbar > 0$, depending only on the odd compression body, with the following property. Let $x \in P$, and let (A, u) be a continuous, piecewise smooth configuration on $(\mathbb{R} \times C) \cup \overline{\mathbb{H}}$: A is a connection on $\mathbb{R} \times \mathbb{E}$ whose restriction to every slice $\{t\} \times \partial C$ is flat, and $u: \overline{\mathbb{H}} \rightarrow \mathbf{M}$ satisfies $u(t) = [A|_{\{t\} \times \Sigma}]$ for $t \in \mathbb{R} = \partial \mathbb{H}$. Suppose that (A, u) equals the constant pair (B_x, x) outside a compact set. Then $\mathcal{Q}(A, u) \in \hbar \mathbb{Z}$.*

Proof. (1) *Invariance.* $\mathcal{Q}$ is unchanged by gauge transformations of $\mathbb{R} \times \mathbb{E}$, whose action on the integrands is pointwise, and by homotopies through configurations of the same kind that are constant outside a fixed compact set. For the latter, Bianchi’s identity and the closedness of ω reduce the variation of $\mathcal{Q}$ to boundary integrals over $\mathbb{R} \times \partial C$ and $\partial \mathbb{H}$, which cancel by the matching condition and Convention 3.1 exactly as in the energy paragraph of the proof of Lemma 6.11; on the torus faces the variations are exact. The slice contributions at the interfaces of the pieces cancel by continuity.

(2) *Removing the section.* Choose T_0 such that $(A, u) = (B_x, x)$ for $|t| \geq T_0$ and outside the half-disk of radius T_0 . For $s \in [0, 1]$ put $u_s(z) = u(z + isT_0)$ and $q_s(t) = u_s(t)$ for $t \in \mathbb{R}$; then $u_1 \equiv x$ and $q_s(\pm T_0) = x$. By (C1) the flat connections on ∂C form a principal bundle over $\mathbf{M}$ whose structure group is the determinant-one gauge group of ∂C modulo its centre, as in the proof of Lemma 6.11. The map $(s, t) \mapsto q_s(t)$ on $[0, 1] \times [-T_0, T_0]$ has the prescribed lift $\alpha(t) = A|_{\{t\} \times \partial C}$ on $\{0\} \times [-T_0, T_0]$ and the constant lift $B_x|_{\partial C}$ on

$[0, 1] \times \{\pm T_0\}$; the union of these three edges is a deformation retract of the square, so the lift extends to flat connections $\alpha_s(t)$ with $[\alpha_s(t)|_\Sigma] = q_s(t)$. With a bounded extension operator $\mathcal{E}$ from tangential one-forms on ∂C to one-forms on C , supported in a collar, the connections $A_s = B(t) + \mathcal{E}(\alpha_s(t) - \alpha(t)) + \Psi dt$, where $A = B(t) + \Psi dt$, together with u_s , form a homotopy as in (1). Hence $\mathcal{Q}(A, u) = \mathcal{Q}(A_1, x)$, and the latter is the Chern–Weil term alone.

(3) *Reduction to a gauge transformation.* The restrictions of A_1 to the slices of $\mathbb{R} \times \partial C$ are flat of class x , so $\alpha_1(t) = h(t)^*(B_x|_{\partial C})$ for a path h in the determinant-one gauge group of ∂C with $h(t) = 1$ for $t \leq -T_0$; for $t \geq T_0$ the path is constant and central, the stabilizer of an irreducible flat connection being the centre. Restriction to ∂C is a fibration of gauge groups, by the homotopy extension property of $(C, \partial C)$, so h lifts to a path $\hat{h}$ of determinant-one gauge transformations of C with $\hat{h}(t) = 1$ for $t \leq -T_0$ and $\hat{h}$ constant for $t \geq T_0$. The gauge transformation $(t, y) \mapsto \hat{h}(t)(y)^{-1}$ carries A_1 to a connection A_2 whose restriction to every slice of $\mathbb{R} \times \partial C$ is $B_x|_{\partial C}$, equal to B_x for $t \leq -T_0$ and to g^*B_x for $t \geq T_0$, where $g = \hat{h}(T_0)^{-1}$ is central on ∂C . By (1), $\mathcal{Q}(A, u) = \mathcal{Q}(A_2, x)$.

(4) *The period.* Let $\mathcal{G}_\partial$ be the group of determinant-one gauge transformations of C that are central on ∂C . For $g \in \mathcal{G}_\partial$ let $\Xi(g) = \mathcal{Q}(A', x)$ for any connection A' as in (3), with slice restrictions $B_x|_{\partial C}$, equal to B_x for $t \ll 0$ and to g^*B_x for $t \gg 0$. This does not depend on A' : two choices are joined by the affine homotopy, which keeps the slice restrictions fixed, and (1) applies. Concatenation and gauge invariance give $\Xi(g_1 g_2) = \Xi(g_1) + \Xi(g_2)$, and Ξ is constant on path components: a path $\lambda \mapsto g_\lambda$ in $\mathcal{G}_\partial$ gives the connection $g_{\lambda(t)}^* B_x$, which is flat on every slice, so its Chern–Weil integrand $F \wedge F$ vanishes, and whose slice restrictions are $B_x|_{\partial C}$. Thus Ξ is a homomorphism from $\pi_0(\mathcal{G}_\partial)$ to $\mathbb{R}$. The subgroup of transformations equal to 1 on ∂C has finite index, and its group of components is $H^3(C, \partial C; \mathbb{Z}) \cong \mathbb{Z}$ by obstruction theory, the fibre S^3 of the group bundle being 2-connected. Hence the image of Ξ is a torsion-free abelian group with a cyclic subgroup of finite index, that is $\hbar_0 \mathbb{Z}$ for some $\hbar_0 \geq 0$. The lattice does not depend on x : for g equal to 1 on ∂C we may, after a homotopy in $\mathcal{G}_\partial$, take $g = 1$ near ∂C , and then $\Xi(g)$ is the Chern–Simons difference of g^*B_x and B_x , whose boundary term vanishes, so $\Xi(g)$ is a fixed multiple of the degree of g in $H^3(C, \partial C; \mathbb{Z})$; the values of Ξ therefore lie in a fixed lattice $(c_\Xi/N)\mathbb{Z}$, N the index of the subgroup, and Ξ , depending continuously on $x \in P$, is locally constant in x . Take $\hbar = \hbar_0$ if $\hbar_0 > 0$ and $\hbar = 1$ otherwise; it is independent of x . $\square$

Lemma 6.14 (seam regularity). *There are $e_0, \alpha, C > 0$, independent of ε and of b , such that for every solution $U = (A, u)$ on X_ε and every seam point b with $E(b, r_0) \leq e_0$:*

- (i) $E(b, r) \leq C E(b, r_0) (r/r_0)^\alpha$ for $\mathfrak{s}_b \leq r \leq r_0$;
- (ii) $|du(z)| \leq C \sqrt{E(b, r_0)} (y + \mathfrak{s}_b)^{\alpha/2-1}$ at distance $y \leq r_0/2$ from the seam, and $d(u(z), u(z')) \leq C \sqrt{E(b, r_0)} |z - z'|^{\alpha/2}$ on $\mathcal{H}_{r_0/2}(b)$;
- (iii) $\text{dist}(u(b), L(C)) \leq C \sqrt{E(b, r_0)} \mathfrak{s}_b^{\alpha/2}$; in the rescaled coordinates on the patch of size $\mathfrak{s}_b$ about b the curvature, the section derivative and all their covariant derivatives are bounded by $C_k \sqrt{E(b, r_0)} \mathfrak{s}_b^{\alpha/2}$; and in gauges on the compression slices over $[b - \mathfrak{s}_b, b + \mathfrak{s}_b]$, $\|B(t) - B_{q(t)}\|_{C^k(C)} \leq C_k \sqrt{E(b, r_0)} \mathfrak{s}_b^{\alpha/2}$ with $q(t) \in L(C)$.

Proof. Rescaled geometry. In the coordinate $w \mapsto (w - b)/\mathfrak{s}_b$ (on I) or Φ_b with $\Phi'_b = 1/\mathfrak{s}$ (on the necks), the collapsing compression and the section have exactly the product mixed geometry of [12, (1.1)] with the fixed surface metric, and unit patches correspond to regions of size $C\mathfrak{s}_b$. Energy is conformally invariant. So the small-energy estimates of [12, Theorem 4.2 and Lemma 4.25], which are stated for that fixed product geometry, and [12, Lemma 4.50],

which allows a C^k -compact family of metrics, apply on every unit patch with constants independent of ε and b : on a patch of energy at most e_0 the curvature, the section derivative and all their covariant derivatives are bounded by $C\sqrt{e}$ in the rescaled coordinates.

Decay above the local scale. Fix r with $\mathfrak{s}_b \leq r \leq r_0$ and write $E(b, r) = G(b, r) + S(b, r)$, with G the gauge part and S the section part; the gauge region of $\mathcal{H}_r(b)$ is $[b - r, b + r] \times C$, and its section region is a half-disk D_r with diameter $[b - r, b + r]$ on the seam.

The endpoint slices. The gauge energy density on the compression is $\mathfrak{s}^{-1} \|F_B\|_{L^2(C)}^2$ per unit time, so the connections $B_\pm$ on the slices $t = b \pm r$ satisfy $\|F_{B_\pm}\|^2 \leq C\lambda_\varepsilon(b \pm r)\partial_r G(b, r) \leq Cr\partial_r G(b, r)$, where λ_ε is the compression scale. When this is small, the proof of Lemma 6.1 gives gauge transformations $g_\pm$ and $q_\pm \in P$ with $B_\pm = g_\pm^*(B_{q_\pm} + \beta_\pm)$, where $\beta_\pm$ lies in the Neumann slice orthogonal to the harmonic fields and $\|\beta_\pm\|_{H^1(C)} \leq C\|F_{B_\pm}\|$; the representatives B_q, q near x , are taken from a smooth local section of flat connections through B_x . The image γ of the semicircle $\partial D_r \setminus [b - r, b + r]$ satisfies $\text{len}(\gamma)^2 \leq Cr\partial_r S(b, r)$ by Cauchy–Schwarz. Its endpoints $u(b \pm r) = [B_\pm|_\Sigma]$ lie within $C\|\beta_\pm\|_{H^1}$ of $q_\pm$, so with $x = q_-$,

$$d(q_+, x) \leq C\delta_b, \quad \delta_b = \text{len}(\gamma) + \|\beta_-\|_{H^1} + \|\beta_+\|_{H^1}.$$

Paths with flat boundary. Let B be a connection on C with flat boundary restriction α , and $\|B - B_x\|_{H^1(C)} = \delta$ small; write $\alpha_x = B_x|_{\partial C}$. Every component of ∂C carries only irreducible flat connections, so the surface Coulomb slice, in which the gauge parameter lies in $H^{3/2}(\partial C)$ (an algebra acting on $H^{1/2}$), and the harmonic chart of the flat connections give $\alpha = \exp(\zeta)^*(\alpha_x + h_\nu + \sigma(\nu))$, where h_ν is the harmonic representative of $\nu \in H^1(\Sigma; \text{ad } \alpha_x)$, σ the quadratic correction of the chart, and $\|\zeta\|_{H^{3/2}} + |\nu| \leq C\delta$. Put $\alpha^\lambda = \exp(\lambda\zeta)^*(\alpha_x + h_{\lambda\nu} + \sigma(\lambda\nu))$ and

$$B^\lambda = B_x + \lambda(B - B_x) + \mathcal{E}(\alpha^\lambda - \alpha_x - \lambda(\alpha - \alpha_x)), \quad 0 \leq \lambda \leq 1,$$

with $\mathcal{E}: H^{1/2}(\partial C) \rightarrow H^1(C)$ a bounded tangential extension. Then $B^0 = B_x$, $B^1 = B$, every B^λ has the flat boundary restriction α^λ , $\sup_\lambda (\|B^\lambda - B_x\|_{H^1} + \|\partial_\lambda B^\lambda\|_{H^1}) \leq C\delta$, and the boundary classes of B^λ form a path of length at most $C\delta$. By the Sobolev embedding $H^1 \subset L^4$ on C , $\|F_{B^\lambda}\|_{L^2} \leq C\delta$, so the path, regarded as a connection on $[0, 1] \times C$ with zero temporal component, has Chern–Weil integral at most $C \int_0^1 \|\partial_\lambda B^\lambda\| \|F_{B^\lambda}\| d\lambda \leq C\delta^2$ in absolute value. A temporal component Ψ does not change this integral, since $\int_C \langle dB \Psi \wedge F_B \rangle = \int_{\partial C} \langle \Psi, F_B|_{\partial C} \rangle = 0$ by Bianchi’s identity and the flatness of the boundary restriction; so Ψ can be chosen to match the temporal component of the solution at the junctions. We use this with $B = B_{q_-} + \beta_-$, for which $\delta = \|\beta_-\|_{H^1}$, and with $B = B_{q_+} + \beta_+$, for which $\delta \leq C\delta_b$, the second path traversed backwards.

The cap. Composing with a fixed representative of the component class in $H^1(C; \mathbb{Z}/2)$, which acts on flat connections of C and preserves $L(C)$, we may take $g_\pm$ of determinant one. The determinant-one gauge group of C is connected, because $H^3(C; \mathbb{Z}) = 0$ for $\partial C \neq \emptyset$ and the fibre S^3 is 2-connected; choose a path $g(t)$, $t \in [b - r, b + r]$, from g_- to g_+ . The gauge transformation $(t, y) \mapsto g(t)(y)^{-1}$ changes neither the topological energy of the solution on $[b - r, b + r] \times C$ nor its boundary classes, keeps all its slice restrictions flat, the torus faces included, and makes its end slices equal to $B_{q_\pm} + \beta_\pm$. Concatenate in t : B_x on $(-\infty, b - r - 1]$, the path from B_x to $B_{q_-} + \beta_-$ on $[b - r - 1, b - r]$, the transformed solution on $[b - r, b + r]$, the path from $B_{q_+} + \beta_+$ to B_x on $[b + r, b + r + 1]$, and B_x on $[b + r + 1, \infty)$. On the section keep u on D_r ; on the half-annulus $\{r \leq |z - b| \leq r + 1, \text{Im } z \geq 0\}$, whose boundary loop consists of γ , of the boundary class paths of the two flat-boundary paths on the seam intervals $[b - r - 1, b - r]$ and $[b + r, b + r + 1]$, and of the outer semicircle mapped to x , use the geodesic

cone from x over this loop; and put $u = x$ elsewhere. The loop has length at most $C\delta_b$ and lies in the ball of radius $C\delta_b$ about x , so the cone has area at most $C\delta_b^2$. The result is a configuration as in Lemma 6.13. Its topological energy is $E(b, r) + Q_{\text{cap}}$, since the solution satisfies the unperturbed equations near the seam by Hypothesis 6.5(X1)–(X2), and

$$|Q_{\text{cap}}| \leq C\delta_b^2 \leq C(\text{len}(\gamma)^2 + \|\beta_-\|_{H^1}^2 + \|\beta_+\|_{H^1}^2) \leq Cr\partial_r E(b, r).$$

By Lemma 6.13, $E(b, r) + Q_{\text{cap}} \in \hbar\mathbb{Z}$.

Conclusion. Take $e_0 < \hbar/2$. There is $c_2 > 0$ such that $r\partial_r E(b, r) \leq c_2$ ensures the smallness required above and $|Q_{\text{cap}}| < \hbar/2$; then $E(b, r) + Q_{\text{cap}}$ lies in $\hbar\mathbb{Z} \cap (-\hbar, \hbar) = \{0\}$, and $E(b, r) \leq |Q_{\text{cap}}| \leq Cr\partial_r E(b, r)$. If $r\partial_r E(b, r) > c_2$, then $E(b, r) \leq e_0 \leq (e_0/c_2)r\partial_r E(b, r)$. Hence $E \leq C_0 r\partial_r E$ for $\mathfrak{s}_b \leq r \leq r_0$, and integration gives (i) with $\alpha = 1/C_0$.

Derivative and Hölder bounds. At distance $y \leq c\mathfrak{s}_b$ from the seam apply the rescaled estimates on patches of radius $C\mathfrak{s}_b$, whose energy is $O(E\mathfrak{s}_b^\alpha)$ by (i); for $y \geq \mathfrak{s}_b$ apply the interior estimate for holomorphic curves on disks of radius $y/2$, whose energy is $O(Ey^\alpha)$. Since $\alpha > 0$ the bound is integrable in y , which gives the Hölder bound. For (iii), part (i) gives energy at most $CE(b, r_0)\mathfrak{s}_b^\alpha$ on the patch of size $R\mathfrak{s}_b$ about b , for a fixed R ; the rescaled small-energy estimates on that patch give the bounds with the factor $\mathfrak{s}_b^{\alpha/2}$, and the spatial Neumann slice about the nearest flat connection, with Lemma 6.1, gives the gauge statement. $\square$

6.8. Compactness and the loss formula.

Proposition 6.15. *Let $\varepsilon_\nu \rightarrow 0$ and $U_\nu = (A_\nu, u_\nu)$ be solutions on X_{ε_ν} with bounded energy. After passing to a subsequence and gauge transformations there are a finite set $Z \subset X_0$ and a solution V on X_0 such that:*

- (i) $U_\nu \rightarrow V$ in C^∞ on compact subsets of $X_0 \setminus (Z \cup I \cup \{z_\pm\})$; here X_{ε_ν} agrees with X_0 on every compact subset of the complement of $I \cup \{z_\pm\}$ for large ν ;
- (ii) on compact subsets of $\overline{S_0}$ away from Z and $z_\pm$, including points of I , the sections converge in C^0 and are uniformly Hölder, and the limit takes values in $L(C)$ on I ;
- (iii) V is smooth up to I , the points of Z are removable, and V extends to a finite-energy solution on X_0 whose tails at the new ends converge exponentially to points $q_\pm \in L(C)$;
- (iv) the original ends break into finitely many trajectories Γ_j .

Proof. (i) On the complement of N and on the part of N at distance at least $2\varepsilon_\nu$ from I the geometry is fixed, and the perturbations have compact support away from the seams, so this is the compactness theorem for the mixed equation [11, Theorem 5.2], [12, Theorem 3 and Theorem 4.44], together with Uhlenbeck and Gromov compactness, with a finite concentration set Z outside of which energy densities converge. Holonomy perturbation terms converge because the loop families avoid the finite concentration set for almost every output point.

(ii) Energy concentration along I occurs at finitely many points, since each concentration point carries at least a fixed quantum: at points of I the threshold e_0 of Lemma 6.14, elsewhere the thresholds of the fixed-geometry theory. Away from them, $E(b, r_0) \leq e_0$ for ν large and r_0 small, and Lemma 6.14 gives uniform Hölder bounds for u_ν up to the seam and $\text{dist}(u_\nu(b), L(C)) \leq C\varepsilon_\nu^{\alpha/2} \rightarrow 0$ on I .

(iii) By (ii), near a point of I away from Z the limit is a continuous J -holomorphic map, J being the complex structure $*_\Sigma$ there by (X1), with boundary values in the embedded Lagrangian $L(C)$ and finite energy. In local coordinates in which $L(C)$ is flat it is of class $W^{1,p}$ for $p > 2$, and boundary regularity for pseudo-holomorphic curves with Lagrangian boundary conditions [27, Theorem B.4.1] gives smoothness. Removal of singularities at interior, seam

and boundary points of Z follows from [12, Theorem 4.44], Uhlenbeck's theorem and [27, Theorem 4.1.2]. At the new ends the energy of V is finite; after rescaling the strip width to that of [11] and passing to temporal gauge, [11, Proposition 5.29] gives exponential decay.

(iv) The chains on the original ends are extracted as in [11, Section 5.4]. $\square$

Proof of Theorem 6.6(ii). The analytic energy of U_ν is $8\pi^2\kappa(U_\nu)$ plus terms that converge along the subsequence. Energy is lower semicontinuous under the convergence of Proposition 6.15, including at the points of Z and along the necks of the original ends, where the action is constant on each connected critical family. Hence the topological energy lost is nonnegative. Glue V and the Γ_j into a configuration with the labels of U_ν ; at a mixed end the incoming limit of Γ_j equals the outgoing limit of the previous piece, and with the convention of (X3) (fixed incoming, free outgoing limit) the index is additive. The topological energy is additive by locality. By part (i) applied on X_{ε_ν} and (X5) on X_0 , both with the constant c , the lost topological energy lies in $\frac{1}{2}\mathbb{Z}_{\geq 0}$ and the lost index is eight times it. This is $d - \text{ind}(V) - \sum_j \text{ind}(\Gamma_j) = 4n$ with $n \geq 0$. If $d \leq 3$, $\text{ind}(V) \geq 0$ and $\text{ind}(\Gamma_j) \geq 1$, then $4n \leq 3$ forces $n = 0$; energy is then preserved and there is no concentration. $\square$

At index zero, Hypothesis 6.5(X3)–(X4) and the loss formula give $\text{ind}(V) = 0$, no broken trajectories, $n = 0$, and energy exhaustion: $Z = \emptyset$, and every compact part of the rescaled endpoint regions and of the rescaled neighbourhood of I carries energy tending to zero.

Corollary 6.16 (convergence on the middle compression). *In this situation fix $r_0 > 0$ so small that the energy of V on every half-disk $\mathcal{H}_{r_0}(b)$ with $b \in I$ is less than $e_0/2$. Then for ν large, uniformly for $b \in I$ at distance at least r_0 from the endpoints and on rescaled unit patches of size ε_ν about b : the curvature and all its derivatives tend to zero; in the spatial Neumann slice the compression connection tends in C^N to the flat lift $B_{q_V(t)}$ of the boundary values q_V of V on I ; and the rescaled time derivatives of the harmonic coefficient v (the component along the harmonic fields h_v , $v \in T_q L(C)$) tend to zero.*

Proof. By energy exhaustion $E(b, r_0) \leq e_0$ for ν large, uniformly in b . By Lemma 6.14(iii) the rescaled quantities are bounded by $C_N \sqrt{e_0} \varepsilon_\nu^{\alpha/2} \rightarrow 0$, and $q(t)$ is within $C\varepsilon_\nu^{\alpha/2}$ of the section value, which converges uniformly to q_V by Proposition 6.15(ii). Near the endpoints, on the necks where $x/\varepsilon_\nu \rightarrow \infty$, Lemma 8.5 gives smallness $C\sqrt{e_0}(\varepsilon_\nu/x)^{\alpha/2} \rightarrow 0$, and on the rescaled cores the convergence to zero comes from the convergence of the rescaled transitions to the flat pairs at $q_\pm$ (Section 8.3). $\square$

7. THE COLLAPSE THEOREM: UNIFORM LINEAR ESTIMATES

This section proves the linear estimates on which the gluing and surjectivity arguments of Section 8 rest. The basic estimates are on a model: a family of compressions at varying small scale coupled to a holomorphic strip along the whole of one boundary line (Sections 7.1–7.4). The endpoint transitions are treated in Section 7.5, and the patched inverse on the actual domain in Section 7.7.

7.1. The model collapsing strip. Let $(C, g_C, \mathbb{E})$ be an odd compression body with a boundary component $\Sigma \subset \partial C$ carrying the section, and let $P = L(C)$ (for simplicity of notation we suppress the other boundary components, whose moduli spaces are points or enter as fixed factors). Fix a compact Lagrangian K in the moduli space $\mathbf{M}$ of Σ . Fix:

- a path $q: \mathbb{R} \rightarrow P$ with $Q = \sup |\dot{q}| < \infty$, flat connections $B(t)$ representing $q(t)$, and $\Psi(t)$ with $\partial_t B - d_B \Psi = h_q$;

- a map $u: S = \mathbb{R} \times [0, 1] \rightarrow \mathbf{M}$ with $u(t, 0) = q(t)$, $u(t, 1) \in K$, bounded in C^1 , and a compatible $J(t, r)$;
- a scale $\mathfrak{s}: \mathbb{R} \rightarrow (0, \bar{\mathfrak{s}}]$ with $|\mathfrak{s}'/\mathfrak{s}| \leq H$; write $h = \mathfrak{s}'/\mathfrak{s}$.

For $U = (p, b)$, a pair of $\text{ad } B(t)$ -valued function and one-form on C depending on t , put $\mathbf{D}_t = \partial_t + [\Psi, \cdot]$, $\mathcal{B}_t(p, b) = (d_B^* b, - * d_B b + d_B p)$ and $P_p(p, b) = (p, 0)$, and define

$$D_{\mathfrak{s}}(U, \nu) = (\mathbf{D}_t U - \mathfrak{s}^{-1} \mathcal{B}_t U + 2h P_p U, D_u \nu),$$

where D_u is the linearized Cauchy–Riemann operator. The domain $\mathcal{D}$ consists of (U, ν) with $b(n) = 0$ on ∂C , closed traces on every component of ∂C , harmonic part of $b|_{\Sigma}$ equal to $\nu(t, 0)$, and $\nu(t, 1) \in T_{u(t, 1)} K$. Norms:

$$\begin{aligned} \|(U, \nu)\|_X^2 &= \int (\mathfrak{s} \|U\|^2 + \mathfrak{s} \|\mathbf{D}_t U\|^2 + \mathfrak{s}^{-1} \|\mathcal{B}_t U\|^2) dt + \|\nu\|_{H^1(S)}^2, \\ \|(f, g)\|_Y^2 &= \int \mathfrak{s} \|f\|^2 dt + \|g\|_{L^2(S)}^2, \end{aligned}$$

all norms on C in the fixed metric g_C . This operator is the linearization of the mixed equation at a configuration that is flat on the compression, in the conformal coordinates of Lemma 6.3; the weight $\mathfrak{s}$ is the Jacobian of the conformal change.

7.2. Hodge theory on a compression body. Let $\mathbb{H}_t = \{h_v : v \in T_{q(t)} P\}$ be the space of absolute harmonic fields and Π_t the $L^2(C)$ -projection onto $\{(0, h) : h \in \mathbb{H}_t\}$. Constants depend only on the odd compression body $(C, g_C, \mathbb{E})$ and on P , which is compact.

Lemma 7.1. *For $U = (p, b)$ in the static domain:*

- (i) $\|\mathcal{B}_t U\|^2 = \|d^* b\|^2 + \|db\|^2 + \|dp\|^2$;
- (ii) $\|U - \Pi_t U\|_{H^1(C)} \leq c \|\mathcal{B}_t U\|$;
- (iii) $\ker \mathcal{B}_t = \{(0, h_v) : v \in T_q P\}$.

Proof. (i) $\langle *db, dp \rangle = \int_C dp \wedge db = \int_{\partial C} p d(b|_{\partial C}) = 0$ since the traces are closed and B is flat. (ii) is the Gaffney–Friedrichs inequality for one-forms with vanishing normal component [39, Corollary 2.1.6 and Theorem 2.1.7], whose kernel is the space of absolute harmonic fields, together with the Poincaré inequality for p , which holds because $H^0(C; \text{ad}) = 0$. (iii) follows from (i), (ii) and (C3). $\square$

We call the harmonic fields h_v the *slow directions*: they span the kernel of the fast operator $\mathfrak{s}^{-1} \mathcal{B}_t$, and their only dynamics is inherited from the boundary value of the strip. The corresponding coefficient is the *slow coefficient*.

Lemma 7.2 (slow directions are tied to the trace). *For U in the static domain write $\Pi_t U = (0, h_w)$ and let ν_0 be the harmonic part of $b|_{\Sigma}$. Then*

$$|\nu_0 - w| \leq c \|\mathcal{B}_t U\|, \quad \|U\|_{L^2} + \|b\|_{H^1(C)} \leq c (\|\mathcal{B}_t U\| + |\nu_0|).$$

Proof. The harmonic part of $h_w|_{\Sigma}$ is w , and the harmonic projection of the trace of $b - h_w$ is bounded by $c \|b - h_w\|_{H^1} \leq c \|\mathcal{B}_t U\|$ by Lemma 7.1(ii). The second inequality follows, since $|w| \leq |\nu_0| + c \|\mathcal{B}_t U\|$ and $\|h_w\| \leq c |w|$. $\square$

Lemma 7.3 (the seam term). *For $U(t)$ smooth in t in the static domains, write $b|_{\Sigma} = \beta + d\xi$ with β harmonic, and $b|_T = d\xi_T$ on the other boundary components. Then*

$$\int_{\partial C} \langle b \wedge \mathbf{D}_t b \rangle = \int_{\Sigma} \langle \beta \wedge \mathbf{D}_t \beta \rangle + R, \quad |R| \leq c |\dot{q}| \|b\|_{H^1(C)}^2.$$

Proof. On Σ , $D_t(d\xi) = d(D_t\xi) + [a_\Sigma, \xi]$ with $a = h\dot{q}$. Expanding, the terms $\int \beta \wedge d(\cdot)$ and $\int d\xi \wedge dD\xi$ vanish by Stokes and flatness, and $\int d\xi \wedge D\beta = \int \xi[a_\Sigma \wedge \beta]$. The remaining terms are bounded by $c|a|_\infty(|\beta| + \|d\xi\|)\|\xi\|$, and $\|\xi\| \leq c\|d\xi\|$ because $H^0(\Sigma; \text{ad}) = 0$. The other components have $H^0 = H^1 = 0$ and are treated the same way. The trace $H^1(C) \rightarrow L^2(\partial C)$ is bounded, and $|a| \leq c|\dot{q}|$. $\square$

7.3. The Gårding estimate. Fix symplectic frames along u as follows: transport a symplectic frame along $t \mapsto u(t, 0)$ and then along each segment $r \mapsto u(t, r)$ by a symplectic connection, and write $\nu = \Phi\xi$ with $\xi: S \rightarrow \mathbb{R}^{2m}$, $\xi_0 = \xi(\cdot, 0)$. Then $D_u\nu = \Phi(\partial_t\xi + \tilde{J}\partial_r\xi + \tilde{S}\xi)$ with $\tilde{J} = \Phi^{-1}J\Phi$ and $\tilde{S}$ bounded by the C^1 norms of u and J ; the derivatives of $\tilde{J}$ contribute zeroth-order terms of the same size. Let $e_i(t)$ be the harmonic representatives of $\Phi(t, 0)\varepsilon_i$. Under Convention 3.1, $\omega(e_i, e_j) = \omega_0(\varepsilon_i, \varepsilon_j)$, and

$$(7.1) \quad \int_\Sigma \langle \beta \wedge D_t\beta \rangle = \omega_0(\xi_0, \partial_t\xi_0) + R', \quad |R'| \leq c(K_u + Q)|\xi_0|^2,$$

where K_u bounds the connection coefficients of the frame. At $r = 1$, $|\omega_0(\xi_1, \partial_t\xi_1)| \leq cK_u|\xi_1|^2$ because K is Lagrangian.

Proposition 7.4 (uniform Gårding estimate). *There are $\mathfrak{s}_1 > 0$ and C , depending on the odd compression body, on Q , H and on the C^1 norms of u and J , but not on $\mathfrak{s}$, $\inf \mathfrak{s}$ or the support of z , such that for $\sup \mathfrak{s} \leq \mathfrak{s}_1$ and every smooth compactly supported $z = (U, \nu) \in \mathcal{D}$,*

$$\|z\|_X \leq C(\|D_{\mathfrak{s}}z\|_Y + \|\nu\|_{L^2(S)}).$$

The same holds for the formal adjoint in the measure $\mathfrak{s} dt dV_C$, whose gauge-fixing component is $-D_t - \mathfrak{s}^{-1}\mathcal{B}_t - h + 2hP_p$, on the same domain.

Proof. The energy identity. Let $\mathcal{A}_t = [D_t, \mathcal{B}_t]$, a zeroth-order operator with $|\mathcal{A}_t U| \leq c|\dot{q}||U|$. By Lemma 6.8 (without the closedness of the traces),

$$\frac{d}{dt} \langle U, \mathcal{B}U \rangle = 2\langle DU, \mathcal{B}U \rangle + \langle U, \mathcal{A}U \rangle + \int_{\partial C} \langle b \wedge D_t b \rangle.$$

For a gauge-fixing component $f = DU - \mathfrak{s}^{-1}\mathcal{B}U + ZU$ with a pointwise endomorphism $|Z_t| \leq z_0 H$ (here $Z = 2hP_p$), expanding $\mathfrak{s}|f|^2$ and integrating the total derivative gives

$$(7.2) \quad \int \mathfrak{s}\|f\|^2 = \int (\mathfrak{s}\|DU + ZU\|^2 + \mathfrak{s}^{-1}\|\mathcal{B}U\|^2) - 2\int \langle ZU, \mathcal{B}U \rangle + \int \langle U, \mathcal{A}U \rangle + \iint_{\partial C} \langle b \wedge D_t b \rangle.$$

No factor $\mathfrak{s}$ multiplies the last three terms, and no derivative of $\mathfrak{s}$ appears: the weight $\mathfrak{s}$ cancels the $\mathfrak{s}^{-1}$ of $\mathcal{B}$ before any integration by parts in t .

For the strip, with $D_0 = \partial_t + \tilde{J}\partial_r$,

$$(7.3) \quad \|D_0\xi\|^2 = \|\partial_t\xi\|^2 + \|\partial_r\xi\|^2 + \int \omega_0(\xi, \partial_t\xi)|_{r=1} dt - \int \omega_0(\xi, \partial_t\xi)|_{r=0} dt,$$

from $-2\omega_0(\partial_t\xi, \partial_r\xi) = \partial_r\omega_0(\xi, \partial_t\xi) - \partial_t\omega_0(\xi, \partial_r\xi)$, and $\|D_u\nu\|^2 \geq \|D_0\xi\|^2 - 2\|\tilde{S}\|\|D_0\xi\|\|\xi\|$.

Cancellation. By Lemma 7.3 and (7.1), the last term of (7.2) is $\int \omega_0(\xi_0, \partial_t\xi_0) dt$ up to an error bounded by $c(Q + K_u)(\int \|\mathcal{B}U\|^2 + \int |\xi_0|^2 + |\xi_1|^2)$. Adding (7.2) and (7.3), the two terms $\pm \int \omega_0(\xi_0, \partial_t\xi_0)$ cancel identically. This is the only point where Convention 3.1 is used.

Absorption. Put $\mathcal{E} = \int \|\mathcal{B}U\|^2 \leq \mathfrak{s}_1 \int \mathfrak{s}^{-1}\|\mathcal{B}U\|^2$ and $\mathcal{J} = \int (|\xi_0|^2 + |\xi_1|^2)$. By Lemma 7.2, $\|U\|$ and the slow coefficient w are bounded pointwise in t by $\|\mathcal{B}U\| + |\nu_0|$, so $|\langle U, \mathcal{A}U \rangle| \leq cQ(\|\mathcal{B}U\|^2 + |\nu_0|^2)$, $2|\langle ZU, \mathcal{B}U \rangle| \leq \frac{1}{4}\mathfrak{s}^{-1}\|\mathcal{B}U\|^2 + cH^2\mathfrak{s}(\|\mathcal{B}U\|^2 + |\nu_0|^2)$, and $\mathfrak{s}\|U\|^2 \leq c\mathfrak{s}_1(\|\mathcal{B}U\|^2 + |\nu_0|^2)$. The one-dimensional trace inequality $|g(0)|^2 \leq$

$\theta \|g'\|^2 + (1 + \theta^{-1}) \|g\|^2$ on $[0, 1]$ bounds $\mathcal{J}$ by a small multiple of $\|\partial_r \xi\|^2$ plus a multiple of $\|\xi\|^2$. Choosing $\mathfrak{s}_1 = c(1 + Q + H)^{-2}$ gives

$$\|z\|_X^2 \leq C(\|D_{\mathfrak{s}} z\|_Y^2 + \|\nu\|_{L^2(S)}^2), \quad C = c(1 + Q + K_u + H)^4.$$

Adjoint. By Lemma 6.8 and the bilinear form of (7.3), $\langle D_{\mathfrak{s}} z, z' \rangle_Y - \langle z, D_{\mathfrak{s}}^* z' \rangle_Y = \int \omega(\nu_0, \mu_0) - \int \omega(\nu_0, \mu_0) = 0$ for $z, z' \in \mathcal{D}$, the term at $r = 1$ vanishing because K is Lagrangian. So the adjoint has the same domain. The reflection $t \mapsto -t$ carries $D_{\mathfrak{s}}^*$ to an operator of the same form with $Z = \mp(2hP_p - h)$, and the same argument applies. $\square$

Remark 7.5. With the opposite sign in Convention 3.1 the estimate fails already for constant $\mathfrak{s}$ and constant q . For a unit vector v and $\zeta = v - iJv$, the pair $b = \chi \operatorname{Re}(e^{i\tau t + \mathfrak{s}\tau(s-1)} \zeta)$ near the seam, where s is the normal coordinate of C with the boundary at $s = 1$, $\nu = \chi \operatorname{Re}(e^{i\tau t - \tau r} \zeta)$ on the strip, with cutoffs χ , lies in the domain and decays away from the seam on both sides. Its ratio $\|z\|_X^2 / (\|D_{\mathfrak{s}} z\|_Y^2 + \|\nu\|^2)$ grows like τ^2 . This is a boundary layer at the seam, where a $\bar{\partial}$ -operator would be glued to a ∂ -operator; it is unrelated to the slow directions.

Remark 7.6 (why no logarithm appears). The slow component $w(t)$ is never estimated through the gauge part. A bound through $\int \mathfrak{s}(|\dot{w}|^2 + |w|^2)$ would require a weighted Hardy inequality, which for $\mathfrak{s} = \bar{\mathfrak{s}}e^{-H|t|}$ costs a factor $\log(1/\inf \mathfrak{s})$. Here w is controlled pointwise in t by ν_0 through Lemma 7.2, and ν_0 by the unweighted strip norm.

7.4. The uniform inverse at a regular limit. Assume now that u is J -holomorphic, converges exponentially to points $x_{\pm} \in P \cap K$, and that J is independent of t near the ends. Assume $D_u: \{\nu \in H^1 : \nu(\cdot, 0) \in TP, \nu(\cdot, 1) \in TK\} \rightarrow L^2$ is onto with kernel $\mathcal{K}$ of dimension d , and fix d smooth functionals Γ of ν with compact support such that $\Gamma|_{\mathcal{K}}$ is an isomorphism.

Lemma 7.7 (ends). *There are $T_0, \mathfrak{s}_1, C$ such that for $\sup \mathfrak{s} \leq \mathfrak{s}_1$ and $z \in \mathcal{D}$ supported in $\{|t| \geq T_0\}$, $\|z\|_X \leq C\|D_{\mathfrak{s}} z\|_Y$, and similarly for $D_{\mathfrak{s}}^*$.*

Proof. For $|t| \geq T_0$ the Lagrangian frames $\Phi^{-1}TP$ and $\Phi^{-1}TK$ at $r = 0, 1$ are close to transverse limits, so $\sup_r |\xi(t, r)| \leq C(\|\partial_r \xi(t, \cdot)\| + \|\mathcal{B}_t U\|)$, using Lemma 7.2 for the distance of ν_0 from TP . Squaring and integrating over t ,

$$\mathcal{J} + \|\xi\|^2 \leq \varsigma \|\partial_r \xi\|^2 + C \int \|\mathcal{B}_t U\|^2 dt$$

with ς as small as we wish once T_0 is large. The proof of Proposition 7.4 on the support of z gives $\|z\|_X^2 \leq C(\|D_{\mathfrak{s}} z\|_Y^2 + \mathcal{J} + \|\xi\|^2) + o(1)\|z\|_X^2$, the error coefficients involving Q and K_u tending to zero as $T_0 \rightarrow \infty$; inserting the displayed inequality absorbs $\mathcal{J}$ and $\|\xi\|^2$, and no L^2 remainder is left. $\square$

Combining Proposition 7.4 on $[-T_0 - 1, T_0 + 1]$ with Lemma 7.7 by a partition of unity in t gives

$$(7.4) \quad \|z\|_X \leq C(\|D_{\mathfrak{s}} z\|_Y + \|\nu\|_{L^2([-T_1, T_1] \times [0, 1])}),$$

with $C, T_1, \mathfrak{s}_1$ independent of $\mathfrak{s}$, and the same for $D_{\mathfrak{s}}^*$.

Proposition 7.8 (uniform inverse without logarithmic loss). *There are $\mathfrak{s}_1, C$ such that for every scale with $\sup \mathfrak{s} \leq \mathfrak{s}_1$ and $|\mathfrak{s}'/\mathfrak{s}| \leq H$, the operator $(D_{\mathfrak{s}}, \Gamma): X \rightarrow Y \oplus \mathbb{R}^d$ is an isomorphism with $\|(D_{\mathfrak{s}}, \Gamma)^{-1}\| \leq C$. Moreover, if $J_{\mathfrak{s}} k = ((0, h_{k(t, 0)}), k)$ for $k \in \mathcal{K}$ and $G_{\mathfrak{s}} k$ is the kernel element of $D_{\mathfrak{s}}$ with $\Gamma G_{\mathfrak{s}} k = \Gamma k$, then $\|G_{\mathfrak{s}} k - J_{\mathfrak{s}} k\|_X \leq C\sqrt{\sup \mathfrak{s}} \|k\|$.*

Proof. Injectivity estimate. Suppose $\|z_j\|_X = 1$ and $\|D_{\mathfrak{s}_j} z_j\|_Y + |\Gamma z_j| \rightarrow 0$ with $\sup \mathfrak{s}_j \rightarrow 0$. By (7.4), $\liminf \|\nu_j\|_{L^2([-T_1, T_1] \times [0, 1])} > 0$. A subsequence of ν_j converges weakly in H^1 and strongly in L^2_{loc} to ν_∞ with $D_u \nu_\infty = 0$, $\nu_\infty(\cdot, 1) \in TK$, and $\nu_\infty(\cdot, 0) \in TP$, the last because $\int \text{dist}(\nu_j(t, 0), TP)^2 \leq c \int \|\mathcal{B}U_j\|^2 \leq c \sup \mathfrak{s}_j$ by Lemma 7.2. Thus $\nu_\infty \in \mathcal{K}$ and $\Gamma \nu_\infty = 0$, so $\nu_\infty = 0$, a contradiction. For the adjoint the limit satisfies $D_u^* \mu_\infty = 0$ with Lagrangian boundary values, hence is orthogonal to the image of D_u , which is everything. The scales $\mathfrak{s}_j$ play no role in the limit, which lives on the strip alone: no convergent subsequence of scales and no lower bound on $\mathfrak{s}_j$ is used.

Surjectivity. The same contradiction argument, applied to $D_{\mathfrak{s}}^*$ (whose limit object is an element of the kernel of D_u^* with Lagrangian boundary values, hence zero), gives $\|y\|_X \leq C\|D_{\mathfrak{s}}^* y\|_Y$. Coercivity of $\langle D_{\mathfrak{s}}^* y, D_{\mathfrak{s}}^* y' \rangle$ and the Lax–Milgram lemma then give, for $f \in Y$, a weak solution $z = D_{\mathfrak{s}}^* y$ of $D_{\mathfrak{s}} z = f$ with $\|z\|_Y \leq C\|f\|_Y$; write $R_{\mathfrak{s}} f = z$, a right inverse of $D_{\mathfrak{s}}$. At fixed $\mathfrak{s}$ weak solutions are strong: the slice operator $(\mathfrak{s}^{-1} \mathcal{B}_t, -J\partial_r)$ with the domain of Section 7.1 is symmetric by Lemma 6.8 and the seam cancellation. It is self-adjoint: an element of the domain of the adjoint is a weak solution, with L^2 right-hand side, of the same mixed boundary problem (vanishing normal component, closed traces, harmonic matching), and the fixed-geometry regularity for exactly this mixed boundary condition [11, Theorem 4.17 and Remark 4.19] places it in the domain. That theorem is stated for the four-dimensional cylinder with one compression body; it applies to a slice operator after extending an element w of the adjoint domain as $\beta(t)w$ on an interval times the slice, to slices with two gauge branches after localizing near each seam and reflecting $\theta \mapsto \pi - \theta$ at the collapsing seam, and the finite-rank harmonic matching does not affect regularity. After a bounded trivialization of the t -dependent domain, regularity of weak solutions is that of [34]. Applying (7.4) to cut-off versions of z gives $z \in X$ with $\|z\|_X \leq C\|f\|_Y$.

Kernel. $J_{\mathfrak{s}} k$ lies in the domain, its strip part is in $\ker D_u$, and $D_{\mathfrak{s}} J_{\mathfrak{s}} k = ((0, D_t h_{k(t, 0)}), 0)$ has Y -norm at most $C\sqrt{\sup \mathfrak{s}}\|k\|$ since k decays exponentially. Hence $z_k = J_{\mathfrak{s}} k - R_{\mathfrak{s}} D_{\mathfrak{s}} J_{\mathfrak{s}} k$ lies in the kernel, Γ is onto on the kernel for small $\sup \mathfrak{s}$, and $(D_{\mathfrak{s}}, \Gamma)$ is an isomorphism. The estimate for $G_{\mathfrak{s}} k - J_{\mathfrak{s}} k$ follows from the uniform bound. $\square$

7.5. The endpoint transition. In this subsection we work with the scale $\mathfrak{s}$ of the collapsing branch of a single slice; on the endpoint neck of X_ε it is $\mathfrak{s} = \varepsilon e^s$. Near an endpoint, with $L = \log(1/\varepsilon)$, in the coordinate $\rho = L - s = \log(1/\mathfrak{s})$ (with s as in Lemma 6.3(ii)), the slice at scale $\mathfrak{s} = e^{-\rho}$ is $(C, g_C) \sqcup [0, \pi] \sqcup (C, \mathfrak{s}^2 g_C)$. On the slice Hilbert space $\mathcal{H}_{\mathfrak{s}}$ with norm $\|z\|_{\mathfrak{s}}^2 = \|U_f\|^2 + \mathfrak{s}\|U_c\|^2 + \|\nu\|^2$, the tangential operator is

$$A_{\mathfrak{s}}(U_f, U_c, \nu) = (\mathcal{B}U_f, \mathfrak{s}^{-1} \mathcal{B}U_c, -J\partial_\theta \nu)$$

on the $\mathfrak{s}$ -independent domain $\mathcal{W}$ given by vanishing normal components, closed traces and harmonic matching at both ends of $[0, \pi]$. The quadratic form $\langle z, A_{\mathfrak{s}} z \rangle_{\mathfrak{s}}$ does not depend on $\mathfrak{s}$. By Lemma 6.8 and Convention 3.1 the seam terms cancel, and $A_{\mathfrak{s}}$ is symmetric. It is self-adjoint by the same argument: with $I_{\mathfrak{s}} = \text{diag}(1, \mathfrak{s}, 1)$, an element of the domain of the adjoint of $I_{\mathfrak{s}} A_{\mathfrak{s}}$ is a weak solution of the mixed boundary problem, hence in the domain by [11, Theorem 4.17 and Remark 4.19].

Proposition 7.9. *There are $\gamma > 0$ and $\mathfrak{s}_0 > 0$ such that for $\mathfrak{s} \leq \mathfrak{s}_0$ and $q \in P$, $\ker A_{\mathfrak{s}} = \{k_v = (h_v, h_v, v) : v \in T_q P\}$ and the spectrum of $A_{\mathfrak{s}}$ meets $(-\gamma, \gamma)$ only in 0. Moreover $\|U_c - h_{\Pi\nu(\pi)}\|_{H^1} \leq C\sqrt{\mathfrak{s}}\|A_{\mathfrak{s}} z\|_{\mathfrak{s}}$, where Π is the projection to $T_q P$.*

Proof. Kernel. If $A_{\mathfrak{s}}z = 0$, then $\mathcal{B}U_f = \mathcal{B}U_c = 0$, so by Lemma 7.1 $U_f = (0, h_{v_f})$ and $U_c = (0, h_{v_c})$; and ν is constant, so the matching gives $v_f = \nu = v_c =: v$, with $v \in T_qP$ because h_v is defined only there.

The complementary operator. The orthogonal complement of T_qP in T_qM is JT_qP , since T_qP is Lagrangian and J is compatible, so $1 - \Pi$ projects onto JT_qP . Let $\mathcal{W}_q^\perp$ be the space of $U = (p, b)$ on C with $b(n) = 0$, closed traces, and harmonic part of $b|_\Sigma$ in JT_qP , and let $\mathcal{B}^\perp$ be $\mathcal{B}$ with domain $\mathcal{W}_q^\perp$. By Lemma 6.8 it is symmetric, JT_qP being Lagrangian. It satisfies

$$(7.5) \quad \|U\|_{H^1(C)} \leq c_1 \|\mathcal{B}U\| \quad (U \in \mathcal{W}_q^\perp) :$$

if $w \in T_qP$ is the bulk harmonic coefficient of U and $\nu_0 \in JT_qP$ the harmonic part of its trace, Lemma 7.2 gives $|w| \leq |\nu_0 - w| \leq c \|\mathcal{B}U\|$, because $\nu_0 \perp w$, and Lemma 7.1(ii) gives the rest. Hence $\mathcal{B}^\perp$ is injective with closed range; it is self-adjoint by the local regularity used above for $A_{\mathfrak{s}}$, so it is an isomorphism $\mathcal{W}_q^\perp \rightarrow L^2(C)$ with inverse bounded by c_1 into H^1 (this is the complementary problem of (7.12)), and $\mathcal{B}^\perp - \mu$ is invertible with $\|(\mathcal{B}^\perp - \mu)^{-1}\|_{L^2 \rightarrow H^1} \leq 2c_1$ for $|\mu| \leq 1/(2c_1)$.

Elimination of the collapsed component. Let A_0 be the tangential operator of the mixed end, on $\mathcal{H}_0 = L^2(C) \oplus L^2([0, \pi])$ with the domain $\mathcal{W}_0$ of pairs (U_f, ν) with $b_f(n) = 0$, closed traces, harmonic part of $b_f|_\Sigma$ equal to $\nu(0)$, and $\nu(\pi) \in T_qP$. It is self-adjoint and its spectrum meets $(-2\gamma, 2\gamma)$ only in 0, uniformly in q [11, (4.20)–(4.24)]; its kernel is $\{(h_v, v) : v \in T_qP\}$. Let $|\lambda| = \gamma$, $\mathfrak{s}\gamma \leq 1/(2c_1)$, and consider $(A_{\mathfrak{s}} - \lambda)z = f$ with $z \in \mathcal{W}$ and $f = (f_f, f_c, f_\nu) \in \mathcal{H}_{\mathfrak{s}}$. Put $\tilde{v} = \Pi\nu(\pi)$ and $\tilde{U}_c = U_c - h_{\tilde{v}}$. The harmonic part of the trace of U_c is $\nu(\pi)$, so that of $\tilde{U}_c$ is $(1 - \Pi)\nu(\pi) \in JT_qP$ and $\tilde{U}_c \in \mathcal{W}_q^\perp$; since $\mathcal{B}h_{\tilde{v}} = 0$, the collapsing equation $\mathcal{B}U_c - \mathfrak{s}\lambda U_c = \mathfrak{s}f_c$ becomes $(\mathcal{B}^\perp - \mathfrak{s}\lambda)\tilde{U}_c = \mathfrak{s}f_c + \mathfrak{s}\lambda h_{\tilde{v}}$. Thus

$$\tilde{U}_c = \mathfrak{s}(\mathcal{B}^\perp - \mathfrak{s}\lambda)^{-1}(f_c + \lambda h_{\tilde{v}}), \quad \|\tilde{U}_c\|_{H^1} \leq C\mathfrak{s}(\|f_c\| + |\tilde{v}|),$$

and the matching condition at $\theta = \pi$ becomes the perturbed Lagrangian condition

$$(7.6) \quad (1 - \Pi)\nu(\pi) = \mathfrak{s}\lambda \Lambda_{\mathfrak{s}\lambda} \tilde{v} + \mathfrak{s}m_f, \quad \Lambda_\mu v = \tau((\mathcal{B}^\perp - \mu)^{-1}h_v), \quad m_f = \tau((\mathcal{B}^\perp - \mathfrak{s}\lambda)^{-1}f_c),$$

where τ is the harmonic part of the trace on Σ . Here $\Lambda_\mu : T_qP \rightarrow JT_qP$ is bounded uniformly, and $|\mathfrak{s}m_f| \leq C\mathfrak{s}\|f_c\| \leq C\sqrt{\mathfrak{s}}\|f\|_{\mathfrak{s}}$ because $\mathfrak{s}\|f_c\|^2 \leq \|f\|_{\mathfrak{s}}^2$. Conversely, a pair (U_f, ν) satisfying the fixed-branch and section equations and (7.6), completed by $U_c = h_{\tilde{v}} + \tilde{U}_c$, is a solution.

Reduction to A_0 and the Neumann series. Fix a cutoff χ_π on $[0, \pi]$, equal to 1 near π and to 0 on $[0, \pi/2]$, and for (U_f, ν) satisfying (7.6) put $\tilde{\nu} = \nu - \chi_\pi(\mathfrak{s}\lambda \Lambda_{\mathfrak{s}\lambda} \tilde{v} + \mathfrak{s}m_f)$. Then $\tilde{\nu}(\pi) = \tilde{v} \in T_qP$ and $\Pi\tilde{\nu}(\pi) = \tilde{v}$, so $y = (U_f, \tilde{\nu}) \in \mathcal{W}_0$, and the fixed-branch and section equations become

$$(A_0 - \lambda)y = (f_f, f_\nu) + \Theta_\lambda y + \mathfrak{s}F_\lambda f_c,$$

with $\Theta_\lambda y = (0, (J\partial_\theta + \lambda)(\chi_\pi \mathfrak{s}\lambda \Lambda_{\mathfrak{s}\lambda} \Pi\tilde{\nu}(\pi)))$ and $\mathfrak{s}F_\lambda f_c = (0, (J\partial_\theta + \lambda)(\chi_\pi \mathfrak{s}m_f))$. Since the trace at π is bounded by the graph norm, $\|\Theta_\lambda y\|_{\mathcal{H}_0} \leq C\mathfrak{s}\|y\|_{\mathcal{W}_0}$, and $\|\mathfrak{s}F_\lambda f_c\| \leq C\sqrt{\mathfrak{s}}\|f\|_{\mathfrak{s}}$. On $|\lambda| = \gamma$ the resolvent $(A_0 - \lambda)^{-1} : \mathcal{H}_0 \rightarrow \mathcal{W}_0$ is bounded by C_γ in the graph norm, so $\|(A_0 - \lambda)^{-1}\Theta_\lambda\|_{\mathcal{W}_0 \rightarrow \mathcal{W}_0} \leq \frac{1}{2}$ for $\mathfrak{s} \leq \mathfrak{s}_0$, and

$$(7.7) \quad y = \sum_{j \geq 0} ((A_0 - \lambda)^{-1}\Theta_\lambda)^j (A_0 - \lambda)^{-1}((f_f, f_\nu) + \mathfrak{s}F_\lambda f_c).$$

Conversely (7.7), followed by $\nu = \tilde{\nu} + \chi_\pi(\mathfrak{s}\lambda \Lambda_{\mathfrak{s}\lambda} \tilde{v} + \mathfrak{s}m_f)$ and $U_c = h_{\tilde{v}} + \tilde{U}_c$, defines a solution $z \in \mathcal{W}$; for $f = 0$ it gives $y = 0$, then $\tilde{v} = 0$, $\nu = 0$ and $U_c = 0$. Hence the circle $|\lambda| = \gamma$ lies

in the resolvent set of $A_{\mathfrak{s}}$. Let $\varpi(U_f, U_c, \nu) = (U_f, \nu)$ and $\iota(U_f, \nu) = (U_f, 0, \nu)$; then $\varpi\iota = 1$, $\iota\varpi = 1 - P_c$, ι is an isometry $\mathcal{H}_0 \rightarrow \mathcal{H}_{\mathfrak{s}}$ and $\varpi = \iota^*$. From (7.7) and the bounds above,

$$\|\varpi(A_{\mathfrak{s}} - \lambda)^{-1}\iota - (A_0 - \lambda)^{-1}\| \leq C\mathfrak{s}, \quad \|P_c(A_{\mathfrak{s}} - \lambda)^{-1}\| + \|(A_{\mathfrak{s}} - \lambda)^{-1}P_c\| \leq C\sqrt{\mathfrak{s}}$$

on $|\lambda| = \gamma$: for the first, $f = \iota g$ has $f_c = 0$ and the series differs from its first term by $O(\mathfrak{s})$, as does ν from $\tilde{\nu}$; for the second, the collapsed component of $(A_{\mathfrak{s}} - \lambda)^{-1}f$ has $\mathcal{H}_{\mathfrak{s}}$ -norm $\sqrt{\mathfrak{s}}\|U_c\| \leq \sqrt{\mathfrak{s}}(C\|\tilde{\nu}\| + \|\tilde{U}_c\|) \leq C\sqrt{\mathfrak{s}}\|f\|_{\mathfrak{s}}$, and a source $f = P_c f$ has $\|f\|_{\mathfrak{s}} = \sqrt{\mathfrak{s}}\|f_c\|$, so its solution is $O(\mathfrak{s}\|f_c\|) = O(\sqrt{\mathfrak{s}}\|f\|_{\mathfrak{s}})$.

Riesz projections. Let $P_{\mathfrak{s}}$ and P_0 be the Riesz projections of $A_{\mathfrak{s}}$ and A_0 for the circle $|\lambda| = \gamma$. Integrating the two estimates, $\|\varpi P_{\mathfrak{s}}\iota - P_0\| \leq C\mathfrak{s}$ and $\|P_c P_{\mathfrak{s}}\| + \|P_{\mathfrak{s}} P_c\| \leq C\sqrt{\mathfrak{s}}$. Since $P_{\mathfrak{s}} - (1 - P_c)P_{\mathfrak{s}}(1 - P_c) = P_c P_{\mathfrak{s}} + P_{\mathfrak{s}} P_c - P_c P_{\mathfrak{s}} P_c$,

$$\|P_{\mathfrak{s}} - \iota P_0 \varpi\|_{\mathcal{H}_{\mathfrak{s}}} \leq \|P_c P_{\mathfrak{s}}\| + \|P_{\mathfrak{s}} P_c\| + \|P_c P_{\mathfrak{s}} P_c\| + \|\iota(\varpi P_{\mathfrak{s}}\iota - P_0)\varpi\| \leq C\sqrt{\mathfrak{s}}.$$

The operator $\iota P_0 \varpi$ is an orthogonal projection on $\mathcal{H}_{\mathfrak{s}}$ of rank $\text{rank } P_0 = n$, and two projections at distance less than one have the same rank; so $P_{\mathfrak{s}}$ has rank n for $\mathfrak{s}$ small. Its range contains the n -dimensional kernel found above. As $A_{\mathfrak{s}}$ is self-adjoint and $\pm\gamma$ are not in its spectrum, $P_{\mathfrak{s}}$ is the spectral projection of $(-\gamma, \gamma)$, and the spectrum of $A_{\mathfrak{s}}$ in $(-\gamma, \gamma)$ is $\{0\}$. Uniformity over q holds because P is compact and the gap of A_0 is uniform [11, (4.24)].

Spatial estimate. Write $U_c = h_{\Pi\nu(\pi)} + \tilde{U}_c$ with $\tilde{U}_c \in \mathcal{W}_q^{\perp}$, as above. By (7.5), $\|\tilde{U}_c\|_{H^1} \leq c_1\|\mathcal{B}\tilde{U}_c\| = c_1\|\mathcal{B}U_c\| \leq c_1\sqrt{\mathfrak{s}}\|A_{\mathfrak{s}}z\|_{\mathfrak{s}}$, since $\|A_{\mathfrak{s}}z\|_{\mathfrak{s}}^2 \geq \mathfrak{s}^{-1}\|\mathcal{B}U_c\|^2$. $\square$

Along the endpoint neck the scale varies, $\mathfrak{s}(\rho) = \mathfrak{s}_b e^{\pm\rho}$. The linearized operator is $\mathcal{D} = \partial_{\rho} - A_{\mathfrak{s}(\rho)} + 2(\log \mathfrak{s})'P_p$, which is $\partial_{\rho} - A_{\mathfrak{s}(\rho)} - 2P_p$ for $\mathfrak{s} = \mathfrak{s}_b e^{-\rho}$; here P_p projects to the scalar of the collapsing component, acting on sections of the bundle of Hilbert spaces $\mathcal{H}_{\mathfrak{s}(\rho)}$. Since the quadratic form of $A_{\mathfrak{s}}$ is independent of $\mathfrak{s}$, the cross term in $\|(\partial_{\rho} - A)z\|^2$ is $-2\int \langle z', Az \rangle_{\mathfrak{s}} = -\int \partial_{\rho} \langle z, Az \rangle_{\mathfrak{s}} d\rho = 0$ for compactly supported z , so

$$\int \|(\partial_{\rho} - A)z\|_{\mathfrak{s}}^2 d\rho = \int (\|z'\|_{\mathfrak{s}}^2 + \|Az\|_{\mathfrak{s}}^2) d\rho,$$

with no derivative of $\mathfrak{s}$. The metric derivative enters only through $\partial_{\rho}\|z\|_{\mathfrak{s}}^2 = 2\langle z', z \rangle_{\mathfrak{s}} + (\log \mathfrak{s})' \|P_c z\|_{\mathfrak{s}}^2$, where P_c projects to the collapsing component, and hence only in weighted estimates, where it is absorbed by Proposition 7.9.

Proposition 7.10. (i) *(Tail estimate.)* Let the scale satisfy $|(\log \mathfrak{s})'| \leq 1$, and put $\|y\|_X^2 = \int (\|y\|_{\mathfrak{s}}^2 + \|\partial_{\rho} y\|_{\mathfrak{s}}^2 + \|A_{\mathfrak{s}} y\|_{\mathfrak{s}}^2) d\rho$ and $\|y\|_Y^2 = \int \|y\|_{\mathfrak{s}}^2 d\rho$. For $0 < |\delta| < \gamma/2$ there is $\mathfrak{s}_0(\delta) > 0$ such that for z supported where $\mathfrak{s}(\rho) \leq \mathfrak{s}_0(\delta)$,

$$\|e^{\delta\rho} z\|_X \leq C_{\delta} \|e^{\delta\rho} \mathcal{D} z\|_Y, \quad \|e^{\delta\rho} z\|_X \leq C_{\delta} \|e^{\delta\rho} \mathcal{D}^* z\|_Y,$$

where $\mathcal{D}^* = -\partial_{\rho} - A_{\mathfrak{s}} - (\log \mathfrak{s})'P_c + 2(\log \mathfrak{s})'P_p$ is the formal adjoint.

(ii) *(Conserved pairing.)* For h in the kernel of the adjoint and $v \in T_q P$, $\langle k_v, h \rangle_{\mathfrak{s}(\rho)}$ is independent of ρ .

(iii) *(Transition.)* Let $\mathcal{T}$ be the complete flat transition: the domain consisting of the rescaled endpoint core (the half-disk of radius 2ε about $z_{\pm}$ rescaled by ε^{-1} , with its fixed conformal structure), continued by the endpoint neck to $\rho = \infty$, with the flat pair (B_q, q) and the scale $\mathfrak{s} \rightarrow 0$ along the neck. With the weight $e^{\delta\rho}$ and fixed limit 0, $\mathcal{D}$ is an isomorphism; with a free limit in $T_q P$ it is onto with kernel $\{k_v\}$ and index n . For $0 < \delta \leq \delta_0$ the fixed-limit inverse satisfies $\|\mathcal{D}^{-1}\| \leq C/\delta$, uniformly in $q \in P$.

Proof. (i) Let $y = e^{\delta\rho}z$, so that $e^{\delta\rho}\mathcal{D}z = (\partial_\rho - A_\mathfrak{s} - \delta + 2(\log \mathfrak{s})'P_\rho)y$. Since the quadratic form of $A_\mathfrak{s}$ does not depend on $\mathfrak{s}$, and $\partial_\rho\|y\|_\mathfrak{s}^2 = 2\langle\partial_\rho y, y\rangle_\mathfrak{s} + (\log \mathfrak{s})'\|P_\rho y\|_\mathfrak{s}^2$,

$$\int \|(\partial_\rho - A_\mathfrak{s} - \delta)y\|_\mathfrak{s}^2 d\rho = \int (\|\partial_\rho y\|_\mathfrak{s}^2 + \|(A_\mathfrak{s} + \delta)y\|_\mathfrak{s}^2 + \delta(\log \mathfrak{s})'\|P_\rho y\|_\mathfrak{s}^2) d\rho.$$

Because $0 < |\delta| < \gamma/2$, Proposition 7.9 gives $\|(A_\mathfrak{s} + \delta)y\|_\mathfrak{s} \geq c_\delta(\|y\|_\mathfrak{s} + \|A_\mathfrak{s}y\|_\mathfrak{s})$ with $c_\delta > 0$. Its spatial estimate, together with $\|h_{\Pi\nu(\pi)}\| \leq C|\nu(\pi)| \leq C(\|y\|_\mathfrak{s} + \|A_\mathfrak{s}y\|_\mathfrak{s})$, gives $\|P_\rho y\|_\mathfrak{s} \leq C\sqrt{\mathfrak{s}}(\|y\|_\mathfrak{s} + \|A_\mathfrak{s}y\|_\mathfrak{s})$, and $\|P_\rho y\|_\mathfrak{s} \leq C\mathfrak{s}\|A_\mathfrak{s}y\|_\mathfrak{s}$ by Lemma 7.1(i) and the Poincaré inequality. For $\mathfrak{s} \leq \mathfrak{s}_0(\delta)$ the last term and the term $2(\log \mathfrak{s})'P_\rho$ are therefore absorbed. For $\mathcal{D}^*$ the conjugated operator is $-\partial_\rho - (A_\mathfrak{s} - \delta) - (\log \mathfrak{s})'P_\rho + 2(\log \mathfrak{s})'P_\rho$, the same computation with $-\partial_\rho$ gives the identity with $A_\mathfrak{s} - \delta$ in place of $A_\mathfrak{s} + \delta$, and the extra term $(\log \mathfrak{s})'P_\rho y$ is absorbed in the same way.

(ii) For h with $\mathcal{D}^*h = -h' - Ah + P_\rho h - 2P_\rho h = 0$, differentiate $\langle k_v, h \rangle_\mathfrak{s}$: the term $P_\rho h$ in the equation cancels the metric derivative, and $Ak_v = P_\rho k_v = 0$.

(iii) *Kernel with fixed limit.* If $\mathcal{D}z = 0$ with decay, the linear energy identity on the core and neck, whose seam terms cancel by Convention 3.1, gives z flat and constant, hence 0. *Cokernel.* A cokernel element h of the free operator pairs trivially with $\mathcal{D}(\beta k_v) = \beta'k_v$ for a cutoff β , so by (ii) its stationary limit is orthogonal to the kernel and h decays; it then vanishes by the argument of Proposition 6.12.

The bound. We use the norms of (i) on the neck, weighted by $e^{\delta\rho}$, together with the fixed H^1 and L^2 norms on the core, and write X_δ, Y_δ . Fix $\bar{\delta} \in (0, \gamma/2)$. By what has been shown, $\mathcal{D}: X_{\bar{\delta}} \rightarrow Y_{\bar{\delta}}$ with fixed limit 0 is an isomorphism; write $\mathcal{D}_{\bar{\delta}}^{-1}$ for its inverse, which is bounded uniformly in q .

Stationary splitting. In the component coordinates the map $k: T_q P \rightarrow \mathcal{H}_\mathfrak{s}$, $v \mapsto k_v$, does not depend on ρ . Let P_ρ^{st} be the $\mathcal{H}_{\mathfrak{s}(\rho)}$ -orthogonal projection onto $\ker A_{\mathfrak{s}(\rho)} = k(T_q P)$, let $P_\rho^n = 1 - P_\rho^{\text{st}}$, and write $P_\rho^{\text{st}}y = k_{\mathfrak{a}(y)}$. With $I_\mathfrak{s} = \text{diag}(1, \mathfrak{s}, 1)$ and the Gram matrix $G_\mathfrak{s} = k^* I_\mathfrak{s} k$, which is uniformly positive, $P^{\text{st}} = k G_\mathfrak{s}^{-1} k^* I_\mathfrak{s}$ and $\mathfrak{a} = G_\mathfrak{s}^{-1} k^* I_\mathfrak{s}$. Since $\partial_\rho I_\mathfrak{s} = (\log \mathfrak{s})' I_\mathfrak{s} P_\rho$,

$$(7.8) \quad \partial_\rho P^{\text{st}} = (\log \mathfrak{s})' P^{\text{st}} P_\rho P^n, \quad \|\partial_\rho P^{\text{st}}\| \leq C|(\log \mathfrak{s})'|\sqrt{\mathfrak{s}},$$

the bound because the collapsed component of k_v has $\mathcal{H}_\mathfrak{s}$ -norm $\sqrt{\mathfrak{s}}\|h_v\|$. Three facts give the structure of the equation: $\mathcal{D}k_a = k_{a'}$ for a function a with values in $T_q P$, since k does not depend on ρ and $A_\mathfrak{s}k_v = P_\rho k_v = 0$; $P^{\text{st}}A_\mathfrak{s} = 0$ and $P^{\text{st}}P_\rho = 0$, since $A_\mathfrak{s}$ is symmetric and annihilates k_v , and k_v has no scalar component; and $P^{\text{st}}\partial_\rho y = -(\partial_\rho P^{\text{st}})y$ when $P^{\text{st}}y = 0$. Consequently, for $z = k_a + z^n$ with $a = \mathfrak{a}(z)$ and $z^n = P^n z$, the equation $\mathcal{D}z = f$ is equivalent to the triangular system

$$(7.9) \quad \mathcal{D}^n z^n = P^n f, \quad a' = \mathfrak{a}(f) - \mathfrak{b}_\mathfrak{s} z^n,$$

where $\mathcal{D}^n = P^n \partial_\rho - A_\mathfrak{s} + 2(\log \mathfrak{s})'P_\rho$ acts on P^n -valued fields and $\mathfrak{b}_\mathfrak{s} y = -(\log \mathfrak{s})' G_\mathfrak{s}^{-1} k^* I_\mathfrak{s} P_\rho y$ satisfies $|\mathfrak{b}_\mathfrak{s} y| \leq C|(\log \mathfrak{s})'|\sqrt{\mathfrak{s}}\|y\|_\mathfrak{s}$. The first equation does not involve a .

The normal equation. Let $\mathcal{Y}$ be the complete line obtained by continuing the neck to $\rho \rightarrow -\infty$ with the slice $(C, g_C) \sqcup [0, \pi] \sqcup (C, \mathfrak{s}_\mathcal{Y}^2 g_C)$ and the flat pair (B_q, q) , where $\mathfrak{s}_\mathcal{Y}$ is smooth and nonincreasing, equals $\mathfrak{s}(\rho)$ for $\rho \geq \rho_1 + 2$, is constant for $\rho \leq \rho_1$, and satisfies $|(\log \mathfrak{s}_\mathcal{Y})'| \leq 1$; here ρ_1 is fixed below, independently of δ and q . All of the above holds on $\mathcal{Y}$ with $\mathfrak{s}_\mathcal{Y}$ in place of $\mathfrak{s}$. For a compactly supported P^n -valued field y in the domain, $P^n \partial_\rho y = \partial_\rho y + (\partial_\rho P^{\text{st}})y$. By the identity displayed before the proposition, the gap $(\|y\|_\mathfrak{s} \leq \gamma^{-1}\|A_\mathfrak{s}y\|_\mathfrak{s})$ for y orthogonal to $\ker A_\mathfrak{s}$, (7.8) and $\|P_\rho y\|_\mathfrak{s} \leq C\mathfrak{s}\|A_\mathfrak{s}y\|_\mathfrak{s}$,

$$\|\mathcal{D}^n y\|_Y \geq (\|\partial_\rho y\|_Y^2 + \|A_\mathfrak{s}y\|_Y^2)^{1/2} - C\mathfrak{s}(\rho_1)^{1/2}\|y\|_X \geq c\|y\|_X$$

once $\mathfrak{s}(\rho_1)$ is small. Conjugation by $e^{\sigma\rho}$ replaces $\mathcal{D}^n$ by $\mathcal{D}^n - \sigma$, so the same bound holds in the norms weighted by $e^{\sigma\rho}$ for $|\sigma| \leq \sigma_0 := c/2$, with a constant independent of σ . The formal adjoint $\mathcal{D}^*$ maps P^n -valued fields to P^n -valued fields: for $P^{\text{st}}y = 0$, $P^{\text{st}}\mathcal{D}^*y = (\partial_\rho P^{\text{st}})y - (\log \mathfrak{s})' P^{\text{st}} P_c y = 0$ by (7.8). The same computation therefore bounds $\|y\|_X$ by the norm of the normal adjoint applied to y , and, weak solutions being strong by the local regularity used above for $A_{\mathfrak{s}}$, $\mathcal{D}^n$ is an isomorphism between the P^n -valued parts of the weighted spaces on $\mathcal{Y}$:

$$(7.10) \quad \|e^{\sigma\rho}y\|_{X(\mathcal{Y})} \leq C_n \|e^{\sigma\rho}\mathcal{D}^n y\|_{Y(\mathcal{Y})}, \quad |\sigma| \leq \sigma_0,$$

with C_n independent of σ and q .

The stationary equation. For $0 < \delta \leq \sigma_0$ and $f \in Y_\delta(\mathcal{Y})$ let $z^n = (\mathcal{D}^n)^{-1} P^n f$, $g = \mathfrak{a}(f) - \mathfrak{b}_{\mathfrak{s}} z^n$ and $a(\rho) = -\int_\rho^\infty g$. By (7.10), $\|e^{\delta\rho}g\|_{L^2} \leq C\|f\|_{Y_\delta}$. Since $e^{\delta\rho}a(\rho) = -\int_0^\infty e^{-\delta u}(e^{\delta\rho}g)(\rho+u) du$ is the convolution of $e^{\delta\rho}g$ with a one-sided kernel of L^1 norm δ^{-1} and L^2 norm $(2\delta)^{-1/2}$,

$$(7.11) \quad \|e^{\delta\rho}a\|_{L^2} \leq \delta^{-1}\|e^{\delta\rho}g\|_{L^2}, \quad \sup_\rho e^{\delta\rho}|a(\rho)| \leq (2\delta)^{-1/2}\|e^{\delta\rho}g\|_{L^2}, \quad a' = g.$$

As $A_{\mathfrak{s}}k_a = 0$, $\partial_\rho k_a = k_{a'}$ and $\|k_v\|_{\mathfrak{s}} \leq C|v|$, the field $S_\delta f = k_a + z^n$ solves $\mathcal{D}S_\delta f = f$ on $\mathcal{Y}$, tends to zero, and satisfies $\|S_\delta f\|_{X_\delta(\mathcal{Y})} \leq C\delta^{-1}\|f\|_{Y_\delta(\mathcal{Y})}$. The factor δ^{-1} arises only in the undifferentiated stationary coefficient; the normal part is solved with no loss.

The core. Fix $\rho_2 \geq \rho_1 + 2$ and cutoffs χ_1, χ_0 of ρ with $\chi_1 = 0$ for $\rho \leq \rho_2$, $\chi_1 = 1$ for $\rho \geq \rho_2 + 1$, $\chi_0 = 0$ for $\rho \leq \rho_2 + 2$ and $\chi_0 = 1$ for $\rho \geq \rho_2 + 3$; on the support of χ_1 the operators of $\mathcal{T}$ and of $\mathcal{Y}$ agree. For $0 < \delta \leq \delta_0 := \min(\sigma_0, \bar{\delta})$ and $f \in Y_\delta$, regard $\chi_0 f$ as a source on $\mathcal{Y}$ and put

$$z_1 = \chi_1 S_\delta(\chi_0 f), \quad r = f - \mathcal{D}z_1 = (1 - \chi_0)f - \chi_1' S_\delta(\chi_0 f).$$

The source r is supported in the fixed compact region $\{\rho \leq \rho_2 + 3\}$ of $\mathcal{T}$, where $e^{\delta\rho}$ and $e^{\bar{\delta}\rho}$ are comparable with constants independent of δ . By (7.11) the stationary part of $S_\delta(\chi_0 f)$ is bounded there by $C\delta^{-1/2}\|f\|_{Y_\delta}$, so $\|r\|_{Y_\delta} \leq C\delta^{-1/2}\|f\|_{Y_\delta}$. Then $z = z_1 + \mathcal{D}_\delta^{-1}r$ solves $\mathcal{D}z = f$ with limit zero, and since $\|\cdot\|_{X_\delta} \leq C\|\cdot\|_{X_{\bar{\delta}}}$ for $\delta \leq \bar{\delta}$,

$$\|z\|_{X_\delta} \leq C\delta^{-1}\|f\|_{Y_\delta},$$

while on every fixed compact subset of $\mathcal{T}$ the field z is bounded by $C\delta^{-1/2}\|f\|_{Y_\delta}$. The fixed-limit kernel is zero at every positive weight, so $z = \mathcal{D}^{-1}f$. The constants are uniform in q , because P is compact and γ , C_n and $\mathcal{D}_\delta^{-1}$ are uniform. The bound is sharp: $z_\delta = k_v \beta(\delta(\rho - \rho_0))$ for a fixed cutoff β has $\|z_\delta\|/\|\mathcal{D}z_\delta\| \geq c/\delta$. $\square$

7.6. Transport along a path in $L(C)$. Let $q(t)$ be a smooth path in $P = L(C)$ with bounded derivatives of every order; in Section 8 it is the path $q_R(t)$ of section boundary values of the approximate solution W_ε . Write $\mathcal{L}_t = T_{q(t)}P$, $j_t v = (0, h_v)$ for the harmonic extension, $\ell_t = (j_t^* j_t)^{-1} j_t^*$ and $\Pi_t = j_t \ell_t$, all bulk operators on $L^2(C)$. A variation $U = (p, b)$, $p = \mathfrak{s}\psi$, splits as $U = j_t v + U_\perp$ with $v = \ell_t U$ and $U_\perp = (1 - \Pi_t)U$. On the domain (vanishing normal component, closed traces) the complementary estimate

$$(7.12) \quad \|U_\perp\|_{H^{r+1}(C)} \leq C_r \|\mathcal{B}_t U\|_{H^r(C)} \quad (r \geq 0)$$

holds: solve the complementary problem with harmonic trace in $J\mathcal{L}_t$, use its elliptic regularity, and remove the finite harmonic kernel by the bulk projection.

Lemma 7.11 (transport). *There is a connection $\nabla_t = D_t + C_t$ along $q(t)$ on the bundle of spatial domains, with $\|C_t\|_{H^r \rightarrow H^r} \leq C_r$ and the same bounds for its time derivatives, which preserves the vanishing of the normal component, the closedness of the traces and the harmonic seam matching, and satisfies*

$$\nabla_t(j_t v) = j_t \nabla_t^{\mathcal{L}} v, \quad \nabla_t^{\mathcal{L}}(\ell_t U) = \ell_t \nabla_t U, \quad [\nabla_t, \Pi_t] = 0, \quad [\nabla_t, P_p] = 0.$$

Consequently the iterated commutators $\mathcal{B}_t^{[0]} = \mathcal{B}_t$, $\mathcal{B}_t^{[j+1]} = [\nabla_t, \mathcal{B}_t^{[j]}]$ satisfy $\mathcal{B}_t^{[j]} j_t = 0$ and $\|\mathcal{B}_t^{[j]} U\|_{H^r} \leq C_{j,r} \|\mathcal{B}_t U\|_{H^r}$. The connection is the derivative of transports $\tilde{T}$ from the domain at q_0 to the domain at q , defined for q near q_0 and bounded on every $H^r(C)$, with $\tilde{T} j_0 = j_q K$, $\ell_q \tilde{T} = K \ell_0$ and $\Pi_q \tilde{T} = \tilde{T} \Pi_0$, where K identifies $\mathcal{L}_{q_0}$ with $\mathcal{L}_q$.

Proof. For q near q_0 , identify $\mathcal{L}_{q_0}$ with $\mathcal{L}_q$ by a smooth map K . On every parallel surface of the boundary collars, identify the exact, coexact and harmonic summands of the Hodge decompositions for the two flat restrictions, blended to the identity in the interior, leaving scalars and normal components unchanged; this gives T^0 , which identifies the domains and their harmonic boundary values by K and is bounded on every $H^r(C)$. Correct it on the harmonic fields: $T^1 = T^0 + (j_q K - T^0 j_0) \ell_0$, which changes no boundary condition and satisfies $T^1 j_0 = j_q K$. For an interior cutoff χ close to one the matrix $M_q = \ell_q \chi j_q$ is invertible; with $e_q = \chi j_q M_q^{-1}$, which has zero boundary trace, put $\tilde{T} = T^1 + e_q (K \ell_0 - \ell_q T^1)$. Then $\tilde{T} j_0 = j_q K$, $\ell_q \tilde{T} = K \ell_0$ and $\Pi_q \tilde{T} = \tilde{T} \Pi_0$. Differentiating these transports along $q(t)$ defines ∇_t , and the identities follow. Since $\mathcal{B}_t j_t = 0$ and ∇_t preserves harmonic extension, induction gives $\mathcal{B}_t^{[j]} j_t = 0$; the commutators have spatial order at most one, so $\|\mathcal{B}_t^{[j]} U\|_{H^r} \leq C \|U\|_{H^{r+1}}$, and (7.12) gives the bound. $\square$

7.7. The inverse on the glued domain. Let V be a regular index-zero solution on X_0 with limits $q_{\pm}$ at the new ends and $\ker D_V = 0$ (on the extended weighted spaces), and let W_{ε} be the approximate solution on X_{ε} of Section 8: V truncated at radius $R = \varepsilon^{1/4}$ about each endpoint, the flat transitions at $q_{\pm}$ inserted, and the flat lift of the section boundary values over $I \times C$. Put $L = \log(1/\varepsilon)$ and $A = 1 + L$.

Norms. On X_{ε} use the complete conformal representatives: on the middle compression $dt^2 + (\varepsilon/r)^2 g_C$ with $dt = dx/r$; on the endpoint necks $ds^2 + g_C$ and the collapsing branch at scale $\mathfrak{s} = \varepsilon e^s$; on the rescaled endpoint cores the fixed transition metric; on the section the corresponding fixed and logarithmic strip metrics. On a small compression with $U = (p, b)$, $p = \mathfrak{s}\psi$, the norms X_{ε} and Y_{ε} are

$$\int (\mathfrak{s}|U|^2 + \mathfrak{s}|D_t U|^2 + \mathfrak{s}^{-1}|\mathcal{B}_t U|^2) dt, \quad \int \mathfrak{s}|f|^2 dt,$$

on the section and the fixed branch the H^1 and L^2 norms, on the cores the fixed mixed H^1 and L^2 norms, patched by uniformly bounded cutoffs, and on the original ends the fixed weights with the free limiting vectors. The norm includes the L^2 part of a stationary field on every finite neck: for a constant stationary vector on a region of length $O(L)$,

$$(7.13) \quad \|k_v\|_{X_{\varepsilon}(\text{region})} \leq C\sqrt{A}|v|.$$

Components. The two *transitions* are the complete flat transitions $\mathcal{T}_{\pm}$ of Proposition 7.10 at $q_{\pm}$. The *auxiliary exterior* is the following complete domain.

Definition 7.12 (auxiliary exterior). Let r be a smooth positive function on I equal to the distance to the nearer endpoint where that distance is at most $x_0/2$, and let t be the complete coordinate on I with $dt = dx/r$, where x is arclength; near $z_{\pm}$ the coordinate t

agrees, up to sign and an additive constant, with $s = -\log r$ of Lemma 6.3(ii). Fix a smooth $F: (0, \infty) \rightarrow (0, 1]$ with $F(y) = y$ for $y \leq \frac{1}{2}$, $F(y) = 1$ for $y \geq 2$ and $0 \leq yF'(y)/F(y) \leq 1$, and put

$$\mathfrak{s}_E = \sqrt{\varepsilon} F(\sqrt{\varepsilon}/r).$$

The *auxiliary exterior* E_ε is X_0 with the gauge piece $I \times C$, carrying the metric $dt^2 + \mathfrak{s}_E^2 g_C$, attached along I : the Lagrangian condition $L(C)$ on I is replaced by the matching condition of Definition 6.4. Its configuration W_ε^E is the truncation V_R of V at $R = \varepsilon^{1/4}$ of Section 8 on X_0 , and on $I \times C$ the flat lift $B_{q_R(t)}$ of the boundary values q_R of V_R on I , with temporal component in the covariant Neumann slice, $\partial_t B - d_B \Psi = h_{q_R}$. The operator D_E is the linearized mixed operator at W_ε^E , with the gauge-fixing component of the metric $dt^2 + \mathfrak{s}_E^2 g_C$ on the compression and of the complete metrics of X_0 elsewhere, as for D_ε in the conformal representative above, on the domain given along I by the conditions of Section 7.1 (vanishing normal component, closed traces, harmonic part of $b|_\Sigma$ equal to the section value) and by the original conditions elsewhere, on the extended weighted spaces at the original mixed ends and at the two new ends.

The scale satisfies $\sup \mathfrak{s}_E \leq \sqrt{\varepsilon}$ and $|\partial_t \log \mathfrak{s}_E| \leq H$ with H independent of ε ; moreover $\mathfrak{s}_E = \varepsilon/r$ where $r \geq 2\sqrt{\varepsilon}$ and $\mathfrak{s}_E = \sqrt{\varepsilon}$ where $r \leq \sqrt{\varepsilon}/2$. Near $z_\pm$, in the coordinate s , the domain E_ε is a half-cylinder with slice $(C, g_C) \sqcup [0, \pi] \sqcup (C, \mathfrak{s}_E(s)^2 g_C)$, where $\mathfrak{s}_E(s) = \varepsilon e^s$ for $s \leq L/2 - \log 2$ and $\mathfrak{s}_E(s) = \sqrt{\varepsilon}$ for $s \geq L/2 + \log 2$. There is a fixed c_R such that W_ε^E coincides with V and its flat lift for $s \leq L/4 - c_R$, and with the flat pair $(B_{q_\pm}, q_\pm)$ on all three branches for $s \geq L/4 + c_R$. On the region $\{r \geq 2\sqrt{\varepsilon}\}$, which contains the whole exterior side of $s = L/2 - \log 2$ and in particular $J_\pm$, the domain, the configuration and the operator coincide with X_ε , W_ε and D_ε in the conformal representative above.

On the new ends of the components, which are not present on X_ε , use the weight $\delta_L = c_0/(1+L)$, bounded above and below on the parts of the components used in the gluing. Each transition overlaps the actual domain on $J_\pm = [\alpha_L, \beta_L]$ with $\alpha_L = L/4 + b_0$, with a fixed $b_0 \geq c_R + 1$ so that W_ε^E is flat on $J_\pm$, $\beta_L = L/2 - \log 2$, of length $\mathbb{T} \sim L/4$, on which the three operators agree under $\rho = L - s$ (Figure 3).

Norms and asymptotic values. Let ϕ_E be a smooth function on E_ε , equal to $\delta_* s$ on the new ends, to $\delta_a s$ on the original end a (the fixed weights of (X3)) and to 0 on a fixed compact core, and depending on the compression only through t . Let $\mathbb{V}$ be the direct sum of $T_{q_-} P \oplus T_{q_+} P$ and of the tangent spaces of the critical families at the original mixed ends. Let $\mathfrak{k}_0(v)$, $v \in \mathbb{V}$, be the cut-off stationary fields of D_V of Proposition 6.9: a fixed cutoff of the end coordinate times the stationary field of the end, in the end chart of V ; at the new ends this is (h_v, v) on the slice $C \sqcup [0, \pi]$. Let $\mathfrak{k}_0^\varepsilon(v)$ be the same fields built with the end charts of V_R ; they coincide with $\mathfrak{k}_0(v)$ where $s \leq L/4 - c_R$ and differ from it by $O(e^{-\beta s}|v|)$ elsewhere. On E_ε put $\mathfrak{k}_E(v) = \mathfrak{k}_0^\varepsilon(v)$ off the compression and $\mathfrak{k}_E(v) = (0, h_{\nu(t)})$ on $I \times C$, where $\nu(t) \in T_{q_R(t)} P$ is the section component of $\mathfrak{k}_0^\varepsilon(v)$ on I . Then $\mathfrak{k}_E(v)$ lies in the domain of D_E . Every element of X_E is $z = \bar{z} + \mathfrak{k}_E(v)$ with $\bar{z}$ decaying, and

$$\begin{aligned} \|z\|_{X_E}^2 &= \|e^{\phi_E} \bar{z}\|_{H^1}^2 + \int_I e^{2\phi_E} (\mathfrak{s}_E \|\bar{U}_c\|^2 + \mathfrak{s}_E \|\mathrm{D}_t \bar{U}_c\|^2 + \mathfrak{s}_E^{-1} \|\mathcal{B}_t \bar{U}_c\|^2) dt + |v|^2, \\ \|f\|_{Y_E}^2 &= \|e^{\phi_E} f\|_{L^2}^2 + \int_I e^{2\phi_E} \mathfrak{s}_E \|f_c\|^2 dt, \end{aligned}$$

where the first terms are the norms of the section and of the gauge fields of X_0 , and the compression integrals use the fixed metric g_C on C . On $J_\pm$ these are the norms X_ε , Y_ε of this section multiplied by the weight. Since $\mathcal{B}_t$ and P_p annihilate harmonic fields, the compression

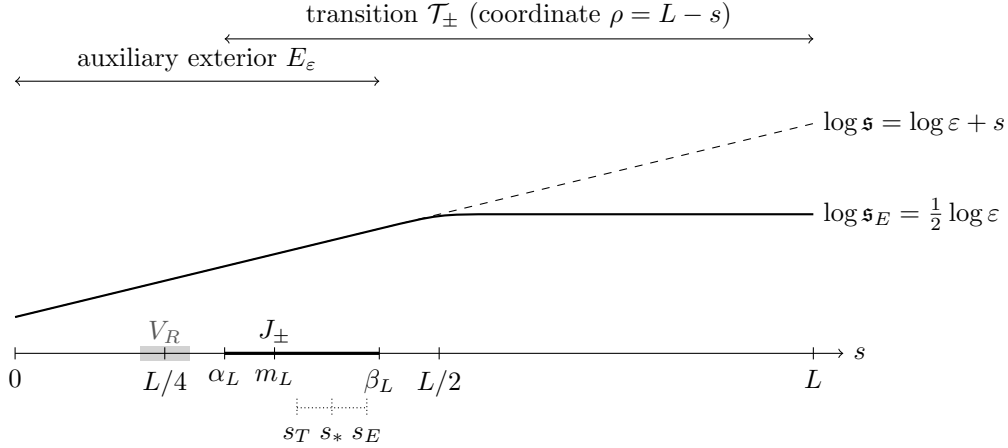


FIGURE 3. An endpoint neck in the exterior coordinate $s \in [0, L]$, $L = \log(1/\varepsilon)$, with $r = e^{-s}$ (not to scale). The dashed line is $\log \mathfrak{s}$ for the scale $\mathfrak{s} = \varepsilon e^s$ of X_ε ; the solid line is the scale $\mathfrak{s}_E$ of the auxiliary exterior, equal to $\mathfrak{s}$ for $s \leq L/2 - \log 2$ and to $\sqrt{\varepsilon}$ for $s \geq L/2 + \log 2$. The shaded interval about $L/4$ ($r = \varepsilon^{1/4}$) is where the truncation V_R of the limit is joined to the flat pair. The two components are used over the arrows and overlap the glued domain on $J_\pm = [\alpha_L, \beta_L]$, with joining point m_L (Section 7.7). The dotted marks are the cuts $s_T < s_E$ of the proof of Proposition 8.12, at distance $C_* + D$ and $C_* - D$ from $L/2$, and their common midpoint $s_* = L/2 - C_*$; for large L they lie between m_L and β_L .

component of $D_E \mathfrak{k}_E(v)$ is $D_t(0, h_{\nu(t)})$; as $\mathfrak{s}_E \leq \sqrt{\varepsilon}$, and $\partial_t \nu$ and $\dot{q}_R$ decay at the rate $\beta > \delta_*$ along the ends,

$$(7.14) \quad \|(D_E \mathfrak{k}_E(v))_c\|_{Y_E} \leq C\varepsilon^{1/4}|v|, \quad \|D_E \mathfrak{k}_E(v)\|_{Y_E} \leq C|v|,$$

and $e^{\phi_E}(\mathfrak{k}_0^\varepsilon(v) - \mathfrak{k}_0(v))$, $e^{\phi_E}(D_{V_R} \mathfrak{k}_0^\varepsilon(v) - D_V \mathfrak{k}_0(v))$ have H^1 and L^2 norms at most $C\varepsilon^{(\beta - \delta_*)/4}|v|$.

Proposition 7.13 (uniform inverse on the auxiliary exterior). *Fix δ_* with $0 < \delta_* < \min(\beta, \gamma/2)$, where β is the decay rate of V and γ the gap of Proposition 7.9. There are $\varepsilon_1 > 0$ and C such that for $0 < \varepsilon < \varepsilon_1$ the operator $D_E: X_E \rightarrow Y_E$ is an isomorphism and*

$$\|z\|_{X_E} \leq C\|D_E z\|_{Y_E} \quad (z \in X_E).$$

In particular the limiting values $v_\pm$ of z at the new ends satisfy $|v_-| + |v_+| \leq C\|D_E z\|_{Y_E}$.

At the weight δ_* , which lies below the gaps of the new mixed ends of X_0 and below β , the operator D_V on the extended weighted spaces is an isomorphism by Hypothesis 6.5(X4), V being regular of index zero.

Proof. Step 1: a graph estimate with compact remainder. Fix s_1 large, to be chosen in (b) independently of ε , and put $r_1 = e^{-s_1}$. We first derive a consequence of Lemma 7.2: at every t , $\mathfrak{s}_E \|U_c\|^2 \leq c \mathfrak{s}_E (\|\mathcal{B}_t U_c\|^2 + |\nu|_I^2) \leq c \sup \mathfrak{s}_E (\mathfrak{s}_E^{-1} \|\mathcal{B}_t U_c\|^2 + |\nu|_I^2)$, and the trace inequality on a collar of I in the section gives

$$(7.15) \quad \int_I e^{2\phi_E} \mathfrak{s}_E \|\bar{U}_c\|^2 dt \leq C\varepsilon^{1/2} \|\bar{z}\|_{X_E}^2.$$

(a) *The middle of I .* On $\{r \geq r_1/4\}$ use conformal coordinates $t' + i\theta'$ on a collar of I in S_0 with $I = \{\theta' = 0\}$, for instance $t' + i\theta' = \log((z - z_-)/(z_+ - z))$ in the half-plane chart, and let $\theta_1 > 0$ be such that $\{r \geq r_1/4, 0 \leq \theta' \leq \theta_1\}$ lies in the collar. There dt'/dt is bounded above and below with all its derivatives, independently of ε . In the time coordinate t' the compression metric is $(dt/dt')^2(dt'^2 + \mathfrak{s}'^2 g_C)$ with $\mathfrak{s}' = \mathfrak{s}_E dt'/dt$. The anti-self-duality equation is conformally invariant, and in dimension four the gauge-fixing component of a metric changed by a conformal factor depending on t' alone differs from the original one by a positive factor and by a zeroth-order term in the temporal component. Hence, on fields supported in $\{r \geq r_1/2\}$ whose section component vanishes for $\theta' \geq \theta_1/2$, $\Upsilon D_E = D_{\mathfrak{s}'} + 2h''P_p$, where Υ multiplies the gauge-fixing component by a positive function of t' bounded above and below with its derivatives, $D_{\mathfrak{s}'}$ is the operator of Section 7.1 with time t' , the strip $\mathbb{R} \times [0, \theta_1]$, the path q_R and the section of V_R , and h'' is bounded independently of ε ; a term $2h''P_p$ is allowed in the proof of Proposition 7.4 as part of the endomorphism Z . Here $\sup \mathfrak{s}' \leq C\varepsilon/r_1$, while $|\partial_{t'} \log \mathfrak{s}'|$, $\sup |\partial_{t'} q_R|$ and the C^1 norms of the section are bounded independently of ε . The Lagrangian condition at the far edge of the strip enters the proof of Proposition 7.4 only through the boundary term $\omega_0(\xi_1, \partial_t \xi_1)$ and the trace term $|\xi_1|^2$, which vanish for these fields, and the proof is unchanged for a strip of fixed width θ_1 . The norms of Section 7.1 and those of X_E, Y_E are uniformly equivalent on this region. Hence, for such fields and ε small,

$$\|\bar{z}\|_{X_E} \leq C(\|D_E \bar{z}\|_{Y_E} + \|\nu\|_{L^2(\{r \geq r_1/2, \theta' \leq \theta_1/2\})}).$$

(b) *The new ends.* For $s \geq s_1 - 1$ the configuration W_ε^E is within $Ce^{-\beta s}$ of the flat pair $(B_{q_\pm}, q_\pm)$ in every C^k norm, with C independent of ε , by the exponential convergence of V (Proposition 6.15(iii)) and the construction of V_R . Identify the slice domains of W_ε^E with those of the flat pair by the map equal on the collapsing branch to the transport $\tilde{T}$ from $q_\pm$ to $q_R(s)$ of Lemma 7.11, and on the fixed branch and the section to an end chart of V_R compatible with the identification K of $\tilde{T}$ at the seam; since $\tilde{T}$ carries h_v to h_{Kv} and harmonic boundary values by K , this map preserves the domain. In this identification $D_E = \mathcal{D}_\pm + \mathfrak{E}$, where $\mathcal{D}_\pm$ is the neck operator of Section 7.5 at the flat pair with scale $\mathfrak{s}_E(s)$, written in the coordinate s as on $J_\pm$, and $\|\mathfrak{E}y\|_{Y_E} \leq Ce^{-\beta s_1} \|y\|_{X_E}$ for y supported in $\{s \geq s_1 - 1\}$. Indeed the only coefficient of $\mathfrak{E}$ carrying $\mathfrak{s}_E^{-1}$ is $\mathfrak{s}_E^{-1}(\mathcal{B}_{q_R(s)}\tilde{T} - \tilde{T}\mathcal{B}_{q_\pm})$; it vanishes on harmonic fields, and it is a first-order operator whose coefficients are $O(e^{-\beta s})$, so by Lemma 7.1(ii) it is bounded by $Ce^{-\beta s} \mathfrak{s}_E^{-1} \|\mathcal{B}_{q_\pm} U_c\|$. By Proposition 7.10(i), which uses of the scale only $|\partial \log \mathfrak{s}| \leq 1$ and a small upper bound, and applies here because $0 \leq \partial_s \log \mathfrak{s}_E \leq 1$ and $\mathfrak{s}_E \leq \sqrt{\varepsilon}$, we have $\|e^{\delta_* s} y\|_X \leq C\|e^{\delta_* s} \mathcal{D}_\pm y\|_Y$ for decaying y supported in $\{s \geq s_1 - 1\}$, with C independent of ε ; by Lemma 7.1 these norms are uniformly equivalent to X_E and Y_E on the ends. Choosing s_1 large, independently of ε , to absorb $\mathfrak{E}$,

$$\|\bar{z}\|_{X_E} \leq C\|D_E \bar{z}\|_{Y_E} \quad \text{for decaying } \bar{z} \text{ supported in } \{s \geq s_1 - 1\}.$$

(c) *The remaining regions.* On the original ends, beyond a fixed start, the weighted end estimates of [11, Section 4.2.1] at the fixed weights give $\|\bar{z}\|_{X_E} \leq C\|D_E \bar{z}\|_{Y_E}$ for decaying $\bar{z}$ supported there. The rest of E_ε is a fixed compact subset of X_0 at positive distance from I and from all ends, on which $W_\varepsilon^E = V$ for ε small; there the local mixed estimates [11, Theorem 4.17 and Remark 4.19] give $\|\bar{z}\|_{H^1} \leq C(\|D_E \bar{z}\|_{L^2} + \|\bar{z}\|_{L^2})$ for $\bar{z}$ supported in it.

(d) *Patching.* Choose a finite partition $\sum_\alpha \chi_\alpha^2 = 1$ subordinate to the regions of (a)–(c), by cutoffs that are functions of t alone on the compression, take the same value on the two sides of the seam over I (on the section the cutoff of (a) is $\chi(t')\psi(\theta')$ with $\psi = 1$ near 0 and $\psi = 0$ for $\theta' \geq \theta_1/2$), and have gradients supported in a fixed compact set. These cutoffs

preserve the domain of D_E , whose conditions are linear and pointwise in t , and $[D_E, \chi_\alpha]$ is multiplication by derivatives of χ_α . Its compression component has Y_E -norm at most $C(\int_I e^{2\phi_E} \mathfrak{s}_E \|\bar{U}_c\|^2 dt)^{1/2} \leq C\varepsilon^{1/4} \|\bar{z}\|_{X_E}$ by (7.15); its other components are bounded by the L^2 norm of $\bar{z}$ over a fixed compact set $\Omega_0 \subset \bar{X}_0$ containing these gradients and the section region of (a). Squaring and summing the estimates of (a)–(c) for $\chi_\alpha \bar{z}$, and absorbing the compression term for ε small, gives

$$(7.16) \quad \|\bar{z}\|_{X_E} \leq C(\|D_E \bar{z}\|_{Y_E} + \|\bar{z}\|_{L^2(\Omega_0)}),$$

where $\|\cdot\|_{L^2(\Omega_0)}$ is the L^2 norm of the section and of the gauge fields of X_0 over Ω_0 , and C and Ω_0 do not depend on ε . The same argument proves (7.16) for the formal adjoint $\hat{D}_E^*$ of the conjugated operator $\hat{D}_E = e^{\phi_E} D_E e^{-\phi_E}$ in the unweighted spaces: Proposition 7.4 contains its adjoint, Proposition 7.10(i) contains $\mathcal{D}^*$ with weights of either sign, and the fixed estimates of (c) hold for adjoints.

Step 2: the injectivity estimate. We claim that $\|z\|_{X_E} \leq C\|D_E z\|_{Y_E}$ for ε small. Otherwise there are $\varepsilon_i \rightarrow 0$ and $z_i = \bar{z}_i + \mathfrak{k}_E(v_i)$ on E_{ε_i} with $\|z_i\|_{X_E} = 1$ and $\|D_E z_i\|_{Y_E} \rightarrow 0$. After passing to a subsequence, $v_i \rightarrow v$, and the restrictions of $e^{\phi_E} \bar{z}_i$ to the section and to the gauge pieces of X_0 , which form a fixed space, converge weakly in H^1 and strongly in L^2 on compact subsets of $\bar{X}_0$, with traces converging strongly in L^2 on compact subarcs of I ; let $\bar{z}_0$ be the limit and $z_0 = \bar{z}_0 + \mathfrak{k}_0(v)$. Lemma 7.2 gives $\text{dist}(\nu_i(t), T_{q_{R_i}(t)} P) \leq c\|\mathcal{B}_t U_{i,c}(t)\|$ at every t , hence $\int_I e^{2\phi_E} \text{dist}(\nu_i, T_{q_{R_i}} P)^2 dt \leq c \sup \mathfrak{s}_E \rightarrow 0$; since $q_{R_i} \rightarrow q_V$ uniformly on compact subarcs, the section component of z_0 has its trace on I in $T_{q_V} L(C)$. The other boundary conditions are closed linear conditions on traces and pass to the limit. On every compact subset of X_0 the coefficients of D_E off the compression equal those of D_V for large i , so $D_V z_0 = 0$ in the interior and up to the boundary. By Fatou's lemma, the convergence of $\mathfrak{k}_0^{\varepsilon_i}(v_i)$ to $\mathfrak{k}_0(v)$ and (7.14), the field z_0 lies in the domain of D_V on the extended weighted space at weight δ_* , with limiting vector v . Since this operator is injective, $z_0 = 0$ and $v = 0$. Then $\|\bar{z}_i\|_{L^2(\Omega_0)} \rightarrow 0$, and $\|D_E \bar{z}_i\|_{Y_E} \leq \|D_E z_i\|_{Y_E} + C|v_i| \rightarrow 0$ by (7.14), so (7.16) gives $\|\bar{z}_i\|_{X_E} \rightarrow 0$; with $|v_i| \rightarrow 0$ this contradicts $\|z_i\|_{X_E} = 1$.

Step 3: surjectivity. By Step 2, D_E is injective with closed range for ε small. In the conjugated coordinates $\hat{z} = e^{\phi_E} \bar{z}$ the operator is $\hat{D}_E$ on the decaying part, and the asymptotic values act by $\hat{F}v = e^{\phi_E} D_E \mathfrak{k}_E(v)$. Suppose that for $\varepsilon_i \rightarrow 0$ there are f_i of unit norm orthogonal to the range. Then $\langle \hat{D}_E \hat{z}, f_i \rangle = 0$ for all decaying $\hat{z}$ in the domain and $\langle \hat{F}v, f_i \rangle = 0$ for all $v \in \mathbb{V}$. The first identity says that f_i is a weak solution of $\hat{D}_E^* f_i = 0$. At the fixed scale ε_i the local regularity of [11, Theorem 4.17 and Remark 4.19], applied with cutoffs, places f_i in the domain, and (7.16) for $\hat{D}_E^*$ bounds f_i in the graph norm by $C\|f_i\|_{L^2(\Omega_0)} \leq C$. After passing to a subsequence, f_i converges weakly on X_0 , and strongly in $L^2(\Omega_0)$, to some f , while its compression component satisfies $\int_I \mathfrak{s}_E \|f_{i,c}\|^2 dt \leq C\varepsilon_i^{1/2}$, by the argument of (7.15). Let ζ be a smooth compactly supported decaying field in the domain of $\hat{D}_V$, and lift it to E_{ε_i} by $\zeta_i = (\zeta, (0, h_{\nu_\zeta(t)}))$, where $\nu_\zeta(t) \in T_{q_V(t)} P$ is the section component of ζ on I . For large i the support of ζ lies where $q_{R_i} = q_V$, so ζ_i lies in the domain of $\hat{D}_E$; off the compression $\hat{D}_E \zeta_i = \hat{D}_V \zeta$, and its compression component $e^{\phi_E} D_t(0, h_{\nu_\zeta})$ has norm $O(\varepsilon_i^{1/4})$. Hence $0 = \langle \hat{D}_E \zeta_i, f_i \rangle \rightarrow \langle \hat{D}_V \zeta, f \rangle$. Likewise $0 = \langle \hat{F}v, f_i \rangle \rightarrow \langle e^{\phi_E} D_V \mathfrak{k}_0(v), f \rangle$, because $\hat{F}v$ converges strongly by (7.14). Smooth compactly supported fields are dense in the decaying domain, so f annihilates the range of the conjugated operator D_V on the extended weighted spaces, which is the whole space; hence $f = 0$. Then $\|f_i\|_{L^2(\Omega_0)} \rightarrow 0$, and the graph bound

gives $\|f_i\| \rightarrow 0$, a contradiction. Thus D_E is onto for ε small, and the bound of Step 2 is the bound of the proposition. $\square$

Lemma 7.14. *The auxiliary exterior operator D_E on the extended weighted spaces at the new ends is an isomorphism at weight δ_L on the new ends, and*

$$\|z_E\|_{X_E} \leq CA\|D_E z_E\|, \quad |v_-| + |v_+| \leq C\sqrt{A}\|D_E z_E\|,$$

where $v_{\pm}$ are the limiting values of z_E at the new ends. A solution of a source supported in $\{s \leq s'\}$ on a new end, with $s' \geq L/4 + c_R$ so that W_{ε}^E is flat beyond s' , differs from its stationary limit, a distance $\mathbb{T}$ beyond s' , by $O(e^{-\sigma_0 \mathbb{T}})$ times the bound, with σ_0 the rate of (7.10).

Proof. At the fixed weight δ_* the operator is an isomorphism with a bound independent of ε by Proposition 7.13. Passing from δ_* to δ_L on the two new ends is the argument of the proof of Proposition 7.10(iii) on each end, with the inverse of Proposition 7.13 in place of the fixed-weight transition inverse: the normal part is solved with no loss, and the stationary coefficient costs $\delta_L^{-1} = CA$ in the weighted L^2 norm, which gives the first bound, and $\delta_L^{-1/2} = C\sqrt{A}$ in the weighted supremum, which bounds the limiting values. On the part $s \leq L/4 + c_R$ of an end the configuration W_{ε}^E is not flat; in the identification of Step 1(b) of the proof of Proposition 7.13 its deviation adds to both equations of the triangular system (7.9) terms bounded by $Ce^{-\beta s}(|a| + \|z^n\|)$, and the scale adds the term $\mathbf{b}z^n$ with coefficient $O(\mathfrak{s}_E^{1/2}|\partial_s \log \mathfrak{s}_E|)$. Measuring a by $\sup_s e^{\delta_L s}|a(s)|$ and z^n in the weighted graph norm, these terms enter the normal equation and the stationary integral with coefficients whose L^1 and L^2 norms over the end are $O(e^{-\beta s_1} + \varepsilon^{1/4})$, and they are absorbed with no further loss in δ_L for s_1 fixed large and ε small; their contribution to the weighted L^2 norm of a is at most $C(1 + \varepsilon^{1/4}L^{1/2})$ times the weighted supremum, hence $O(\delta_L^{-1/2})$. The homogeneous-tail statement is the exponential decay of the normal part and the integration of the stationary equation from the source-free region. $\square$

Proposition 7.15. *For ε small, the linearization D_{ε} of the mixed equation at W_{ε} is invertible, with $\|D_{\varepsilon}^{-1}f\|_{X_{\varepsilon}} \leq CA\|f\|_{Y_{\varepsilon}}$, and $\text{ind } D_{\varepsilon} = 0$.*

Proof. Source removal. On each overlap put $u = (s - \alpha_L)/\mathbb{T}$ and fix cutoffs θ, ω of u with $\theta\omega = \omega$, θ supported in $(1/8, 7/8)$ and equal to one on $[1/4, 3/4]$, ω supported in $(1/4, 3/4)$ and equal to one on $[1/3, 2/3]$. Let S be the inverse of the stationary line problem on the overlap, extended to a complete line of constant scale beyond it, at weight $e^{\delta_L(s - \alpha_L)}$; by Proposition 7.10(iii), $\|S(\omega f)\| \leq CA\|f\|$. Put $z_{\text{loc}} = \sum_{\pm} \theta S(\omega f)$ and $f' = f - D_{\varepsilon} z_{\text{loc}} = (1 - \omega)f - [D_{\varepsilon}, \theta]S(\omega f)$. Since the derivative of θ is $O(\mathbb{T}^{-1}) = O(A^{-1})$, $\|f'\| \leq C\|f\|$, and $f' = 0$ on $1/3 \leq u \leq 2/3$.

Components. Split f' into sources f_E on the exterior and $f_{\pm}$ on the transitions, with the partition changing inside the band $1/3 \leq u \leq 2/3$ on which $f' = 0$. Solve $z_E = D_E^{-1}f_E$ by Lemma 7.14, and on each transition $z_{\pm} = k_{v_{\pm}} + \mathcal{D}^{-1}f_{\pm}$ with the limiting values $v_{\pm}$ of z_E . This imposes two separate equalities between the exterior and each transition; no equality between v_- and v_+ , and no rank condition on the exterior evaluation, is imposed. Both component solutions are homogeneous a distance at least $\mathbb{T}/6 - O(1)$ before the joining point $m_L = (\alpha_L + \beta_L)/2$, so by the homogeneous-tail statements, on a fixed collar of m_L , $\|z_E - z_{\pm}\| \leq CAe^{-\sigma_0 \mathbb{T}/6}\|f\|$; the stationary parts cancel because the limiting values agree.

Patching. With cutoffs β_E and $\beta_{\pm} = 1 - \beta_E$ of fixed slope at m_L , put $\mathcal{P}(f) = z_{\text{loc}} + \beta_E z_E + \sum \beta_{\pm} z_{\pm}$. The boundary conditions are linear, so $\mathcal{P}(f)$ lies in the domain, and

$D_\varepsilon \mathcal{P}(f) = f + \sum [D_\varepsilon, \beta_E](z_E - z_\pm)$, whose error has norm at most $CAe^{-\sigma_0 T/6} \|f\|$. Converting the component norms to X_ε with (7.13), the stationary pieces cost $C\sqrt{A}(|v_-| + |v_+|) \leq CA\|f\|$, so $\|\mathcal{P}(f)\| \leq CA\|f\|$. For large L the Neumann series gives a right inverse $\mathcal{Q}_\varepsilon = \mathcal{P}(1 + \mathcal{E}_\mathcal{P})^{-1}$, where $\mathcal{E}_\mathcal{P} = D_\varepsilon \mathcal{P} - 1$ has norm at most $CAe^{-\sigma_0 T/6}$, with $\|\mathcal{Q}_\varepsilon\| \leq CA$.

Injectivity. Let $D_\varepsilon z = 0$. On each overlap write $z = k_a + w$ with w normal; w decays exponentially toward m_L in terms of its collar norms, and $|a(m_L)| \leq C\|z\|$. Cut z at m_L into component fields completed by the stationary value $k_{a(m_L)}$; their sources are cutoff derivatives times $z - k_{a(m_L)}$, of norm $Ce^{-\sigma_0 T/6} \|z\|$. The component inverses and (7.13) give $\|z\| \leq CAe^{-\sigma_0 T/6} \|z\|$, so $z = 0$ for large L , since $T \sim L/4$ and $A = 1 + L$.

Index. D_ε is invertible, so its index is 0, in agreement with Theorem 6.6(i), since W_ε is a configuration in the class of V . $\square$

The loss of one factor A is sharp in these norms: a stationary field k_v multiplied by a cutoff of slope δ_L has norm ratio c/δ_L . It is harmless because the approximation error of W_ε is a positive power of ε (Section 8).

7.8. The scalar Laplacian on the glued domain. Let g_ε be the complete conformal metric on the gauge domain of X_ε used in Section 7.7. Choose a smooth $\lambda_\varepsilon > 0$ equal to $\mathfrak{s}$ on the compression, equal to one on the unchanged pieces and away from the ends of the transition cores, interpolated on fixed buffers and constant in the normal direction at the boundary, and put $\hat{g}_\varepsilon = \lambda_\varepsilon^{-2} g_\varepsilon$, $d\mu_\varepsilon = \lambda_\varepsilon^2 dV_{\hat{g}_\varepsilon}$. On the compression $\hat{g}_\varepsilon = dT^2 + g_C$ with $dT = dt/\mathfrak{s}$; the rescaled metric has uniformly bounded local geometry, and its large parts are product cylinders with compact cross-section joined through finitely many fixed cores. In dimension four the physical norm of a one-form is $\|a\|_{L^2(g_\varepsilon)}^2 = \int |a|_{\hat{g}_\varepsilon}^2 d\mu_\varepsilon$, which is $\int \mathfrak{s}|(p, b)|^2 dt dV_C$ on the compression, with $p = \mathfrak{s}\psi$.

Lemma 7.16 (scalar inverse). *Let $\mathcal{L}_\varepsilon = \lambda_\varepsilon^2 d_\mathbb{A}^{*, g_\varepsilon} d_\mathbb{A}$ with Neumann condition, for a reference $\mathbb{A}$ within a fixed small C^k -distance of W_ε in the rescaled metric. Then $\|\chi\|_{L^2(\mu_\varepsilon)} \leq C_P \|d_\mathbb{A} \chi\|_{L^2(\mu_\varepsilon)}$ and $\|\mathcal{L}_\varepsilon^{-1} f\|_{H_{\text{uloc}}^{m+1}(\hat{g}_\varepsilon)} \leq C_m \|f\|_{H_{\text{uloc}}^{m-1}(\hat{g}_\varepsilon)}$, with constants independent of ε ; the same holds with the weights of the original ends.*

Proof. The quadratic form of $\mathcal{L}_\varepsilon$ is $\int |d_\mathbb{A} \chi|^2 d\mu_\varepsilon$. On each product piece of the compression the Poincaré inequality is the spatial Neumann gap, uniform over P because every flat connection on C is irreducible, integrated in time. The transition cores have compact Neumann inequalities once enlarged to contain an irreducible slice (on each core λ_ε is comparable with one), the tails of the original ends have their spatial gaps, and a fixed compact core of the unchanged gauge domain, enlarged to contain an irreducible slice, has its Neumann inequality. Each of these regions R_k therefore carries an inequality $\|\chi\|_{L^2(R_k)} \leq C \|d_\mathbb{A} \chi\|_{L^2(R_k)}$ with no mean-value term, since $\mathbb{A}$ is irreducible on every slice, and the regions cover with overlap at most N_0 , so

$$\|\chi\|_{L^2(\mu_\varepsilon)}^2 \leq \sum_k \|\chi\|_{L^2(R_k)}^2 \leq C \sum_k \|d_\mathbb{A} \chi\|_{L^2(R_k)}^2 \leq CN_0 \|d_\mathbb{A} \chi\|_{L^2(\mu_\varepsilon)}^2.$$

For the inverse, cover by unit rescaled patches Q_j ; the number of patches in a distance shell grows polynomially. Conjugating the form by $e^{\gamma d_j}$, with d_j a smoothed distance from Q_j and γ small, and using the variational inverse and local elliptic regularity gives

$$\|\mathcal{L}_\varepsilon^{-1} f_j\|_{H^{m+1}(Q_\ell)} \leq C_m e^{-\gamma d(Q_\ell, Q_j)} \frac{\lambda_\varepsilon(Q_j)}{\lambda_\varepsilon(Q_\ell)} \|f_j\|_{H^{m-1}(Q_j)}$$

for f_j supported in Q_j . Since $\partial_T(\mathfrak{s}^{-1}) = -h$, the function λ_ε^{-1} is uniformly Lipschitz in the rescaled distance and bounded below, so the ratio is at most $C(1 + d(Q_\ell, Q_j))$, which is absorbed by half the exponential. Writing $f = \sum_j f_j$ with f_j supported in Q_j and $\|f_j\|_{H^{m-1}} \leq C\|f\|_{H^{m-1}(Q'_j)}$ for a fixed enlargement Q'_j ,

$$\begin{aligned} \|\mathcal{L}_\varepsilon^{-1}f\|_{H^{m+1}(Q_\ell)} &\leq C_m \sum_j e^{-\gamma d(Q_\ell, Q_j)/2} \|f_j\|_{H^{m-1}(Q_j)} \\ &\leq C_m \left(\sup_{n \geq 0} \#\{j : n \leq d(Q_\ell, Q_j) < n+1\} e^{-\gamma n/4} \right) \sum_{n \geq 0} e^{-\gamma n/4} \|f\|_{H_{\text{uloc}}^{m-1}}, \end{aligned}$$

and the supremum is finite, uniformly in ℓ and ε , because the shell counts grow polynomially in n . This is the inverse bound; uniqueness in the uniformly local space follows from the quadratic identity with a small decreasing exponential weight. The weights of the original ends, whose logarithmic gradients are smaller than γ , are handled in the same way. $\square$

8. THE COLLAPSE THEOREM: GLUING AND SURJECTIVITY

Throughout, V is a regular index-zero solution on X_0 with limits $q_\pm$ at the new ends, $L = \log(1/\varepsilon)$ and $A = 1 + L$. Constants C, c depend on the fixed data and on the finite set $\mathcal{M}^0(X_0)$ but not on ε , on neck lengths, or on the minimum of the compression scale.

The section has two parts. Existence (Section 8.1 and Proposition 8.3): the approximate solution W_ε is corrected by a contraction in strong norms, whose inverse loses the factor A^{m+1} , so the solution is unique only in a ball of radius shrinking like a power of $1/A$. Surjectivity (Proposition 8.12): an arbitrary solution near V is first shown, by cutting it along the necks and comparing the pieces with the components, to lie within a positive power of ε of W_ε , by estimates that lose no logarithm; only then does it enter the uniqueness ball.

8.1. Strong norms and the strong inverse. *Strong norms.* On the small compression, with w the fixed end weight (equal to one on the region $N \cap X_\varepsilon$ where X_ε differs from X_0 , Definition 6.2(iv)), put

$$\begin{aligned} \|f\|_{Y_\varepsilon^m}^2 &= \sum_{i+r \leq m} \|\nabla_t^i(w\mathfrak{s}^{1/2}f)\|_{L_t^2 H^r(C)}^2, \\ \|z\|_{X_\varepsilon^m}^2 &= \|wv\|_{H_t^m}^2 + \sum_{i=0}^m \|w\mathfrak{s}^{-1/2}\nabla_t^i W\|_{L_t^2 H^{m+1-i}(C)}^2 + \|w\mathfrak{s}^{1/2}\nabla_t^{m+1}U\|_{L_t^2 L^2(C)}^2, \end{aligned}$$

and on the other gauge regions and the section weighted H^m and H^{m+1} , patched by cutoffs of bounded overlap, with the free limiting vectors at the original ends added. For $m = 0$ this is equivalent to X_ε . On the region $N \cap X_\varepsilon$

$$(8.1) \quad \|U\|_{H^m} + \|\mathfrak{s}^{1/2}U\|_{H^{m+1}} + \|\mathfrak{s}^{-1/2}U_\perp\|_{H^m} + \|\mathfrak{s}^{-1/2}\nabla_C U_\perp\|_{H^m} \leq C_m \|z\|_{X_\varepsilon^m},$$

and Sobolev embedding on unit patches gives unscaled C^1 bounds for the spatial coefficients, p and the section variation.

Proposition 8.1 (strong inverse). $D_\varepsilon: X_\varepsilon^m \rightarrow Y_\varepsilon^m$ is invertible with $\|D_\varepsilon^{-1}\| \leq C_m A^{m+1}$.

Proof. Let $\mathcal{V}$ be a global tangential differential operator equal to ∇_t where $\mathfrak{s} \leq \mathfrak{s}_0/2$ on the compression, extended over the adjacent section collar by the corresponding boundary derivative, cut off where $\mathfrak{s} \geq \mathfrak{s}_0$ and in the collar interior, and vanishing on the other regions

and on the supports of all perturbations. By Lemma 7.11, $\mathcal{V}$ preserves the linear domain. We prove by induction on $j \leq m$ that

$$(8.2) \quad \|\mathcal{V}^j z\|_{X_\varepsilon} \leq C_j \mathbf{A}^{j+1} \|D_\varepsilon z\|_{Y_\varepsilon^j}.$$

For $j = 0$ this is Proposition 7.15. Commuting D_ε through $\mathcal{V}^j$ gives

$$D_\varepsilon(\mathcal{V}^j z) = \mathcal{V}^j D_\varepsilon z + \sum_{i < j} (\mathfrak{s}^{-1} a_{ij}(t) \mathcal{B}_t^{[j-i]} \mathcal{V}^i U + b_{ij} \mathcal{V}^i z),$$

where a_{ij} and b_{ij} are bounded, the factor $\mathfrak{s}^{-1}$ comes from the derivatives of the scale, and b_{ij} collects the terms of order zero. By Lemma 7.11, $\mathcal{B}_t^{[j-i]}$ annihilates the harmonic fields and $\|\mathcal{B}_t^{[j-i]} U\| \leq C \|\mathcal{B}_t U\|$, so no factor $\mathfrak{s}^{-1}$ multiplies a slow term and

$$\|\mathfrak{s}^{-1} a_{ij} \mathcal{B}_t^{[j-i]} \mathcal{V}^i U\|_{Y_\varepsilon} \leq C \|\mathfrak{s}^{-1/2} \mathcal{B}_t \mathcal{V}^i U\| \leq C \|\mathcal{V}^i z\|_{X_\varepsilon}.$$

Applying Proposition 7.15 to $\mathcal{V}^j z$, which costs one factor $\mathbf{A}$, and the induction hypothesis to the terms with $i < j$,

$$\|\mathcal{V}^j z\|_{X_\varepsilon} \leq C \mathbf{A} \left(\|D_\varepsilon z\|_{Y_\varepsilon^j} + \sum_{i < j} C_i \mathbf{A}^{i+1} \|D_\varepsilon z\|_{Y_\varepsilon^i} \right) \leq C_j \mathbf{A}^{j+1} \|D_\varepsilon z\|_{Y_\varepsilon^j},$$

which is (8.2). The spatial derivatives cost nothing: on the compression they are recovered from the equation by (7.12); for the section, the derivative in the direction normal to the seam is recovered from the Cauchy–Riemann equation and the tangential derivative, and the seam trace of $\nabla_t^j \nu$ is controlled through the harmonic matching, which ∇_t preserves by Lemma 7.11; both bounds are uniform in $\mathfrak{s}$. In the regions where the geometry is uniformly nondegenerate, ordinary mixed elliptic estimates give H^{m+1} control by $\|D_\varepsilon z\|_{Y^m} + \|z\|_{X_\varepsilon}$. Summing gives the bound with $j = m$. $\square$

The exact matching chart. Along $q(t)$, let $B_t(v)$ be the flat connection in the spatial Neumann Coulomb slice of $B_t = B_{q(t)}$ with bulk harmonic coordinate v : $F_{B_t(v)} = 0$, $d_{B_t}^*(B_t(v) - B_t) = 0$, vanishing normal component, $\ell_t(0, B_t(v) - B_t) = v$, $D_v B_t(0) = j_t$. Let e_t be a section chart whose restriction to $\mathcal{L}_t$ is the boundary class of $B_t(v)$. For $U = j_t v + U_\perp$ with $U_\perp = (p, w)$ put

$$\Phi_t(U) = B_t + (B_t(v) - B_t + T_{t,v} w + \mathcal{R}_t(v, w)) + (\Psi_t + p/\mathfrak{s}) dt,$$

where $T_{t,v}$ is the transport of the complementary field to $B_t(v)$ by the surface Hodge decompositions on the parallel surfaces of a fixed collar of ∂C , blended to the identity in the interior and leaving the normal component unchanged, with the harmonic summands identified so that the class of the trace $(T_{t,v} w)|_\Sigma$ is $de_t(v) \tau_t w$; and $\mathcal{R}_t(v, w)$ is the surface correction defined below, which is tangential, is supported in the collar, makes the trace flat with class $e_t(v + \tau_t w)$, and satisfies $\mathcal{R}_t(v, 0) = 0$ and $D_w \mathcal{R}_t(v, 0) = 0$. On the section use e_t at the seam extended into a collar. In this chart the matching and flatness conditions hold exactly, and the mixed operator $\mathfrak{B}_t(v, U_\perp)$ satisfies

$$(8.3) \quad \mathfrak{B}_t(v, 0) = 0, \quad D_{U_\perp} \mathfrak{B}_t(0, 0) = \mathcal{B}_t,$$

so the quadratic part vanishes on the slow directions.

The surface correction. Let $[0, r_1) \times \partial C$ be a product collar of ∂C with normal coordinate r , write $\Sigma_r = \{r\} \times \partial C$, and let χ be a cutoff of r equal to 1 near 0 and to 0 for $r \geq r_1/2$. For given t, v and r let $\alpha = \alpha_{t,v,r}$ be the restriction of $B_t(v)$ to Σ_r . It is flat and, by (C1), irreducible on every component, so $\Delta_\alpha = d_\alpha d_\alpha^*$ is invertible on two-forms and $\ker d_\alpha^*$ on one-forms is the sum of the harmonic space $\mathcal{H}_\alpha$ and $\text{im } d_\alpha^*$. Let $\mathfrak{i}_{t,v,r}: T_{q(t)} \mathbf{M} \rightarrow \mathcal{H}_\alpha$ be

the isomorphism given by $de_t(v)$ and the identification of the cohomology of Σ_r with that of Σ by the flat connection $B_t(v)$ on the collar (on the torus faces $\mathcal{H}_\alpha = 0$). For a tangential one-form ϕ on Σ_r small in $H^{1/2}$ and $\vartheta \in T_{q(t)}\mathbf{M}$ small, let $\zeta = \mathcal{S}_\alpha(\phi, \vartheta)$ be the small solution in $\ker d_\alpha^*$ of the *surface flatness equation*

$$(8.4) \quad d_\alpha \zeta + \frac{1}{2}[(\phi + \zeta) \wedge (\phi + \zeta)] = 0, \quad \text{harmonic part of } \zeta = \mathbf{i}_{t,v,r} \vartheta.$$

Equivalently $\zeta = \mathbf{i}_{t,v,r} \vartheta + \mathbf{K}_\alpha(\phi + \zeta, \phi + \zeta)$ with $\mathbf{K}_\alpha(a, b) = -\frac{1}{2}d_\alpha^* \Delta_\alpha^{-1}[a \wedge b]$, a bilinear operator of order -1 on Σ_r ; since $[\cdot \wedge \cdot]$ maps $H^{1/2} \times H^{1/2}$ to L^2 on a surface, the contraction principle gives $\mathcal{S}_\alpha$, smooth in (ϕ, ϑ) and in the parameters, with values in H^1 and $\mathcal{S}_\alpha(0, 0) = 0$. By construction $F_{\alpha+\phi+\zeta} = d_\alpha \phi$; in particular $\alpha + \phi + \mathcal{S}_\alpha(\phi, \vartheta)$ is flat whenever $d_\alpha \phi = 0$. Let $\phi_{t,r}$ be the tangential part of $T_{t,v}w$ on Σ_r . On $\Sigma_0 = \partial C$ it is closed, because w has closed traces and $T_{t,v}$ preserves closedness. Let $\vartheta = \vartheta_t(v, w)$ be the small solution of the finite-dimensional equation

$$(8.5) \quad [\alpha_{t,v,0} + \phi_{t,0} + \mathcal{S}_{\alpha_{t,v,0}}(\phi_{t,0}, \vartheta)] = e_t(v + \tau_t w),$$

which exists by the implicit function theorem, since $\vartheta \mapsto [\alpha_{t,v,0} + \mathcal{S}_{\alpha_{t,v,0}}(0, \vartheta)]$ is the harmonic chart of $\mathbf{M}$ at $e_t(v)$, with derivative $de_t(v)$ at 0; the class map is smooth on flat connections of class $H^{1/2}$ near $\alpha_{t,v,0}$. Define

$$\mathcal{R}_t(v, w) = \chi(r) \mathcal{S}_{\alpha_{t,v,r}}(\phi_{t,r}, \vartheta_t(v, w)).$$

Then $\mathcal{R}_t$ is tangential, the trace $\alpha_{t,v,0} + \phi_{t,0} + \mathcal{R}_t|_{\partial C}$ is flat with class $e_t(v + \tau_t w)$, and $\mathcal{R}_t(v, 0) = 0$. At $w = 0$ the derivative in w of the left side of (8.5) is the class of $\phi_{t,0}(\dot{w})$ plus $de_t(v)\dot{\vartheta}$, and that of the right side is $de_t(v)\tau_t \dot{w}$, which is the class of $\phi_{t,0}(\dot{w})$ by the normalization of $T_{t,v}$. Hence $D_w \vartheta_t(v, 0) = 0$ and $D_w \mathcal{R}_t(v, 0) = 0$. The correction is computed surface by surface, with r as a parameter, and the only boundary quantity it uses is the finite-dimensional vector $\vartheta_t(v, w)$.

Lemma 8.2 (local quadratic estimate). *Let $Q = [0, 2] \times C$, $m \geq 4$, and let q be a path in P with $\|q\|_{C^{m+1}} \leq K$. Write $\|w\|_{H_+^m} = \|w\|_{H^m(Q)} + \|\nabla_C w\|_{H^m(Q)}$. There are r_0, C depending on m, K and the odd compression body such that for $\|v\|_{H^m} + \|U_\perp\|_{H_+^m} \leq r_0$,*

$$\|\mathfrak{B}(v, U_\perp) - \mathcal{B}W\|_{H^m(Q)} \leq C(\|v\|_{H^m} + \|U_\perp\|_{H^m})\|U_\perp\|_{H_+^m},$$

with the corresponding difference estimate, and the surface correction satisfies

$$\|\mathcal{R}(v, w)\|_{H_+^m} \leq C\|w\|_{H^m}\|w\|_{H_+^m},$$

$$\|D\mathcal{R}(v, w)[\dot{v}, \dot{w}]\|_{H_+^m} \leq C(\|w\|_{H_+^m}(\|\dot{v}\|_{H^m} + \|\dot{w}\|_{H^m}) + \|w\|_{H^m}\|\dot{w}\|_{H_+^m}).$$

Proof. Let $Q_1 = [0, 2] \times [0, r_1] \times \partial C$ be the collar part of Q . We use the following fact. (S) Let $\mathbf{L}_{t,r}$ be a family of operators on sections over ∂C of order $-k \leq 0$, that is, bounded $H^\sigma(\partial C) \rightarrow H^{\sigma+k}(\partial C)$ for $0 \leq \sigma \leq m+1$, whose coefficients are smooth functions of (t, r) and of $v(t)$. Then $f \mapsto \mathbf{L}_{t,r}f(t, r)$ is bounded on $H^m(Q_1)$ and on $H_+^m(Q_1)$, with a norm depending on $\|v\|_{H^m}$; for $k \geq 1$ so is $f \mapsto \nabla_{\partial C} \mathbf{L}_{t,r}f(t, r)$. Indeed $\partial_t^i \partial_r^j (\mathbf{L}_{t,r}f) = \sum \binom{i}{i'} \binom{j}{j'} (\partial_t^{i'} \partial_r^{j'} \mathbf{L}_{t,r})(\partial_t^{i-i'} \partial_r^{j-j'} f)$; each $\partial_t^{i'} \partial_r^{j'} \mathbf{L}_{t,r}$ is again of order $-k$, with operator norm bounded for $i' \leq m-1$ and square integrable in t for $i' = m$, because $v \in H^m \subset C^{m-1}$; surface derivatives are applied after the operator; and integration over (t, r) gives the claim, the factor with $i' = m$ being paired with $f \in L_{t,r}^\infty L^2(\partial C)$ for the H^m bound and with $f \in L_{t,r}^\infty H^1(\partial C)$ for the H_+^m bound; both hold for $m \geq 3$ by the Sobolev inequality in the two variables (t, r) . No trace of f on a surface Σ_r is taken: r is a parameter throughout. The algebra property of $H^m(Q_1)$, $m \geq 4 > \frac{1}{2} \dim Q_1$, is used for products.

The transport. $T_{t,v}$ is of order zero in the sense of (S), and $T_{t,v} - 1$ has norm $O(\|v\|_{H^m})$ on H^m and H_+^m . Hence $\|\phi\|_{H_+^m(Q_1)} \leq C\|w\|_{H_+^m}$ and $\|\phi\|_{H^m(Q_1)} \leq C\|w\|_{H^m}$.

The finite-dimensional parameter. The left side of (8.5) is a smooth function of v , ϑ and $\phi_{t,0} \in H^{1/2}(\partial C)$, and $\vartheta_t(v, w)$ vanishes to second order in $\phi_{t,0}$ at $w = 0$. By the trace theorem, for $i \leq m-1$, $\|\partial_t^i \phi_{t,0}\|_{L_t^2 H^{1/2}} \leq C\|\partial_t^i w\|_{L_t^2 H^1(C)} \leq C\|w\|_{H^m}$, while for $i = m$ the bound is $C\|\partial_t^m w\|_{L_t^2 H^1(C)} \leq C\|w\|_{H_+^m}$. In the chain rule for $\partial_t^m \vartheta$ the m -th derivative of $\phi_{t,0}$ occurs only linearly, multiplied by a coefficient that vanishes at $\phi_{t,0} = 0$. Hence

$$\begin{aligned} \|\vartheta\|_{H^m} &\leq C\|w\|_{H^m}\|w\|_{H_+^m}, \\ \|D\vartheta[\dot{v}, \dot{w}]\|_{H^m} &\leq C(\|w\|_{H_+^m}(\|\dot{v}\|_{H^m} + \|\dot{w}\|_{H^m}) + \|w\|_{H^m}\|\dot{w}\|_{H_+^m}). \end{aligned}$$

This is the only place where a boundary trace enters; it enters through a finite-dimensional quantity, and the normal derivative contained in H_+^m controls it.

The surface correction. Solve $\zeta = \mathfrak{i}\vartheta + K_\alpha(\phi + \zeta, \phi + \zeta)$ in $H_+^m(Q_1)$. By (S) with $k = 1$ and the algebra property, $\|K_\alpha(a, b)\|_{H^m} + \|\nabla_{\partial C} K_\alpha(a, b)\|_{H^m} \leq C\|a\|_{H^m}\|b\|_{H^m}$, and the normal derivative $\partial_r K_\alpha(a, b) = (\partial_r K_\alpha)(a, b) + K_\alpha(\partial_r a, b) + K_\alpha(a, \partial_r b)$ gives $\|K_\alpha(a, b)\|_{H_+^m} \leq C(\|a\|_{H^m}\|b\|_{H_+^m} + \|a\|_{H_+^m}\|b\|_{H^m})$. Since $\|\mathfrak{i}\vartheta\|_{H_+^m} \leq C\|\vartheta\|_{H^m}$, the contraction principle gives, for $\|\phi\|_{H_+^m} + \|\vartheta\|_{H^m}$ small, $\|\zeta\|_{H_+^m} \leq C(\|\vartheta\|_{H^m} + \|\phi\|_{H^m}\|\phi\|_{H_+^m})$, and the derivative of ζ in (ϕ, ϑ) satisfies the corresponding bilinear bound; the dependence on v through α is smooth, and $\mathcal{S}_\alpha(0, 0) = 0$ for every α . With the bounds for ϕ and ϑ , this proves the two estimates for $\mathcal{R} = \chi\zeta$.

The quadratic estimate. Write $U_\perp = (p, w)$ and let $a = B_t(v) - B_t + T_{t,v}w + \mathcal{R}_t(v, w)$ be the spatial increment of the chart. The spatial part of the mixed operator in the chart is $\mathfrak{B}_t(v, U_\perp) = \mathcal{B}_t(p, a) + (0, -\frac{1}{2}[a \wedge a] + [a, p])$, with the conventions of Section 7.1. As $B_t(v)$ is flat and in the Coulomb slice, $\mathfrak{B}_t(v, 0) = 0$, and

$$\begin{aligned} \mathfrak{B}_t(v, U_\perp) - \mathcal{B}_t U_\perp &= \mathcal{B}_t(0, (T_{t,v} - 1)w + \mathcal{R}_t(v, w)) \\ &\quad + (0, -[b_v \wedge (T_{t,v}w + \mathcal{R}_t)] - \frac{1}{2}[(T_{t,v}w + \mathcal{R}_t) \wedge (T_{t,v}w + \mathcal{R}_t)] + [a, p]), \end{aligned}$$

where $b_v = B_t(v) - B_t = O(|v|)$. The first term is bounded in $H^m(Q)$ by $C\|v\|_{H^m}\|w\|_{H_+^m} + C\|\mathcal{R}\|_{H_+^m}$, and the bracket terms by $C(\|v\|_{H^m} + \|U_\perp\|_{H^m})\|U_\perp\|_{H^m}$, by the algebra property. With the bound for $\mathcal{R}$ this is the first estimate. The difference estimate follows by applying the same argument to the derivative along the segment joining $(v, U_\perp)$ and $(v', U'_\perp)$, using the bound for $D\mathcal{R}$. $\square$

The same surface-by-surface argument gives the estimates used at first order and for the temporal component. For $\|\phi\|_\infty$ and $|\vartheta|$ small, $\|\mathcal{S}_\alpha(\phi, \vartheta)\|_{H^1(\Sigma_r)} \leq C(|\vartheta| + \|\phi\|_\infty\|\phi\|_{L^2})$ and $|\vartheta_t(v, w)| \leq C\|w(t)\|_{H^1(C)}^2$. The temporal component of the chart nonlinearity, consisting of $\partial_t(B_t(v) - B_t - j_t v)$, $\partial_t((T_{t,v} - 1)w)$, $\partial_t \mathcal{R}_t(v, w)$ and $\mathfrak{s}^{-1}[a, p]$, carries the weight $\mathfrak{s}^{1/2}$ with no factor $\mathfrak{s}^{-1}$ on slow terms, and its top time derivatives are controlled by $\mathfrak{s}^{1/2}\|\nabla_t^{m+1}U\|$. At the level of first-order norms this gives $\|N(z)\|_Y \leq C\mathbf{d}(z)\|z\|_X$. Here, for a field z on any of the domains X_ε , E_ε or $\mathcal{T}_\pm$, the coefficient size $\mathbf{d}(z)$ is the supremum over the unit patches of that domain of $\|v\|_{C^1} + \|U_\perp\|_{C^0}$ on patches of the compression, where $z = (v, U_\perp)$ in the exact matching chart, of $\|a\|_{C^0}$ on the other gauge patches, and of $\|\nu\|_{C^0}$ on section patches (the patches of $N \cap X_\varepsilon$ are described next). The same argument on the other patches, where the nonlinearity is the quadratic term of the ordinary mixed equation, gives the bound there. This is the form used in Lemmas 8.6 and 8.9 and in the proof of Proposition 8.12.

On the region $N \cap X_\varepsilon$ cover by unit patches Q_k in the conformal cylinder coordinate; all logarithmic derivatives of $\mathfrak{s}$ are bounded, so $\mathfrak{s}/\mathfrak{s}(k) \in [e^{-2H}, e^{2H}]$ on Q_k , and multiplication by $(\mathfrak{s}/\mathfrak{s}(k))^{\pm 1/2}$ is bounded on $H^m(Q_k)$ uniformly. Freezing the weights at $\mathfrak{s}(k) \leq 1$ and summing Lemma 8.2 over patches gives the quadratic estimate $\|N_\varepsilon(z) - N_\varepsilon(z')\|_{Y^m} \leq C(\|z\|_{X^m} + \|z'\|_{X^m})\|z - z'\|_{X^m}$ with constants independent of ε and of the neck lengths.

8.2. The approximate solution and existence. Let $R = \varepsilon^{1/4}$. Near each endpoint, for $x \leq R$, replace V by its limiting flat pair $(B_{q_\pm}, q_\pm)$ using a cutoff of fixed slope in the logarithmic coordinate; since V converges to its limits at rate $\beta > 0$ (Proposition 6.15(iii)), this changes V by $O(R^\beta)$ in every C^k norm. Over $I \times C$ use the flat lift $B_{q_R(t)}$ in the covariant Neumann slice of the boundary values $q_R(t)$ of the truncated section, with temporal component chosen so that $\partial_t B - d_B \Psi = h_{\dot{q}_R}$. On the necks and transitions use the flat constant pair. The only errors are the truncation error and the horizontal velocity of the flat lift, whose weighted size is $\mathfrak{s}^{1/2}|\dot{q}_R| \leq C\varepsilon^{1/2}r^{\beta-1/2}$ on the compression; since ∇_t preserves the harmonic extension, the same bound holds for all its derivatives. Integrating,

$$(8.6) \quad \|\mathcal{F}_\varepsilon(W_\varepsilon)\|_{Y_\varepsilon^m} \leq C_m e_\varepsilon, \quad e_\varepsilon = R^\beta + \sqrt{\varepsilon K_\beta(R)}, \quad K_\beta(R) = 1 + \int_R^{r_0} r^{2\beta-2} dr,$$

and $e_\varepsilon \leq C\varepsilon^{\sigma_1}$ for some $\sigma_1 > 0$.

Proposition 8.3 (existence). *Let $\varrho \leq r_0$ be a radius below which the summed quadratic and difference estimates hold. For ε small there is a unique solution $\mathcal{G}_\varepsilon(V) = W_\varepsilon + z^*$ with $\|z^*\|_{X^m} \leq \varrho/(C_m \mathbf{A}^{m+1})$, and in fact $\|z^*\|_{X^m} \leq C_m \mathbf{A}^{m+1} e_\varepsilon$. It is regular.*

Proof. In the exact matching chart the equation is $\mathcal{F}_\varepsilon(W_\varepsilon) + D_\varepsilon z + N_\varepsilon(z) = 0$. By Proposition 8.1, the quadratic estimate and (8.6), the map $z \mapsto -D_\varepsilon^{-1}(\mathcal{F}_\varepsilon(W_\varepsilon) + N_\varepsilon(z))$ is a contraction of the ball of radius $\varrho/(C_m \mathbf{A}^{m+1})$ for small ε , since $\mathbf{A}^{2m+2}e_\varepsilon \rightarrow 0$. Regularity follows from surjectivity of the linearization at nearby points. $\square$

8.3. The endpoint necks. On each endpoint neck, at a time s , the restriction of a configuration to the slice is a triple $c = (B_f, B_c, \nu)$ of a connection on the fixed copy of C , a connection on the collapsing copy, and a path in the moduli space of Σ with $\nu(0)$ and $\nu(\pi)$ the restrictions of B_f and B_c . The critical family is $\mathcal{F} = \{(B_q, B_q, q) : q \in P\}$. Gluing the triple gives one connection on the double of C along Σ with a collar of length π inserted, and the mixed action becomes the Chern–Simons functional of that three-manifold. For $C = C_1$ it has two torus boundary components carrying the flat condition; since the flat connection on the odd bundle over T^2 is isolated with $H^1(T^2; \text{ad}) = 0$, the functional is still single valued on a gauge-invariant neighbourhood of $\mathcal{F}$, and its critical set there is $\mathcal{F}$, of Morse–Bott type. Since $\mathcal{F}$ is connected, the action is constant on it, and we normalize it to vanish there.

Proposition 8.4 (neck decay). *(i) There are $\delta, C, \mathfrak{s}_2 > 0$ such that for $\mathfrak{s} \in (0, \mathfrak{s}_2]$ and all triples within δ of $\mathcal{F}$ in the H^1 norm of the fixed metric g_C on both branches, $|a(c)| \leq C\|\nabla_{\mathfrak{s}} a(c)\|_{\mathfrak{s}}^2$, where the gradient is taken in the metric weighting the collapsing component by $\mathfrak{s}$.*

(ii) Let $c(s)$, $s \in [s_0, s_1]$, solve the neck equation with $\mathfrak{s}(s) = \mathfrak{s}_b e^{\pm s} \leq \mathfrak{s}_2$, all slices within δ of $\mathcal{F}$, and energy e . Let $p(s) \in P$ be the projection determined by the flat projection of the fixed branch B_f . Then the energy on $[s_0 + r, s_1 - r]$ is at most $e e^{-\gamma r}$ and

$$\text{length}_P(p|_{[s_0+r, s_1-r]}) \leq C\sqrt{e} e^{-\gamma r/2},$$

with C, γ independent of $s_1 - s_0$ and of $\mathfrak{s}_b$.

Proof. (i) Let $\mathfrak{s}_2$ be the constant $\mathfrak{s}_0$ of Proposition 7.9. The action $a(c)$ does not depend on $\mathfrak{s}$, and the squared gradient norm $\|F_{B_f}\|^2 + \mathfrak{s}^{-1}\|F_{B_c}\|^2 + \|\partial_\theta \nu\|^2$ (up to the slice terms) is nonincreasing in $\mathfrak{s}$. The neighbourhood in the statement does not depend on $\mathfrak{s}$, so the inequality at $\mathfrak{s} = \mathfrak{s}_2$ implies it for all $\mathfrak{s} \leq \mathfrak{s}_2$. At $\mathfrak{s} = \mathfrak{s}_2$ write c as the image of x under the exact matching chart (8.3) centred at a point $c_0 \in \mathcal{F}$, with x in the Coulomb slice and in the linear domain of $A_{\mathfrak{s}_2}$, and orthogonal to $T_{c_0}\mathcal{F}$; in this chart the matching and flatness conditions hold exactly, and the decomposition exists for δ small because P is compact. The Hessian at c_0 is $A_{\mathfrak{s}_2}$, whose kernel is $T_{c_0}\mathcal{F}$ and whose spectrum meets $(-\gamma_0, \gamma_0)$ only in 0, with γ_0 the gap of Proposition 7.9; with the elliptic estimate at the fixed scale $\mathfrak{s}_2$ this gives $\|x\|_{H^1} \leq C\|A_{\mathfrak{s}_2}x\|$. Taylor expansion, with the multiplication $H^1 \times H^1 \rightarrow L^2$ in dimension three and the chart bounds of Lemma 8.2, gives

$$|a(c) - \frac{1}{2}\langle A_{\mathfrak{s}_2}x, x \rangle| \leq C\|x\|_{H^1}^3, \quad \|\nabla_{\mathfrak{s}_2}a(c) - A_{\mathfrak{s}_2}x\| \leq C\|x\|_{H^1}^2.$$

Hence $\|x\|_{H^1} \leq C\|\nabla_{\mathfrak{s}_2}a(c)\| + C\|x\|_{H^1}^2$, so $\|x\|_{H^1} \leq 2C\|\nabla_{\mathfrak{s}_2}a(c)\|$ once $\|x\|_{H^1} \leq 1/(2C)$, and

$$|a(c)| \leq C\|x\|_{H^1}^2 \leq C'\|\nabla_{\mathfrak{s}_2}a(c)\|^2.$$

(ii) Along a solution $a' = -\sigma^2$ with $\sigma^2 = \|\xi_f\|^2 + \mathfrak{s}\|\xi_c\|^2 + \|\partial_s \nu\|^2$, the kinetic energy density. By (i), $a' \leq -a/C$ from both ends of the neck, which gives the exponential decay of the energy. The projection p determined by the fixed branch satisfies $|p'| \leq C\sigma$, because the fixed branch is measured in the unweighted metric; integrating $\sigma \leq \sqrt{-a'}$ over dyadic intervals gives the length bound with no factor depending on the length of the neck. $\square$

The choice of projection matters: with the nearest-point projection of the collapsing branch in the unscaled metric, the method only gives the bound multiplied by $(\inf \mathfrak{s})^{-1/2}$.

8.4. Convergence to the flat family on the necks. Let U_ν be index-zero solutions on X_{ε_ν} converging to V as in Proposition 6.15. By Theorem 6.6(ii) there is no energy loss and no breaking, so the energy e_ν of U_ν on the part of each endpoint neck at distance at least s_0 from the exterior tends to the tail energy of V , which is small for s_0 large, and on windows $[S, L - S]$ with $S \rightarrow \infty$ it tends to zero.

Lemma 8.5 (convergence to the flat family). *Fix $e_0 > 0$ small. For ν large, on every interval of the neck on which the energy is at most e_0 , the following hold uniformly in the position and in ν :*

- (i) *the spatial curvatures satisfy $\sup_s \|F_{B_j(s)}\|_{L^2(C)} \leq C\sqrt{e}$, $j = f, c$, and there are gauges in which $B_j(s) = B_{q_j(s)} + \beta_j(s)$ with β_j in the Neumann slice orthogonal to the harmonic fields and $\|\beta_j(s)\|_{H^1} \leq C\sqrt{e}$;*
- (ii) *on the collapsing seam the conclusions of Lemma 6.14 hold with $E(b, r_0) \leq e$; in particular the section is uniformly Hölder on a fixed collar, and $\sup_\theta d(u(s, \theta), q_f(s)) + d_P(q_c(s), q_f(s)) \leq C\sqrt{e}$.*

Proof. (i) Near a seam center b use the holomorphic coordinate Φ_b with $\Phi'_b(s) = 1/\mathfrak{s}(s)$; in it the collapsing compression and the section have exactly the fixed product mixed geometry, and fixed patches correspond to regions of size $C\mathfrak{s}_b$. The energy on each such patch is at most e , by conformal invariance. The mixed mean-value inequality [12, Theorem 4.2], the interior estimate [12, Lemma 4.50] and the derivative bounds of [12, Lemma 4.25] give the curvature bound after integration over a finite cover; the spatial slice and a Hodge bootstrap give the gauges.

(ii) The hypothesis of Lemma 6.14 is the energy bound $E(b, r_0) \leq e \leq e_0$. Its part (ii) gives the Hölder bound, and integrating the derivative bound across the section interval from the fixed branch, together with its part (iii), gives the last statement. $\square$

Lemma 8.5 is applied with a fixed small bound e_0 rather than with vanishing energy. It therefore holds on the whole neck $[s_0, L - s_0]$ for a fixed s_0 , and gives a uniform Hölder bound for the section on unit patches of the complete section geometry, including the region between the exterior and the windows. With vanishing energy on the windows $[S, L - S]$ it gives convergence into any prescribed flat neighbourhood there, and by Proposition 8.4 and Proposition 6.15, the rescaled transitions converge to the flat pairs at $q_{\pm}$.

8.5. First-order estimates on the necks and cuts. On a neck interval $[a, b]$ with midpoint s_* , let q_m be the fixed-branch projection at s_* and $B_* = (B_{q_m}, B_{q_m}, q_m)$ the flat pair at q_m , with temporal coefficient zero. Let $\|z\|_s^2 = \|U_f\|^2 + \mathfrak{s}\|U_c\|^2 + \|\nu\|^2$ and A_s be as in Section 7.5, and define $\|z\|_{X^0(J)}^2 = \int_J (\|z\|_s^2 + \|z_s\|_s^2 + \|A_s z\|_s^2)$. Put $d(s) = \min(s - a, b - s)$ and $\mathbf{w} = e^{cd(s)}$ with $2c < \gamma$.

Lemma 8.6 (first-order neck estimate). *For e and the neighbourhood of Lemma 8.5 small, there is a reference Coulomb gauge on each branch of the neck at B_* in which the exact matching coordinate z satisfies*

$$\|\mathbf{w}z\|_{X^0([a+3, b-3])} \leq C\sqrt{e}, \quad \|z\|_{X^0([a+r, b-r])} \leq C\sqrt{e}e^{-cr}.$$

Proof. Gauge. Start from the gauges of Lemma 8.5(i), in which $B_j(s) = B_{q_j(s)} + \beta_j(s)$ with β_j in the spatial Neumann slice. Differentiating the slice condition in s gives $d_B^* d_B \psi_j = d_B^* (\partial_s B_j)$ for the temporal coefficients, so the spatial Neumann gap gives $\|\psi_j(s)\| \leq C\sigma(s)$, where σ is the kinetic energy density of Proposition 8.4(ii). Since the energy on $[a + r, b - r]$ is at most $e e^{-\gamma r}$ and $2c < \gamma$, summation over unit intervals gives $\|\mathbf{w}\psi\| \leq C\sqrt{e}$ in the physical measure $\int (\|\psi_f\|^2 + \mathfrak{s}\|\psi_c\|^2) ds$, and likewise $\sup_s \mathbf{w} d_P(q_j(s), q_m) \leq C\sqrt{e}$ by the length bound. For the bound on ψ_j : in the Coulomb form of the local section $q \mapsto B_q$, $d_B^* \partial_s B_j = O(|\beta_j|(|q'_j| + \|\partial_s \beta_j\|))$, and $|q'_j| + \|\partial_s \beta_j\| \leq C(\sigma + \|\psi_j\|_{H^1})$, so the term is absorbed by the smallness of β_j . Let χ solve the reference Coulomb equation $d_{B_*}^* ((\exp \chi)^* A - B_*) = 0$ on each branch, in the physical metric, with Neumann conditions on ∂C and at $s = a, b$; it exists and is small on the finite neck by the Neumann gap and the implicit function theorem. Test it against $\mathbf{w}^2 \chi$ in the physical one-form measure. The spatial slice parts pair trivially with $d_C \chi$ up to the difference of the references $B_{q_j(s)} - B_*$, which is bounded by $C d_P(q_j(s), q_m)$, and the remaining linear source is the temporal coefficient. Since $|\partial_T \log \mathbf{w}| \leq c$ in the conformal time, absorption gives $\|\mathbf{w} d\chi\| \leq C\sqrt{e}$.

Equation. In the exact matching chart the equation is $D_s z + N_s(z) = 0$ with $D_s = \partial_s - A_s + 2hP_p$ and $\|N_s(z)\|_s \leq C d(z)(\|z_s\|_s + \|A_s z\|_s)$, where the coefficient size $d(z)$ bounds the spatial coefficients and the C^0 norm of the section. For compactly supported v , $\|(\partial_s - A_s)v\|^2 = \|v_s\|^2 + \|A_s v\|^2$, since $I_s A_s$ is independent of s and the Green pairings cancel; hence $\|v\|_{X^0} \leq C(\|D_s v\| + \|v\|)$, the L^2 term being kept for the stationary kernel. Apply this to $v = \mathbf{w}\zeta z$ with a cutoff ζ , and use $\|\mathbf{w}z\|_{L^2} \leq C\sqrt{e}$, which follows from Proposition 8.4(ii) and the gauge estimate. $\square$

Cut positions (Figure 3). Fix D larger than all cutoff buffers and choose $C_* > D + 1 + \log 2$. On each neck, of length L in the exterior cylindrical coordinate s , cut the exterior piece near $s_E = L/2 - C_* + D$ and the transition piece near $s_T = L/2 - C_* - D$, and use the same midpoint $s_* = L/2 - C_*$ to define the stationary parameter of both pieces. The parts of

U_ν used for the two pieces therefore overlap on a band of fixed width $2D$ on which both representatives are the actual solution, and both cuts, with their cutoff bands of unit width, lie in $\{s \leq L/2 - \log 2\}$, where E_ε coincides with X_ε . By Lemma 8.6 each exact matching cut has first-order trace of size $C\sqrt{\varepsilon}e^{-c(L/2-O(1))}$ at the cut, since each cut lies at distance $L/2 - O(1)$ from both ends of the neck $[s_0, L - s_0]$, where $L = \log(1/\varepsilon)$. With component end weights $\delta \leq c/4$ on the new ends, the cut error in the weighted first-order norms is

$$(8.7) \quad \tau_{\text{cut}} \leq C\sqrt{\varepsilon}e^{-cL/2}\varepsilon^{\delta L/2} \leq C\varepsilon^{c/4}.$$

8.6. Comparison on the transitions and the exterior. The comparison estimates are proved in Coulomb gauges on the two kinds of components. These gauges are constructed with the scalar inverse of Lemma 7.16.

Let U_ν^E be the *cut exterior piece* of U_ν : equal to U_ν on the exterior side of the exterior cut s_E on each neck, and completed for $s \geq s_E + 1$ by the flat pair at the midpoint value q_m of that neck, by a cutoff of fixed slope in s applied in the gauge of Lemma 8.6. Likewise the *cut transition* is U_ν on the transition side of s_T , completed by the flat pair at q_m .

Lemma 8.7 (component Coulomb gauges). *Lemma 7.16 holds, with the same proof, on the auxiliary exterior E_ε with the metric of Definition 7.12, the weight e^{ϕ_E} and λ equal to $\mathfrak{s}_E$ on the compression, and on the complete flat transition $\mathcal{T}$ with the weight $e^{\delta\rho}$. For ν large there is a gauge transformation of U_ν^E , trivial at the original ends in the limit, after which $U_\nu^E = W_\varepsilon^E + \alpha_E$ with*

$$\lambda^2 d_{W_\varepsilon^E}^* \alpha_E = 0, \quad \alpha_E(n) = 0, \quad \|\alpha_E\|_{H_{\text{uloc}}^N} \rightarrow 0,$$

and $\alpha_E - \mathfrak{k}_E(v_*)$ decaying at the new ends, where v_* records the difference between q_m and $q_\pm$. The same holds for the cut transition relative to the flat pair at q_m , with fixed limit zero.

Proof. The rescaled geometry of E_ε and of $\mathcal{T}$ is again a union of product cylinders with compact cross-section joined through finitely many fixed cores, λ^{-1} is Lipschitz in the rescaled distance because $|\partial_T \mathfrak{s}_E^{-1}| = |\partial_t \log \mathfrak{s}_E| \leq H$, and every slice is irreducible, so the proof of Lemma 7.16 applies verbatim. Take the local gauges of Proposition 6.15(i) on compact subsets of X_0 , of Corollary 6.16 on the middle compression and of Lemma 8.6 on the necks. On an overlap of two of them the two representatives are related by a gauge transformation g ; on each slice both are close to the same irreducible flat connection, so after a choice of central sign $g = \exp \chi$ with χ small, and the gauge equation bounds χ by the two connection differences. Interpolate by $\exp(\theta\chi)$ with a fixed cutoff θ whose derivative in the product time is bounded; the overlaps form a tree, so the central signs can be chosen consistently. This gives a representative $W_\varepsilon^E + \alpha$ with $\|\alpha\|_{H_{\text{uloc}}^N}$ small. The reference Coulomb equation $\lambda^2 d_{W_\varepsilon^E}^* ((\exp \chi)^*(W_\varepsilon^E + \alpha) - W_\varepsilon^E) = 0$, $\nabla_n \chi = 0$, has derivative $\lambda^2 d^* d$ at $\chi = 0$, which is invertible on uniformly local spaces by the first part; the implicit function theorem gives a small solution. On the new ends the stationary field k_{v_*} is harmonic, coclosed and normal-free, so it contributes no Coulomb source, and the gauge parameter decays. $\square$

Lemma 8.8 (transition comparison). *In the Coulomb gauge of Lemma 8.7, which retains the midpoint value q_m as a free stationary parameter, the cut transition differs from the flat pair at q_m by at most $C\tau_{\text{cut}}$ in the weighted first-order norm.*

Proof. The flat pair at q_m is a solution. Write the cut transition as the flat pair plus z in the exact matching chart. Then

$$\mathcal{D}z = -N(z) + e_{\text{cut}}, \quad \|N(z)\|_{Y_\delta} \leq C d(z) \|z\|_{X_\delta}, \quad \|e_{\text{cut}}\|_{Y_\delta} \leq C\tau_{\text{cut}},$$

where e_{cut} is the error of the cutoff to the flat pair and $d(z)$, the size of the coefficients, tends to zero; the gauge-fixing component carries no source because z is in Coulomb gauge. The fixed-limit inverse of Proposition 7.10(iii) at weight δ gives $\|z\|_{X_\delta} \leq C\delta^{-1}(d(z)\|z\|_{X_\delta} + C\tau_{\text{cut}})$, and for $d(z)$ small the first term is absorbed. $\square$

Lemma 8.9 (exterior comparison). *Let z be the exact matching coordinate of $W_\varepsilon^E + \alpha_E$, the cut exterior piece U_ν^E in the Coulomb gauge of Lemma 8.7, relative to the configuration W_ε^E of the auxiliary exterior (Definition 7.12), with a weight $\delta_* \leq c/4$ on the new ends. Then*

$$\|z\|_{X_E} \leq C(\tau_{\text{cut}} + \varepsilon^{1/4} + \varepsilon^{(\beta-\delta_*)/4}),$$

where X_E is the norm of Proposition 7.13, and the free end coordinates of z , which are the midpoint parameters of the two necks, obey the same bound.

Proof. Write $\mathcal{F}_E(z) = f_E + D_E z + \mathcal{N}_E(z)$. The reference error is $\|f_E\|_{Y_E} \leq C(\varepsilon^{1/4} + \varepsilon^{(\beta-\delta_*)/4})$: the first term is the horizontal velocity $\mathfrak{s}_E^{1/2}|\dot{q}_R|$ of the flat lift, with $\mathfrak{s}_E \leq \sqrt{\varepsilon}$; the second is the truncation of V at $R = \varepsilon^{1/4}$, which is supported where $s = L/4 + O(1)$, has size $O(R^\beta)$ there, and carries the weight $e^{\delta_* s} = O(R^{-\delta_*})$. The nonlinear term with comparison coordinate zero satisfies $\|\mathcal{N}_E(z)\|_{Y_E} \leq C d(z)\|z\|_{X_E}$, because the spatial remainder vanishes on the exact flat family, the temporal coefficient derivative is quadratic, and the stationary fields are exact flat members. Here $d(z) \rightarrow 0$ is the coefficient size, which tends to zero by Proposition 6.15, Corollary 6.16 and Lemma 8.5. The operator D_E is inverted from Y_E to X_E with a bound independent of ε by Proposition 7.13; the gauge-fixing component of z vanishes by the choice of gauge, and the cut error, which comes from the cutoff to the flat pair at q_m and is bounded by (8.7), is measured in the same norms. Absorption gives the bound. The end coordinate of z is the midpoint parameter because the same midpoint is used for both cuts. $\square$

8.7. The global gauge and the uniform estimate on the glued domain. *Zeroth-order norm.* We use the rescaled metric $\widehat{g}_\varepsilon$ and measure μ_ε of Section 7.8. A variation x on the glued domain is written $x = \bar{x} + \mathfrak{e}_\varepsilon(v_{\text{orig}})$, where v_{orig} is the vector of limiting values at the original ends and $\mathfrak{e}_\varepsilon$ a fixed extension supported in the tails of the original ends, with zero temporal and normal components and $\|w D_\varepsilon \mathfrak{e}_\varepsilon(v)\| \leq C|v|$. Put

$$\|x\|_{Z_\varepsilon} = \|w\bar{x}\|_{L^2(\mu_\varepsilon)} + \|w\bar{x}_{\text{sec}}\|_{L^2} + |v_{\text{orig}}|,$$

the physical zeroth-order norm of gauge field and section, and use the same expression for nonlinear configurations relative to W_ε . It contains the stationary fields along the entire finite necks, which on a neck of length comparable to L contribute up to $\sqrt{L}$ times their pointwise size.

Lemma 8.10 (global gauge). *In the situation of Proposition 8.12, for large ν there is a gauge transformation of U_ν on the glued domain, trivial at the original ends in the limit, after which $U_\nu = W_\varepsilon + x$ with x in the reference Coulomb gauge of W_ε , x of finite X_ε norm, and*

$$\|x\|_{Z_\varepsilon} \leq C\sqrt{A}(\tau_{\text{cut}} + \varepsilon^{1/4} + R^{\beta-\delta_*}), \quad \|x\|_{H_{\text{uloc}}^N(\widehat{g}_\varepsilon)} \rightarrow 0.$$

Proof. Components. By Lemmas 8.8 and 8.9 the cut transitions and the cut exterior differ from the pieces of W_ε by at most $C(\tau_{\text{cut}} + \varepsilon^{1/4} + R^{\beta-\delta_*})$ in first-order norms, and the change from V to its truncation costs CR^β . A stationary parameter of size ρ on a neck of length $O(L)$ has physical L^2 norm $C\sqrt{A}\rho$; the compression contribution is no larger. This gives the Z_ε bound on the part of each piece that is used. The qualitative H_{uloc}^N bound follows from Proposition 6.15, Corollary 6.16 and Lemma 8.5.

Joining. On an overlap band the two representatives represent the same solution, $A_T = g^* A_E$. On each slice both are close to the same irreducible flat connection, so g may be chosen near the identity, $g = \exp \chi$, and the spatial gauge equation with the Neumann gap gives $\|\chi(T)\|_{H^1(C)} \leq C(\|(A_E - B_q)_{\text{sp}}\| + \|(A_T - B_q)_{\text{sp}}\|)$; the temporal part of the gauge equation controls $\partial_T \chi$ in the same way. Interpolate by $\exp(\theta \chi)$ with a fixed cylindrical cutoff θ , whose product derivative on the compression is $\mathfrak{s}\theta'$ and hence bounded. The result is an actual gauge-equivalent representative $W_\varepsilon + \alpha$ with the two stated bounds. The pieces overlap along a tree of bands, so the choices of central sign on the bands can be made consistently with the marked bundle.

Coulomb gauge. The reference Coulomb equation $\lambda_\varepsilon^2 d_{A^0}^{*,g_\varepsilon}((\exp \chi)^*(A^0 + \alpha) - A^0) = 0$, $\nabla_n \chi = 0$, with A^0 the connection of W_ε , has derivative $\mathcal{L}_\varepsilon$ at $\chi = 0$; by Lemma 7.16 and the small H_{uloc}^N norm of α it has a small solution with $\|\chi\|_{H_{\text{uloc}}^{m+1}} \leq C\|\alpha\|_{H_{\text{uloc}}^m}$, which belongs to the weighted space at the original ends. To estimate it in the physical norm, write $(\exp \chi)^*(A^0 + \alpha) - A^0 = J(\chi)d_0\chi + \text{Ad}_{e^{-\chi}}\alpha$ with $J(0) = 1$, and test the equation against $w^2\chi$:

$$0 = \int \langle d_0(w^2\chi), J(\chi)d_0\chi + \text{Ad}_{e^{-\chi}}\alpha \rangle d\mu_\varepsilon.$$

The main term is at least $(1 - o(1))\|wd_0\chi\|^2$; the weight terms are of relative size δ , the maximal weight exponent of the original ends. The stationary part $\mathfrak{e}_\varepsilon(v_{\text{orig}})$ of α is integrated by parts onto its divergence, bounded by $C|v_{\text{orig}}|$, and $\text{Ad}_{e^{-\chi}}\alpha - \alpha$ is small times χ . Absorption gives $\|wd_0\chi\| \leq C(\|w\bar{\alpha}\| + |v_{\text{orig}}|)$, and the same bound for the change of the connection. The section is unchanged. Finite X_ε norm holds for each fixed ν since the region $N \cap X_\varepsilon$ is compact. $\square$

Proposition 8.11 (uniform estimate on the glued domain). *Uniformly for small ε , $\|x\|_{X_\varepsilon} \leq C(\|D_\varepsilon x\|_{Y_\varepsilon} + \|x\|_{Z_\varepsilon})$ for all x in the domain, with no normalization of the asymptotic values.*

Proof. Write $D_\varepsilon = P_\varepsilon + K_\varepsilon$, with P_ε the differential part and K_ε the perturbation term. Let $\mathfrak{s}_1$ be the threshold of Proposition 7.4. On a unit patch of $N \cap X_\varepsilon$ on which $\mathfrak{s} \leq \mathfrak{s}_1$, the configuration W_ε is the flat lift of the path $q(t)$ and P_ε is the operator of Proposition 7.4 (linearization at the flat family along $q(t)$, scale with $|\mathfrak{s}'/\mathfrak{s}| \leq H$); since $\|\nu\|_{L^2(S)} \leq \|z\|_{L^2}$, that proposition gives

$$(8.8) \quad \|z\|_{X_\varepsilon}^2 \leq C(\|P_\varepsilon z\|_{Y_\varepsilon}^2 + \|z\|_{L^2}^2) \quad (\text{supp } z \text{ in one patch}).$$

On patches where $\mathfrak{s} \geq \mathfrak{s}_1/2$ the rescaled geometry lies in a fixed compact family, and the ordinary local mixed estimates give (8.8) there as well.

Use a partition $\sum \chi_\alpha^2 = 1$ of bounded overlap by cutoffs that depend on the complete time coordinate only on the compression (constant on each fibre C), share the same value on the two sides of every seam, and are extended constantly across a fixed section collar; a cutoff in $\log(\mathfrak{s}/\mathfrak{s}_1)$ has derivative bounded by $C|h|$. A spatial cutoff on C would have commutator $\mathfrak{s}^{-1}d_C\chi$, which is not bounded uniformly; these cutoffs preserve the domain and have localized commutator bounds $\|[P_\varepsilon, \chi_\alpha]z\|_Y \leq C\|z\|_{L^2(\text{supp } d\chi_\alpha)}$. Applying (8.8) to each $\chi_\alpha z$ and summing,

$$\|z\|_{X_\varepsilon}^2 \leq C \sum_\alpha \|\chi_\alpha z\|_{X_\varepsilon}^2 \leq C \sum_\alpha (\|\chi_\alpha P_\varepsilon z\|_{Y_\varepsilon}^2 + \|[P_\varepsilon, \chi_\alpha]z\|_{Y_\varepsilon}^2 + \|\chi_\alpha z\|_{L^2}^2) \leq C(\|P_\varepsilon z\|_{Y_\varepsilon}^2 + \|z\|_{L^2}^2),$$

the first inequality because the cutoffs have bounded derivatives and are constant on the fibres, and the last because the supports of the χ_α and of the $d\chi_\alpha$ have bounded overlap. This is the claim for the differential part P_ε , with constants depending on the overlap and not

on the number of patches. The perturbation term has output and dependence region in the unchanged geometry and is bounded on L^2 . Conjugation by the weights of the original ends adds bounded zeroth-order terms, and subtracting $\mathfrak{e}_\varepsilon(v_{\text{orig}})$ reduces to decaying fields. $\square$

8.8. Surjectivity of the gluing map.

Proposition 8.12 (surjectivity of gluing). *Let $\varepsilon_\nu \rightarrow 0$ and let U_ν be index-zero solutions on X_{ε_ν} converging, in the sense of Proposition 6.15, to the regular index-zero solution V . Then $U_\nu = \mathcal{G}_{\varepsilon_\nu}(V)$ for all large ν .*

Proof. We write $\varepsilon = \varepsilon_\nu$; the sequence enters only through thresholds.

Step 1: global gauge. By Lemma 8.10, after a gauge transformation $U_\nu = W_\varepsilon + x$ with x of finite X_ε norm, $\|x\|_{Z_\varepsilon} \leq C\sqrt{A}(\tau_{\text{cut}} + \varepsilon^{1/4} + R^{\beta-\delta_*})$ and coefficient size $\mathfrak{d}(x) \rightarrow 0$.

Coordinates. Lemma 8.10 produces x as an affine connection difference in the reference Coulomb gauge, while Proposition 8.3 works in the exact matching chart. The two are related by the chart map Φ_t of Section 8.1: the affine difference is $\Phi_t(U) - W_\varepsilon$, whose spatial part differs from the chart coordinate by $B_t(v) - B_t - j_tv + (T_{t,v} - 1)w + \mathcal{R}_t(v, w)$, which is quadratic in (v, w) with the bounds of Lemma 8.2, and whose temporal part is affine. On small coefficient neighbourhoods the change of coordinates and its inverse are therefore bounded in Z_ε uniformly in ε (the difference being pointwise quadratic, of size at most $C\mathfrak{d}\|x\|_{Z_\varepsilon}$) and preserve finiteness of the X_ε norm at fixed ε ; and the gauge-fixing component of $\mathcal{F}_\varepsilon$ in the chart is the affine Coulomb condition composed with this change. We use the chart coordinate below and keep the name x .

Step 2: a priori estimate in the first-order norm. Write $\mathcal{F}_\varepsilon(W_\varepsilon + x) = f_\varepsilon + D_\varepsilon x + N_\varepsilon(x)$ with $\|f_\varepsilon\| \leq Ce_\varepsilon$ by (8.6) and $\|N_\varepsilon(x)\|_{Y_\varepsilon} \leq C\mathfrak{d}(x)\|x\|_{X_\varepsilon}$: the pure harmonic spatial term vanishes on the exact flat family by (8.3), every remaining singular term contains a complementary field measured by its graph weight, and the section uses only C^0 smallness. Proposition 8.11 and absorption of $\mathfrak{d}(x) \rightarrow 0$ give

$$\|x\|_{X_\varepsilon} \leq C(e_\varepsilon + \sqrt{A}(\tau_{\text{cut}} + \varepsilon^{1/4} + R^{\beta-\delta_*})) \leq C\varepsilon^{\sigma_c}$$

for some $\sigma_c > 0$, since τ_{cut} , e_ε and $R^{\beta-\delta_*}$ are positive powers of ε by (8.7) and (8.6).

Step 3: interpolation. On four-dimensional unit patches of the conformal geometry, which are copies of one fixed compact manifold, interpolation between the local H^N bounds, which hold uniformly by the convergence of Proposition 6.15, Corollary 6.16 and Lemma 8.5 and are preserved by the gauges of Step 1, and the first-order bound of Step 2 gives $\|U_\perp\|_{C^k} \leq C\varepsilon^{\sigma_c\theta_1}$ for the complementary gauge field, using $\int \mathfrak{s}^{-1}\|U_\perp(t)\|_{H^1}^2 dt = \int \|U_\perp(T)\|_{H^1}^2 dT$ with $dt = \mathfrak{s} dT$, so that no power of the minimal scale is lost. The slow coefficient is not interpolated on product patches; its supremum is obtained from the section through the matching identity $v = \Pi\nu|_{\text{seam}} - \Pi\tau_t(U_\perp|_\Sigma)$. The section is interpolated on two-dimensional patches between the uniform Hölder bound of Lemma 8.5 and the L^2 bound. Hence the coefficient size satisfies $\mathfrak{d}(x) \leq C\varepsilon^\sigma$ for some $\sigma > 0$.

Step 4: uniqueness. The norm X_ε on the glued domain carries only the original end weights; the artificial weights of the component problems do not enter. Let $z^* = \mathcal{G}_\varepsilon(V) - W_\varepsilon$, so that $\|z^*\|_{X^M} \leq a := C_MA^{M+1}e_\varepsilon$ by Proposition 8.3. The difference $y = x - z^*$ satisfies $D_\varepsilon y = N_\varepsilon(z^*) - N_\varepsilon(x)$. The difference estimate holds in the asymmetric form

$$\|N_\varepsilon(x) - N_\varepsilon(z^*)\|_{Y_\varepsilon} \leq C(\mathfrak{d}(x) + a)\|x - z^*\|_{X_\varepsilon},$$

where $\mathfrak{d}(x)$ is the unweighted coefficient size of x from Step 3 and the weighted suprema $\mathfrak{s}^{-1/2}|p_{z^*}|$, $\mathfrak{s}^{-1/2}|\nabla_C(U_\perp)_{z^*}|$ needed by the singular terms, such as $\mathfrak{s}^{-1}[p_{z^*}, h_{v_x-v_{z^*}}]$, are placed

on the factor z^* , where they are bounded by a through (8.1). The inverse of Proposition 7.15 then gives

$$\|y\| \leq CA(d(x) + a)\|y\|.$$

Both $A d(x)$ and $A a$ are positive powers of ε multiplied by powers of A , hence tend to zero, and $y = 0$ for large ν . $\square$

The unknown solution never has to enter the logarithmically shrinking ball of Proposition 8.3 directly. It first reaches a positive power of ε by estimates free of logarithms, and only then meets the single factor A of the inverse on the glued domain.

8.9. Proof of Theorem 6.6(iii). The order of choices is: the data outside N and the bound b ; the neck lengths of the ordinary necks, for which Hypothesis 6.5(X4) holds; the finite set $\mathcal{M}^0(X_0)$, which fixes the decay rates β_V and the weights; and finally ε . Negative indices: if solutions of negative index existed for a sequence $\varepsilon_\nu \rightarrow 0$, a subsequence would converge by Theorem 6.6(ii); the energy bound of part (i) leaves finitely many negative indices, and the loss formula with Hypothesis 6.5(X4) excludes each. For index zero, $\mathcal{G}_\varepsilon$ is defined by Proposition 8.3 and is injective for small ε : $\mathcal{M}^0(X_0)$ is finite, and $\mathcal{G}_\varepsilon(V) \rightarrow V$ as $\varepsilon \rightarrow 0$ in the sense of Proposition 6.15, since $\|\mathcal{G}_\varepsilon(V) - W_\varepsilon\|_{X^m} \rightarrow 0$ and W_ε agrees with V away from shrinking neighbourhoods of the rectangle. If it were not surjective for a sequence $\varepsilon_\nu \rightarrow 0$, a sequence of solutions outside its image would converge to some $V \in \mathcal{M}^0(X_0)$ by Theorem 6.6(ii), contradicting Proposition 8.12. Regularity of the image is part of Proposition 8.3. $\square$

9. TWO-HANDLES

9.1. Heegaard triples and the three Lagrangians. Let $(W_2, E, \nu): Z_0 \rightarrow Z_1$ be a two-handle cobordism along a framed link L of n components, with E trivial at both ends (Proposition 4.2), with a filling frame (Definition 4.6) of lifting class ζ . Choose a subordinate Heegaard triple $(\Sigma; H_0, H_1, H_2)$ of genus g adapted to the ball (Section 4.3), and take the boundary sum with the fixed torus: $C_i = H_i \natural ([0, 1] \times T^2)$, $h = g + 1 \geq 2$ (after a stabilization in a small ball of the standard collar, away from the bouquet, if necessary). Put $r = g - n$. The transition ρ_ζ of the filling frame on Σ induces a symplectomorphism τ_ζ of M_h , and the three Lagrangians are

$$(9.1) \quad (\tau_\zeta L_{h_0}(C_0), L(C_1), L_{h_2}(C_2)) =: (P_0, P_1, P_2),$$

where the primary perturbation of C_1 is zero, that of C_2 is the height function th_Q of the next subsection, and that of C_0 is small and chosen so that P_0 is transverse to P_1 and P_2 and misses $P_1 \cap P_2$. Since ρ_ζ is an automorphism of the $SO(3)$ bundle, it conjugates every operator in the problem; in the trivialization of the filling frame over $e_0 \times H_0$ the boundary problem of C_0 is literally the untwisted one, and ζ enters only through the global topology of the triangle.

9.2. The auxiliary pair. The manifold $Y_{12} = H_1 \cup H_2$ is $\#^r(S^1 \times S^2)$ and, for vanishing primary perturbations of C_1 and C_2 , the pair $(L(C_1), L(C_2))$ intersects cleanly in $Q \cong SU(2)^r$, with $TQ = TL(C_1) \cap TL(C_2) = H^1(Y_{12}^\#; \text{ad})$. Here the flat connections of $Y_{12}^\#$ are determined by the holonomies $b_j \in SU(2)$ of the r free generators of π_1 , with lifts fixed by the global frame of $H_1 \cup H_2$; the torus summand contributes the fixed quaternion pair, and the nonzero element of ϕ_M identifies its two sign choices. The primary perturbation of C_2 is the height function th_Q , $h_Q = \sum_j (1 - \frac{1}{2} \text{tr } b_j)$, with $t > 0$ small, and the primary perturbation of C_1 is zero; transversality of (P_0, P_2) and empty triple intersection are obtained by perturbing C_0

alone. The critical points are the 2^r points a_S with $b_j = -1$ for $j \in S$ and $b_j = 1$ otherwise; a_S has Morse index $m(a_S) = 3|S|$, and there is a unique maximum $q_* = a_{\{1, \dots, r\}}$.

Lemma 9.1. *Use the conventions under which the anti-self-duality equation on an outgoing end $[0, \infty) \times Y$ is the downward gradient flow of the Chern–Simons functional perturbed by $+t h_Q$ with $t > 0$.*

- (i) *For the gauge filling $F_r = \natural^r(S^1 \times D^3) \natural(T^2 \times D^2)$ of $Y_{12}^\#$ with outgoing limit a , on the component of the flat fillings, $\text{ind} = 3r - m(a)$; other components have index $3r - m(a) + 8\kappa$.*
- (ii) *For a gauge, mixed or ordinary configuration from a to x , with κ normalized to vanish on constant configurations, $\text{ind} = m(a) - m(x) + 8\kappa$, and the energy is $8\pi^2\kappa + t(h_Q(a) - h_Q(x))$ plus secondary terms.*
- (iii) *If $t \max h_Q$ plus the size of all secondary terms is less than the smallest positive value of $8\pi^2\kappa$, every counted configuration has $\kappa \geq 0$.*

Reversing one of the three conventions (the sign of the flow, the sign of t , outgoing versus incoming end) replaces $3r - m(a)$ by $m(a)$ and the maximum by the minimum throughout; reversing two restores the statement.

Proof. (i) At a flat connection A on F_r with the unperturbed end, the operator $-d_A^* \oplus d_A^+$ on decaying spaces is injective and surjective: a decaying kernel element is closed and represents a class in the image of $H_c^1(F_r; \text{ad}) \cong H_3(F_r; \text{ad}) = 0$, hence is exact, $a = d_A f$, and $f = 0$ by $H^0(Y_{12}^\#; \text{ad}) = 0$; a cokernel element has vanishing function part by $H^0(F_r; \text{ad}) = 0$ and a self-dual part that is exact by $H^2(F_r; \text{ad}) = H^2(T^2; \text{ad}) = 0$, hence decays and vanishes by $\|\omega\|^2 = \int \omega \wedge \omega = 0$. So the decaying index is 0. The asymptotic operator on the tangent space TQ of the critical manifold moves from $-\delta$ to $t \text{Hess}_a h_Q$, which has $m(a)$ negative and $3r - m(a)$ positive eigenvalues; for an outgoing end each eigenvalue crossing from negative to positive increases the index by one. Hence $\text{ind} = 3r - m(a)$. Equivalently, for small t the solutions are flat fillings A_p , $p \in Q$, followed on the end by the flow $\dot{p} = -t \nabla h_Q$ ending at a , and they form the stable manifold of a , of dimension $3r - m(a)$. Other components differ by the index of an instanton class. (ii) By additivity: a filling to a followed by a configuration from a to x is a filling to x , and constant mixed pairs have index zero, so the same function m serves on the gauge and symplectic sides. The energy formula is the index-energy law with the perturbation term. (iii) The energy is nonnegative and the lattice of topological energies is discrete. $\square$

With Scaduto's $\mathbb{Z}/4$ grading, $\text{gr} = m - r$ modulo four on every generator [38, Section 7.3]; in particular q_* has degree $2r$, the degree of the relative invariant of $\natural^r(S^1 \times D^3)$ [38, Sections 7.7–7.8].

Lemma 9.2 (top generator). *Let C, C' be finite complexes over $\mathbb{F}_2$ with bases indexed by the same set, graded by m , and let $N: C \rightarrow C'$ be an isomorphism whose matrix entries $N_{a,x}$ vanish unless $m(x) = m(a) + 4k$ with $k \geq 0$. Then N maps the unique generator of maximal m to itself.*

Proof. If a has maximal m , then $N(a)$ is a multiple of a , and the multiple is nonzero because N is injective. $\square$

The hypothesis $k \geq 0$ cannot be replaced by a congruence modulo four: for $r = 4$ the isomorphism $N = I + E_{\{1,2,3,4\}, \emptyset}$, which adds the minimum to the image of the maximum, satisfies the congruence and violates the conclusion.

Definition 9.3. On $CF(P_1, P_2)$ define the relative grading m by $m(a) - m(x) = \text{ind}(u) - 8\kappa(u)$ for any strip class u from a to x ; it is independent of u by monotonicity, and for t small it is the Morse index of h_Q . Let $\theta_S \in CF(P_1, P_2)$ be the unique generator of maximal m ; it is the maximum q_* of h_Q . Let $\theta_G \in C_G(Y_{12})$ be the relative invariant of the filling F_r : the count of index-zero solutions on F_r with cylindrical end $Y_{12}^\#$, with lifts by even gauge transformations, projected to the framed complex.

Proposition 9.4. $\theta_G = q_*$ and $\theta_S = q_*$ as chains; both are cycles, their classes are nonzero, and $N_{12}\theta_G = \theta_S$.

Proof. By Lemma 9.1 an index-zero filling has $\kappa = 0$ and $m(a) = 3r$, and for small t the flat fillings flowing to q_* form the stable manifold of the maximum, a single point, which is regular because the operator of part (i) of that lemma is invertible. So $\theta_G = q_*$. (This agrees with [38, Sections 7.7–7.8], where the relative invariant of $\natural^r(S^1 \times D^3)$ is a basis element of the free group $I^\#(\#^r(S^1 \times S^2))$ of rank 2^r , by the boundary-sum identity [38, Section 7.2] and [21, Proposition 5.4] at the connected cut T^3 .) A trajectory from q_* has index $3r - m(x) + 8\kappa$ with $\kappa \geq 0$, which is never one, so q_* is a cycle on both sides. By Lemma 9.1 the map N_{12} of Theorem 3.6 satisfies the hypothesis of Lemma 9.2, so $N_{12}q_* = q_*$. The filtrations on the two sides are defined by the same function m , so the gauge filling class and the direction of triangularity of N_{12} cannot come apart under a change of conventions. $\square$

9.3. The torus-summed triangle and its lifting group. Let $X^\#$ be the torus-summed triangle: the union of $\Delta \times \Sigma_h$ and $\Delta \times T^2$ (the two surface sheets) with the edge pieces $e_i \times C_i$, where each torus compression body C_i is glued to both sheets along $e_i \times (\Sigma_h \sqcup T^2)$. Its ends are $Y_{ij}^\# = C_i \cup (-C_j)$. By Proposition 4.3 and the construction of Section 2, removing the tube from W_2 and inserting $T^\circ \times [-1, 1]$ gives $W_2^\# = X^\# \cup_{Y_{12}^\#} F_r$ with its bundle. The classes of $H^1(X^\#; \mathbb{F}_2)$ that vanish on both surface sheets restrict to zero on each compression piece, since $H^1(C_i) \rightarrow H^1(\Sigma_h)$ is injective, so they come from the incidence graph of the torus region, a pair of pants with three boundary circles: this group is $\Phi = \mathbb{F}_2^3 / \langle (1, 1, 1) \rangle$, whose restriction to the end $Y_{ij}^\#$ is $s_{ij} = t_i + t_j$. Each class is realized by an automorphism supported on the torus region and equal to the identity near its joining tree.

9.4. The gauge two-handle map as a triangle map. Let $E^\#$ be the bundle on the torus-summed triangle $X^\#$ of Section 9.3, and $\Phi = \mathbb{F}_2^3 / \langle (1, 1, 1) \rangle$ its lifting group, with $s_{ij} = t_i + t_j$. Let μ_G^E be the Φ -framed gauge triangle map $C_G(Y_{01}) \otimes C_G(Y_{12}) \rightarrow C_G(Y_{02})$.

Proposition 9.5. Let $F_G(W_2, E)$ be the map of [21, Section 5] of $(W_2^\#, E^\#)$ with the subgroup ϕ_{W_2} , between the complexes of $(Z_i^\#, E^\#|_{Z_i^\#})$ with the markings of the filling frame. Then $F_G(W_2, E) = [\mu_G^E(\cdot \otimes \theta_G)]$ on homology.

Proof. By Proposition 4.3, $W_2^\# = X^\# \cup_{Y_{12}^\#} F_r$, where F_r is the gauge filling of Lemma 9.1(i), with the tube in the product region. Write $W_2^\# = X^\# \circ W'$ with $W' = (Y_{01}^\# \times [0, 1]) \sqcup F_r$, cut along the disconnected manifold $Y_{01}^\# \sqcup Y_{12}^\#$. The restriction $H^0(W') \oplus H^0(X^\#) = \mathbb{F}_2^2 \oplus \mathbb{F}_2 \rightarrow H^0(Y_{01}^\# \sqcup Y_{12}^\#) = \mathbb{F}_2^2$ is $(a, b, c) \mapsto (a + c, b + c)$, which is onto, so the connecting image ψ of [21, Proposition 5.4] vanishes. The subgroup of $W_2^\#$ restricts to the index-two subgroup $\{s_{12} = 0\} \subset \Phi$ on $X^\#$ and to 0 on F_r , since the torus meridian class does not extend over $T^2 \times D^2$. By [21, Proposition 5.4] and the Künneth isomorphism of [21, Section 5.2], $F_G(W_2, E) = \mu^{\{s_{12}=0\}}(\cdot \otimes \theta_G)$, where θ_G is the relative invariant of F_r (as in the definition

of θ_G), computed modulo even gauge transformations only. Finally let $g \in \Phi$ with $s_{12} = 1$, let ι be its action on the critical points of $Y_{12}^\#$, and let π be the projection to the Φ -quotient complex. With Φ -invariant data, g maps solutions with Y_{12} -limit b bijectively to those with limit ιb , preserving the orbits at the other two ends, whose restriction of $\{s_{12} = 0\}$ is already onto. So g acts freely on the lifts at Y_{12} and trivially on the orbits at the other ends, and each Φ -orbit of solutions contains exactly one solution for each lift of b . Hence $\mu^{\{s_{12}=0\}}(a \otimes b) = \mu^\Phi(a \otimes \pi b)$ with no factor two, and $\mu^{\{s_{12}=0\}}(\cdot \otimes \tilde{\theta}_G) = \mu_G^E(\cdot \otimes \theta_G)$. Nontrivial E is covered by [21, Section 5.1], which allows an arbitrary bundle. $\square$

9.5. The mixed triangle.

Definition 9.6. The *mixed triangle* is the following domain for the mixed equation. Let Δ be a triangle with incoming vertices v_{01}, v_{12} , outgoing vertex v_{02} and edges e_i , and let γ be a proper arc from the interior of e_0 to the interior of e_2 separating v_{02} from the incoming vertices, dividing Δ into G (containing v_{01}, v_{12}) and S . The gauge part is the restriction of $X^\#$ over G : the two surface sheets over G and the edge pieces over $e_i \cap G$. The section is a map $S \rightarrow M_h$, with boundary conditions P_0 on $e_0 \cap S$ and P_2 on $e_2 \cap S$, matched along γ to both sheets (the torus sheet matching the one-point moduli space). At $\gamma \cap e_0$ and $\gamma \cap e_2$ the domain has the mixed ends of C_0 and C_2 . The ends are the gauge inputs $Y_{01}^\#, Y_{12}^\#$, the strip output at $P_0 \cap P_2$, and the two mixed ends. Metrics are products near the seams, and the section almost complex structure is the complex structure $*_\Sigma$ near γ . Let $B \subset G$ be the rectangle joining a compact subinterval I of the middle edge e_1 to an arc $J \subset \gamma$; the collapse of Theorem 6.6 takes place over B .

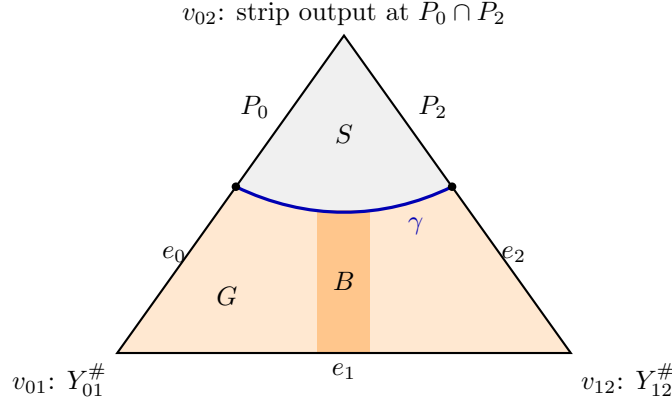


FIGURE 4. The mixed triangle of Definition 9.6. The gauge part lies over G and the section over S ; they are matched along the arc γ , whose endpoints are the mixed ends of C_0 and C_2 . The rectangle B joins a subinterval of the middle edge e_1 to γ . Collapsing B (Theorem 6.6) and stretching the ordinary necks gives the symplectic triangle composed with $N_{01} \otimes N_{12}$; stretching along the closed three-manifold $Y_{02}^\#$, which by Lemma 9.11 passes through B , gives N_{02} composed with the gauge triangle.

Lemma 9.7 (vertex identification). *Let D be any domain of DFL type for a pair (P, P') covered by Theorem 3.6 and Remark 3.7, with the same end types as the special quintuple of [11, Section 3.1], and regular invariant data. The translation-invariant data on the mixed*

ends may differ from those of the quintuple. Then its index-zero count is chain homotopic to the vertex map of Theorem 3.6 for that pair.

Proof. The space of such domains and data, with fixed end types, is connected: the conformal structure of the section region, the placement of the seam, the metrics near the mixed ends and the neck lengths vary in convex sets, and the changes are compactly supported. Theorem 5.2 gives a chain homotopy along a path, and Theorem 3.6(ii) handles the choices of f , χ and η . A change of the translation-invariant data on a mixed end (the polar geometry of Lemma 6.3(ii) against the product geometry of the quintuple, matched along the seam) is a family whose parameter derivative is independent of the end coordinate there; the flat references on C and the constant sections do not depend on the end metric, and the mixed ends have a uniform gap on the complement of the stationary kernel, so Lemma 10.14 gives the chain homotopy. $\square$

The DFL pieces produced by the degenerations below are such domains, so the maps they count are the vertex maps up to chain homotopy.

9.6. Critical points on the twisted inputs and the gauge triangle.

Lemma 9.8. *Let $ij \in \{01, 12, 02\}$, with the ρ_ζ -marking on C_0 when $i = 0$, and let α be a critical point of the primary equation $*F_B + \nabla \mathfrak{h}_{ij}(B) = 0$ on $Y_{ij}^\#$. Then α is flat on a collar of a copy of Σ_h disjoint from the primary dependence regions, w_2 is nonzero on Σ_h , α is irreducible, an automorphism fixing α whose lifting class vanishes on Σ_h is the identity, the nonzero element of the lifting group acts freely on the critical set, and the critical set corresponds bijectively to $P_i \cap P_j$ and is nondegenerate. Moreover end perturbations as in Proposition 3.4 can be chosen so that the critical set of $\text{CS}_{\mathfrak{h}_{ij}} + f_{ij}$ equals that of $\text{CS}_{\mathfrak{h}_{ij}}$.*

Proof. The pair (C_0, C_j) with the ρ_ζ -composed boundary marking is a pair of the type of [11, Section 2.3], and its Lagrangian is $\tau_\zeta L(C_0)$. Flatness near Σ_h , irreducibility and the correspondence with $P_i \cap P_j$ with nondegeneracy are [11, Lemmas 2.43, 2.45, 2.47], whose proofs use only flatness and irreducibility on the overlap and Mayer–Vietoris. The restriction of $E_{ij}^\#$ to a push-off of Σ_h into C_j is the odd surface bundle, unchanged by ρ_ζ . Freeness follows from Theorem 5.4(i). For the last statement, the twisted-marked pairs are not presentations, so Proposition 3.4 is not applied as stated; its proof is repeated for each pair ij , as follows. If a sequence f_n with vanishing C^0 norms of gradients and of their derivatives on fixed balls had critical points B_n not gauge equivalent to critical points of $\text{CS}_{\mathfrak{h}}$, then Uhlenbeck compactness on the closed three-manifold and elliptic regularity would give a limit critical point α of $\text{CS}_{\mathfrak{h}}$, near which the inverse function theorem for the gauge-fixed critical equation, with invertible extended Hessian, allows only the zero α itself, since f_n vanishes near α . $\square$

Proposition 9.9. *On $X^\#$ with the bundle $E^\#$ and the three closed ends of Lemma 9.8, arbitrarily small Φ -invariant secondary perturbations give: no solutions of negative index, a finite regular index-zero space, and regular one-dimensional spaces whose boundary strata are an index-zero solution with one index-one trajectory at one end. The count μ_G^E is a chain map.*

Proof. The ends are nondegenerate and irreducible by Lemma 9.8, so the gauge-fixed operator is Fredholm, the gauge-fixing component having positive gaps. The index-energy law $\text{ind}(A) = 8\kappa(A) + c(\alpha_0, \alpha_1, \alpha_2)$ holds on the pure gauge triangle: for two connections A, A' with the same limits, glue to both the same connections on compact fillings of the three

ends (every $\mathrm{SO}(3)$ bundle over a closed three-manifold extends over some compact four-manifold); since the ends are acyclic, additivity of the index [13, Proposition 3.9] reduces $\mathrm{ind}(A) - \mathrm{ind}(A')$ to the difference on the closed manifold, where the Atiyah–Singer formula gives $8(\kappa(A) - \kappa(A'))$. Compactness is Uhlenbeck compactness with point concentrations of energy $8\pi^2$ in the normalization of Section 3.6 and index loss eight; a small three-sphere about a concentration point carries no nonzero lifting class, so the limiting bundle is identified with $E^\#$ without any unitary lift. Surjectivity in the invariant class is Theorem 5.4(ii), with the loops supplied by an incoming end: on $Y_{12}^\#$ the bundle admits a $\mathrm{U}(2)$ lift because H^3 of a closed oriented three-manifold is torsion free, the critical limit is flat on a collar of Σ_h with odd bundle, and loops on that surface, which have zero lifting class in Φ , are transported into the interior along a path; holonomies are conjugated by parallel transport, which preserves a basis. Finite selection after compactness at each index and freeness of Φ complete the proof. $\square$

9.7. The triangle identity.

Theorem 9.10. *With the mixed maps N_{01} , N_{12} , N_{02} of Theorem 3.6 for the three pairs of (9.1) and the symplectic triangle map μ_S^E , which counts holomorphic triangles only,*

$$N_{02} \circ \mu_G^E \simeq \mu_S^E \circ (N_{01} \otimes N_{12}) \quad \text{over } \mathbb{F}_2.$$

The proof compares two degenerations of the mixed triangle of Definition 9.6, with the regions G , S , the arc γ and the rectangle B fixed there.

Lemma 9.11. *Every proper arc in G from the interior of e_0 to the interior of e_2 meets B . Consequently the closed cut $Y_{02}^\#$ passes through the collapsing rectangle, and a product collar of the cut exists only when the compression scale over I is of order one.*

Proof. B meets the frontier of G only in I and J , and $G \setminus B$ has two components, one containing each input vertex. The arcs $e_0 \cap G$ and $e_2 \cap G$ lie in different components, so an arc joining them meets B . Over B the metric is $dt^2 + \lambda_\varepsilon^2(d\rho^2 + g_\Sigma)$, whose surface factor is scaled by ε , so no neighbourhood of the cut is a product with a fixed closed metric until λ is raised to one. $\square$

Lemma 9.12. *There is a smooth path F_τ , $\tau \in [0, 1]$, of data satisfying (H1)–(H5) and Hypothesis 6.5 on one marked domain with the bundle E , equal to the data of the collapse (Theorem 6.6) at small scale ε with the ordinary necks of Theorem 5.1 for τ near 0 and to the data in which a long product collar of the closed cut $Y_{02}^\#$ is inserted for τ near 1, such that: the scales stay in a compact positive range (the compression scale over I is at least ε and the cut collar has bounded length); all holonomy loop families are fixed and only their forms vary, the forms being supported away from the seam collars and the ends while the loops themselves may pass through them; $\partial_\tau F_\tau$ is supported in a compact set disjoint from all ends; and every F_τ is invariant under Φ .*

Proof. The path has five stages, each inside a convex set of such data: (A) raise the scale over I to one by the convex combination $\mu_s^2 = (1-s)\lambda_\varepsilon^2 + s$ of metrics of the form $dt^2 + \mu^2 g_C$ and of the matching section metrics; (B) interpolate linearly to a metric which is a product $a^2 du^2 + g_{02}$ on a collar of the cut, the two metrics agreeing with every prescription on every overlap because on the compression pieces the collar coordinate is the edge coordinate and g_{02} restricts to the compression metrics, and on B it is the stage (A) metric; (C) lengthen the collar; (D) switch on the end perturbation of $Y_{02}^\#$ on the collar; (E) interpolate the remaining metrics,

section structures and forms to the data with the collar of the cut inserted. Compatible almost complex structures are joined by $J_k = (-A_k^2)^{-1/2} A_k$ for the convex combination k of the associated metrics, where $\omega(u, v) = k(A_k u, v)$; this equals the Hodge structure wherever both ends do. Forms are replaced by their self-dual projections in the varying metric. The bundle over the collar is a product because all transitions are constant along the edges, so stretching the collar lifts to E with no clutching. Invariance holds because every class of Φ vanishes on the surface sheets and hence, by injectivity of $H^1(C_i) \rightarrow H^1(\Sigma_h)$, on the compression pieces, and every loop has zero lifting class in Φ . $\square$

Proof of Theorem 9.10. Endpoint 1: the outgoing cut. Stretch along the closed three-manifold $Y_{02}^\#$ separating the gauge triangle from the DFL domain for the ρ_ζ -marked pair (P_0, P_2) . We verify the hypotheses of Theorem 5.1 for these two pieces and this connected cut. (H1): on the gauge piece the index-energy law is part of Proposition 9.9; on the DFL piece it is [11, Proposition 3.20, Lemma 3.15] for the marked pair, and on the joined problem it follows by additivity across the cut, the constant pair at each output generator having index zero, so that the constant of the joined problem is the sum of the two constants. (H2): μ_G^E is regular by Proposition 9.9, and N_{02} is regular in the invariant class by Theorem 5.4, with the loops supplied from its incoming end $Y_{02}^\#$, whose critical limits are flat on a collar of an odd copy of Σ_h by Lemma 9.8; the section part of a cokernel element is killed by harmonic traces on the matching arc, and the two compression ends are conjugate to the untwisted ones by τ_ζ . The cross-section at the cut is invertible because the critical points of $Y_{02}^\#$ are nondegenerate and irreducible (Lemma 9.8), so the extended Hessian, including the gauge-fixing component, has a spectral gap. (H3)–(H5) hold by construction. Theorem 5.1 gives the count $N_{02} \circ \mu_G^E$ for long cut collars, the DFL piece being identified with the vertex map by Lemma 9.7, and a finite change of collar length is a compact family (Theorem 5.2). The lifting groups glue by the fibre product: restriction $\Phi \rightarrow \mathbb{F}_2^2$ to the two incoming ends is an isomorphism, the output group of the DFL piece restricts isomorphically to the lifting group of the cut, and the connecting H^0 term at the connected cut vanishes.

Endpoint 0: the interval collapse. Collapsing B by Theorem 6.6, the limit domain is a disk with a physical P_1 boundary interval and two new mixed ends; cutting off two DFL disks along ordinary strip ends leaves a triangle domain with boundary (P_0, P_1, P_2) . The triangle piece is regular for a generic domain-dependent almost complex structure supported away from the neighbourhood N of the collapsed arc, where Hypothesis 6.5(X1) fixes the Hodge structure: there are no constant triangles because the triple intersection is empty, so every solution is nonconstant on an open set away from N by unique continuation, and the standard transversality argument applies there. Choose the ordinary neck lengths first: by Theorem 5.1 the zero-scale domain is then regular in index at most zero with index-zero count $\mu_S^E(N_{01} \otimes N_{12})$, the two DFL pieces being identified with the vertex maps by Lemma 9.7; this is Hypothesis 6.5(X4), and (X5) holds by Lemma 6.7. Then choose ε small; by Theorem 6.6(iii) the count at scale ε equals the zero-scale count. Hypotheses (X1)–(X3) hold because the middle primary perturbation vanishes, the secondary perturbations are supported away from B , and the original ends are regular.

The family. Apply Theorem 5.2 to the path of Lemma 9.12. Its hypotheses: the index-energy law on every fibre with the constant of the twisted endpoint, since a reference configuration on one fibre is a reference on every fibre and the topological energy involves no metric (so the index of a fixed reference is constant in τ by continuity of the Fredholm family, and the law at $\tau = 0$ is Theorem 6.6(i)); an energy bound independent of τ , the terms $\int \chi' f$ being bounded by the C^0 norms of the end perturbations; family compactness, with weak limits

identified with the prescribed bundle E by modifying bundle isomorphisms only on small balls around the concentration points joined by arcs to an incoming slice, which introduces no lifting class; family regularity in the invariant class by Theorem 5.4, with holonomy terms built from local unitary lifts, which exist on neighbourhoods of the loops and are unique up to tensoring with a flat real line, so that each term is defined up to sign and the span is unaffected; and no internal neck breaks, since the scales stay in a compact positive range. The breaks of the parametrized one-dimensional space therefore occur only at the original ends: the two closed inputs and the ordinary output. At an incoming break the index -1 piece may carry the other even lift of that input; a unique class of Φ aligns it, and by invariance of the whole family the aligned piece is already counted. The boundary count gives the chain homotopy. $\square$

9.8. The two-handle square. Let p_i^ϕ be the admissible presentations of $Z_0 = Y_{01}$ and $Z_1 = Y_{02}$ given by the splittings (H_0, H_1) and (H_0, H_2) with the global frames ϕ_i^{gl} of Section 4.4. Let $P_{\phi_i}^G$ and $P_{\phi_i}^S$ be the transport of complexes from the filling frame to the global frame: on the gauge side the pushforward by the bundle isomorphism, and on the symplectic side the pushforward by the symplectomorphism τ_{a_1} (respectively $\tau_{a'_2}$) of M_h . By naturality of every construction under bundle isomorphisms, $N_{p_i^\phi} P_{\phi_i}^G = P_{\phi_i}^S N_{0i}$ on chains, and by the definition of the morphisms of [21], which include the identifications at the ends, $I^\#(W_2, E; \phi^{\text{gl}}) = P_{\phi_1}^G F_G(W_2, E)(P_{\phi_0}^G)^{-1}$.

Theorem 9.13. *Define the symplectic map*

$$S_{W_2, E} = P_{\phi_1}^S \mu_S^E(\cdot \otimes \theta_S) (P_{\phi_0}^S)^{-1} : HF(Z_0, p_0^\phi) \rightarrow HF(Z_1, p_1^\phi).$$

Then $H(N_{p_1^\phi}) \circ I^\#(W_2, E; \phi_0^{\text{gl}}, \phi_1^{\text{gl}}) = S_{W_2, E} \circ H(N_{p_0^\phi})$.

Proof. Insert $x \otimes \theta_G$ into the chain homotopy of Theorem 9.10. Since θ_G is a cycle, $N_{02} \mu_G^E(x \otimes \theta_G)$ is homologous to $\mu_S^E(N_{01} x \otimes N_{12} \theta_G) = \mu_S^E(N_{01} x \otimes \theta_S)$ by Proposition 9.4. By Proposition 9.5, $N_{02} F_G(W_2, E) = \mu_S^E(\cdot \otimes \theta_S) N_{01}$ on homology. Conjugating by the transports gives the claim. $\square$

The map $S_{W_2, E}$ is defined by two pushforwards by symplectomorphisms and a triangle count; it does not involve gauge theory.

Remark 9.14. For E nontrivial the gauge map can vanish for grading reasons: for the (-1) -framed unknot with the nontrivial bundle, $\chi = 1$, $\sigma = -1$, $P(E) = 1$, and the map $I^\#(S^3) \rightarrow I^\#(S^3)$ has degree two by Remark 2.6, hence is zero [38, Section 7.6]. The theorem then says that the symplectic triangle map $S_{W_2, E}$ vanishes on homology.

10. EXCISION ALONG A TORUS FOR THE MIXED EQUATION

The one-handle comparison of Section 11 passes through a mixed domain whose gauge input is a connected sum along a torus. On the gauge side such a manifold is split by Floer's excision theorem [5, 22], in the form of [21, Theorem 5.6]. This section proves the corresponding mixed statement, Theorem 10.2.

10.1. The folded splittings. Let M have presentation data with torus compression bodies U_0, U_1 of genus $h \geq 2$, and let $C = C_{h,g}$, $g = h + k$, be a compression body from Σ_h to Σ_g adding k handles. Put $V_i = C \circ U_i$, the enlarged torus compression bodies. Then $V_0 \cup (-V_1)$ is the torus-summed object $(M^+)^\#$ of $M^+ = M \#^k(S^1 \times S^2)$: every added meridian disk occurs on both sides, and the two disks form a nonseparating sphere. Let $L_C \subset M_h^- \times M_g$ be the Lagrangian correspondence of C . For a Lagrangian $P \subset M_h$ transverse to the lower projection, the composition $P \circ L_C$ is embedded, because the lower projection of L_C is a submersion and its upper projection an embedding: L_C is the flat moduli space of C , which fibres over the moduli space of Σ_h with fibre $\mathrm{SU}(2)^k$, and restriction to Σ_g is injective since $\pi_1(\Sigma_g) \rightarrow \pi_1(C)$ is onto [11, Proposition 2.21].

Height Lagrangians. For $t > 0$ small, perturb the primary data of V_i by t times a height function whose restriction to the clean intersection $L(V_i) \cap (L(U_i) \circ L_C) \cong \mathrm{SU}(2)^{g-1}$ has the 2^{g-1} nondegenerate critical points of Section 9; call the resulting upper Lagrangians $R_i \subset M_g$ the *height Lagrangians* at height t . The top generator q_i^* of each auxiliary pair below is defined intrinsically, by the grading m of Section 11.3; it is a critical point of the height function, its maximum or its minimum according to whether the perturbed Lagrangian is the first or the second entry of the pair. *Lower Lagrangians.* Then choose small primary data $\tilde{P}_i$ of U_i , in a neighbourhood of $L(U_i)$ that is small relative to t (the neighbourhoods $\mathcal{N}(t)$ fixed after Proposition 11.10). In $X = M_h^- \times M_g$ put

$$A_0 = \tilde{P}_0^- \times R_0, \quad A_1 = \tilde{P}_1^- \times R_1,$$

consider the triple (A_0, L_C, A_1) , and require $A_0 \pitchfork L_C$, $L_C \pitchfork A_1$, $A_0 \pitchfork A_1$ and $A_0 \cap L_C \cap A_1 = \emptyset$. These are generic conditions: the triple intersection has expected dimension $3d - 2 \cdot 2d = -d < 0$, where $d = (3h - 3) + (3g - 3)$ is the dimension of each of the three Lagrangians in the manifold X of dimension $2d$, and the universal restriction map for primary perturbations is a submersion [11, Section 2.3]. (With unperturbed data the triple intersection would contain the fibres of L_C over $L(U_0) \cap L(U_1)$.) The pairs (A_0, L_C) and (L_C, A_1) unfold to $(R_0, \tilde{P}_0 \circ L_C)$ and $(\tilde{P}_1 \circ L_C, R_1)$, whose intersections are the critical sets of the height functions.

The gauge pieces are $D_i = (-U_i) \cup_{T^2} V_i$, with $\partial D_i = (-\Sigma_h) \sqcup \Sigma_g$, and C . Flat connections on D_i restrict bijectively to pairs of flat connections on U_i and V_i : across the odd torus the gluing is unique, since the flat connection on the odd torus has central determinant-one stabilizer and $H^*(T^2; \mathrm{ad}) = 0$; so $L(D_i) = L(U_i)^- \times L(V_i)$, perturbed to A_i . The closed cut of the pair (A_0, A_1) is

$$Z = D_0 \cup (-D_1) \cong (-M) \# M^+ \# T^3,$$

with one torus factor. For a split almost complex structure $J_- \times J_+$ on X , folding gives an isomorphism of complexes $\mathcal{K}: CF_X(A_0, A_1) \rightarrow CF(\tilde{P}_0, \tilde{P}_1)^\vee \otimes CF(R_0, R_1)$, and a cycle $\sum x_i^\vee \otimes y_i$ on the right is the chain map $v \mapsto \sum x_i^\vee(v) y_i$.

The metrics are chosen as follows: U_0, U_1 share their metric near Σ_h , V_0, V_1 near Σ_g , and all torus collars carry the flat metric induced from the fixed metric on T^3 , with the fixed odd bundle.

10.2. The torus region and the domains. Let $T = \mathbb{R}^2/\Lambda$ be the flat torus, P_T the odd bundle and α its flat connection, unique up to gauge. The adjoint bundle splits into three flat real line bundles with nontrivial characters, so $H^*(T; \mathrm{ad} \alpha) = 0$, the twisted Laplacian satisfies $\Delta_\alpha \geq \mu_T^2$ in every degree with $\mu_T = \pi \min\{|\xi| : \xi \in \Lambda^* \setminus 2\Lambda^*\}$, and the determinant-one stabilizer of α is the centre.

Consider the special quintuple X_D of [11, Section 3.1] for a pair of torus compression bodies (Y, Y') with $\partial Y = \Sigma' \sqcup T$. Its gauge part contains $\mathbb{U}_- \times T$, where $\mathbb{U}_-$ is a planar region with

one gauge input arm, two mixed-end arms $\theta \rightarrow \pm\infty$ and the matching line $\mathbb{U}_\partial = \{0\} \times \mathbb{R}_\theta$; the metric on $\mathbb{U}_-$ is flat near the matching line. Hence X_D contains the flat product region

$$N_X = T \times [-1, 0]_s \times \mathbb{R}_\theta,$$

whose face $\{s = 0\}$ is the torus component of the matching surface, and for $|\theta| \geq 2$ this region lies in the mixed ends. Since the moduli space of the odd torus is a point, the matching condition there says only that the restriction of the connection to each fibre $T \times \{(0, \theta)\}$ is flat; we call it the *flat-fibre condition*. Its linearization is $\zeta(\partial_s) = 0$, $d_\alpha \zeta_\theta = 0$ on the face, with the longitudinal component $\zeta(\partial_\theta)$ free.

- Definition 10.1.** (a) For $0 \leq R < \infty$, the *collared domain* is $W_R = X_D \cup (T \times [0, R]_r \times \mathbb{R}_\theta)$, glued along the torus face, with the flat product metric and no perturbation on the collar and the flat-fibre condition at $r = R$. Its mixed-end slices are $Y_R = Y \cup ([0, R] \times T)$ and Y'_R . It is diffeomorphic to X_D by a map ψ_R that stretches $s \in [-3/4, -1/4]$ and is the identity outside N_X , hence on the supports of all perturbations, on the gauge and strip ends, near the other matching surface and on the section.
- (b) The *completed domain* is $W_\infty = X_D \cup (T \times [0, \infty)_r \times \mathbb{R}_\theta)$, with no torus face and mixed slices Y_∞, Y'_∞ .
- (c) For the two sides $X_{D,-}$ (the reflected presentation of the lower end M of the one-handle cobordism) and $X_{D,+}$ (the enlarged presentation of M^+) and $\lambda > 0$, the *joined domain* W_λ^J is $W_{-, \lambda} \cup W_{+, \lambda}$ glued along their torus faces by the marked bundle isometry reversing the torus orientation together with the normal coordinate. It has the bridge $T \times [-\lambda, \lambda] \times \mathbb{R}_\theta$, two gauge inputs $(-M)^\#$ and $(M^+)^\#$, a section $\mathbb{U}_+ \rightarrow \mathbf{M}_h \times \mathbf{M}_g$ with the split structure $J_- \times J_+$ on all of $\mathbb{U}_+$, strip output at $A_0 \cap A_1$, and mixed-end slices $D_{0, \lambda} = Y_- \cup_{[-\lambda, \lambda] \times T} Y_+$ and $D_{1, \lambda} = Y'_- \cup_{[-\lambda, \lambda] \times T} Y'_+$: on each arm the slices of the two sides are joined through the bridge.

Weights are functions $\tau \geq 0$ with $|d\tau| \leq 1$ and $|\nabla d\tau| \leq 1$, equal to the end coordinate on the ends and, on the torus region, to a function $\tau_0(\theta)$ of θ alone.

The critical manifold of the joined mixed ends is $P_i \times Q_i$ for the two separate critical manifolds, because the two flat connections glue uniquely across the odd torus. The lifting group of W_λ^J is $\Phi_J \cong \mathbb{F}_2^2$, restricting isomorphically to $\Phi_- \oplus \Phi_+$, with lifting class $s_Z = s_- + s_+$ at the cut Z ; its elements are lifting classes that vanish on every piece (they come from the graph with a vertex for each surface sheet and each compression piece and an edge for each contact region), since restriction of H^1 from each compression body to its full boundary is injective.

Counting index-zero solutions defines: the separate maps $N_\pm^{(R)}$ on W_R , which for $R = 0$ are $N_\pm$; the completed maps $N_{\pm, \infty}$ on W_∞ ; and the joined maps $N_\lambda^J: C_G((-M)^\#) \otimes C_G((M^+)^\#) \rightarrow CF_X(A_0, A_1)$.

10.3. Statement. Let $N_-: C_G((-M)^\#) \rightarrow CF(\tilde{P}_0, \tilde{P}_1)^\vee$ be the mixed map of the reflected presentation of the lower end M (Section 11.2), $N_+: C_G((M^+)^\#) \rightarrow CF(R_0, R_1)$ that of the enlarged presentation with the height Lagrangians, and $N_Z: C_G(Z) \rightarrow CF_X(A_0, A_1)$ the mixed map of the pair (A_0, A_1) (Remark 3.7) with the data at width λ_0 fixed in Section 10.9. Let $\mathbf{E}: C_G(Z) \rightarrow C_G((-M)^\#) \otimes C_G((M^+)^\#)$ be the excision map of [21, Theorem 5.6], with the end-perturbation data f_Z of the same data on Z , and $\bar{\mathbf{E}}$ the reverse excision map. For other data on Z both sides change by the same gauge continuation map.

Theorem 10.2 (mixed torus excision). *On homology over $\mathbb{F}_2$,*

$$[\mathcal{K}][N_Z] = ([N_-] \otimes [N_+]) [\mathbf{E}].$$

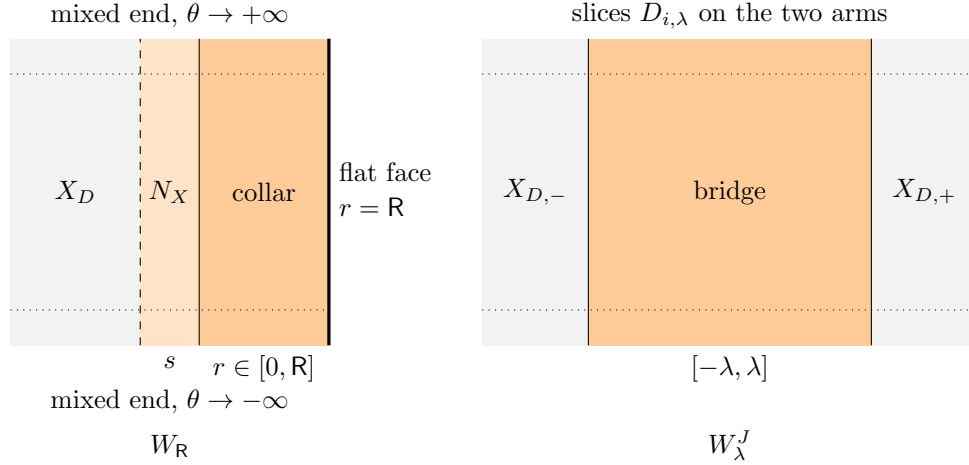


FIGURE 5. The torus region, drawn in the base coordinates (normal coordinate horizontally, the end coordinate θ vertically); every point carries a torus T with the flat odd connection. Left: the collared domain W_R of Definition 10.1(a), with the flat product region N_X of X_D , the collar and the flat face at $r = R$; for $|\theta| \geq 2$ (dotted lines) the region lies in the mixed ends. The completed domain W_∞ has $R = \infty$. Right: the joined domain W_λ^J , in which the collars of the two sides are glued along their flat faces into a bridge of width 2λ . No closed hypersurface separates the two sides, since the torus region runs into the mixed ends.

The gauge half. Z contains two parallel tori separating the two summands; cutting along them and regluing crosswise gives $Z' = (-M)^\# \sqcup (M^+)^\#$. The bundle is odd on these tori and every component of Z and Z' meets one. The excision cobordism $\mathcal{E}: Z \rightarrow Z'$ is $T^2 \times P$, P a pair of pants with input c and outputs $c_\pm$, with the products of the punctured summands inserted along two arcs from c to $c_\pm$; its lifting group is $\Phi_{\mathcal{E}} = \{(s, s_-, s_+) : s = s_- + s_+\}$, dual to the four hypersurfaces of the construction of [21, Section 5.3]. By [21, Theorem 5.6], $[\mathcal{E}]$ is an isomorphism with inverse $[\bar{\mathcal{E}}]$. The theorem is stated over $\mathbb{Z}$; the maps $\mathcal{E}$, $\bar{\mathcal{E}}$ and the chain homotopies in its proof are defined over $\mathbb{Z}$, so the conclusion holds after reduction to $\mathbb{F}_2$, and the Künneth identification for Z' is over the field $\mathbb{F}_2$.

Plan of the proof. Only the mixed statement is new. It is not an instance of neck-stretching along a closed hypersurface: the torus region runs into the mixed ends, so no closed three-manifold separates the two sides. The proof has three steps:

- (a) (closed cut) $\mathcal{K}N_{\lambda_0}^J \simeq \mathcal{K}N_Z \bar{\mathcal{E}}$ at a fixed width λ_0 (Proposition 10.20, including the data change of Lemma 10.19);
- (b) (width) $N_b^J \simeq N_a^J$ for $0 < a \leq b < \infty$ (Proposition 10.16);
- (c) (completion) for large λ and large R , $\mathcal{K}N_\lambda^J = N_{-, \infty} \otimes N_{+, \infty}$ and $N_\pm^{(R)} = N_{\pm, \infty}$ as matrices, and $N_\pm^{(R)} \simeq N_\pm$ (Theorem 10.30, Proposition 10.15).

Then $[\mathcal{K}N_Z][\bar{\mathcal{E}}] = [N_- \otimes N_+]$, and since $[\bar{\mathcal{E}}] = [\mathcal{E}]^{-1}$ the theorem follows.

10.4. The linearized operator at the odd flat torus. The constant $\mu_T > 0$ of Section 10.2, the square root of the lowest eigenvalue of the twisted Laplacian of α , is the spectral gap that makes the following operators invertible.

Proposition 10.3 (the operator at the odd flat torus). *Let B be an oriented surface with boundary of bounded geometry (a half-plane, a strip, a convex domain, or $\mathbb{R}^2$), and let $D = (d_\alpha^*, \sqrt{2}d_\alpha^+)$ act on ad-valued one-forms on $T \times B$ with the domain $a(\nu) = 0$ on $T \times \partial B$ and d_α -closed torus restriction on $T \times \partial B$. Then D and its adjoint are invertible, with $\|D^{-1}\|_{L^2 \rightarrow L^2} = 1/\mu_T$ independently of B . The same holds with weights $e^{\delta\tau}$, $|\delta\tau| \leq 1$, τ constant on the torus fibres, for $|\delta| < \mu_T$, with $\|D^{-1}\| \leq (\mu_T - |\delta|)^{-1}$. For an anti-self-dual A on $T \times \{s \geq 0\}$ with flat torus restriction on the boundary,*

$$\int_T \langle \nabla_s F_A, F_A \rangle = 4 \int_T \langle [F_{sx}, F_{sy}], u \rangle, \quad d_{A_t} u = \iota_{\partial_t} F_A|_T,$$

so the boundary flux is cubic in the curvature.

Proof. Splitting along the torus, D becomes $d + d^*$ on forms on B with absolute boundary conditions plus a zeroth-order term Λ_T , of size at least μ_T , that anticommutes with it, so $\|Da\|^2 = \|\mathfrak{D}u\|^2 + \|\Lambda_T u\|^2 \geq \mu_T^2 \|u\|^2$, where $\mathfrak{D} = d + d^*$ on the base. The two boundary conditions remove the first-order cross term and the torus symplectic term: closedness of the fibre restriction kills the former, and closedness together with $H^1(T; \text{ad}) = 0$ kills the latter; boundary curvature enters only the H^1 constant. By Green's formula with the domain of D (free normal component a_t and exact fibre restriction a_T), the adjoint has domain $\omega(\nu, \tau_B) = 0$ for the tangential base direction τ_B and $d_\alpha^*((\iota_\nu \omega)|_T) = 0$, with no condition on the scalar, and satisfies the same identity. The weighted version conjugates by $e^{\delta\tau}$, which commutes with the fibrewise operators because τ is constant on fibres. The flux identity follows from the Bianchi identity and flatness on the boundary: in the gauge with fibre restriction α and $A_s = 0$, the linear terms in ∂_t cancel after integration over T because $[D_x, D_y] = F_\alpha = 0$. $\square$

Corollary 10.4 (the stationary operator at the odd flat torus). *Let $I \subset \mathbb{R}$ be a compact interval, a closed half-line or $\mathbb{R}$, with coordinate r . On pairs $\zeta = (\varphi, b)$ of an ad-valued function and one-form on $T \times I$, with the product metric and the pulled-back flat connection α , let*

$$\mathcal{A}_\alpha(\varphi, b) = (d_\alpha^* b, *d_\alpha b + d_\alpha \varphi),$$

with the domain given by $b(\partial_r) = 0$ and $d_\alpha(b|_{T \times \{r\}}) = 0$ at every endpoint r of I , and no condition on φ . Then $\mathcal{A}_\alpha$ is self-adjoint and $\|\mathcal{A}_\alpha \zeta\| \geq \mu_T \|\zeta\|$. If $\tau = \tau(r)$ satisfies $|\tau'| \leq 1$ and $|\delta| < \mu_T$, then $\|e^{\delta\tau} \mathcal{A}_\alpha \zeta\| \geq (\mu_T - |\delta|) \|e^{\delta\tau} \zeta\|$.

Proof. Let $\zeta = (\varphi, b)$ lie in the domain and have compact support, and let $\beta \in C_c^\infty(\mathbb{R})$ be a nonzero cutoff. On $T \times B$ with $B = \mathbb{R}_t \times I$, a plane, a half-plane or a strip, put $a_N = \beta(t/N)(-\varphi dt + b)$. It satisfies the boundary conditions of Proposition 10.3: $a_N(\partial_r) = \beta(t/N)b(\partial_r) = 0$, and its restriction to each boundary torus fibre is closed. For a one-form $a = \psi dt + c$ on $T \times B$ one has $d^*a = -\partial_t \psi + d^*c$ and $\sqrt{2}d^+a$ corresponds, under the identification of Λ^+ with the one-forms of the three-dimensional slice used in [11, (4.16)], to $\partial_t c + *d_\alpha c - d_\alpha \psi$, isometrically. Hence $\|Da_N\| = \|(\partial_t + \mathcal{A}_\alpha)(\beta(t/N)\zeta)\|$, and

$$\|Da_N\|^2 = N\|\beta\|^2 \|\mathcal{A}_\alpha \zeta\|^2 + N^{-1}\|\beta'\|^2 \|\zeta\|^2,$$

because the cross term is a multiple of $\int (\beta^2)' = 0$. Proposition 10.3 gives $\|Da_N\|^2 \geq \mu_T^2 \|a_N\|^2 = \mu_T^2 N \|\beta\|^2 \|\zeta\|^2$. Dividing by N and letting $N \rightarrow \infty$ gives the inequality for compactly supported ζ , and the general case follows by approximation. The weight $\tau(r)$ is constant on the torus fibres and independent of t , so the weighted version of Proposition 10.3 gives the weighted inequality in the same way. For ζ_1, ζ_2 in the domain, Green's formula leaves the boundary terms $\pm \int_{T \times \partial I} \langle b_1 \wedge b_2 \rangle$ and terms containing $b_i(\partial_r)$; the latter vanish, and the

former vanish because the torus restrictions are closed, hence exact since $H^1(T; \text{ad } \alpha) = 0$, and $\int_T \langle d_\alpha u \wedge d_\alpha v \rangle = \int_T d \langle u, d_\alpha v \rangle = 0$. The boundary condition is elliptic in the sense of [6], the closed and the exact conditions on T coinciding, so $\mathcal{A}_\alpha$ is self-adjoint; with the lower bound it is invertible. $\square$

Theorem 10.5 (regularity at the flat torus boundary). *For $\rho > 0$ let $D_\rho^+ = \{(s, t) : s \geq 0, s^2 + t^2 < \rho^2\}$ and $Q_\rho = T \times D_\rho^+$. Let A be an L_1^p connection on Q_2 , $p > 2$, anti-self-dual, with weakly flat restriction to $T \times \{z\}$ for z on the boundary diameter.*

- (a) *A is gauge equivalent to a connection smooth up to the boundary.*
- (b) *There are $\hbar > 0$ and C_k , depending only on T and k , such that if $\epsilon^2 = \int_{Q_2} |F_A|^2 \leq \hbar$ then $\sup_{Q_1} |\nabla_A^k F_A| \leq C_k \epsilon$, and in a Coulomb–Neumann gauge $A = \alpha + a$ with $\|a\|_{C^k(Q_1)} \leq C_k \epsilon$.*
- (c) *Suppose that A satisfies the flat-fibre condition and has finite energy on the half-space $T \times \{s \geq 0\}$. Then*

$$\sup_{T \times D_1(z)} |\nabla_A^k F_A| \longrightarrow 0 \quad (|z| \rightarrow \infty).$$

Proof. (a) is the regularity theorem of [12, Theorem 1] with the torus as matching surface; its proof uses the Lagrangian boundary condition only through local charts, and applies to the one-point moduli space. We prove (b) in three steps.

Curvature bound. Put $f(s, t) = \int_{T \times \{(s, t)\}} |F_A|^2$ on D_2^+ . For an anti-self-dual connection on the flat product the Weitzenböck formula reads $\nabla_A^* \nabla_A F_A = -\{F_A, F_A\}$, with no curvature term; integrating over the torus fibres gives $(\partial_s^2 + \partial_t^2) f \geq -C \sup |F_A| f$ in the interior. On the boundary diameter, $\partial_s f = 2 \int_T \langle \nabla_s F_A, F_A \rangle$, which by the flux identity of Proposition 10.3 is cubic: u is determined fibrewise by $d_{A_t} u = \iota_{\partial_t} F_A|_T$, and since $H^0(T; \text{ad } \alpha) = 0$ and A_t is close to α , $\|u\| \leq C \|F_A\|$ on each fibre; so $|\partial_s f| \leq C \sup |F_A| f$. The mean-value inequality for functions on a half-disk satisfying these two inequalities [41], with Heinz’s argument for the factor $\sup |F_A|$, gives $\sup_{D_{3/2}^+} f \leq C \epsilon^2$ once $\epsilon^2 \leq \hbar$, and interior estimates on T times small disks turn this into $\sup_{Q_{3/2}} |F_A|^2 \leq C \epsilon^2$.

Gauge. For ϵ small the restrictions of A to the fibres are C^0 -close to α up to gauge. The Neumann Laplacian of α is invertible because $H^0(T; \text{ad } \alpha) = 0$, so the implicit function theorem, with compactness below the threshold, gives a gauge on $Q_{3/2}$ in which $A = \alpha + a$ with $d_\alpha^* a = 0$, $a(\nu) = 0$ on the face and $\|a\|_{L_1^2(Q_{3/2})} \leq C \epsilon$.

Derivative bounds. In this gauge the equations read $Da = -\sqrt{2}(a \wedge a)^+$, with $D = (d_\alpha^*, \sqrt{2}d_\alpha^+)$ as in Proposition 10.3, and the flat-fibre condition reads $d_\alpha a_T = -\frac{1}{2}[a_T \wedge a_T]$ on the face, whose linear part is the boundary condition of that proposition. This boundary problem is elliptic [6], so local regularity up to the face gives

$$\|a\|_{H^{k+1}(Q_1)} \leq C_k (\|(a \wedge a)^+\|_{H^k(Q_{3/2})} + \|[a_T \wedge a_T]\|_{H^{k+1/2}(\partial Q_{3/2})} + \|a\|_{L^2(Q_{3/2})}),$$

and bootstrapping from $\|a\|_{L_1^2} \leq C \epsilon$, with the quadratic right side, gives $\|a\|_{C^k(Q_1)} \leq C_k \epsilon$ and the bounds on $\nabla_A^k F_A$.

(c) Given $\epsilon > 0$, choose a compact set outside which the energy is less than $\min(\epsilon^2, \hbar)$; for $|z|$ large $T \times D_2(z)$ lies outside it, and (b), or its interior version (Remark 10.6), gives the bound. $\square$

Remark 10.6. Part (b) of Theorem 10.5 holds, with constants depending only on T and k , on interior boxes $T \times D_\rho(z)$ disjoint from the face and with no boundary condition. Below the

threshold the curvature bounds are Uhlenbeck's interior estimates [13, Section 2.3], applied on balls of a fixed radius, and the gauge statement follows from the contradiction argument of (b), since a flat connection on $T \times D_2(z)$ is gauge equivalent to α . The local gauges may be taken even: the stabilizer of α is the Klein four-group, whose three nontrivial elements are odd and realize the three nonzero classes of $H^1(T; \mathbb{F}_2)$, so composing with one of them makes any local gauge even without moving α . Below, *small-energy regularity* refers to (b) and to this interior version.

10.5. The stationary problems and uniform decay.

Proposition 10.7 (uniform stationary gap). *Let $q \in P$ and $R \in [0, \infty]$. The stationary operator $\mathcal{A}_{q,R}(\varphi, b, \nu) = (d^*b, (*d_B + \text{Hess } h)b + d\varphi, J\partial_\sigma\nu)$ on Y_R and the section interval, with the matching conditions at the surface, the Lagrangian condition at the end of the interval and the flat-fibre condition at $r = R < \infty$, is self-adjoint with kernel T_qP , and its nonzero spectrum avoids $(-\gamma, \gamma)$ for a $\gamma > 0$ independent of q and R . The same holds for the joined slices $D_{i,\lambda}$, with kernel $T(P_i \times Q_i)$, uniformly in $\lambda \geq \lambda_0$. The stationary representatives B_q may be taken equal to α on the collar.*

Proof. Self-adjointness uses the Green cancellation at the matching surface (Convention 3.1) and the elliptic boundary theory of [6]. The kernel is computed as in Proposition 7.9. For uniformity, suppose that there are $q_n \in P$, $R_n \in [0, \infty]$ and ζ_n orthogonal to the kernel with $\|\zeta_n\| = 1$ and $\|\mathcal{A}_{q_n, R_n}\zeta_n\| \rightarrow 0$. Let χ be a cutoff of r , equal to 0 for $r \leq 1$ and to 1 for $r \geq 2$; it commutes with the primary Hessian, which is supported away from the collar. On the collar the operator is the stationary operator $\mathcal{A}_\alpha$ at the odd flat torus on $T \times [0, R_n]$, so Corollary 10.4 gives

$$\mu_T \|\chi\zeta_n\| \leq \|\mathcal{A}_\alpha(\chi\zeta_n)\| \leq \|\mathcal{A}_{q_n, R_n}\zeta_n\| + \|\chi'\zeta_n\|_{L^2(1 \leq r \leq 2)}.$$

Hence $\|\zeta_n\|_{L^2(K)} \geq 1/C$ for the fixed compact region $K = \{r \leq 2\}$ of the slice together with the section interval. Local elliptic estimates give a bound in $H^1(K)$, and after passing to a subsequence $q_n \rightarrow q$, $R_n \rightarrow R \in [2, \infty]$, and ζ_n converges weakly to a limit $\zeta \neq 0$, strongly in $L^2(K)$, with $\mathcal{A}_{q,R}\zeta = 0$ and the limiting boundary conditions. The kernel elements satisfy $\mathcal{A}_\alpha\zeta = 0$ on the collar and therefore decay like $e^{-\mu_T r}$ there, by Corollary 10.4, uniformly in q ; so the kernels converge strongly and orthogonality to the kernel passes to the limit; ζ is a nonzero kernel element orthogonal to the kernel, a contradiction. $\square$

Proposition 10.8 (uniform decay on growing mixed ends). *There are e_0, c, C_m , independent of $R \in [0, \infty]$ and of $\lambda \geq \lambda_0$, such that a solution of the autonomous mixed equation on a half-cylinder over Y_R (or $D_{i,\lambda}$) with kinetic energy less than e_0 converges to a stationary solution (B_q, q) in the sense of (10.1) below: its kinetic energy beyond θ decays at rate $2c$, and, in a temporal gauge, its difference from the stationary limit, section included, has H^m norm on unit intervals $[\theta, \theta + 1]$ of the whole slice decaying at rate c .*

Proof. *Convergence into a neighbourhood of the critical family.* On each unit interval $[\theta, \theta + 1]$ the kinetic energy is less than e_0 . On the compact part of the slice local compactness applies, and on the collar Theorem 10.5(b), summed over a cover of $T \times [0, R]$ by half-disks and disks with bounded overlap, bounds the curvature linearly by the square root of the local energy. Hence for e_0 below a threshold independent of R every slice lies in a fixed neighbourhood of the critical family.

Gradient inequality. On this neighbourhood the local action $\mathcal{F}$ is well defined, with critical set the critical family, which is Morse–Bott with the uniform gap γ of Proposition 10.7. Expanding $\mathcal{F}$ about the nearest critical point, in the slice orthogonal to the kernel, gives

$|\mathcal{F} - \mathcal{F}_{\text{crit}}| \leq C\|\nabla\mathcal{F}\|^2$ with C depending only on γ ; the flat-fibre condition removes the boundary term at $r = R$ in the Hessian, so C is independent of R .

Decay. Along a solution $\partial_\theta \mathcal{F} = -\|\nabla\mathcal{F}\|^2 \leq -(\mathcal{F} - \mathcal{F}_{\text{crit}})/C$, so the kinetic energy $\mathcal{F}(\theta) - \mathcal{F}_{\text{crit}}$ beyond θ decays at the rate $2c = 1/C$, and the length of the path of slices over $[\theta, \infty)$ is at most the sum over unit intervals of the square roots of their kinetic energies, which decays at the rate c ; so the solution converges to a stationary solution (B_q, q) . In a temporal gauge the difference from π^*B_q solves the linearized equation with coefficients bounded by the same quantity, and local elliptic estimates on $[\theta - 1, \theta + 2]$ times the slice, uniform in R by Theorem 10.5 on the collar, give the H^m bound on $[\theta, \theta + 1]$ at the rate c . $\square$

10.6. Decay along the infinite torus collar. A gauge transformation of the odd bundle over a region of $T \times \mathbb{R}^2$ is *even* if it lifts to the determinant-one group. The even stabilizer of α is trivial (Section 10.2); the full stabilizer is the Klein four-group, whose three nontrivial elements are odd. We use Proposition 10.8 in the following form: if $E(\theta_0) < e_0$ is the kinetic energy of a solution beyond θ_0 on a half-cylinder over Y_R (or $D_{i,\lambda}$), then in a temporal gauge, for $\theta \geq \theta_0 + 1$,

$$(10.1) \quad E(\theta) \leq E(\theta_0)e^{-2c(\theta-\theta_0)}, \quad \|U - \pi^*B\|_{H^m([\theta, \theta+1] \times Y_R)} \leq C_m E(\theta_0)^{1/2} e^{-c(\theta-\theta_0)},$$

where π^*B is the stationary limit and the section is included in the norm. For a connection A on $T \times \Omega$, $\Omega \subset \mathbb{R}^2$ with coordinates $z = (z_1, z_2)$, write $F_{jT} = (\iota_{\partial_{z_j}} F_A)|_{T \times \{z\}}$ and F_T for the restriction of F_A to a torus fibre.

Lemma 10.9 (torus slice). *There are $e_T > 0$ and, for each $k \geq 0$, constants C_k depending only on T , with the following property. Let $\Omega \subset \mathbb{R}^2$ be a simply connected domain and A a smooth connection on $T \times \Omega$ such that every $z \in \Omega$ has a neighbourhood N and an even gauge transformation on $T \times N$ after which $\|A - \alpha\|_{C^1(T \times N)} \leq e_T$. Then there is a unique even gauge transformation g of $T \times \Omega$ with*

$$g^*A = \alpha + b + \phi_1 dz_1 + \phi_2 dz_2, \quad d_\alpha^* b(z) = 0, \quad \|b(z)\|_{C^0(T)} \leq C_0 e_T \quad (z \in \Omega),$$

where $b(z)$ is a one-form and the $\phi_j(z)$ are sections of ad , both on the fibre T . If moreover $\|F_A\|_{C^{k+1}(T \times \Omega)} \leq e_T$, then for every $z \in \Omega$

$$(10.2) \quad \sum_{|\beta|+j \leq k+1} (\|\partial_z^\beta b(z)\|_{H^j(T)} + \|\partial_z^\beta \phi(z)\|_{H^j(T)}) \leq C_k \sum_{i \leq k+1} \|(\nabla_A^i F_A)|_{T \times \{z\}}\|_{L^2(T)}.$$

The fibrewise equations gain derivatives only in the torus directions, so the top base derivatives of ϕ require the curvature derivative of order $k+1$. We call g^*A the torus slice of A .

Proof. The fibre slice. Consider the map $(\xi, b) \mapsto \exp(\xi)^*(\alpha + b)$ from $\Omega^0(T; \text{ad}) \times \ker d_\alpha^*$ to connections on the odd bundle over T , in $C^{1,\beta} \times C^{0,\beta}$. Its derivative at $(0,0)$ is $(\xi, b) \mapsto d_\alpha \xi + b$, an isomorphism onto $C^{0,\beta}$ one-forms, because $\Omega^1(T; \text{ad } \alpha) = \text{im } d_\alpha \oplus \ker d_\alpha^*$ and d_α is injective ($H^0 = H^1 = 0$). By the implicit function theorem every connection C^1 -close to α is $\exp(\xi)^*(\alpha + b)$ with $d_\alpha^* b = 0$ and ξ, b small. The decomposition is unique among even gauge transformations: if g is even and $g^*(\alpha + b) = \alpha + b'$ with b, b' small in C^0 , then $dg = g(\alpha + b') - (\alpha + b)g$ bounds g in C^1 , and as $b, b' \rightarrow 0$ every limit of g lies in the stabilizer of α and is even, hence is the identity; so for e_T small g is close to the identity and equals the gauge transformation given by the implicit function theorem.

The global gauge. By hypothesis and the fibre slice, each fibre $T \times \{z\}$ carries a unique even $g(z)$ putting $A|_{T \times \{z\}}$ into the slice; by uniqueness and the implicit function theorem with parameters, $g(z)$ depends smoothly on z . The resulting family is fibrewise even, and it

lifts to the determinant-one group over $T \times \Omega$ because the obstruction lies in $H^1(T \times \Omega; \mathbb{F}_2) = H^1(T; \mathbb{F}_2)$ and vanishes on a fibre. Uniqueness of g follows from the fibrewise uniqueness.

Estimates. On a fibre, $F_T = d_\alpha b + \frac{1}{2}[b \wedge b]$. Since $H^1(T; \text{ad } \alpha) = 0$, the elliptic estimate for $d_\alpha + d_\alpha^*$ gives $\|b\|_{H^{j+1}(T)} \leq C\|d_\alpha b\|_{H^j(T)} \leq C(\|F_T\|_{H^j(T)} + \|b\|_{C^0}\|b\|_{H^{j+1}(T)})$, and the last term is absorbed. For the base components, $F_{jT} = \partial_{z_j} b - d_{\alpha+b}\phi_j$. Applying d_α^* and using $d_\alpha^* \partial_{z_j} b = \partial_{z_j} d_\alpha^* b = 0$ gives

$$\Delta_\alpha \phi_j + d_\alpha^*[b, \phi_j] = -d_\alpha^* F_{jT},$$

and $\Delta_\alpha \geq \mu_T^2$ yields $\|\phi_j\|_{H^{j'+1}(T)} \leq C\|F_{jT}\|_{H^{j'}(T)}$. Then $\partial_{z_j} b = F_{jT} + d_{\alpha+b}\phi_j$ controls the first base derivatives of b . Differentiating these identities in z , and writing $\nabla_A = \nabla_\alpha + [b + \phi dz, \cdot]$, an induction on $|\beta|$ gives (10.2): for $|\beta| = k + 1$ the equation for $\partial_z^\beta \phi_j$ has source $d_\alpha^* \partial_z^\beta F_{jT}$, a curvature derivative of order $k + 1$, and the lower-order products are absorbed because $\|F_A\|_{C_A^{k+1}} \leq e_T$. $\square$

Proposition 10.10 (decay along the completed tail). *Let $V = (A, u)$ be a finite-energy solution on W_∞ (or on $W_{\pm, \infty}$). Let $\gamma_T = \mu_T/4$, fix m , and let $0 \leq \delta \leq \delta_T$, where*

$$\delta_T = \min\{\mu_T/4, \mu_T^2/8, c/2\},$$

c is the rate in (10.1), and the weight on the torus region is $\tau_0(\theta)$ with $|\tau'_0| \leq 1$ and $|\tau''_0| \leq 1$. There are $S_0 \geq 3$ and C_m such that:

- (i) $\sup_{T \times D_1(z)} |\nabla_A^k F_A| \leq C_k e^{-\gamma_T(r(z) - S_0)}$ for every z with $r(z) \geq S_0 + 2$, uniformly in $\theta(z)$;
- (ii) Lemma 10.9 applies on $\Omega = (S_0, \infty) \times \mathbb{R}_\theta$, and in the torus slice $A = \alpha + a_V$ with

$$\|a_V\|_{H_\delta^m(T \times [S, \infty) \times \mathbb{R})} \leq C_m e^{-\gamma_T(S - S_0)} \quad (S \geq S_0 + 2),$$

where H_δ^m is weighted by $e^{\delta\tau_0(\theta)}$.

Proof. Smallness. The energy of V on $T \times [S, \infty) \times \mathbb{R}$ tends to zero as $S \rightarrow \infty$, since these regions decrease to the empty set and carry no perturbation. By small-energy regularity (Remark 10.6) applied on the boxes $T \times D_2(z)$ with $r(z) \geq S_0 - 1$, for S_0 large we have $\sup_{r \geq S_0 - 1} |\nabla_A^k F_A| \leq e_T$ for $k \leq m$, together with local gauges in which $\|A - \alpha\|_{C^1} \leq e_T$. So Lemma 10.9 applies on $(S_0 - 1, \infty) \times \mathbb{R}$, and on each fibre $A = \alpha + b$ with $\|b\|_{C^0} \leq C e_T$.

A differential inequality. Put $e(z) = \int_{T \times \{z\}} |F_A|^2$. On the flat product the anti-self-dual curvature satisfies the Weitzenböck identity $\nabla_A^* \nabla_A F_A = -\{F_A, F_A\}$, with no curvature term and $|\langle \{F, F\}, F \rangle| \leq c_W |F|^3$. Summing $\partial_i^2 |F|^2$ over the four coordinate directions and integrating over T , where the torus directions integrate to zero,

$$(\partial_{z_1}^2 + \partial_{z_2}^2)e \geq 2 \int_T |\nabla_{A,T} F_A|^2 - 2c_W \sup |F_A| e.$$

The components of F_A in the parallel frame of $\Lambda^2(\mathbb{R}^4)$ are sections of ad over the fibre, and $\int_T |\nabla_T^\alpha w|^2 = \langle \Delta_\alpha w, w \rangle \geq \mu_T^2 \|w\|^2$ for such sections. With $A|_T = \alpha + b$ and $\|b\|_{C^0}$ small this gives $\int_T |\nabla_{A,T} F_A|^2 \geq \frac{3}{4} \mu_T^2 e$ for e_T small, and therefore

$$(10.3) \quad (\partial_{z_1}^2 + \partial_{z_2}^2)e \geq \mu_T^2 e \quad \text{on } r \geq S_0 - 1.$$

Integrability in θ . On each mixed end the slice is Y_∞ , and by Proposition 10.8 and (10.1) the curvature of V on $[\theta, \theta + 1] \times Y_\infty$ has H^2 norm at most $C e^{-c|\theta|}$; by the trace and Sobolev inequalities on unit boxes, $e + |\partial_\theta e| \leq C e^{-2c|\theta|}$ uniformly in r for $|\theta|$ large. Since $2\delta \leq c < 2c$, the function $f(r) = \int_{\mathbb{R}} e^{2\delta\tau_0(\theta)} e(r, \theta) d\theta$ is finite and of class C^2 for $r \geq S_0 - 1$, the integrations by parts below have no boundary terms, and $\int_{S_0}^\infty f(r) dr < \infty$.

Decay. By (10.3),

$$f'' = \int e^{2\delta\tau_0} ((\partial_r^2 + \partial_\theta^2)e) d\theta - \int (e^{2\delta\tau_0})'' e d\theta \geq (\mu_T^2 - 4\delta^2 - 2\delta)f \geq \frac{1}{2}\mu_T^2 f,$$

using $(e^{2\delta\tau_0})'' = (4\delta^2\tau_0'^2 + 2\delta\tau_0'')e^{2\delta\tau_0}$ and $\delta \leq \delta_T$. So f is convex, nonnegative and integrable on $[S_0, \infty)$, hence nonincreasing with limit 0. Put $\gamma'_T = \mu_T/\sqrt{2}$ and $h(r) = f(r) - f(S_0)e^{-\gamma'_T(r-S_0)}$. Then $h'' \geq \gamma_T'^2 h$, $h(S_0) = 0$ and $h \rightarrow 0$; at a positive interior maximum $h'' \leq 0 < \gamma_T'^2 h$, which is impossible. Hence

$$f(r) \leq f(S_0)e^{-\gamma'_T(r-S_0)} \quad (r \geq S_0).$$

Higher norms. By small-energy regularity, which is linear in the square root of the local energy, $\|\nabla_A^i F_A\|_{L^2(T \times Q)} \leq C_i \|F_A\|_{L^2(T \times 2Q)}$ for every unit square Q with $r \geq S_0 + 1$ on Q ; the weight varies by at most the factor $e^{2\delta}$ on $2Q$. Summing over a cover by unit squares with bounded overlap,

$$\sum_{i \leq m} \int_{r \geq S} e^{2\delta\tau_0} \|\nabla_A^i F_A\|_{L^2(T)}^2 dz \leq C \int_{S-2}^\infty f(r) dr \leq C f(S_0) e^{-\gamma'_T(S-2-S_0)}.$$

Integrating (10.2) with $k = m - 1$ against the weight gives (ii), since $\gamma'_T/2 > \gamma_T$. For (i), small-energy regularity on $T \times D_2(z)$ bounds $\sup_{T \times D_1(z)} |\nabla_A^k F_A|$ by $C_k (\int_{r(z)-2}^{r(z)+2} f)^{1/2}$, because $e^{2\delta\tau_0} \geq 1$. $\square$

For each mixed end we use the stationary representatives B_q of Proposition 10.7, equal to α on the torus collar. For q near a given point of the critical manifold we fix a smooth local family $q \mapsto B_q$ with this property, and in temporal gauge on product collars of the genus boundary surfaces: the flat restrictions to a product collar of T inside the compression body are gauge equivalent to α by gauge transformations depending smoothly on q , and these are extended over the compression body by cutoffs of their logarithms; the primary perturbation is supported away from the collars, where the connections are flat, so the temporal gauge along a collar is available. On Y_R and on $D_{i,\lambda}$ the family is independent of R and λ away from the collar, and equal to α on the collar (on the bridge in the joined case). Its tangent vectors are supported away from the torus collar and have vanishing normal components at the genus boundary.

Lemma 10.11 (adapted representatives). *Let V be a finite-energy solution on W_∞ with limits q_e at the mixed ends e . There are $S_V \geq S_0 + 3$, Θ_V and a representative V^{ad} of V , obtained from V by an even gauge transformation, such that:*

- (i) *on $T \times [S_V, \infty) \times \mathbb{R}_\theta$, $V^{\text{ad}} = \alpha + a_V$ is the torus slice, so Proposition 10.10(ii) applies to a_V ;*
- (ii) *on each mixed end e , with outward end coordinate θ , $V^{\text{ad}} - \pi^* B_{q_e}$ (and the section minus the constant q_e) lies in $H_\delta^m(\{\theta \geq \Theta_V\})$, and its norm on $\{\theta \geq \Theta\}$ tends to 0 as $\Theta \rightarrow \infty$;*
- (iii) *on the original gauge and ordinary ends V^{ad} has the standard exponential asymptotics at nondegenerate ends [13, Section 4.1], [33].*

The same holds on $W_{\pm, \infty}$, for pairs on $W_{-, \infty} \sqcup W_{+, \infty}$, and with α replaced by the joined stationary connections $B_p \cup \alpha \cup B_q$ on the slices $D_{i,\lambda}$ when applied to finite joined configurations.

Proof. On each mixed end e choose a temporal gauge in which V converges to a stationary pair. The stationary limit is gauge equivalent to $\pi^* B_{q_e}$ by a gauge transformation independent

of θ ; composing with it, the limit is exactly $\pi^*B_{q_e}$, and (10.1) gives (ii) for this representative V_e^E on $\{\theta \geq \Theta_V - 1\}$ once Θ_V is beyond the threshold of Proposition 10.8, because $\delta < c$. On $T \times [S_V - 1, \infty) \times \mathbb{R}$ let V^T be the torus slice of Proposition 10.10. On the overlap $\{\theta \geq \Theta_V - 1, r \geq S_V - 1\}$ both representatives are α plus a term that is small in C^1 and tends to zero with θ : the first because $B_{q_e} = \alpha$ on the collar, the second by Proposition 10.10(i). The relative gauge transformation is even and maps a connection close to α to one close to α , so it is close to the even stabilizer of α , which is trivial; it is $\exp \varphi$ with φ small, and the torus-slice argument (fibrewise $d_\alpha \varphi$ and the base derivatives of φ are controlled by the two connection differences) gives $\|\varphi\|_{H_\delta^{m+1}(\{\theta \geq \Theta, r \geq S_V - 1\})} \rightarrow 0$ as $\Theta \rightarrow \infty$. Replace V_e^E on the end by $\exp(\chi(r)\varphi)^*V_e^E$, with $\chi(r) = 0$ for $r \leq S_V - 1$ and $\chi(r) = 1$ for $r \geq S_V$; it agrees with V^T for $r \geq S_V$ and with V_e^E for $r \leq S_V - 1$, and it still satisfies (ii). Finally extend over the core. The even gauge transformation relating V to these representatives on the outer region $O = \{r \geq S_V\} \cup \{|\theta| \geq \Theta_V\}$ and on the original ends (in this section, the ends of W_∞ other than the torus region) lifts to the determinant-one group, so the obstruction to extending it over the core K lies in $H^4(K, K \cap O; \mathbb{Z})$, which vanishes because K meets the free genus matching faces; the normalization of the temporal gauge at each mixed end is even for the same reason, the critical manifolds being quotients by the determinant-one group. The extension changes V only on compact sets disjoint from $\{r \geq S_V\} \cup \{\theta \geq \Theta_V\}$, after enlarging S_V and Θ_V . $\square$

10.7. Energy, index and the loss formula.

Theorem 10.12. *(i) For every $R \in [0, \infty]$ and every configuration c on W_R with labels $\mathbf{a} = (a, x)$, the linearized operator on the extended weighted spaces is Fredholm and $\text{ind } D_c = 8\kappa(c) + \mathfrak{c}(x) - \mathfrak{c}(a)$, with $2\kappa(c) - 2\kappa_0(\mathbf{a}) \in \mathbb{Z}$, the constants being those of [11, Proposition 3.20] for X_D .*

(ii) For every $\lambda > 0$ and every configuration c on W_λ^J with labels $(\mathbf{a}_-, \mathbf{a}_+)$, the same holds with the sum of the two constants, and $\text{ind } D_c = \text{ind } D_{c_-} + \text{ind } D_{c_+}$, where $c_\pm$ are obtained by first deforming c , with fixed labels and hence fixed κ and index, to a configuration equal to α on a slice $T \times \{r_0\} \times \mathbb{R}$ of the bridge, then cutting there and completing by α .

(iii) The constants depend only on the labels, not on R or λ .

Proof. (i) For $R < \infty$, pull back by ψ_R : the pulled-back configuration is a DFL configuration on X_D with the metric changed on $T \times [-\frac{3}{4}, -\frac{1}{4}] \times \mathbb{R}_\theta$, a region that runs into both mixed ends and changes their slices from Y to Y_R . Along the family of metrics g_s , $s \in [0, R]$, the operator on the extended weighted spaces stays Fredholm, because Proposition 10.7 gives a gap uniform in the collar length; so the index is constant along the family and equals the index on X_D , given by [11, Proposition 3.20 and Lemma 3.15]. The topological energy is independent of the metric, and the boundary term in Stokes' theorem at the flat face vanishes, the fibre restriction being flat and $H^1(T; \text{ad}) = 0$. For $R = \infty$, deform c to equal α far out on the tail and compare W_∞ with W_R by excision across a band of the collar on which all operators are the operator of Proposition 10.3: the operators $D_{W_\infty} \oplus M_R$ and $D_{W_R} \oplus M_\infty$, where M are the invertible models of Proposition 10.3 on the tail pieces, have equal index once the band is longer than a constant depending only on the Fredholm estimate of D_{W_∞} ; that estimate, for D_{W_∞} and its adjoint, is the compact-remainder estimate of Step 2 of the proof of Proposition 10.25 with an infinite collar. The tail contributes nothing to the topological energy. (ii) The energy is local and the section splits, so $\kappa(c) = \kappa(c_-) + \kappa(c_+)$; the index

splits by the same excision with two bands, the plane models having index zero, and the asymptotic values add. (iii) holds because every constant was fixed on X_D . $\square$

Proposition 10.13 (loss formula). *Let U_j be solutions of index d on W_{R_j} with $R_j \rightarrow \infty$, or on $W_{\lambda_j}^J$ with $\lambda_j \rightarrow \infty$. After passing to a subsequence there are solutions V_β on the completed domains and broken trajectories Γ_ℓ on the original ends with $d - \sum_\beta \text{ind}(V_\beta) - \sum_\ell \text{ind}(\Gamma_\ell) = 4n$, $n \in \mathbb{Z}_{\geq 0}$. If all $\text{ind}(V_\beta) \geq 0$, all $\text{ind}(\Gamma_\ell) \geq 1$ and $d \leq 3$, then $n = 0$ and no energy is lost: after gauge transformations U_j converges to the limit pieces in C^∞ on compact subsets, and for every $\varsigma > 0$ there are a compact set K in the completed domain and j_0 such that the energy of U_j outside K is less than ς for $j \geq j_0$. The same statements hold for sequences of solutions on a fixed completed domain W_∞ or $W_{-, \infty} \sqcup W_{+, \infty}$.*

Proof. Energy can be lost through interior instantons, through finite-energy instantons on $T \times \mathbb{R}^2$ or on T times a half-plane with flat-fibre boundary, through concentration on a boundary torus fibre, through seam and section bubbles, and through escape along the mixed ends, possibly spread along the lengthening slices. No classification is needed. Extract limits V_β on compact subsets of the completed domains, using fixed-geometry compactness and Theorem 10.5 at the flat face, and chains Γ_ℓ at the original ends. Build a reference configuration $\mathcal{R}_j$ on the same finite geometry as U_j , with the labels of U_j , as follows. First make each V_β and each Γ_ℓ exactly stationary near the attaching regions on the original ends, by cutoffs along paths of configurations with fixed limits; this changes neither the topological energy nor the index. For a separate sequence, pass to the adapted representative $V^{\text{ad}} = \alpha + a_V$ on $r \geq S_V$ (Lemma 10.11) and truncate its torus tail: put $\alpha + \chi_j(r)a_V$ on $T \times [S_V, R_j] \times \mathbb{R}$, with $\chi_j = 1$ for $r \leq R_j/2$, $\chi_j = 0$ for $r \geq R_j/2 + 1$ and $|\chi_j'| \leq 2$. The result equals α near $r = R_j$, so it satisfies the flat-fibre condition and is a configuration on W_{R_j} . For a joined sequence truncate the two tails in the same way at $r_\pm = \lambda_j/2$ and join the flat parts by the bridge identification, which carries α to α . Finally attach the chains Γ_ℓ at the original ends by the cutoff construction of [11, Propositions 4.65 and 4.69]. The path $s \mapsto \alpha + (1 - s + s\chi_j)a_V$ consists of configurations on W_∞ with fixed labels, so by the Stokes argument in the proof of Theorem 10.12 the topological energy is constant along it, and the truncated configuration has the same energy on W_{R_j} because it is flat beyond $r = R_j/2 + 1$. Hence $\kappa(\mathcal{R}_j) = \sum \kappa(V_\beta) + \sum \kappa(\Gamma_\ell)$ exactly, and by Theorem 10.12(i)–(iii), with additivity of the constants $\mathfrak{c}$ along the original ends, $\text{ind } D_{\mathcal{R}_j} = I = \sum \text{ind } V_\beta + \sum \text{ind } \Gamma_\ell$. By Theorem 10.12, $d - I = 4n$ and $\kappa(U_j) - \kappa(\mathcal{R}_j) = \frac{1}{2}n$. The analytic energy is $8\pi^2\kappa$ plus a correction from the perturbations that converges, by strong L^p convergence of the holonomy sources away from the concentration set; lower semicontinuity of the analytic energy gives $n \geq 0$. If $n = 0$, the limit of the analytic energies equals the sum of the analytic energies of the limit pieces. The energy on a compact set converges to that of the limit on the same set, by the smooth convergence away from the concentration set, and every concentration point carries a fixed quantum; so there is no concentration and the energy outside a compact set K converges to that of the limit pieces outside K , which is small for K large. For a fixed completed domain the same argument applies with $\mathcal{R}_j$ replaced by a fixed truncation. $\square$

10.8. Families with stationary ends.

Lemma 10.14. *Let F_λ , $\lambda \in [0, 1]$, satisfy the hypotheses of Theorem 5.2 except compact support of $\partial_\lambda F$, and instead: on each mixed end the critical set is connected, and it and a smooth family of stationary references R_q are independent of λ , with $F_\lambda(R_q) = 0$; there $\partial_\lambda F_\lambda$ is independent of the end coordinate, and it vanishes on the gauge and ordinary ends; and*

the stationary operators have a gap and the decay of Proposition 10.8 holds uniformly along the family. Then the conclusions of Theorem 5.2 hold.

Proof. On a mixed end, in the chart $R_q + z$ with z in the weighted domain (weight $e^{\delta\tau}$ in the end coordinate τ), differentiating $F_\lambda(R_q) = 0$ in λ gives $\partial_\lambda F_\lambda(R_q) = 0$, since R_q does not depend on λ . Hence

$$\begin{aligned} \partial_\lambda F_\lambda(R_q + z) &= \int_0^1 D(\partial_\lambda F_\lambda)(R_q + sz) z \, ds, \\ \|e^{\delta\tau} \partial_\lambda F_\lambda(R_q + z)\|_{L^2} &\leq \sup_{s \in [0,1]} \|D(\partial_\lambda F_\lambda)(R_q + sz)\| \|e^{\delta\tau} z\|_{L^2_1}, \end{aligned}$$

and the operator norm is bounded uniformly along the end because $\partial_\lambda F_\lambda$ is independent of the end coordinate and $\|z\|_{C^0}$ is small there. So the parameter derivative lies in the weighted target, $\lambda \mapsto F_\lambda$ is C^1 as a map into it, and the augmented operator $(\dot{\lambda}, z) \mapsto \dot{\lambda} \partial_\lambda F_\lambda + D_\lambda z$ is a rank-one extension of a Fredholm operator. Breaks occur only at the gauge and ordinary ends. A finite-energy trajectory on a mixed end runs from the connected critical family to itself. Its energy is a period of the action; if it is zero the trajectory is constant, and otherwise $\kappa > 0$, so by the lattice condition (H1) its index is at least four. Such a trajectory cannot split off a family of fibre index at most one whose other components, being regular, have index at least -1 . The preglued configuration agrees with the core on the mixed end, so the parameter derivative vanishes on the joined end and neck. Family compactness uses the uniform decay and the λ -independent constants of Theorem 10.12. $\square$

Proposition 10.15. *Changing the length of a separate torus collar from ℓ_0 to ℓ_1 , with the outer flat-fibre boundary present throughout, changes the regular separate map by a chain homotopy.*

Proof. Apply Lemma 10.14. The stationary references are perturbed-flat and equal to α on the collar; the perturbed flatness equation involves the metric only on the support of the perturbation, so the references are independent of the collar length; the pulled-back reference satisfies the equation on the collar because it is flat there; and the variation of the metric is supported in a region of the collar independent of the end coordinate. $\square$

Proposition 10.16 (width homotopy). *For $0 < a \leq b < \infty$, $N_b^J \simeq N_a^J$.*

Proof. Change the bridge length by a family of metrics supported on the bridge. The exact stationary references $R_{\lambda,q}$ on the joined mixed ends, flat and equal to α on the bridge, depend on the width only through the length of the flat region; differentiating $F_\lambda(R_{\lambda,q}) = 0$ shows that the nondecaying part of the parameter derivative vanishes. After pulling back by width-changing diffeomorphisms that are the identity outside the bridge, the hypotheses of Lemma 10.14 hold; on the compact range $[a, b]$ each slice has a gap, depending continuously on the width, so the gap and the decay are uniform along the family. $\square$

10.9. The closed cut. We describe the gauge part of W_λ^J in the coordinates of the special quintuple [11, Section 3.1]. For each side, $\mathbb{U}_- \subset \mathbb{C}$ is the region of [11, Figure 1]: it contains the input arm $(-\infty, -3] \times [-1, 1]$, the arms $[-2, 0] \times [2, \infty)$ and $[-2, 0] \times (-\infty, -2]$ and the square $[-1, 0] \times [-2, 2]$, its boundary consists of the matching line $\mathbb{U}_\partial = i\mathbb{R}$ and two curves η_- , η'_- , and the gauge domain is $\mathbb{R} \times Y_0 \cup \mathbb{U}_- \times \Sigma \cup \mathbb{R} \times (-Y'_0)$, where $\mathbb{R} \times \partial Y_0$ is glued to $\eta_- \times \Sigma$ through the parameter t of η_- and the collar coordinate of a neighbourhood $\mathbb{R} \times (\frac{1}{2}, 1]$ of η_- , and similarly for η'_- . Along the input arm, $t = \text{Re}$, η_- is the line $\text{Im} = 1$ and η'_- the line $\text{Im} = -1$. For the two sides of W_λ^J write $Y_\pm$, $Y'_\pm$ for the compression bodies ($Y_- = -U_0$,

$Y'_- = -U_1$, $Y_+ = V_0$, $Y'_+ = V_1$), $\Sigma_- = \Sigma_h$, $\Sigma_+ = \Sigma_g$, and $\mathbb{U}_-^{(\pm)}$, $\eta_-^{(\pm)}$, $\eta_-'^{(\pm)}$ for the planar data. The torus part of the gauge domain of W_λ^J is $T \times \tilde{\mathbb{U}}_\lambda$ with

$$\tilde{\mathbb{U}}_\lambda = \mathbb{U}_-^{(-)} \cup ([0, 2\lambda] \times \mathbb{R}) \cup \varrho_\lambda(\mathbb{U}_-^{(+)}), \quad \varrho_\lambda(z) = 2\lambda - \bar{z},$$

where the bridge coordinate of Definition 10.1 is $\operatorname{Re} z - \lambda$, and the torus factors of the two sides are identified by the marked bundle isometry. The genus parts are $\mathbb{U}_-^{(-)} \times \Sigma_h$ and $\mathbb{U}_-^{(+)} \times \Sigma_g$.

The cut. Let $\gamma_0 \subset \tilde{\mathbb{U}}_\lambda$ be the arc consisting of the segment $\{-3\} \times [\frac{1}{2}, 1]$, the segment $[-3, 2\lambda + 3] \times \{\frac{1}{2}\}$ and the segment $\{2\lambda + 3\} \times [\frac{1}{2}, 1]$, with its two corners rounded, and γ_1 its mirror image in the real axis. The endpoints of γ_0 are the points of $\eta_-^{(-)}$ and $\varrho_\lambda(\eta_-^{(+)})$ with parameter $t = -3$, and the arc leaves them along the collar coordinate; similarly for γ_1 . Let $\mathbf{a}_\pm = \{-3\} \times [-1, 1]$ in the input arm of $\mathbb{U}_-^{(\pm)}$. Put

$$Z_{\text{cut}} = \bigcup_{\pm} (\{-3\} \times (Y_\pm)_0 \cup \{-3\} \times (-Y'_\pm)_0 \cup \mathbf{a}_\pm \times \Sigma_\pm) \cup (\gamma_0 \sqcup \gamma_1) \times T.$$

Let $\tilde{\mathbb{U}}_{\text{up}}$, $\tilde{\mathbb{U}}_{\text{mid}}$, $\tilde{\mathbb{U}}_{\text{down}}$ be the closures of the components of $\tilde{\mathbb{U}}_\lambda \setminus (\gamma_0 \cup \gamma_1)$ containing the upper arm, the two input arms, and the lower arm.

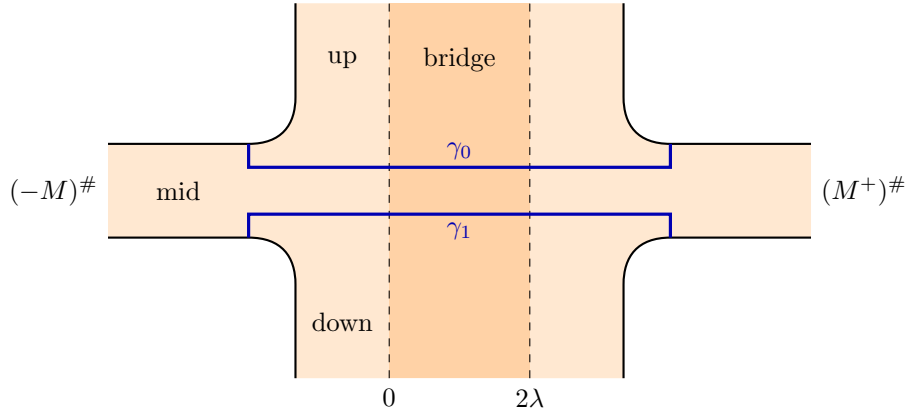


FIGURE 6. The torus base $\tilde{\mathbb{U}}_\lambda$ of the joined domain W_λ^J (Section 10.9): the planar region $\mathbb{U}_-$ of the lower side on the left, its reflection for the upper side on the right, and the bridge between the two matching lines (dashed). The arcs γ_0 and γ_1 run from the boundary curves of one input arm to those of the other. The closed cut Z_{cut} consists of $(\gamma_0 \sqcup \gamma_1) \times T$ together with the slices of the input arms at $t = -3$; it separates the part containing the matching lines (regions up and down, a domain for the pair (A_0, A_1)) from the part containing the two gauge inputs and the middle of the bridge (region mid, Kronheimer and Mrowka’s reverse excision cobordism).

Lemma 10.17 (the closed cut in the joined domain). *(i) Z_{cut} is a closed connected hypersurface in the gauge part of W_λ^J , disjoint from the matching lines, the section and the supports of the secondary perturbations, and there is an orientation-preserving diffeomorphism $Z_{\text{cut}} \cong Z = D_0 \cup (-D_1)$, covered by a bundle isomorphism, which is the identity on the pieces $(Y_\pm)_0$, $(-Y'_\pm)_0$ and the genus collars, and carries $\gamma_i \times T$ onto the torus neck of D_i .*

- (ii) Z_{cut} separates the gauge part into two pieces. The piece (I) containing the matching lines, together with the section $\mathbb{U}_+$ and a cylindrical end $(-\infty, 0] \times Z$ attached along Z_{cut} , is a domain of DFL type for the pair (A_0, A_1) with compression bodies D_0, D_1 , with the end types of the special quintuple of (D_0, D_1) : its genus parts are $(\mathbb{U}_\pm^{(\pm)} \cap \{\text{Re} \geq -3\}) \times \Sigma_\pm$, its compression pieces $[-3, \infty) \times (Y_\pm)_0$ and $[-3, \infty) \times (-Y'_\pm)_0$ together with $(\tilde{\mathbb{U}}_{\text{up}} \sqcup \tilde{\mathbb{U}}_{\text{down}}) \times T$, its gauge input slice is Z_{cut} , and its mixed-end slices are $D_{0,\lambda}$ and $D_{1,\lambda}$, which are D_0 and D_1 with torus necks of length $2\lambda + 4$.
- (iii) The other piece (II), with a cylindrical end attached along Z_{cut} , is the reverse excision cobordism $\overline{W}: Z' \rightarrow Z$ of [21, Section 5.3], with $Z' = (-M)^\# \sqcup (M^+)^\#$ and cylindrical input ends. Here the two tori of Theorem 5.6 of [21] are $T_a = T \times \{x_a\}$ and $T_b = T \times \{x_b\}$ for points $x_a \in \gamma_0, x_b \in \gamma_1$, identified through the product structure of $T \times \tilde{\mathbb{U}}_\lambda$; the product part $T \times U$ of [21, Figure 4] is $T \times \tilde{\mathbb{U}}_{\text{mid}}$, truncated on the input arms; and the bundle is odd on T_a and T_b .
- (iv) Let $d_1, d_2, d_3, d_4 \in H^1(W_\lambda^J; \mathbb{F}_2)$ be dual to the hypersurfaces $\eta_-^{(-)} \times T, \eta_-'^{(-)} \times T, \varrho_\lambda(\eta_-^{(+)}) \times T, \varrho_\lambda(\eta_-'^{(+)}) \times T$. Then $d_1 = d_2, d_3 = d_4$ and $\Phi_J = \langle d_1, d_3 \rangle$. Restricted to (II), Φ_J is the subgroup $\phi_{\overline{W}}$ of [21, Section 5.3] generated by the duals of the four copies of an interval times T over the dotted edges of U , and its image at the ends is $\Phi_{\overline{E}} = \{(s, s_-, s_+) : s = s_- + s_+\}$. Restricted to (I), Φ_J is the lifting group of the domain of the pair (A_0, A_1) , generated by the class that restricts to s_Z on Z ; and Φ_J is the fibre product of the two restrictions over $\phi_Z = \langle s_Z \rangle$.
- (v) After a change of the metric of $\tilde{\mathbb{U}}_\lambda$ supported in a compact neighbourhood of the rounded corners of $\gamma_0 \cup \gamma_1$, disjoint from the bridge ends, the arms and the matching lines, Z_{cut} has a neighbourhood $(-1, 1) \times Z_{\text{cut}}$ with the product metric.

Proof. (i) Near the input arms the gauge domain is the product of $t = \text{Re}$ with the slice $\{t\} \times (Y_\pm)_0 \cup \{t\} \times [-1, 1] \times \Sigma_\pm \cup \{t\} \times (-Y'_\pm)_0$ in the genus part and with the collar segments $\{t\} \times (\frac{1}{2}, 1]$ in the torus part. Reading Z_{cut} across: the arc γ_0 joins the torus boundary of $(Y_-)_0$ at $\eta_-^{(-)}(-3)$ to that of $(Y_+)_0$ at $\varrho_\lambda(\eta_-^{(+)})(-3)$, so $(Y_-)_0 \cup \gamma_0 \times T \cup (Y_+)_0$ is $Y_- \cup_T Y_+ = (-U_0) \cup_T V_0 = D_0$ with a torus neck of length equal to the length of γ_0 ; likewise $(-Y'_-)_0 \cup \gamma_1 \times T \cup (-Y'_+)_0 = -D_1$; and $\mathbf{a}_- \times \Sigma_h, \mathbf{a}_+ \times \Sigma_g$ glue D_0 to $-D_1$ along Σ_h and Σ_g . So $Z_{\text{cut}} \cong D_0 \cup (-D_1) = Z$. The bundle is the given one on each piece and the pulled-back odd bundle on $\gamma_i \times T$, so the identification is a bundle isomorphism. The primary perturbations of the pieces $\mathbb{R} \times (Y_\pm)_0$ and $\mathbb{R} \times (-Y'_\pm)_0$ are independent of t and restrict on Z_{cut} to the primary perturbations of D_0 and D_1 . Connectedness is clear. The matching lines lie in $\text{Re} \in \{0, 2\lambda\}$, and the section and the secondary supports lie in $\{t > -3\}$ by the choice of data. The end-perturbation terms $\chi_\pm f_\pm$ on the two gauge inputs are chosen to be independent of t on $\{t < -4\}$, so that Z_{cut} meets them where they equal the end data $f_\pm$.

(ii) The component of the complement containing the matching lines consists of the pieces listed. The region $\tilde{\mathbb{U}}_{\text{up}}$ is bounded by γ_0 , by the parts $t \geq -3$ of $\eta_-^{(-)}$ and $\varrho_\lambda(\eta_-^{(+)})$, and by the end of the upper arm, whose cross-section is $[-2, 2\lambda + 2]$; it is diffeomorphic to $[-3, \infty) \times [0, 1]$ by a diffeomorphism that is an isometry onto a product for large t and that matches the collar coordinates at the two η -curves. Hence $[-3, \infty) \times (Y_-)_0 \cup \tilde{\mathbb{U}}_{\text{up}} \times T \cup [-3, \infty) \times (Y_+)_0$ is diffeomorphic to $[-3, \infty) \times (D_0)_0$, and similarly for D_1 . With the genus parts this is the special quintuple of (D_0, D_1) restricted to $t \geq -3$, except that the metric of the compression pieces is induced from $\tilde{\mathbb{U}}_\lambda$: on $\{t \leq t_1\}$ for a fixed t_1 it differs from a product in a compact set, and for $t \geq t_1$ it is the product with the slices $D_{i,\lambda}$, whose torus necks have length $2\lambda + 4$

rather than the length of γ_i . Its section is $\mathbb{U}_+$ with the Lagrangians A_0, A_1 , the perturbed Lagrangians of D_0, D_1 being $\tilde{P}_i^- \times R_i = A_i$ (Section 10.1).

(iii) The other component is

$$(-\infty, -3] \times ((Y_-)_0 \sqcup (-Y'_-)_0 \sqcup (Y_+)_0 \sqcup (-Y'_+)_0) \\ \cup (\{\text{Re} \leq -3\}\text{-parts of the input arms}) \times \Sigma_{\pm} \cup \tilde{\mathbb{U}}_{\text{mid}} \times T.$$

The first two parts form $(-\infty, -3] \times Z'_{\text{cut}}$, where $Z'_{\text{cut}} = (Y_- \cup_{\Sigma_h} (-Y'_-)) \sqcup (Y_+ \cup_{\Sigma_g} (-Y'_+))$ is $Z' = (-M)^{\#} \sqcup (M^+)^{\#}$ cut along its two tori, with four boundary tori. Truncate the input arms at $\text{Re} = -T_1$ and $\text{Re} = 2\lambda + T_1$. Then $U := \tilde{\mathbb{U}}_{\text{mid}} \cap \{-T_1 \leq \text{Re} \leq 2\lambda + T_1\}$ is a disk with eight corners, whose edges in cyclic order are: γ_0 ; the part $t \in [-T_1, -3]$ of $\varrho_{\lambda}(\eta_-^{(+)})$; the end of the right input arm; the corresponding part of $\varrho_{\lambda}(\eta_-'^{+})$; γ_1 ; the parts of $\eta_-'^{(-)}$ and, after the end of the left input arm, of $\eta_-^{(-)}$. The four η -edges (the dotted edges of [21, Figure 4]) times T are glued to $[-T_1, -3] \times \partial Z'_{\text{cut}}$; the two arm ends times T are two-sided collars of the tori of Z' ; and $\gamma_0 \times T, \gamma_1 \times T$ are two-sided collars of T_a, T_b in Z . This is the description of the standard cobordism in the proof of [21, Theorem 5.6]: the complement of $T \times U$ is a product of an interval with $Z'_{\text{cut}} = Z_{\text{cut}} \setminus (\text{collars of } T_a \cup T_b)$, and $T \times U$ is a product. The cobordism runs from Z' at $\text{Re} = -T_1$ to Z , so it is the reverse cobordism $\overline{W}$. The bundle is the product of the odd bundle with U on $T \times U$ and the product bundle of Z'_{cut} elsewhere; the identification of T_a with T_b required in [21, Section 5.3] is the product structure.

(iv) The hypersurfaces $\eta_-^{(-)} \times T$ and $\eta_-'^{(-)} \times T$, together with the matching face $U_{\partial}^{(-)} \times \Sigma_h$ and parts of the ends, bound the region $\mathbb{R} \times (Y_-)_0 \cup U_-^{(-)} \times \Sigma_h \cup \mathbb{R} \times (-Y'_-)_0$; so their duals agree in $H^1(W_{\lambda}^J; \mathbb{F}_2)$, which is Lefschetz dual to H_3 relative to the boundary and the ends. The same holds on the $+$ side. The class d_1 restricts to the dual of the torus in $(-M)^{\#}$ and to zero in $(M^+)^{\#}$, and d_3 the other way round, so $\langle d_1, d_3 \rangle$ maps isomorphically onto $\Phi_- \oplus \Phi_+$, which is Φ_J . On Z , both d_1 and d_3 restrict to the dual of T_a , which equals that of T_b since the two tori cobound Z'_{cut} -pieces; this is s_Z . The restrictions to (II) are the duals of the dotted-edge hypersurfaces of (iii), which generate $\phi_{\overline{W}}$, with image $\{s = s_- + s_+\}$ at the ends. In (I) the two hypersurfaces $\eta_-^{(-)}|_{t \geq -3} \times T$ and $\varrho_{\lambda}(\eta_-^{(+)})|_{t \geq -3} \times T$ cobound $\tilde{\mathbb{U}}_{\text{up}} \times T$ relative to the boundary, so d_1 and d_3 restrict to the same class, which vanishes on the matching surfaces and restricts to s_Z on Z . Since Z is connected, restriction $H^1(W_{\lambda}^J) \rightarrow H^1(\text{I}) \oplus H^1(\text{II})$ is injective with image the pairs agreeing on Z .

(v) Choose coordinates $(u, s) \in (-\frac{1}{4}, \frac{1}{4}) \times \gamma_i$ near γ_i , so that the collars of γ_0 and γ_1 are disjoint and lie in the domain, equal to the Euclidean coordinates (t, Im) near the endpoints and to $(\text{Im} \mp \frac{1}{2}, \text{Re})$ along the horizontal segment, and replace the metric by $du^2 + ds^2$ near the rounded corners, interpolating in a compact region. Near the endpoints this is the product structure of the input arms, so Z_{cut} acquires a product collar. $\square$

Remark 10.18. Reversing the direction of the interval factors, the same description gives the forward cobordism $\mathcal{E}: Z \rightarrow Z'$ as $T \times U \cup [0, 1] \times Z'_{\text{cut}}$. Since Z'_{cut} is the union of $(T \times I_-)^{\#}(-M)$ and $(T \times I_+)^{\#}M^+$, this is $T \times P$ with $P = U \cup ([0, 1] \times I_-) \cup ([0, 1] \times I_+)$, a pair of pants, and with the products of the punctured summands inserted along two arcs from the input circle to the two output circles; this is the model used in Lemma 11.1(ii).

Data at the fixed width. Let $\mathfrak{d}_J$ be the regular joined data at width λ_0 : the primary data of Section 10.1, the end perturbations $f_{\pm}$ on the two gauge inputs, a split invariant secondary perturbation $\eta_- + \eta_+$ with each term supported in the core of its own side, and

the split structure $J_- \times J_+$ on $\mathbb{U}_+$. For $\ell \geq 0$ let $\mathfrak{d}_Z(\ell)$ be the *closed-cut data*: the metric of Lemma 10.17(v) with $[-\ell, \ell] \times Z$ inserted along Z_{cut} ; the end perturbation f_Z of the data of N_Z at width λ_0 , multiplied by a cutoff of the neck coordinate supported in $[-\ell, \ell]$ and equal to one on $[-\ell+1, \ell-1]$; the regular invariant secondary data of N_Z on (I) and of $\bar{E}$ on (II), supported away from the neck; the same primary data, end perturbations $f_{\pm}$ and split structure $J_- \times J_+$. Here N_Z is the mixed map of Remark 3.7 for (D_0, D_1) with the metrics induced on Z_{cut} (the *data at width* λ_0), and $\bar{E}$ is computed at the same end data $f_{\pm}$ and f_Z .

Lemma 10.19 (data change at fixed width). *For every $\ell \geq 0$ for which $\mathfrak{d}_Z(\ell)$ is regular in index at most one, the joined counts at $\mathfrak{d}_J$ and at $\mathfrak{d}_Z(\ell)$ are chain homotopic over $\mathbb{F}_2$.*

Proof. Let ψ_ℓ be a diffeomorphism of $W_{\lambda_0}^J$ onto the domain with the inserted neck, equal to the identity outside a compact neighbourhood of Z_{cut} and stretching a collar of Z_{cut} . Pull back $\mathfrak{d}_Z(\ell)$ by ψ_ℓ . The two sets of data on $W_{\lambda_0}^J$ then differ only in a compact set K of the gauge domain disjoint from all ends: in the metric (near Z_{cut}), in the term f_Z (supported in the neck) and in the secondary perturbations. On the gauge inputs, the strip output and the mixed ends they coincide, with the same critical data. Interpolate linearly in the metrics and in the coefficients of the holonomy terms. The family is invariant under Φ_J : the metrics and secondary terms are invariant by choice, and f_Z is invariant under ϕ_Z and under the determinant-one group, while Φ_J restricts on Z into ϕ_Z (Lemma 10.17(iv)). The parameter derivative is supported in K . Theorem 5.2 applies and gives the chain homotopy. $\square$

Proposition 10.20. *There is $\lambda_0 > 0$ such that $\mathcal{K}N_{\lambda_0}^J \simeq \mathcal{K}N_Z\bar{E}$ over $\mathbb{F}_2$, where $N_{\lambda_0}^J$ is the joined map at the data $\mathfrak{d}_J$ and N_Z is taken with the data at width λ_0 .*

Proof. Fix any $\lambda_0 > 0$. By Lemma 10.17, $W_{\lambda_0}^J$ with the data $\mathfrak{d}_Z(\ell)$ is the union, along a neck $[-\ell, \ell] \times Z$, of the DFL domain (I) for the pair (A_0, A_1) and the gauge cobordism (II) $= \bar{W}$, and its lifting group is the fibre product of theirs over ϕ_Z . The cut Z is connected, its critical points for the perturbed functional with f_Z are nondegenerate and irreducible, since the odd torus kills the stabilizers, the connecting H^0 term vanishes, and the framed groups glue by the fibre product [21, Proposition 5.4]. Hypotheses (H1)–(H5) hold, (H1) by Theorem 10.12(ii). By Theorem 5.1, for $\ell \geq \ell_0$ the data $\mathfrak{d}_Z(\ell)$ are regular in index at most zero, and a further invariant perturbation supported in the pieces, small enough not to change the index-zero sets, makes them regular in index one; the joined count is then the composite of the count of (I) and that of (II). By Lemma 9.7, whose mixed-end clause covers the passage from the slices D_{i, λ_0} of (I) to the ends of the quintuple of (D_0, D_1) with the metrics of Z_{cut} , the count of (I) is chain homotopic to N_Z . The count of (II) is chain homotopic to $\bar{E}$, cobordism maps being independent of the interior data [21, Section 5.1]. With Lemma 10.19 this gives $N_{\lambda_0}^J \simeq N_Z\bar{E}$, and composing with $\mathcal{K}$ gives the claim. $\square$

10.10. Torus completion. *Conventions.* Throughout this subsection L stands for R on the separate collared domains $W_L = W_{\pm, R}$ and for λ on the joined domains $W_L = W_{\lambda}^J$, and W_{∞} stands for $W_{\pm, \infty}$ or $W_{-, \infty} \sqcup W_{+, \infty}$. The index e runs over the mixed ends; the critical manifold of the end e is $\mathcal{C}_e$ (P or Q on a separate side, $P_i \times Q_i$ on a joined end) and its slice is $Y_{e, L}$ (Y_R , Y'_R or $D_{i, \lambda}$). Fix $m \geq 4$. The weight τ of Definition 10.1 is chosen with $\tau = \tau_0(\theta)$ on N_X and on the torus region, $|\tau'_0| \leq 1$, $|\tau''_0| \leq 1$, $\tau_0(\theta) = |\theta|$ for $|\theta| \geq 2$ and $\tau \geq 0$. Fix δ with

$$0 < \delta \leq \delta_T, \quad \delta < \gamma/2,$$

where δ_T is as in Proposition 10.10 and γ is the gap of Proposition 10.7, and with δ below the weights of the original mixed ends in [11, Definition 3.9]. Constants C , C_{inv} , C_{quad} and r_1 depend on the fixed data and on the finite sets of completed solutions, not on L .

Lemma 10.21 (regular completed pieces). *For generic secondary perturbations in the invariant class of Theorem 5.4, supported in a fixed compact subset of the core of X_D away from the torus region, the matching lines and the ends, the completed domains carry no solutions of negative index, and all solutions of index at most one are regular. The index-zero sets are finite.*

Proof. By Theorem 10.12 and the energy identity, the index of a solution with given labels is bounded below. Proceed upward from the least index d that occurs. A sequence of solutions of index d on the fixed completed domain has, by Proposition 10.13 for a fixed completed domain, a limit of total index $d - 4n$ with $n \geq 0$ and nonnegative trajectory contributions; since no lower index occurs, the limit has index d , there are no chains and $n = 0$, so the space is compact. Theorem 5.4(ii) gives universal surjectivity with invariant holonomy directions supported in the prescribed region, and Theorem 5.4(iii) gives finitely many directions and a generic small choice making index d regular; a negative index then does not occur. Repeating up to index one, with the smallness preserved at each step, proves the lemma. Finiteness at index zero follows from compactness and regularity. $\square$

We fix such data on $W_{\pm, \infty}$ and use the same secondary perturbations on $W_{\pm, R}$ and W_{λ}^J (split in the joined case). The primary data, end perturbations and almost complex structures are those of Section 10.1.

Asymptotic values. Let P be a smooth configuration on W_L ($L \leq \infty$) which on each mixed end e converges to $\pi^* B_{q_e}$ and the constant section q_e , and which equals α on a collar of the flat face in the separate case. For each e fix an exponential map $\exp_{q_e}: T_{q_e} \mathcal{C}_e \rightarrow \mathcal{C}_e$ of a metric in which the Lagrangian boundary conditions are totally geodesic [11, Lemma 4.4], and a cutoff $\beta_e(\theta)$ of the outward end coordinate, equal to 0 for $\theta \leq \theta_e$ and to 1 for $\theta \geq \theta_e + 1$, with θ_e fixed. Put

$$k_e(v) = \left(\pi^* \frac{d}{ds} \Big|_{s=0} B_{\exp_{q_e}(sv)}, v \right), \quad v \in T_{q_e} \mathcal{C}_e,$$

the tangent vector of the stationary family together with the constant section vector, transported along the section strip. Its connection part is supported away from the torus collar, it has vanishing normal component on the genus matching faces, and its restriction to Σ' is closed with class v , so $\beta_e k_e(v)$ satisfies the linearized matching condition.

Definition 10.22 (gluing spaces). Let P be as above. Let $\mathcal{X}_L^m$ be the space of pairs $w = (\zeta, \nu)$ with $\zeta \in H_{\delta}^m(W_L; \Lambda^1 \otimes \text{ad})$ and $\nu \in H_{\delta}^m(\mathbb{U}_+; u_P^* TM)$, where $\|\zeta\|_{H_{\delta}^m} = \|e^{\delta\tau} \zeta\|_{H^m}$ with derivatives taken with respect to the Levi-Civita connection and the connection of P , and similarly for ν , subject to the linearized boundary conditions: on the genus matching faces $\mathbb{U}_{\partial} \times \Sigma'$ the normal component of ζ vanishes, its restriction ζ_{θ} to each surface is $d_{A_{\theta}}$ -closed and $[\zeta_{\theta}] = \nu(0, \theta)$ [11, Definition 3.9(iii)]; on the Lagrangian arcs ν is tangent to the Lagrangians; and on the flat face (separate case) $\zeta(\partial_r) = 0$ and $d_{\alpha}(\zeta|_{T \times \{(L, \theta)\}}) = 0$. Put

$$X_L = \mathcal{X}_L^m \oplus \bigoplus_e T_{q_e} \mathcal{C}_e, \quad \|(w, p)\|_{X_L} = \|w\|_{\mathcal{X}_L^m} + |p|,$$

$$Y_L = H_{\delta}^{m-1}(W_L; (\Lambda^0 \oplus \Lambda^+) \otimes \text{ad}) \oplus H_{\delta}^{m-1}(\mathbb{U}_+; \Lambda^{0,1} \otimes u_P^* TM),$$

$$D_L(w, p) = \left(d_{A_P}^* \zeta, \mathfrak{L}_P(w + \sum_e \beta_e k_e(p_e)) \right),$$

where $\mathfrak{L}_P$ is the linearization at P of the mixed operator $\mathfrak{M}$ (the perturbed anti-self-duality operator and the section operator).

The image $\mathfrak{L}_P(\beta_e k_e(v))$ of a cut-off stationary field equals $\sigma(d\beta_e)k_e(v)$ plus $\beta_e(\mathfrak{L}_P - \mathfrak{L}_{\pi^* B_{q_e}})k_e(v)$, because the stationary operator annihilates $k_e(v)$; the first term is supported in a fixed compact band and the second decays like $P - \pi^* B_{q_e}$. So $D_L: X_L \rightarrow Y_L$ is bounded, with the part acting on asymptotic values bounded by $C|p|$. The operator D_L differs from the operator on the extended weighted spaces of Theorem 10.12, whose asymptotic values are represented by the cut-off Coulomb kernel representatives, by an operator that vanishes on $\mathcal{X}_L^m$; this difference has finite rank, so D_L is Fredholm with the index of Theorem 10.12. The gauge-fixing component of D_L acts on the decaying part ζ only. This differs from [11, Proposition 4.6], where $d_A^* \zeta = 0$ is imposed on the whole field including its asymptotic part: here the Coulomb condition is imposed on the decaying coordinate of the chart (Lemma 10.27).

Lemma 10.23 (preglued references). *For $V \in \mathcal{M}^0(W_\infty)$ let $V^{\text{ad}} = \alpha + a_V$ on $r \geq S_V$ be as in Lemma 10.11. For $L \geq 2S_V + 2$ define $P_L(V)$ on W_L to be V^{ad} outside $T \times [S_V, L] \times \mathbb{R}$ and $\alpha + \chi_L(r)a_V$ on $T \times [S_V, L] \times \mathbb{R}$, where $\chi_L = 1$ for $r \leq L/2$, $\chi_L = 0$ for $r \geq L/2 + 1$ and $|\chi'_L| \leq 2$. For a pair (V_-, V_+) define $P_\lambda(V_-, V_+)$ on W_λ^J by the same truncation of both tails at $r_\pm = \lambda/2$, with α on the middle of the bridge, joined by the bridge identification. Then, with $P = P_L$:*

- (i) P_L is a configuration with the labels of V (of the pair), equals α near the face, and on each mixed end converges to $\pi^* B_{q_e}$ on $Y_{e,L}$ (to $B_p \cup \alpha \cup B_q$ on $D_{i,\lambda}$);
- (ii) $\mathfrak{M}(P_L)$ is supported in $T \times [L/2, L/2 + 1] \times \mathbb{R}$ (on both sides of the bridge) and $\|\mathfrak{M}(P_L)\|_{Y_L} \leq Ce^{-\gamma T L/2}$;
- (iii) $\text{ind } D_L = 0$;
- (iv) $\|P_L - \alpha\|_{C^m(T \times [S_V, L] \times \mathbb{R})} \leq Ce^{-\gamma T(S - S_0)}$ for $S \geq S_V$; on each end $\sup_L \|P_L - \pi^* B_{q_e}\|_{H_\delta^m(\{\theta \geq \Theta\})} \rightarrow 0$ as $\Theta \rightarrow \infty$; and $P_L = V^{\text{ad}}$ on $\{r \leq L/2\}$.

Proof. (i) is immediate from Lemma 10.11. (ii) Only the truncation band carries an error. There the equation is the unperturbed anti-self-duality equation, and since $F_{\alpha+a_V}^+ = 0$,

$$F_{\alpha+\chi a_V}^+ = (d\chi \wedge a_V)^+ + (\chi^2 - \chi)(a_V \wedge a_V)^+,$$

whose H_δ^{m-1} norm is at most $C\|a_V\|_{H_\delta^m(r \geq L/2)}$, bounded by Proposition 10.10(ii); the section is unchanged. (iii) The path $\alpha + (1 - s + s\chi_L)a_V$ shows, as in the proof of Proposition 10.13, that $\kappa(P_L) = \kappa(V)$ (the sum $\kappa(V_-) + \kappa(V_+)$ in the joined case). By Theorem 10.12(i)–(iii) the index is $8\kappa + \mathfrak{c}(x) - \mathfrak{c}(a)$ with constants independent of L , so $\text{ind } D_{P_L} = \text{ind } D_V = 0$ (the sum of the two indices, by Theorem 10.12(ii)). (iv) follows from Propositions 10.10(i) and 10.10(ii) and Lemma 10.11(ii). $\square$

Lemma 10.24 (uniform scalar inverse). *There are r_0 , L_0 and C such that for $L \in [L_0, \infty]$ and every connection A on W_L with $\|A - A_{P_L(V)}\|_{C^m} \leq r_0$ (for $L = \infty$, with $P_\infty(V) = V^{\text{ad}}$), the Neumann problem*

$$\xi \longmapsto (d_A^* d_A \xi, \nabla_n \xi|_{\partial W_L}), \quad H_\delta^{m+1}(W_L; \text{ad}) \rightarrow H_\delta^{m-1}(W_L; \text{ad}) \oplus H_\delta^{m-1/2}(\partial W_L; \text{ad}),$$

where ∂W_L consists of the genus matching faces and the flat face, is an isomorphism with inverse bounded by C .

Proof. It suffices to treat $A = A_{P_L(V)}$; write $\Delta_A = d_A^* d_A$. For ξ with vanishing normal derivative, the estimate $\|\xi\|_{H_\delta^1} \leq C(\|\Delta_A \xi\|_{H_\delta^{-1}} + \|\xi\|_{L^2(K)})$, with a fixed compact K , is proved by the partition of the proof of Proposition 10.25 below, with the following models.

On the torus region, where A is within $Ce^{-\gamma T(S-S_0)}$ of α , the Neumann Laplacian of α satisfies $\langle \Delta \xi, \xi \rangle = \|d\xi\|^2 \geq \|d_T \xi\|^2 \geq \mu_T^2 \|\xi\|^2$ on any base, and the weight changes it by terms of order $\delta \leq \mu_T/4$. On the mixed ends the slice Neumann Laplacian of B_{q_e} on $Y_{e,L}$ has a gap $\gamma_s > 0$ uniform in L : the collar part is controlled by the torus mass, and a normalized counterexample has a nonzero weak limit on $Y_{e,\infty}$ which is a parallel section, hence zero, because B_{q_e} is irreducible; so $-\partial_\theta^2 + \Delta_{Y_{e,L}}$ is invertible on weighted spaces for $\delta^2 < \gamma_s^2$. This holds for the weights used here, since $\gamma \leq \gamma_s$: on pairs $(\varphi, d_B \varphi')$ the stationary operator squares to the Neumann Laplacian, because the primary Hessian annihilates infinitesimal gauge directions at a critical point, and $\delta < \gamma/2$. The original ends have acyclic slices, and the core is compact. Higher derivatives follow from local elliptic estimates on unit patches. If the uniform bound failed, a normalized sequence would have a nonzero limit ξ_∞ on W_∞ with $\Delta_V \xi_\infty = 0$, Neumann boundary values and weighted decay; then $\|d_V \xi_\infty\|^2 = 0$, so ξ_∞ is parallel and vanishes on a torus fibre, where $H^0(T; \text{ad } \alpha) = 0$; hence $\xi_\infty = 0$, a contradiction. In the joined case the limit lives on the two sides and the same argument applies to each. The operator is formally self-adjoint, and its weighted index is zero because δ lies below all the gaps; so it is an isomorphism, and the boundary data are handled by a translated lifting of fixed width. A perturbation of A of size r_0 in C^m changes the operator by a term of norm Cr_0 , which is absorbed for r_0 small. $\square$

Proposition 10.25 (uniform inverse). *There are L_0 and C_{inv} such that for $L \geq L_0$ and every V (every pair) in the finite completed index-zero sets, $D_L: X_L \rightarrow Y_L$ at $P = P_L(V)$ is an isomorphism with $\|D_L^{-1}\| \leq C_{\text{inv}}$.*

Proof. By Lemma 10.23(iii) and the remark after Definition 10.22, D_L is Fredholm of index zero, so it suffices to prove $\|(w, p)\|_{X_L} \leq C_{\text{inv}} \|D_L(w, p)\|_{Y_L}$. Write $D_L^{\text{dec}} w = D_L(w, 0)$.

Step 1: models. Fix parameters S, Θ, T_0, ℓ , chosen below in this order.

(T) The torus region $\{r \geq S\}$ (the part of the bridge at distance at least S from both pieces in the joined case), including its corners inside the mixed ends. There $D_L^{\text{dec}} = D_\alpha + \mathfrak{a}$, where D_α is the operator of Proposition 10.3 and $\mathfrak{a}$ is of order zero with $|\mathfrak{a}| \leq Ce^{-\gamma T(S-S_0)}$ by Lemma 10.23(iv); no perturbation acts there, and the corners of the torus region inside the mixed ends carry only the gauge field. The base is a strip $[S, L] \times \mathbb{R}$ with the flat-fibre condition at $r = L$, or a strip in the bridge; on it Proposition 10.3 with the weight $e^{\delta\tau(\theta)}$ gives $\|e^{\delta\tau}\zeta\| \leq (\mu_T - \delta)^{-1} \|e^{\delta\tau} D_\alpha \zeta\|$ for ζ vanishing near $r = S$. For a straight boundary the Bochner identity for $d + d^*$ on the base with absolute boundary conditions has no boundary term, so the identity in the proof of Proposition 10.3 also bounds $\|\nabla \zeta\|$ by $\|D_\alpha \zeta\|$. Hence $\|\zeta\|_{H_\delta^1} \leq C_T \|D_\alpha \zeta\|_{L_\delta^2}$ with C_T depending only on μ_T and δ .

(E) The mixed end e , beyond Θ , restricted to $r \leq S + \ell$. There $D_L^{\text{dec}} = \sigma(\partial_\theta + \mathcal{A}_{q_e, L}) + \mathfrak{b}$, where $\mathcal{A}_{q_e, L}$ is the stationary operator of Proposition 10.7 on the whole slice $Y_{e, L}$ [11, Section 4.2.1, (4.16)] and $\|\mathfrak{b}\zeta\|_{L^2} \leq \eta(\Theta) \|\zeta\|_{H^1}$ with $\eta(\Theta) \rightarrow 0$ uniformly in L by Lemma 10.23(iv). A field supported in $\{\theta \geq \Theta\}$ and extended by zero is in the domain of the translation-invariant operator on $\mathbb{R} \times Y_{e, L}$. Since $\mathcal{A} = \mathcal{A}_{q_e, L}$ is self-adjoint with spectrum meeting $(-\gamma, \gamma)$ only in 0, and $0 < \delta < \gamma$, Fourier transform in θ gives $\|(\partial_\theta + \mathcal{A} - \delta)^{-1}\| \leq \min(\delta, \gamma - \delta)^{-1}$; the identity $\|\partial_\theta \xi\|^2 + \|(\mathcal{A} - \delta)\xi\|^2 = \|(\partial_\theta + \mathcal{A} - \delta)\xi\|^2$ and the elliptic estimate $\|\xi\|_{H^1(Y_{e, L})} \leq C(\|\mathcal{A}\xi\| + \|\xi\|)$, which is uniform in L because it is local, $Y_{e, L}$ has bounded geometry and on the collar $\mathcal{A} = \mathcal{A}_\alpha$ (Corollary 10.4), give $\|e^{\delta\theta}\xi\|_{H^1} \leq C_E \|e^{\delta\theta}(\partial_\theta + \mathcal{A})\xi\|$ with C_E independent of L .

(O) The original gauge and ordinary ends beyond T_0 : translation-invariant operators with invertible asymptotic operators, as in Theorem 5.1, perturbed by terms that are small for T_0 large.

(K) A fixed compact region K' containing the rest: the local elliptic estimates of [11, Theorem 4.17 and Remark 4.19] at the matching faces and interior estimates give $\|\chi w\|_{H^1} \leq C(\|D_L^{\text{dec}} w\|_{L^2(K'')} + \|w\|_{L^2(K'')})$ for a cutoff χ supported in K' and a slightly larger compact K'' .

Step 2: the compact-remainder estimate. Let χ be a smooth function on $\mathbb{R}$ with $\chi = 0$ on $(-\infty, 0]$ and $\chi = 1$ on $[1, \infty)$. Put $\psi_T = \chi((r-S)/\ell)$ on the torus region (in the joined case, the product of the two such functions of r_- and r_+), $\psi_{E,e} = (1 - \psi_T)\chi(\theta - \Theta)$ on the end e , cutoffs ψ_G, ψ_O of fixed slope on the original ends beyond T_0 , and $\psi_K = 1 - \psi_T - \sum_e \psi_{E,e} - \psi_G - \psi_O$, which has compact support. The gradient of ψ_T is at most C/ℓ and is supported in $\{S \leq r \leq S + \ell\}$; the gradient of $\psi_{E,e}$ is at most C/ℓ on the corner band $\{\theta \geq \Theta, S \leq r \leq S + \ell\}$, and elsewhere is supported in the compact set $K = \{\Theta \leq \theta \leq \Theta + 1, r \leq S + \ell\}$ together with the fixed parts of the slices. The corner band, where the torus tail enters the mixed ends, is the only noncompact region of transition; there both models (T) and (E) apply, and the cutoffs vary in r with slope $1/\ell$. For $i \in \{T, E\}$,

$$\|\psi_i w\|_{H_\delta^1} \leq C_i(\|\psi_i D_L^{\text{dec}} w\|_{L_\delta^2} + \|\sigma(d\psi_i)w\|_{L_\delta^2} + \|\mathfrak{a}\psi_i w\|_{L_\delta^2}),$$

with $\mathfrak{b}$ in place of $\mathfrak{a}$ for E . Choose S and Θ so large that $C_T \sup |\mathfrak{a}| \leq \frac{1}{2}$ and $C_E \eta(\Theta) \leq \frac{1}{2}$; these choices do not depend on ℓ , since C_T and C_E do not. Then choose T_0 similarly and sum:

$$\|w\|_{H_\delta^1} \leq \sum_i \|\psi_i w\|_{H_\delta^1} \leq C(\|D_L^{\text{dec}} w\|_{L_\delta^2} + \ell^{-1} \|w\|_{L_\delta^2} + \|w\|_{L^2(K_1)}),$$

where K_1 is a compact set containing K , K'' and the compact transition regions. Choosing ℓ with $C\ell^{-1} \leq \frac{1}{2}$ absorbs the middle term. The constants C_T , C_E and the constants of (O) do not depend on ℓ , and the constant of (K) multiplies only terms supported in K_1 ; so

$$(10.4) \quad \|w\|_{H_\delta^1} \leq C(\|D_L^{\text{dec}} w\|_{L_\delta^2} + \|w\|_{L^2(K_1)}) \quad (L \geq L_0),$$

with $L_0 = 2(S + \ell) + 2$, so that $K_1 \subset \{r \leq L/2\}$ and $P_L = V^{\text{ad}}$ on K_1 . In the joined case $K_1 = K_{1,-} \sqcup K_{1,+}$, one compact set in the core region of each side, at a fixed position relative to that side.

Step 3: higher order and asymptotic values. Cover W_L by unit patches of finitely many isometry types: flat patches $T \times Q$ in the torus region, flat half-patches at the face, translates of fixed patches along the mixed ends and the original ends, and fixed patches of the core. The coefficients of D_L are bounded in C^m uniformly in L (Lemma 10.23(iv)). On the flat patches the local estimate is that of $D_\alpha + \mathfrak{a}$ with the flat-fibre domain at the face: the tangential derivatives commute with D_α and preserve the domain, and the normal derivatives are recovered from the equation, as in the proof of Theorem 10.5(b); on the other patches it is [11, Theorem 4.17 and Remark 4.19]. The weight varies by a bounded factor on each patch. Summing with bounded overlap and using (10.4),

$$\|w\|_{\chi_L^m} \leq C(\|D_L^{\text{dec}} w\|_{H_\delta^{m-1}} + \|w\|_{L^2(K_1)}).$$

Since the part of D_L acting on asymptotic values is bounded by $C|p|$,

$$(10.5) \quad \|(w, p)\|_{X_L} \leq C(\|D_L(w, p)\|_{Y_L} + \|w\|_{L^2(K_1)} + |p|).$$

Step 4: the limit operator is injective. Let $D_V: X_\infty \rightarrow Y_\infty$ be defined as in Definition 10.22 on W_∞ with $P = V^{\text{ad}}$. We show that it is injective. Let $D_V(w, p) = 0$, $w = (\zeta, \nu)$. Write

$k_e = (k_e^A, v_e)$ and let $h_e(v) = (h_e^A(v), v)$ be the Coulomb kernel representative of v used in Theorem 10.12. Then $k_e^A(v) - h_e^A(v) = d_{B_{q_e}} \psi_e(v)$ for a function $\psi_e(v)$ on the slice with vanishing normal derivative, because both are tangent to the stationary family at q_e , represent the same vector v and have vanishing normal components. The variation

$$\zeta_1 = \zeta + \sum_e \beta_e k_e^A(p_e) - d_V \left(\sum_e \beta_e \psi_e(p_e) \right), \quad \nu_1 = \nu + \sum_e \beta_e v_e(p_e),$$

has connection part equal to $\sum \beta_e h_e^A(p_e)$ plus a decaying field, and $\mathfrak{L}_V(\zeta_1, \nu_1) = 0$ because V is a solution and the equation and the matching conditions are gauge invariant. The divergence $d_V^* \zeta_1$ is decaying, because $d_V^* \zeta = 0$, h_e is coclosed on the slice and the remaining terms contain $d\beta_e$ or $V - \pi^* B_{q_e}$. By Lemma 10.24 on W_∞ there is a decaying ξ with $d_V^*(\zeta_1 + d_V \xi) = 0$ and vanishing normal components. Then $(\zeta_1 + d_V \xi, \nu_1)$ lies in the kernel of the operator on the extended weighted spaces of Theorem 10.12 at V , which is injective because it is onto and of index zero. So $p = 0$, $\zeta_1 + d_V \xi = 0$ and $\nu = \nu_1 = 0$; with $p = 0$ this reads $\zeta = -d_V \xi$, and $d_V^* \zeta = 0$ gives $d_V^* d_V \xi = 0$, hence $\xi = 0$ by Lemma 10.24, and $\zeta = 0$. In the joined case D_V is the direct sum of the operators of V_- and V_+ : the section structure is $J_- \times J_+$, the Lagrangians $A_i = \tilde{P}_i^- \times R_i$ are products, the Σ_h -matching couples only the $-$ side and the Σ_g -matching only the $+$ side, and the stationary fields $k_e(v_-, v_+) = k_{e,-}(v_-) + k_{e,+}(v_+)$ split because the joined stationary family is $B_p \cup \alpha \cup B_q$. So it is injective as well.

Step 5: contradiction. Suppose $\mathbb{L}_j \rightarrow \infty$ and $(w_j, p_j) \in X_{\mathbb{L}_j}$ with $\|(w_j, p_j)\| = 1$ and $D_{\mathbb{L}_j}(w_j, p_j) \rightarrow 0$, for a fixed V (the sets are finite). By (10.5), $\liminf_j (\|w_j\|_{L^2(K_1)} + |p_j|) \geq 1/C$. Pass to a subsequence with $p_j \rightarrow p_\infty$. Every compact subset of W_∞ embeds in $W_{\mathbb{L}_j}$ for large j , isometrically and with $P_{\mathbb{L}_j} = V^{\text{ad}}$ on it, and w_j is bounded in H^m there; a diagonal subsequence converges weakly in H_{loc}^m and strongly in L_{loc}^2 to w_∞ , with $\|w_\infty\|_{\mathcal{X}_\infty^m} \leq 1$ by lower semicontinuity along an exhaustion. The linear boundary conditions pass to the limit. On each compact set the operators $D_{\mathbb{L}_j}$ and their actions on asymptotic values coincide with those of D_V for large j , so $D_V(w_\infty, p_\infty) = 0$, and Step 4 gives $(w_\infty, p_\infty) = 0$. But $\|w_\infty\|_{L^2(K_1)} + |p_\infty| \geq 1/C$ by strong convergence on K_1 , a contradiction. In the joined case the exhaustions are taken separately on the two sides: the part of $W_{\lambda_j}^J$ within distance $\lambda_j/2$ of the $\pm$ core is identified with the corresponding part of $W_{\pm, \infty}$, the bridge between them escapes to infinity, and the limit is a pair $(w_{\pm, \infty}, p_{\pm, \infty})$ on $W_{-, \infty} \sqcup W_{+, \infty}$, which vanishes by Step 4, while $K_1 = K_{1,-} \sqcup K_{1,+}$ retains mass. This proves the uniform bound, and invertibility follows from index zero. $\square$

Lemma 10.26 (the flat-fibre chart). *Let $s = m - \frac{1}{2}$ and let H_δ^s denote the weighted Sobolev space of ad-valued torus one-forms on the face $T \times \mathbb{R}_\theta$, with weight $e^{\delta\tau_0(\theta)}$. Let Π_{cl} be the fibrewise projection onto d_α -closed forms and $Q = d_\alpha^*(d_\alpha d_\alpha^*)^{-1}$ on fibrewise two-forms.*

- (a) *There are r_3, C such that for every closed b with $\|b\|_{H_\delta^s} \leq r_3$ the equation $c = -\frac{1}{2}Q[(b+c) \wedge (b+c)]$ has a unique coexact solution $c(b)$ with $\|c\|_{H_\delta^s} \leq r_3$. Then $\alpha + b + c(b)$ is flat on every fibre, $c(0) = 0$, $Dc(0) = 0$, and $\|c(b) - c(b')\|_{H_\delta^s} \leq C(\|b\| + \|b'\|)\|b - b'\|$. Conversely, if $\alpha + \tilde{b}$ is flat on every fibre and $\|\tilde{b}\|_{H_\delta^s} \leq r_3/C$, then $\tilde{b} = b + c(b)$ with $b = \Pi_{\text{cl}} \tilde{b}$.*
- (b) *There is a linear map $\text{Ext}_\delta: H_\delta^s(T \times \mathbb{R}) \rightarrow H_\delta^m(W_{\mathbb{L}})$, with norm bounded independently of $\mathbb{L}$, whose values are torus one-forms supported in the collar $T \times [\mathbb{L} - 1, \mathbb{L}] \times \mathbb{R}$ with vanishing dr and $d\theta$ components, and whose restriction to the face is the identity.*
- (c) *The map $\Psi^{\text{face}}(\zeta) = \zeta + \text{Ext}_\delta(c(\zeta_T|_{r=\mathbb{L}}))$, defined on the ball of radius r_3/C of fields satisfying the linear flat-fibre condition, takes values in fields x for which $\alpha + x$ satisfies*

the nonlinear flat-fibre condition and $x(\partial_r) = 0$ at the face; it is inverted by $x \mapsto x - \text{Ext}_\delta(c(\Pi_{\text{cl}}x_T|_{r=\text{L}}))$, and both maps have first and second derivatives bounded independently of L .

Proof. (a) The operators Π_{cl} and $\mathbf{Q}$ act fibrewise and are Fourier multipliers in the eigenmodes of the flat bundles $\text{ad } \alpha$ on T ; they commute with ∂_θ and with multiplication by functions of θ , in particular with $e^{\delta\tau_0}$. Since $\Delta_\alpha \geq \mu_T^2$ on each fibre, $\mathbf{Q}$ is bounded on the isotropic space $H^s(T \times \mathbb{R})$ (its symbol is bounded by μ_T^{-1} in the torus frequencies and does not involve the θ -frequency). As $s > \frac{3}{2}$, $H^s(T \times \mathbb{R})$ is an algebra, and $\|e^{\delta\tau_0}fg\|_{H^s} \leq C\|e^{\delta\tau_0}f\|_{H^s}\|e^{\delta\tau_0}g\|_{H^s}$ because $\tau_0 \geq 0$ and $e^{-\delta\tau_0}$ has bounded derivatives. Hence the right side of the equation is a contraction of the ball of coexact forms of radius r_3 for r_3 small, and the contraction estimates give the stated bounds. Flatness: $F_{\alpha+b+c} = d_\alpha c + \frac{1}{2}[(b+c) \wedge (b+c)] = 0$, because $d_\alpha b = 0$ and $d_\alpha \mathbf{Q}$ is the identity on two-forms, $H^2(T; \text{ad } \alpha)$ being zero. Conversely, write $\tilde{b} = b + \tilde{c}$ with $b = \Pi_{\text{cl}}\tilde{b}$ and $\tilde{c}$ coexact; flatness gives $d_\alpha \tilde{c} = -\frac{1}{2}[\tilde{b} \wedge \tilde{b}]$, and $\mathbf{Q}d_\alpha$ is the identity on coexact forms, so $\tilde{c}$ solves the equation and equals $c(b)$ by uniqueness. (b) Let $\Lambda = (1 + \Delta_{T \times \mathbb{R}})^{1/2}$, acting on torus one-forms on $T \times \mathbb{R}$ with the connection α , and $E_0(c)(\rho) = \chi(\rho)e^{-\rho\Lambda}c$, $\rho = \text{L} - r$, with a fixed cutoff χ supported in $[0, 1]$ and equal to one near 0. Mode by mode $\int_0^\infty |\partial_\rho^j e^{-\rho\Lambda}|^2 d\rho = \mu^{2j-1}/2$, so E_0 is bounded from $H^{k-1/2}$ to H^k of the half-space for integers k , and by interpolation from H^s to H^m . Put $\text{Ext}_\delta(c) = e^{-\delta\tau_0}E_0(e^{\delta\tau_0}c)$. The collar is a translate of a fixed product collar, so the norm does not depend on L . (c) By (b) and the trace theorem, $\zeta_T|_{r=\text{L}}$ lies in H_δ^s with norm at most $C\|\zeta\|_{H_\delta^m}$; the torus restriction of $\zeta + \text{Ext}_\delta(c)$ at the face is $\zeta_T|_{r=\text{L}} + c(\zeta_T|_{r=\text{L}})$, flat by (a) since $\zeta_T|_{r=\text{L}}$ is closed, and the normal component is unchanged. For the inverse, if $\alpha + x$ satisfies the nonlinear condition, (a) gives $x_T|_{r=\text{L}} = b + c(b)$ with $b = \Pi_{\text{cl}}x_T|_{r=\text{L}}$, so $x - \text{Ext}_\delta(c(b))$ has closed restriction b . The derivative bounds follow from those of c and the boundedness of Ext_δ , the trace and Π_{cl} . $\square$

The chart. For $z = (w, p) \in X_{\text{L}}$ small, $w = (\zeta, \nu)$, let $\zeta' = \zeta + \sum_e \beta_e k_e(p_e)$ and $\nu' = \nu + \sum_e \beta_e v_e(p_e)$ be the fields with the stationary fields added ($v_e(p_e)$ being the section part of $k_e(p_e)$). Let $\Psi_{\text{L}}(z)$ be the configuration with connection

$$A_P + \zeta' + \rho(s)(E(\cdot, \zeta') - \zeta') + \sum_e \varphi_e(\pi^* B_{\exp_{q_e} p_e} - \widehat{B}_{e, p_e}) + \text{Ext}_\delta(c(\zeta_T|_{r=\text{L}}))$$

and section $\exp_{u_P}(\nu')$ taken with the metrics of [11, Lemma 4.4]. Here the terms involving E and $\widehat{B}_{e, p}$ are those of the chart of [11, Section 4.1]: the correction $\rho(s)(E(\cdot, \zeta') - \zeta')$ is built from the maps E_τ of [11, (4.3) and Lemma 4.4] for the component Σ' only, on a fixed collar of the genus matching faces, and imposes the nonlinear matching and Lagrangian conditions; $\widehat{B}_{e, p} = \pi^* B_{q_e} + k_e^A(p) + \rho(s)(E_1(\cdot, k_e^A(p)) - k_e^A(p))$, with k_e^A the connection part of k_e , and φ_e is a cutoff of θ equal to one where $\beta_e = 1$ and supported in $\{\theta \geq \theta_e\}$, so that the limit at the end e is exactly $\pi^* B_{\exp_{q_e} p_e}$. The last term is the face chart of Lemma 10.26, absent in the joined case. All terms except ζ and the face term are supported away from the torus region, because k_e , the families B_q and the maps E_τ for Σ' are. Put $P^{(p)} = \Psi_{\text{L}}(0, p)$, the reference with its mixed-end limits moved to $\exp_{q_e} p_e$; it equals P on the torus region. Define

$$\mathcal{F}_{\text{L}}(z) = \left(d_{A_P}^* \zeta, \mathfrak{M}(\Psi_{\text{L}}(z)) \right) \in Y_{\text{L}}.$$

Lemma 10.27 (chart and quadratic estimate). *There are L_0 , r_1 and C_{quad} such that for $\text{L} \geq \text{L}_0$ and $P = P_{\text{L}}(V)$:*

- (a) $\Psi_L(z)$ satisfies the nonlinear matching, Lagrangian and flat-fibre conditions, and zeros of $\mathcal{F}_L$ are solutions of the mixed equation. Conversely, if U satisfies these conditions and $\|U - P^{(p)}\|_{H_\delta^m} + |p| \leq r_1/C_{\text{quad}}$, then there are ξ with $\|\xi\|_{H_\delta^{m+1}} \leq C_{\text{quad}}\|U - P^{(p)}\|_{H_\delta^m}$ and w with

$$\exp(\xi)^*U = \Psi_L(w, p), \quad d_{A_P}^*\zeta = 0, \quad \|w\|_{\mathcal{X}_L^m} \leq C_{\text{quad}}(\|U - P^{(p)}\|_{H_\delta^m} + |p|).$$

- (b) $\mathcal{F}_L(0) = (0, \mathfrak{M}(P_L))$ and $D\mathcal{F}_L(0) = D_L$;
(c) $\mathcal{F}_L$ is of class C^2 on the ball of radius r_1 in X_L , with $\|D^2\mathcal{F}_L\| \leq C_{\text{quad}}$. Consequently $\mathcal{F}_L(z) = \mathcal{F}_L(0) + D_L z + N_L(z)$ with

$$\|N_L(z) - N_L(z')\|_{Y_L} \leq C_{\text{quad}}(\|z\|_{X_L} + \|z'\|_{X_L})\|z - z'\|_{X_L}.$$

Proof. (a) The flat-fibre condition holds by Lemma 10.26(c), because $\Psi_L(z) - \zeta - \text{Ext}_\delta(\cdot)$ equals α on the collar of the face; the matching and Lagrangian conditions hold by the construction of [11, Section 4.1], which involves only the fixed collars of the genus faces, the section and the parts of the mixed ends away from the torus collar. For the converse, first invert the chart at fixed p : at the face by Lemma 10.26(c), and at the genus faces by the inverse function theorem for the maps E_τ on the collar [11, Section 4.1], whose estimates are uniform because this region is fixed or, along the ends, translation invariant; the two collars are disjoint. This writes $U = \Psi_L(\tilde{w}, p)$ with $\|\tilde{w}\| \leq C(\|U - P^{(p)}\| + |p|)$. Then solve $d_{A_P}^*(\zeta(\exp(\xi)^*U)) = 0$, with vanishing normal components on the faces, for ξ , where $\zeta(\cdot)$ is the decaying coordinate of the inverted chart: the derivative in ξ at $\xi = 0$, $U = P$, $p = 0$ is the Neumann problem of Lemma 10.24, the second derivatives are bounded uniformly since $m \geq 4$ and W_L has bounded geometry, and the implicit function theorem applies. (b) is immediate: the gauge-fixing component does not involve p , the corrections $\rho(s)(E - \cdot)$ and $B_{\exp p} - \hat{B}_{e,p}$ are of second order, and $D\Psi_L(0)(w, p) = w + \sum_e \beta_e k_e(p_e)$.

(c) We list the terms of $\mathfrak{M}(\Psi_L(z))$. On the torus region the equation is F^+ , quadratic in the connection, and the face term is controlled by Lemma 10.26; since W_L is a union of unit patches of finitely many isometry types and H^m is an algebra in dimension four for $m \geq 3$, $\|e^{\delta\tau}fg\|_{H^{m-1}} \leq C\|e^{\delta\tau}f\|_{H^m}\|g\|_{H^m}$ with C independent of L . The genus matching charts and the section are those of [11, Section 4.1] on a region that is fixed or, along the mixed ends, translation invariant; their estimates are uniform by translation invariance. The primary holonomy terms on the mixed ends are autonomous and act slice by slice; their first and second derivatives are bounded on H^m of unit patches by [11, Proposition 6.1], uniformly by translation. The secondary terms have compact support. It remains to see that the stationary fields do not produce nondecaying terms. Where $\beta_e = \varphi_e = 1$, $\Psi_L(z) = \pi^*B_q + y$ with $q = \exp_{q_e} p_e$ and $y = (A_P - \pi^*B_{q_e}) + \zeta + (\text{the difference of the } E\text{-corrections at } (P, \zeta') \text{ and at } (\pi^*B_{q_e}, k_e(p_e)))$, which is decaying and vanishes when $P = \pi^*B_{q_e}$ and $\zeta = 0$. Since $\mathfrak{M}(\pi^*B_q) = 0$ for every q , each derivative of $\mathfrak{M}(\Psi_L)$ of order at least one in p is the difference of the corresponding derivatives at $\pi^*B_q + y$ and at π^*B_q , bounded by $C\|y\|_{H_\delta^m}$ in Y_L . On the band where β_e or φ_e varies, which is compact and independent of L , all terms are bounded. Hence the second derivatives are bounded uniformly, and the estimate for N_L follows from the mean value theorem. $\square$

Proposition 10.28 (existence and local uniqueness). *Let $r = \min\{r_1, (4C_{\text{inv}}C_{\text{quad}})^{-1}\}$. There is L_1 such that for $L \geq L_1$ and every V (every pair) in the completed index-zero sets there is a unique $z_L(V)$ in the ball of radius r in X_L with $\mathcal{F}_L(z_L(V)) = 0$. It satisfies $\|z_L(V)\| \leq 2C_{\text{inv}}\|\mathfrak{M}(P_L)\|_{Y_L} \leq Ce^{-\gamma\tau L/2}$, and $\mathcal{G}_L(V) := \Psi_L(z_L(V))$ is a regular solution.*

Proof. By Proposition 10.25 and Lemma 10.27, the map $z \mapsto -D_L^{-1}(\mathcal{F}_L(0) + N_L(z))$ sends the ball of radius r into itself when $C_{\text{inv}}\|\mathcal{F}_L(0)\| \leq r/2$, which holds for $L \geq L_1$ by Lemma 10.23(ii), and it is a contraction with constant $2C_{\text{inv}}C_{\text{quad}}r \leq \frac{1}{2}$. If z, z' are two zeros in the ball, then $z - z' = -D_L^{-1}(N_L(z) - N_L(z'))$, so $\|z - z'\| \leq \frac{1}{2}\|z - z'\|$ and $z = z'$; uniqueness holds in the whole ball of fixed radius r . The derivative $D\mathcal{F}_L(z_L) = D_L + DN_L(z_L)$ with $\|DN_L(z_L)\| \leq 2C_{\text{quad}}r \leq (2C_{\text{inv}})^{-1}$ is invertible. It is the linearization of the mixed equation at $\mathcal{G}_L(V)$ in the chart, with the Coulomb condition on the decaying coordinate and the stationary fields k_e ; since this slice and the Coulomb slice of the solution itself are both transverse to the gauge orbits, surjectivity of the linearization does not depend on the slice, and $\mathcal{G}_L(V)$ is regular. $\square$

A gauge transformation is *end-even* if it is even on the torus region and on all ends.

Proposition 10.29 (convergence into the gluing chart). *Let $L_j \rightarrow \infty$ and let U_j be index-zero solutions on W_{L_j} which, in the sense of Proposition 10.13, converge without loss to V (to a pair (V_-, V_+)) in the completed index-zero sets. Then for large j there are end-even gauge transformations g_j with $g_j^*U_j = \Psi_{L_j}(z_j)$, $\mathcal{F}_{L_j}(z_j) = 0$ and $\|z_j\|_{X_{L_j}} \rightarrow 0$. In particular $[U_j] = [\mathcal{G}_{L_j}(V)]$ for large j .*

Proof. We treat the separate case; the joined case is identical, with two pieces, the bridge in place of the tail and the slices $D_{i,\lambda}$. Let V^{ad} , S_V , Θ_V be as in Lemma 10.11, and let $q_{j,e}$ be the limits of U_j at the mixed ends. Fix $\Theta \geq \Theta_V + 2$ and $S \geq S_V + 2$, to be taken large, in this order. Consider the regions

$$R_E^e = \{\theta \geq \Theta\} \times Y_{e,L_j} \text{ (with the section strip)}, \quad R_T = T \times [S, L_j] \times [-\Theta - 1, \Theta + 1],$$

the compact core region R_K , consisting of the part of W_{L_j} with $r \leq S + 1$, $|\theta| \leq \Theta + 1$ and the original ends up to a fixed T_0 , and the original ends R_O beyond $T_0 - 1$. These regions cover W_{L_j} ; the overlaps $R_E^e \cap R_T = \{\Theta \leq |\theta| \leq \Theta + 1, r \geq S\}$ are bands of unit width and growing length, and all other overlaps are compact.

Energy exhaustion. By Proposition 10.13, for every $\varsigma > 0$ there are a compact set K_ς and j_ς with the energy of U_j outside K_ς at most ς for $j \geq j_\varsigma$. The energy density outside the fixed secondary supports and the regions where the cutoffs $\chi_\pm$ are nonconstant is nonnegative: the curvature density on the torus region and the kinetic energy on the ends. We write $E_j(\Omega)$ for the energy of U_j on a region Ω and $o_j(1)$ for quantities tending to zero as $j \rightarrow \infty$ with Θ and S fixed.

The mixed ends. The kinetic energy of U_j beyond $\Theta - 1$ converges to that of V , which is at most $Ce^{-2c(\Theta-1-\Theta_V)}$ by (10.1); for Θ large it is below the threshold of Proposition 10.8, which is uniform in L . In a temporal gauge on R_E^e , normalized by a gauge transformation independent of θ so that the limit is exactly $\pi^*B_{q_{j,e}}$, the representative U_j^E satisfies, by (10.1) summed over unit intervals with the weight $e^{\delta\theta}$ and $\delta < c$,

$$\|U_j^E - \pi^*B_{q_{j,e}}\|_{H_\delta^m(R_E^e)} \leq Ce^{\delta\Theta}E_j(\{\theta \geq \Theta - 1\})^{1/2} \leq Ce^{-(c-\delta)\Theta} + o_j(1).$$

The slices Y_{e,L_j} include the whole torus collar, so this controls the corners $\{\theta \geq \Theta, r \geq S\}$ of the torus region. The same estimate for V , the comparison of the sections at $\theta = \Theta$ and the smooth convergence on compact sets give $d(q_{j,e}, q_e) \leq Ce^{-c\Theta} + o_j(1)$, so the coordinates $p_{j,e} = \exp_{q_e}^{-1} q_{j,e}$ satisfy $|p_j| \rightarrow 0$ along the diagonal choice below. On R_E^e , where $\beta_e = \varphi_e = 1$, the moved reference is $P_{L_j}^{(p_j)} = \pi^*B_{q_{j,e}} + (P_{L_j} - \pi^*B_{q_e}) + O(|p_j| |P_{L_j} - \pi^*B_{q_e}|)$, the last term

coming from the E -corrections, so

$$\|U_j^E - P_{\mathbb{L}_j}^{(p_j)}\|_{H_\delta^m(R_E^e)} \leq \|U_j^E - \pi^* B_{q_{j,e}}\|_{H_\delta^m(R_E^e)} + C \sup_{\mathbb{L}} \|P_{\mathbb{L}} - \pi^* B_{q_e}\|_{H_\delta^m(\{\theta \geq \Theta\})},$$

and both terms are small for Θ large, uniformly in j large, by the display and Lemma 10.23(iv).

The torus strip. Let $R'_T = T \times [S - 2, \mathbb{L}_j] \times [-\Theta - 3, \Theta + 3]$. For $\varsigma > 0$ let K_ς, j_ς be as in the energy exhaustion; if $S - 2$ exceeds the largest value of r on K_ς , then R'_T is disjoint from K_ς for every Θ , so $E_j(R'_T) \leq \varsigma$ for $j \geq j_\varsigma$. We write $\varsigma(S)$ for the least such ς ; it tends to zero as $S \rightarrow \infty$, independently of Θ . By small-energy regularity at the face and in the interior (Theorem 10.5(b), Remark 10.6), for S and j large the curvature of U_j and its covariant derivatives up to order m are small on R_T and there are local gauges with $\|U_j - \alpha\|_{C^1} \leq e_T$. Lemma 10.9 on the simply connected base $[S - 1, \mathbb{L}_j] \times [-\Theta - 2, \Theta + 2]$ gives the torus slice $U_j^T = \alpha + a_j$, and (10.2), integrated over unit patches with small-energy regularity, gives

$$\|a_j\|_{H^m(R_T)} \leq C E_j(R'_T)^{1/2} \leq C \varsigma(S)^{1/2} \quad (j \geq j_\varsigma(S)).$$

The weight on R_T is at most $e^{\delta(\Theta+1)}$. Since $P_{\mathbb{L}_j} = \alpha + \chi_{\mathbb{L}_j} a_V$ on R_T ,

$$\|U_j^T - P_{\mathbb{L}_j}^{(p_j)}\|_{H_\delta^m(R_T)} \leq e^{\delta(\Theta+1)} (C \varsigma(S)^{1/2} + C e^{-\gamma T(S-S_0)}),$$

which is small once S is chosen large after Θ .

The core and the original ends. On R_K , which is compact and contained in $\{r \leq \mathbb{L}_j/2\}$ for large j , there are end-even gauge transformations with $\|U_j^K - V^{\text{ad}}\|_{C^{m+1}(R_K)} = o_j(1)$, by the convergence of Proposition 10.13; and $P_{\mathbb{L}_j}^{(p_j)} - V^{\text{ad}} = O(|p_j|)$ there. On R_O the standard exponential decay at nondegenerate ends, with constants uniform for small end energy [13, Section 4.1], [33], gives $\|U_j^O - P_{\mathbb{L}_j}\|_{H_\delta^m(R_O)} \leq \varsigma(T_0) + o_j(1)$ in temporal gauges normalized at the critical points.

Overlap logarithms. On $R_E^e \cap R_T$ both U_j^E and U_j^T are α plus a term that is small in C^1 , the first because $B_{q_{j,e}} = \alpha$ on the collar. The relative gauge transformation is even, so it is close to the even stabilizer of α , which is trivial, and it equals $\exp \varphi_j$ with φ_j small. On each fibre $d_\alpha \varphi_j$ equals the difference of the two torus components up to quadratic terms, and the base derivatives of φ_j equal the difference of the base components up to quadratic terms; as in the proof of Lemma 10.9,

$$\|\varphi_j\|_{H_\delta^{m+1}(R_E^e \cap R_T)} \leq C (\|U_j^E - \alpha\|_{H_\delta^m(R_E^e \cap R_T)} + \|U_j^T - \alpha\|_{H_\delta^m(R_E^e \cap R_T)}),$$

with C independent of the length of the band. On the compact overlaps with R_K and R_O the representatives are close to V^{ad} , respectively to the same stationary or critical configuration, and V^{ad} is irreducible on each overlap, which contains torus fibres; so the relative gauge transformations are close to the identity after a choice of central sign, and their logarithms are small in H^{m+1} .

Patching. First join U_j^T to the two representatives U_j^E across the two bands, by $\exp(\chi(\pm\theta - \Theta)\varphi_j)$ with a fixed cutoff χ of θ ; the overlap graph of R_T and the two R_E^e is a tree. The union of these regions meets R_K in a connected compact set, and R_K meets R_O in compact connected sets; patch in this order, each time with one choice of central sign and a cutoff of fixed slope. The result is an end-even gauge transformation g_j of $W_{\mathbb{L}_j}$, which on R_K differs by a small gauge transformation from the gauge transformations of the convergence. Since all cutoffs have bounded derivatives and all logarithms are small,

$$\|x_j\|_{H_\delta^m(W_{\mathbb{L}_j})} \leq \eta(\Theta, S, T_0) + o_j(1), \quad x_j = g_j^* U_j - P_{\mathbb{L}_j}^{(p_j)},$$

where η tends to zero when Θ , then S , then T_0 tend to infinity. A diagonal choice of Θ, S, T_0 along the sequence gives $\|x_j\|_{H_\delta^m} + |p_j| \rightarrow 0$.

Slice and chart. Gauge transformations preserve the flat-fibre, matching and Lagrangian conditions, so $g_j^*U_j$ satisfies them. By Lemma 10.27(a) there are ξ_j with $\|\xi_j\|_{H_\delta^{m+1}} \leq C\|x_j\|_{H_\delta^m}$ and w_j with $\exp(\xi_j)^*g_j^*U_j = \Psi_{L_j}(w_j, p_j)$, $d_{A_P}^*\zeta_j = 0$ and $\|z_j\| \leq C(\|x_j\| + |p_j|) \rightarrow 0$, where $z_j = (w_j, p_j)$; since U_j is a solution, $\mathcal{F}_{L_j}(z_j) = 0$. For large j , $\|z_j\| < r$, and Proposition 10.28 gives $z_j = z_{L_j}(V)$. $\square$

The weighted norm of the difference from the reference is estimated on each region in a frame in which the stationary limit of that region is exact: on the mixed ends, including the corners of the torus region, in the normalized temporal frame, and on the torus strip, which has bounded θ -extent, in the torus slice. Since the weight multiplies only decaying fields, and the torus region is controlled by the spectral gap μ_T of the fibre rather than by a Poincaré inequality on the base, the constants do not depend on L .

Theorem 10.30 (torus completion). *Choose the secondary data of the completed domains by Lemma 10.21 and use them on $W_{\pm, R}$ and W_λ^J . There are R_0 and λ_1 such that for $R \geq R_0$ and $\lambda \geq \lambda_1$:*

- (i) *there are no solutions of negative index on $W_{\pm, R}$ and W_λ^J ;*
- (ii) *gluing defines bijections*

$$\mathcal{G}_R: \mathcal{M}^0(W_\infty) \rightarrow \mathcal{M}^0(W_R), \quad \mathcal{G}_\lambda: \mathcal{M}^0(W_{-, \infty}) \times \mathcal{M}^0(W_{+, \infty}) \rightarrow \mathcal{M}^0(W_\lambda^J)$$

onto sets of regular points, preserving labels, relative classes and markings.

In particular, the index-zero counts satisfy $N_\pm^{(R)} = N_{\pm, \infty}$ and $\mathcal{K}N_\lambda^J = N_{-, \infty} \otimes N_{+, \infty}$ as matrices.

Proof. Negative indices. By Theorem 10.12 and the energy identity there are finitely many negative indices at which solutions can occur for given labels. If solutions U_j of a fixed negative index d existed on W_{L_j} with $L_j \rightarrow \infty$, Proposition 10.13 would give $d = \sum \text{ind } V_\beta + \sum \text{ind } \Gamma_\ell + 4n$ with $n \geq 0$, completed pieces of nonnegative index (Lemma 10.21) and chains of index at least one, which is impossible.

The map. For $L \geq \max\{L_0, L_1\}$, Proposition 10.28 defines $\mathcal{G}_L(V)$ for each representative V . It is well defined on classes and equivariant under the lifting groups. If $V' = h^*V$ for an end-even gauge transformation h of W_∞ , trivial at the original ends in the limit, then V'^{ad} and V^{ad} differ by an end-even gauge transformation h' which on $\{r \geq S_V\}$ preserves the torus slice, hence is constant there with value in the stabilizer of α ; it extends over the truncated tail and the collar as the same constant, which fixes α , so $P_L(V') = h'^*P_L(V)$, and the chart, the equation and the slice condition are equivariant, so $\mathcal{G}_L(V') = h'^*\mathcal{G}_L(V)$. The labels are those of V , and the relative class and bundle marking are those of $P_L(V)$, which equal those of V by Lemma 10.23(iii) and the construction. In the joined case the lifting group is $\Phi_J \cong \Phi_- \oplus \Phi_+$, the even stabilizer of α on the bridge is trivial, and the joined reference of a pair is determined up to Φ_J by the classes of $V_\pm$, so no gluing parameter occurs.

Injectivity. $\mathcal{G}_L(V)$ converges to V on compact subsets as $L \rightarrow \infty$, since $\|z_L(V)\| \rightarrow 0$. If $\mathcal{G}_{L_j}(V)$ and $\mathcal{G}_{L_j}(V')$ were gauge equivalent for $L_j \rightarrow \infty$, the gauge transformations would be bounded on compact sets by the equation for gauge transformations, and a limit would give $[V] = [V']$. The completed sets are finite, so injectivity holds for L large.

Surjectivity. If for $L_j \rightarrow \infty$ there were index-zero solutions U_j on W_{L_j} outside the image, Proposition 10.13 would give a subsequence converging without loss to some V (a pair) of

index zero, by the index balance and the regularity of Lemma 10.21, and Proposition 10.29 would give $U_j = \mathcal{G}_{L_j}(V)$ for large j , a contradiction.

Counts. The coefficient of $N_{\pm}^{(R)}$ at labels (a, x) is the number of index-zero solutions with these labels, which by (ii) equals that of $N_{\pm, \infty}$. In the joined case the strip output is at $A_0 \cap A_1 = (\tilde{P}_0 \cap \tilde{P}_1) \times (R_0 \cap R_1)$ and the labels of a joined solution are the pairs of labels of its two components, so the coefficient of N_{λ}^J at $(a_- \otimes a_+, (x_-, x_+))$ is the product of the coefficients of $N_{-, \infty}$ and $N_{+, \infty}$; composing with the identification $\mathcal{K}$ of generators gives the last identity. $\square$

10.11. Proof of Theorem 10.2. *Order of choices.* (1) Fix the separate mixed maps $N_{\pm}$ on $X_{D, \pm}$. (2) Fix λ_0 , the data of N_Z at width λ_0 and the data of $\bar{E}$ (Proposition 10.20), and regular joined data $\mathfrak{d}_J$ at width λ_0 . (3) Choose regular data on the completed domains (Lemma 10.21). (4) Choose $R \geq R_0$ and $\lambda \geq \lambda_1$ (Theorem 10.30). (5) At these finite lengths, add an invariant secondary perturbation supported in the cores, regular in index one and small enough that the finite regular index-zero sets and the absence of negative indices persist; by compactness of the index-zero sets and the implicit function theorem the index-zero counts are unchanged.

By Proposition 10.20, which contains Lemma 10.19; by Proposition 10.16 with $a = \lambda_0$ and $b = \lambda$, applied to the data $\mathfrak{d}_J$ changed by a small generic invariant secondary perturbation that is regular at both widths (Theorem 5.4), followed, at width λ , by Theorem 5.2 for the compactly supported change from these data to those of steps (3) and (5), both regular; and by Theorem 10.30,

$$[\mathcal{K}N_Z][\bar{E}] = [\mathcal{K}N_{\lambda_0}^J] = [\mathcal{K}N_{\lambda}^J] = [N_{-, \infty}] \otimes [N_{+, \infty}] = [N_-^{(R)}] \otimes [N_+^{(R)}] = [N_-] \otimes [N_+],$$

the last step by Proposition 10.15 and Theorem 5.2 for the compactly supported change of secondary data between $N_{\pm}$ and $N_{\pm}^{(R)}$. All structures on the section are split, so no change of the section structure occurs. Since $[\bar{E}] = [E]^{-1}$, the theorem follows. $\square$

11. ONE- AND THREE-HANDLES

We keep the notation of Section 10.1.

11.1. The relative invariant of the one-handle cobordism. Let $W_k: M \rightarrow M^+$ be the trace of k one-handles attached away from the tube, and let $B_k = (W_k \setminus \text{int } \nu(\gamma)) \natural (T^2 \times D^2)$, where $\nu(\gamma)$ is the tube; the two end balls of the tube turn the boundary of the first factor into $(-M) \# M^+$, and its boundary sum with the torus filling gives Z .

Lemma 11.1. (i) $\mathcal{W} \cup_{Z_0} F_0 \cup_{Z_1} F_1 \cong B_k$ by an orientation-preserving diffeomorphism covered by a bundle isomorphism, where $\mathcal{W}$ is the gauge triangle assembled from D_0, C, D_1 over the two surface sheets $\Delta \times \Sigma_h, \Delta \times \Sigma_g$ and $F_i = V_i \times [0, 1]$ fill its auxiliary boundaries $Z_i = V_i \cup (-V_i)$. The diffeomorphism restricts on the boundary to the identification $Z = D_0 \cup (-D_1) \cong (-M) \# M^+ \# T^3$ of Section 10.1, carries the complement of the torus parts onto the complement of the tube $\nu(\gamma)$ in W_k , and the lifting group of the union is zero, as is that of B_k .

(ii) $B_k \cup_Z \mathcal{E} \cong W_k^{\#}$, with the incoming end $-M$ read as outgoing, and the lifting group of the union is ϕ_{W_k} .

Proof. (i) We use four-manifolds built from a planar base. Let $\mathbb{D}$ be a closed disk and $c_0, c_m, c_1 \subset \mathbb{D}$ three disjoint chords; they cut $\mathbb{D}$ into three outer half-disks G_0, G_m, G_1 (with

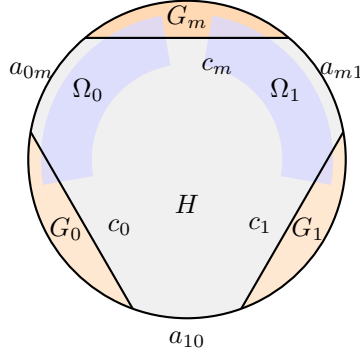


FIGURE 7. The planar base of the proof of Lemma 11.1(i). The fibre is Σ_h over the hexagon H , Σ_g over G_m and the torus T over G_0 and G_1 ; over the chords c_0, c_m, c_1 the pieces U_0, C, U_1 are inserted. Over the bands Ω_0 and Ω_1 the four-manifold is $F_0 = [0, 1] \times V_0$ and $F_1 = [0, 1] \times V_1$; the boundary over $\partial\mathbb{D}$ is Z .

$\partial G_e \setminus \partial\mathbb{D} = c_e$) and a hexagon H , whose boundary consists alternately of c_0 , an arc $a_{0m} \subset \partial\mathbb{D}$, c_m , an arc a_{m1} , c_1 and an arc a_{10} . Put

$$\mathfrak{X} = H \times \Sigma_h \cup G_m \times \Sigma_g \cup G_0 \times T \cup G_1 \times T \cup c_0 \times U_0 \cup c_m \times C \cup c_1 \times U_1,$$

where $c_e \times Y$ is glued to the two adjacent regions along $c_e \times \partial Y$, with $\partial U_i = \Sigma_h \sqcup T$ and $\partial C = \Sigma_h \sqcup \Sigma_g$, corners rounded in product collars, and the bundle given by the bundles of U_i, C and their boundary restrictions. The boundary $\partial\mathfrak{X}$ lies over $\partial\mathbb{D}$; reading it along $\partial\mathbb{D}$ from G_0 through a_{0m}, G_m, a_{m1}, G_1 and a_{10} gives $(T \times I) \cup U_0 \cup_{\Sigma_h} C \cup_{\Sigma_g} (-C) \cup_{\Sigma_h} (-U_1) \cup_T (T \times I) \cup U_1 \cup_{\Sigma_h} (-U_0)$, that is $V_0 \cup (-V_1)$ and $(-U_0) \cup U_1$ glued along two tori, which is $Z = D_0 \cup (-D_1)$.

Step 1: $\mathscr{W} \cup F_0 \cup F_1 \cong \mathfrak{X}$. Let $\Omega_0 \subset \text{int } \mathbb{D}$ be a closed rectangle $[0, 1]_y \times [0, 1]_x$ near a_{0m} that crosses, in the x -direction and in this order, G_0, c_0, H, c_m and G_m , with $c_0 \cap \Omega_0 = [0, 1] \times \{\frac{1}{4}\}$ and $c_m \cap \Omega_0 = [0, 1] \times \{\frac{3}{4}\}$; let Ω_1 be the analogous rectangle near a_{m1} crossing G_m, c_m, H, c_1, G_1 . Over Ω_0 the fibre across x is T -collar, U_0 , Σ_h -collar, C , Σ_g -collar, so $\mathfrak{X}|_{\Omega_0} \cong [0, 1] \times (U_0 \cup_{\Sigma_h} C) = [0, 1] \times V_0 = F_0$, and likewise $\mathfrak{X}|_{\Omega_1} \cong F_1$; the boundary $\partial\Omega_i$ carries Z_i . On $\mathbb{P} = \mathbb{D} \setminus \text{int}(\Omega_0 \cup \Omega_1)$, a disk with two holes, the chords are cut into the following arcs: two arcs of c_0 from $\partial\mathbb{D}$ to $\partial\Omega_0$, a short one near a_{0m} and a long one; three arcs of c_m , from $\partial\mathbb{D}$ to $\partial\Omega_0$, from $\partial\Omega_0$ to $\partial\Omega_1$, and from $\partial\Omega_1$ to $\partial\mathbb{D}$; and two arcs of c_1 from $\partial\Omega_1$ to $\partial\mathbb{D}$. Let $\mathcal{N}_0 \subset \mathbb{P}$ be a band from $\partial\mathbb{D}$ to $\partial\Omega_0$ containing the long and the short arc of c_0 , the region $G_0 \setminus \Omega_0$ between them, the thin region of H between the short arc of c_0 and the first arc of c_m , and that arc. Across $\mathcal{N}_0$ the fibre is a Σ_h -collar, U_0 traversed from Σ_h to T , a T -collar, U_0 traversed from T to Σ_h , a Σ_h -collar, C and a Σ_g -collar, that is $(-U_0) \cup_T U_0 \cup_{\Sigma_h} C = (-U_0) \cup_T V_0 = D_0$ with collars of its boundary; so $\mathfrak{X}|_{\mathcal{N}_0} \cong e_0 \times D_0$. Likewise a band $\mathcal{N}_1$ from $\partial\Omega_1$ to $\partial\mathbb{D}$ gives $e_1 \times D_1$, and a neighbourhood $\mathcal{N}_m$ of the middle arc of c_m gives $e_m \times C$. The complement of $\mathcal{N}_0 \cup \mathcal{N}_m \cup \mathcal{N}_1$ in $\mathbb{P}$ consists of two hexagons, the rest of H with fibre Σ_h and the rest of G_m with fibre Σ_g , each with its three edges on $\mathcal{N}_0, \mathcal{N}_m, \mathcal{N}_1$ and its three vertex arcs on $\partial\mathbb{D}, \partial\Omega_0, \partial\Omega_1$. This is the assembly of $\mathscr{W}$ over the two sheets, with the output vertex at $\partial\mathbb{D}$ and the inputs at $\partial\Omega_0, \partial\Omega_1$. The identifications are the identity on the pieces U_i, C and on the surfaces, so the bundles correspond.

Step 2: the Heegaard pieces. Write $M = H_0 \cup_\Sigma H_1$ for the Heegaard splitting of genus $h-1$ of the presentation, and $U_i = H_i \cup_D ([0, 1] \times T)$, where the sum disk $D \subset \partial H_i$ is identified with a disk $D \subset \{0\} \times T$; by Definition 3.3(ii), D is the equatorial disk of the ball $b(B^3)$, and we write o' for its centre. So $\Sigma_h = (\Sigma \setminus D) \cup_c T^\circ$, $T^\circ = T \setminus D$, $c = \partial D$. The compression body C is $C'' \cup_{c \times [0, 1]} (T^\circ \times [0, 1])$, where C'' is obtained from $(\Sigma \setminus D) \times [0, 1]$ by attaching k three-dimensional one-handles to $(\Sigma \setminus D) \times \{1\}$ away from ∂D , and $\Sigma_g = (\Sigma^+ \setminus D) \cup_c T^\circ$ with Σ^+ the surgered surface. Accordingly $\mathfrak{X} = \mathfrak{X}_M \cup \mathfrak{X}_T$, where $\mathfrak{X}_T$ collects the torus parts: T° over H and G_m , $T^\circ \times [0, 1]$ over c_m , the factor $[0, 1] \times T$ of U_i over c_i , and T over G_i ; and $\mathfrak{X}_M$ collects the rest: $\Sigma \setminus D$ over H , $\Sigma^+ \setminus D$ over G_m , C'' over c_m and H_i over c_i . They meet along $H' \times c \cup (c_0 \sqcup c_1) \times D$, where $H' = H \cup G_m \cup (\text{band of } c_m)$.

Step 3: $\mathfrak{X}_T$. Thicken each c_i into a band $b_i = c_i \times [0, 1]$, the factor $[0, 1]$ being the interval factor of $[0, 1] \times T \subset U_i$, and similarly c_m . Then $\mathbb{D}' = H' \cup b_0 \cup G_0 \cup b_1 \cup G_1$ is a disk, H' is a band across it separating $Q_i = b_i \cup G_i$, and the fibres are T° over H' and T over $Q_0 \cup Q_1$. Hence

$$\mathfrak{X}_T = (\mathbb{D}' \times T) \setminus \text{int}(H' \times D),$$

and $H' \times D$ is a tubular neighbourhood of the arc $\beta' = \beta_0 \times \{o'\}$, where β_0 crosses the band H' from a_{10} to the opposite boundary arc and $o' \in D$ is the centre; the interface with $\mathfrak{X}_M$ is its lateral boundary $H' \times c \cup (c_0 \sqcup c_1) \times D \cong S^2 \times [0, 1]$. The arc β' is a chord of the disk $\mathbb{D}' \times \{o'\}$, so it cobounds a half-disk δ with an arc of $\partial \mathbb{D}' \times \{o'\}$. A regular neighbourhood B of δ is a four-ball meeting $\partial(\mathbb{D}' \times T)$ in a three-ball, and we may take $H' \times D \subset B$ with $\nu(\beta') \cap \partial B = \nu(\beta') \cap \partial(\mathbb{D}' \times T)$. Then $(\mathbb{D}' \times T) \setminus \text{int } B \cong D^2 \times T^2$, and $B \setminus \text{int } \nu(\beta')$ is the complement of a trivial arc in a four-ball: in the model $B = D^3 \times [0, 1]$, $\beta' = \{0\} \times [0, 1]$, $\nu(\beta') = D_{1/2}^3 \times [0, 1]$, it is $S^2 \times [\frac{1}{2}, 1] \times [0, 1]$, with the lateral boundary $S^2 \times \{\frac{1}{2}\} \times [0, 1]$. Hence

$$\mathfrak{X}_T \cong (T^2 \times D^2) \natural (S^2 \times [\tfrac{1}{2}, 1] \times [0, 1]),$$

the boundary sum being taken along the three-ball $\partial B \cap \text{int}(\mathbb{D}' \times T)$, and the interface with $\mathfrak{X}_M$ is $S^2 \times \{\frac{1}{2}\} \times [0, 1]$.

Step 4: $\mathfrak{X}_M$. Refill the disk: $\mathfrak{X}'_M = \mathfrak{X}_M \cup (H' \times D) \cup (c_0 \times \mathcal{V}_0 \cup c_1 \times \mathcal{V}_1)$, where $\mathcal{V}_i \cong D \times [0, 1]$ is a collar of D in H_i removed from $\mathfrak{X}_M$ beforehand (which does not change its diffeomorphism type). Then $\mathfrak{X}'_M = H \times \Sigma \cup c_0 \times H_0 \cup c_1 \times H_1 \cup c_m \times C' \cup G_m \times \Sigma^+$, with $C' = C'' \cup D \times [0, 1]$ the compression body from Σ to Σ^+ , and $\nu = H' \times D \cup c_0 \times \mathcal{V}_0 \cup c_1 \times \mathcal{V}_1$ is a tubular neighbourhood of β' . Identify H with $[-1, 1]_u \times [0, 1]_t$ so that $c_0 = \{u = -1\}$, $c_1 = \{u = 1\}$, $a_{10} = \{t = 0\}$ and $c_m = \{t = 1\} \times [u_1, u_2]$ with $0 < u_1 < u_2 < 1$. Then $H \times \Sigma \cup c_0 \times H_0 \cup c_1 \times H_1 = [0, 1]_t \times (H_0 \cup [-1, 1]_u \times \Sigma \cup H_1) = [0, 1] \times M$. The part over c_m and G_m is, after absorbing the collar $G_m \times \Sigma^+$ of the upper boundary, $[u_1, u_2] \times C'$ attached along $[u_1, u_2] \times \Sigma = \{1\} \times [u_1, u_2] \times \Sigma \subset \{1\} \times M$. As $C' = \Sigma \times [0, 1] \cup h_1 \cup \dots \cup h_k$, this is a collar on $\{1\} \times M$ together with the pieces $[u_1, u_2] \times h_j = [u_1, u_2] \times D^1 \times D^2 \cong D^1 \times D^3$ attached along $\partial D^1 \times (D^2 \times [u_1, u_2])$, two three-balls in the top boundary straddling Σ : these are four-dimensional one-handles. Hence $\mathfrak{X}'_M \cong W_k$, with incoming end $-M$ over a_{10} and outgoing end M^+ over the opposite arc. Choosing $\beta_0 = \{0\} \times [0, 1]$, which ends on a_{0m} , the arc β' becomes $\gamma = [0, 1] \times \{q\}$ with $q = (0, o') \in [-1, 1] \times \Sigma \subset M$, the centre of $b(B^3)$, which is the product path, and ν becomes the tube $\nu(\gamma)$ of the decorated cobordism; the one-handles are attached away from it. So $\mathfrak{X}_M \cong W_k \setminus \text{int } \nu(\gamma)$, and the interface with $\mathfrak{X}_T$ is the lateral boundary $S^2 \times [0, 1]$ of the tube.

Step 5: assembly. By Steps 3 and 4, $\mathfrak{X}$ is obtained from $W_k \setminus \text{int } \nu(\gamma)$ by attaching $S^2 \times [\frac{1}{2}, 1] \times [0, 1]$ along $S^2 \times \{\frac{1}{2}\} \times [0, 1]$, which is a collar and does not change the diffeomorphism type, boundary summed with $T^2 \times D^2$ at a ball in the lateral part of the boundary. Since

$\partial(W_k \setminus \text{int } \nu(\gamma)) = (-M) \# M^+$ is connected, the boundary sum does not depend on the ball, and $\mathfrak{X} \cong B_k$. With Step 1 this proves the diffeomorphism. The bundle of $\mathfrak{X}$ is trivial on $\mathfrak{X}_M$ and the pullback of the odd bundle P_T under $\mathfrak{X}_T \subset \mathbb{D}' \times T \rightarrow T$ on $\mathfrak{X}_T$, glued along the interface by the fixed trivialization of P_T over D ; the summand $S^2 \times [\frac{1}{2}, 1] \times [0, 1]$ lies over D , where P_T is trivialized. So the diffeomorphism is covered by an isomorphism with the bundle of B_k , trivial on $W_k \setminus \nu(\gamma)$ and pulled back from T^2 on $T^2 \times D^2$, boundary summed along the trivialization over the ball. On the boundary, the pieces over a_{10} and over the opposite arc are $-M \setminus B$ and $M^+ \setminus B$, and the torus part over $\partial \mathbb{D}'$ is T^3 minus the two balls at the ends of $\nu(\beta')$, which are the connected-sum balls, in the identification of Section 10.1; the cut tori T_a, T_b of Lemma 10.17 are the fibres over points of the arcs $\partial G_0 \cap \partial \mathbb{D}$ and $\partial G_1 \cap \partial \mathbb{D}$.

For the lifting groups: restriction $H^1(\mathcal{W} \cup F_0 \cup F_1; \mathbb{F}_2) \rightarrow H^1(\mathcal{W}) \oplus H^1(F_0) \oplus H^1(F_1)$ is injective, since $H^0(\mathcal{W}) \oplus H^0(F_0) \oplus H^0(F_1) \rightarrow H^0(Z_0) \oplus H^0(Z_1)$, $(a, b, c) \mapsto (a + b, a + c)$, is onto. The caps carry the determinant-one group, so a lifting class of the union vanishes on Z_0 and Z_1 . A lifting class of $\mathcal{W}$ vanishing on the surface sheets restricts to zero on C and, on each D_i , to a multiple of the dual of its neck torus: H^1 of C , of U_i and of V_i injects into H^1 of Σ_g , of Σ_h and of Σ_g respectively, and Mayer–Vietoris across the neck torus of $D_i = (-U_i) \cup_T V_i$ leaves only the dual of that torus. In addition, the incidence graph of $\mathcal{W}$ (two sheets, three edge pieces and six contact regions) has first Betti number two, and the two corresponding classes restrict to zero on every piece. These two classes restrict nontrivially and independently to Z_0 and Z_1 , each being the dual of the surface sheet in the vertex slice. A lifting class vanishing on the sheets is therefore determined by its restrictions to Z_0 and Z_1 , where the neck-torus duals and the incidence classes are nonzero, and so it vanishes. The lifting group of the union is therefore zero, which is the group of B_k . (ii) Model $\mathcal{E}$ as $T^2 \times P$ with the summand products inserted along two arcs. Attaching the disk base of $T^2 \times D^2$ in B_k caps the input circle c of the pair of pants and restores the product annulus, with the identity parametrizations at both exterior circles and no Dehn twist; across the cap the two sum collars join to the lateral collar of the original tube, with the prescribed normal frame. The complement of the tube and its bundle are unchanged. The cut Z is connected, so the lifting group of the union consists of the classes vanishing on B_k and restricting into $\Phi_{\mathcal{E}}$; compatibility on Z forces $s = 0$ and $s_- = s_+$, which is the torus lifting group ϕ_{W_k} . $\square$

Lemma 11.2. $[E](\Theta_{B_k}) = \text{ten}(I^\#(W_k))$ on homology over $\mathbb{F}_2$, where $\Theta_{B_k} = [\mu_{\mathcal{W}}](\theta_0, \theta_1)$ is the relative invariant of B_k , $\mu_{\mathcal{W}}$ is the gauge triangle map of $\mathcal{W}$ modulo the gauge group enlarged by Φ , $\theta_i \in C_G(Z_i)$ is the relative invariant of the filling F_i , defined as θ_G in Section 9, and $\text{ten}(F) = \sum_i a_i \otimes b_i$ for $F(v) = \sum_i \langle a_i, v \rangle b_i$, the pairing being the gauge Kronecker pairing between $-M$ and M .

Proof. The relative invariant. By Lemma 11.1(i) and [21, Proposition 5.4] with two caps, whose connecting images vanish because the auxiliary cuts are connected and meet the triangle, together with the projection argument of Proposition 9.5 for each input, $\Theta_{B_k} = [\mu_{\mathcal{W}}](\theta_0, \theta_1)$.

Gluing. For a long neck along Z with nondegenerate data, the index-zero moduli spaces of $B_k \cup_Z \mathcal{E}$ are, by Theorem 5.1, the fibre products over the critical points z_σ of Z (with their two even lifts σ) of those of B_k and of $\mathcal{E}$; the gluing parameter is a point since the determinant-one stabilizer $\{\pm 1\}$ extends over both pieces and the connecting image vanishes.

Counting. Let $e_{\sigma\tau}$ be the number of index-zero solutions on $\mathcal{E}$ from z_σ to the output lifts (a_0, b_τ) for fixed labels. Restriction of $\Phi_{\mathcal{E}}$ to the lifts of z and of the $-M$ output is an isomorphism, so the coefficient of $E([z])$ at $[a] \otimes [b]$ is $e_{00} + e_{01}$. The element of $\Phi_{\mathcal{E}}$ acting on the z and M^+ lifts gives a bijection between solutions from z_1 to b_τ and from z_0 to $b_{\tau+1}$, so

$e_{1\tau} = e_{0,\tau+1}$. If β_σ are the coefficients of Θ_{B_k} at z_σ , the glued count at $(a_0, \cdot)$, summed over the output lifts, is $\sum_\tau \sum_{z,\sigma} \beta_\sigma e_{\sigma\tau} = \sum_z (\beta_0 + \beta_1)(e_{00} + e_{01})$, the coefficient of $E(\Theta_{B_k})$. Each orbit of lifts is counted once, because the lifting group acts freely and the counts are taken in the quotient.

Reversal. By Lemma 11.1(ii) the union is $W_k^\#$ with the incoming end read as outgoing; the same moduli spaces, with the incoming lift fixed and the output lifts summed, compute the framed coefficient $\langle I^\#(W_k)[a], [b] \rangle$ [38, Section 7.2], the pairing between $C_G((-M)^\#)$ and $C_G(M^\#)$ being the reversal pairing [38, Section 7.4]. Independence of data holds on homology by functoriality under continuation. $\square$

11.2. Reflection. For an object (M, b) with equatorial disk $D = \{x_3 = 0\}$ in the ball, put $\bar{b} = b \circ r_B$ with $r_B(x_1, x_2, x_3) = (x_1, x_2, -x_3)$; this is an orientation-preserving parametrization of a ball in $-M$ that fixes D and exchanges the two half-balls. On the torus $T^3 = T^2 \times S_c^1$ use $\rho_T(x, y, c) = (x, y, -c)$, which reverses orientation, preserves the odd two-torus with its bundle, the class α_T and the flat metric. Transporting the torus attachment by r_B and ρ_T defines the attachment of the *reflected object* $(-M, \bar{b})$, with $(-M)^\# = -(M^\#)$ as marked bundles; reflecting twice is the identity. For a presentation $p = (C_0, C_1)$ put $\bar{p} = (C_1, C_0)$, with all spatial data transported; ordinary strip data are reflected by $u(s, t) \mapsto u(-s, 1 - t)$, $J \mapsto J(-s, 1 - t)$, $H \mapsto -H(-s, 1 - t)$, with the full one-form transformation.

Reflection gives pairings of complexes

$$C_G((-M)^\#, \bar{p}) = C_G(M^\#, p)^\vee, \quad CF(\bar{p}) = CF(p)^\vee,$$

under which the gauge and ordinary continuation maps are transposed: $\text{cont}_G(\bar{P}, \bar{Q}) = \text{cont}_G(Q, P)^\vee$ and $\text{cont}_S(\bar{P}, \bar{Q}) = \text{cont}_S(Q, P)^\vee$. The mixed completion of a reflected presentation is *not* reflected: mixed solutions are not carried to mixed solutions. We choose independent regular invariant data (end perturbation and secondary perturbation) defining $N_{\bar{p}}$; by Theorem 3.6(ii) its homology class does not depend on the choice, and in particular $[N_{\bar{p}}] = [N_P]$.

11.3. Continuations and composition. The Lagrangians $\tilde{P}_i$ and R_i are simply connected with $\pi_2 = 0$. The correspondence L_C is an $\text{SU}(2)^k$ -bundle over M_h , simply connected with $\pi_2(L_C) = \mathbb{Z}$; spheres in L_C have zero area and zero Chern number, so every pair and triangle used below is monotone with minimal Maslov number 4.

Continuation between primary data. For two admissible choices of primary data with the same splitting and metric data, the Lagrangians differ by Hamiltonian isotopies. The continuation map $\text{cont}_S(P, P')$ is defined by pulling back by the Hamiltonian flow: a strip with moving boundary on $\phi_t(L)$ is the same as a strip with fixed boundary on L for the transformed almost complex structure and the Hamiltonian one-form $H dt$, whose restriction to the boundary vanishes [31, Section 14.4]. In this form the continuation is Wehrheim and Woodward's data-change map on fixed Lagrangians, and it is a chain map, equals the identity for constant data, is compatible with composition and is independent of the path up to chain homotopy [44, Theorems 3.9, 3.13, 4.1, 4.2 and Example 2.6]. Both boundary Lagrangians may move. In small convex neighbourhoods of exact-graph primitives the continuation depends only on the endpoints.

The auxiliary pairs. The pair (A_0, L_C) in X unfolds to the pair $(R_0, \tilde{P}_0 \circ L_C)$ in M_g , and (L_C, A_1) to $(\tilde{P}_1 \circ L_C, R_1)$; their intersections are the critical sets of the height functions of Section 10.1. On $CF_X(A_0, L_C)$ and $CF_X(L_C, A_1)$ define the relative grading m by $m(a) - m(x) = \text{ind}(u) - 8\kappa(u)$ for any strip class u from a to x ; it is independent of u by monotonicity.

For t small it coincides with the Morse index of the height function (for $i = 0$) or with the Morse index of its negative (for $i = 1$), and there is a unique generator q_i^* of maximal m .

Definition 11.3. The *compression cup* of a k -fold one-handle, at height Lagrangians, is

$$U_C = \mathcal{BK}(\mu_X(q_0^*, q_1^*)): HF(\tilde{P}_0, \tilde{P}_1) \rightarrow HF(R_0, R_1),$$

where $\mu_X: CF_X(A_0, L_C) \otimes CF_X(L_C, A_1) \rightarrow CF_X(A_0, A_1)$ is the triangle map in X , $\mathcal{K}$ is the folding isomorphism and $\mathcal{B}$ turns a cycle $\sum x_i^\vee \otimes y_i$ into the map $v \mapsto \sum x_i^\vee(v)y_i$. At other admissible presentations it is conjugated by continuation maps. The *compression cap* $V_C: HF(R_0, R_1) \rightarrow HF(\tilde{P}_0, \tilde{P}_1)$ of a three-handle is the transpose of the cup of the reflected one-handle, $V_C = (U_{\tilde{C}})^\vee$ under the pairings of Section 11.2.

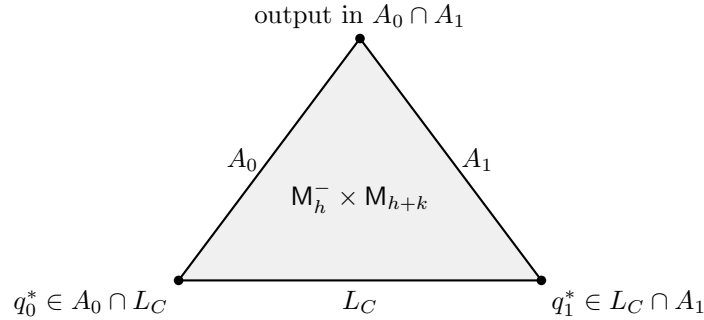


FIGURE 8. The folded triangle defining the compression cup (Definition 11.3). The inputs are the top generators q_0^* and q_1^* of the auxiliary pairs, and the output lies in $CF_X(A_0, A_1)$, which the folding isomorphism $\mathcal{K}$ identifies with $CF(\tilde{P}_0, \tilde{P}_1)^\vee \otimes CF(R_0, R_1)$, that is, with maps from $CF(\tilde{P}_0, \tilde{P}_1)$ to $CF(R_0, R_1)$. Unfolded, the triangle is a quilted surface whose seam carries the correspondence L_C (Remark 11.5).

The cup is a count of holomorphic triangles, parallel to the two-handle map $\mu_S(\cdot \otimes \theta_S)$; its definition involves no gauge theory and no geometric composition.

Lemma 11.4. For $k = 0$, $L_C = \Delta \subset M_h^- \times M_h$ and U_C equals the continuation map $\text{cont}_S(\tilde{P}, R)$ on homology.

Proof. Let S be the triangle with boundary arcs labelled A_0, Δ, A_1 and D the disk obtained by doubling S along its Δ -arc. For data (J_D, K_D) on D with K_D vanishing on the doubling arc, put $J_X(z) = (-J_D(\bar{z})) \times J_D(z)$ on S . The map $v \mapsto (v \circ \bar{\cdot}, v)|_S$ is a bijection from solutions on D to solutions on S with boundary on (A_0, Δ, A_1) : a solution on S glues to a Lipschitz map on D that solves the equation weakly across the arc, hence is smooth; linearized operators correspond. The two ends of S adjacent to the Δ -arc double to strip ends of D labelled $(\tilde{P}_i, R_i)$, and the output end doubles to one incoming and one outgoing end, which is the folding $\mathcal{K}$ followed by $\mathcal{B}$. So $U_C(x)$ is the count of disks with boundary arcs labelled $R_1, \tilde{P}_1, \tilde{P}_0, R_0$ with x at the incoming end and q_0^*, q_1^* at the two auxiliary ends. The top generator q_i^* of $CF(\tilde{P}_i, R_i)$ is the image of the unit cap, defined as follows. In a Weinstein neighbourhood of $\tilde{P}_i$ let H_i be a function equal to t times the height function pulled back to the neighbourhood, so that its Hamiltonian flow ϕ^s carries $\tilde{P}_i$ to R_i at $s = 1$ up to a Hamiltonian isotopy supported away from the intersection points, and the top generator q_i^* is

a critical point of H_i lying on $\tilde{P}_i \cap \phi^s(\tilde{P}_i)$ for every s . As in Section 11.3, pull back by ϕ^s : the cap becomes a disk with one outgoing end, the fixed boundary label $\tilde{P}_i$, and a Hamiltonian one-form βH_i whose restriction to the boundary vanishes, which is the surface $S^\cup$ of [44] with perturbation data. A cap with output x has index $m(q_i^*) - m(x) + 8\kappa$: by additivity of the index, gluing the constant cap at q_i^* to a strip from q_i^* to x gives the grading difference of Section 11.3, and the constant cap has index zero. By Lemma 9.1(ii)–(iii), in the form of Lemma 11.6, counted caps have $\kappa \geq 0$, so an index-zero cap has $\kappa = 0$ and output q_i^* ; it has energy zero and is constant. The constant map at q_i^* is a regular solution, since $dH_i(q_i^*) = 0$ and q_i^* is a nondegenerate critical point. Gluing the two unit caps into the auxiliary ends [44, Theorem 3.13] produces a strip with moving boundary from $(\tilde{P}_0, \tilde{P}_1)$ to (R_0, R_1) , whose map is the continuation [44, Theorem 4.1, Example 2.6]. $\square$

Remark 11.5. By the unit-insertion formula of Wehrheim and Woodward [44, Corollary 5.5], U_C agrees with the morphism map of the quilted correspondence L_C [45] followed by the geometric composition isomorphisms $\tilde{P}_i \circ L_C$ and continuation to R_i . That formula rests on the strip-shrinking theorem [44, Theorem 5.1], whose proof there is given in outline. The composition isomorphism itself is also established by Lekili and Lipyanskiy without strip shrinking [25, Theorem 2 and Corrigendum, Remark 1(1)], whose hypotheses hold here: the targets are monotone with the same constant, the Lagrangians $\tilde{P}_i$ and L_C are compact, connected, orientable and simply connected with minimal Maslov number 4, and the compositions are embedded. Neither statement is used in this paper.

Lemma 11.6. *The statements of Lemma 9.1 hold for the fillings $F_i = V_i \times [0, 1]$ of $Z_i = V_i \cup (-V_i) = (\#^{g-1} S^1 \times S^2) \# T^3$ and the pairs (A_0, L_C) and (L_C, A_1) , with $r = g - 1$, the height functions defining R_i , and the top generators q_i^* .*

Proof. The proof of Lemma 9.1 uses only that the filling retracts onto a wedge of r circles and a torus, so that $H_c^1 = H^2 = 0$ with twisted coefficients, and that the critical manifold is $SU(2)^r$ with the height function; both hold for F_i . The mixed maps of the pairs (A_0, L_C) and (L_C, A_1) are those of Remark 3.7. $\square$

11.4. The folded triangle.

Theorem 11.7. *Let N_0, N_1 be the mixed maps of the pairs (A_0, L_C) and (L_C, A_1) (after unfolding, $(R_0, \tilde{P}_0 \circ L_C)$ and $(\tilde{P}_1 \circ L_C, R_1)$), with gauge inputs $Z_0 = D_0 \cup (-C)$ and $Z_1 = C \cup (-D_1)$, both diffeomorphic to $V_i \cup (-V_i) = (\#^{g-1} S^1 \times S^2) \# T^3$, and N_Z the mixed map of (A_0, A_1) with the data at width λ_0 of Theorem 10.2. Then*

$$N_Z \circ \mu_{\mathcal{W}} \simeq \mu_X \circ (N_0 \otimes N_1) \quad \text{over } \mathbb{F}_2,$$

where $\mu_{\mathcal{W}}$ is the gauge triangle of the pieces D_0, C, D_1 . Moreover $N_i \theta_i = q_i^*$.

Proof. The proof is that of Theorem 9.10 with the following inputs. *Collapse.* The middle compression body $C = C_{h,g}$ is an odd compression body (Section 6.1), with $L(C) = L_C \subset M_h^- \times M_g$ and both boundary surfaces matched. Hypotheses (X1)–(X3) hold because the middle primary perturbation vanishes and all other perturbations are supported in the interiors of the two exterior gauge pieces. The triangle piece in X is regular for a generic domain-dependent almost complex structure supported away from N , as in the proof of Theorem 9.10. For (X4), stretch the two ordinary necks cutting off the DFL disks of the pairs (A_0, L_C) and (L_C, A_1) from the triangle domain of (A_0, L_C, A_1) ; Theorem 5.1 gives regularity in index at most zero with count $\mu_X(N_0 \otimes N_1)$. The linear inputs of Section 7 hold in this instance with

the two-trace slice space $H^1(\Sigma_h; \text{ad}) \oplus H^1(\Sigma_g; \text{ad})$: the gap is uniform in the scale because no harmonic field on C has vanishing harmonic trace ($H^1(C, \partial C; \text{ad}) = 0$), and the reversed lower sheet enters the Green identity with the sign of Convention 3.1. The period lattice of capped configurations is the image of $\mathbb{Z}^2/\mathbb{Z}(1,1)$, of index step four, as in Section 3.6. *Outgoing cut.* Stretch along the closed cut $Z \cong (-M) \# M^+ \# T^3$, which is connected. Its critical points are nondegenerate and irreducible, the connecting H^0 term vanishes, and the framed groups glue by the fibre product; Theorem 5.1 gives $N_Z \circ \mu_{\mathcal{W}}$. *Family.* The path of data has the five stages of Lemma 9.12: (A) raise the scale of $C = C_{h,g}$ from ε to one; (B) interpolate to a metric that is a product $a^2 du^2 + g_Z$ on a collar of the cut Z , where the cut passes through the former collapsing region of C , the collar coordinate there is the edge coordinate of the compression piece, and g_Z restricts to the metrics of D_0 , C and D_1 ; (C) lengthen the collar; (D) switch on the end perturbation of Z on the collar; (E) interpolate the remaining data to the data with the collar of the cut inserted. Each stage lies in a convex set of data satisfying the hypotheses of Theorem 5.2, the parameter derivative is compactly supported away from the ends, and invariance holds because the classes of the lifting group vanish on $\Sigma_h \sqcup \Sigma_g$. Theorem 5.2 applies. *Classes.* The pair $(R_i, \tilde{P}_i \circ L_C)$ is the auxiliary pair of a height Lagrangian with unique top generator q_i^* , and Lemmas 11.6 and 9.2 give $N_i \theta_i = q_i^*$. $\square$

Proposition 11.8 (cup formula). *Put $N_P^\dagger = N_-^\vee: CF(\tilde{P}_0, \tilde{P}_1) \rightarrow C_G(M^\#)$, the transpose of the mixed map $N_- = N_{\tilde{P}}$ of the reflected presentation, so that $N_P^\dagger = N_{\tilde{P}}^\vee$, defined by $\langle a, N_P^\dagger x \rangle_G = (N_- a)(x)$. Then on homology*

$$U_C = N_+ \circ I^\#(W_k) \circ N_P^\dagger.$$

For $k = 0$ this reads $\text{cont}_S(\tilde{P}, R) = N_R \circ \text{cont}_G(\tilde{P}, R) \circ N_P^\dagger$, with cont_G the gauge continuation.

Here and below the mixed map and its transpose are indexed by the primary data when these vary: for a presentation p with primary data P we write $N_P = N_p$ and $N_P^\dagger = N_p^\dagger$.

Proof. Insert $\theta_0 \otimes \theta_1$ into Theorem 11.7. By Definition 11.3, $N_i \theta_i = q_i^*$ and Theorem 10.2,

$$U_C = \mathcal{BK} \mu_X(q_0^*, q_1^*) = \mathcal{BK} N_Z \mu_{\mathcal{W}}(\theta_0, \theta_1) = \mathcal{B}(N_- \otimes N_+) \mathbf{E}(\Theta_{B_k}),$$

and $\mathbf{E}(\Theta_{B_k}) = \text{ten}(I^\#(W_k)) = \sum a_i \otimes b_i$ by Lemma 11.2. Hence

$$U_C(x) = \sum_i (N_- a_i)(x) N_+ b_i = \sum_i \langle a_i, N_P^\dagger x \rangle N_+ b_i = N_+ I^\#(W_k) N_P^\dagger x. \quad \square$$

The factor $N_P^\dagger$ is not yet known to be N_P^{-1} ; that is proved below, after naturality, in this order.

11.5. Order of choices. In each application of Proposition 11.8: first the upper height t and the upper Lagrangians R_i ; then the lower Lagrangians $\tilde{P}_i$ in a neighbourhood small relative to t , subject to the generic incidence conditions of Section 10.1 (expected dimension $-(3h + 3g - 6) < 0$); then the data at width λ_0 of Theorem 10.2. A finite change of torus collar length returns the torus factors to the fixed data by Proposition 10.15, and the secondary data are then replaced by the prescribed ones by Theorem 3.6(ii). Hence the maps N appearing in one application are the vertex maps on homology. Squares proved with the data of one application are moved to other data at the vertices by Proposition 11.9, whose own quantifiers are ordered in the same way: height first, then the neighbourhood $\mathcal{U}(t)$, then the lower Lagrangians, and last a generic height Lagrangian.

11.6. The one- and three-handle squares. A *height Lagrangian at height t* for $k = 0$ is an upper Lagrangian R of Section 10.1 for the product compression body: the primary data of U perturbed by t times the height function, with its unique maximal generator q^* . A further perturbation of R supported away from the critical set, of C^1 -size at most $\delta(t)$ for a function δ depending only on the splitting, the metric data and the height function, keeps it a height Lagrangian at height t ; the generic perturbations used below can be taken that small.

Proposition 11.9 (small-primary naturality). *Fix a splitting, metric data and a finite family of primary perturbations containing the height function. For every sufficiently small height $t > 0$ there is a neighbourhood $\mathcal{U}(t)$ of zero in the family such that for any two regular data P, Q in $\mathcal{U}(t)$, $N_Q \circ \text{cont}_G(P, Q) = \text{cont}_S(P, Q) \circ N_P$ on homology.*

Proof. The quantifiers are chosen in this order. First fix the height t . Let $\mathcal{U}(t)$ be the set of primary data whose Lagrangians, and those of their reflections, lie in the neighbourhoods $\mathcal{N}(t)$ fixed after Proposition 11.10; the condition is symmetric under reflection, which exchanges the two Lagrangians of the pair. Then, given regular $P, Q \in \mathcal{U}(t)$, choose one height Lagrangian A at height t for the reflected presentation which, by a small generic perturbation supported away from its critical set, satisfies the incidence conditions of Section 10.1 with respect to both $\bar{P}$ and $\bar{Q}$; this is possible because the universal restriction map for primary perturbations is a submersion. The product case of Proposition 11.8 now applies to $(\bar{P}, A)$ and $(\bar{Q}, A)$ at once and gives $\text{cont}_S(\bar{P}, A) = N_A \text{cont}_G(\bar{P}, A) N_{\bar{P}}^\vee$ and similarly for $\bar{Q}$; with $[N_{\bar{P}}] = [N_P]$ (Section 11.2) this is $B_{\bar{P}} = N_A G_{\bar{P}} N_{\bar{P}}^\vee$, $B_{\bar{Q}} = N_A G_{\bar{Q}} N_{\bar{Q}}^\vee$, where G and B are the gauge and symplectic continuations to A . Eliminating the invertible N_A gives $G_{\bar{Q}}^{-1} G_{\bar{P}} N_{\bar{P}}^\vee = N_{\bar{Q}}^\vee B_{\bar{Q}}^{-1} B_{\bar{P}}$. The composites are the transposes of $\text{cont}_G(Q, P)$ and $\text{cont}_S(Q, P)$ under reflection; transposing gives the claim. $\square$

Proposition 11.10 (mixed duality). *Let t and $\mathcal{U}(t)$ be as in Proposition 11.9. There is a neighbourhood $\mathcal{U}'(t) \subset \mathcal{U}(t)$ such that $N_{\bar{P}} = (N_P^{-1})^\vee$ on homology for every regular $P \in \mathcal{U}'(t)$.*

Proof. Choose a second height $t' \leq t$ so small that every height Lagrangian at height t' lies in $\mathcal{U}(t)$, and put $\mathcal{U}'(t) = \mathcal{U}(t) \cap \mathcal{U}(t')$. Given regular $P \in \mathcal{U}'(t)$, let R be a height Lagrangian at height t' for the presentation of P , generic with respect to P as in the proof of Proposition 11.9. The product case of Proposition 11.8 at height t' gives $\text{cont}_S(P, R) = N_R \text{cont}_G(P, R) N_{\bar{P}}^\vee$, and Proposition 11.9 at height t , applied to the primary data $P, R \in \mathcal{U}(t)$, gives $N_R \text{cont}_G(P, R) = \text{cont}_S(P, R) N_P$. Hence $\text{cont}_S(P, R) = \text{cont}_S(P, R) N_P N_{\bar{P}}^\vee$; cancelling the isomorphism $\text{cont}_S(P, R)$ gives $N_P N_{\bar{P}}^\vee = 1$. $\square$

The admissible neighbourhoods. The conditions that define $\mathcal{U}(t)$ and $\mathcal{U}'(t)$ in the proofs of Propositions 11.9 and 11.10 concern only the Lagrangians of the primary data and of their reflections, which must be lower Lagrangians small relative to t and t' ; the height thresholds depend only on the splitting, the metric data and the height functions of Section 10.1, which are fixed with the splitting; the incidence conditions are imposed on the height Lagrangians, not on the primary data. For each splitting, metric data and t fix an open C^1 -neighbourhood $\mathcal{N}(t)$ of the pair $(L(U_0), L(U_1))$, and the corresponding neighbourhood of the reflected pair, so small that the unfolding and grading statements of Section 11.3, Theorem 11.7 and Proposition 11.8 hold for every lower pair in them and every height Lagrangian at height t satisfying the incidence conditions; this is possible because the height Lagrangians at height t form a C^1 -bounded class. Take $\mathcal{U}(t)$ to be the set of primary data whose Lagrangians, and those of their reflections, lie in these neighbourhoods, and $\mathcal{U}'(t) = \mathcal{U}(t) \cap \mathcal{U}(t')$; these sets satisfy

the requirements in the proofs above. For each splitting and metric data fix once and for all a height t_* below these thresholds, and for each finite family F as in Definition 3.3 let $\mathcal{U}(F) = \mathcal{U}'(t_*) \cap \text{span } F$. Then $\mathcal{U}(F) = \mathcal{U}(F') \cap \text{span } F$ for $F \subset F'$, and in particular if $P \in \mathcal{U}(F)$ and G is a finite family, then $P + g \in \mathcal{U}(F \cup G)$ for all sufficiently small $g \in \text{span } G$. These propositions use Theorem 3.6 only for presentation data, so this defines $\mathcal{U}(F)$. Two admissible choices of primary data in $\mathcal{U}(F_1)$ and $\mathcal{U}(F_2)$ are joined through intermediate choices in $\mathcal{U}(F_1) \cap \mathcal{U}(F_1 \cup F_2)$ and $\mathcal{U}(F_2) \cap \mathcal{U}(F_1 \cup F_2)$, and the continuation along such a chain is the direct continuation, by the composition law of Section 11.3.

Theorem 11.11 (one-handle square). *Let $W_k: M \rightarrow M^+$ be a k -fold one-handle cobordism with the product bundle, p an admissible presentation adapted to its attaching balls, and $C \circ p$ the enlarged presentation with height Lagrangians, with primary data in the neighbourhoods of Proposition 11.10 at p and at $C \circ p$. Then $N_{C \circ p} \circ I^\#(W_k) = U_C \circ N_p$ on homology.*

Proof. By Propositions 11.8 and 11.10, $U_C = N_{C \circ p} I^\#(W_k) N_p^\dagger = N_{C \circ p} I^\#(W_k) N_p^{-1}$ with primary data in the neighbourhoods fixed above. At other small data in the same neighbourhoods, conjugate by the continuations and apply Proposition 11.9 at both ends. $\square$

Theorem 11.12 (three-handle square). *Let $W_3: M^+ \rightarrow M$ be a three-handle cobordism, read as the one-handle cobordism $\overline{W}_3: -M \rightarrow -M^+$ of the reflected presentation. Then $N_p \circ I^\#(W_3) = V_C \circ N_{C \circ p}$ on homology, with primary data in the neighbourhoods of Proposition 11.10.*

Proof. Reading the same oriented four-manifold from the opposite ends gives $I^\#(W_3) = I^\#(\overline{W}_3)^\vee$: the anti-self-duality equation on W_3 with ends read in reflected charts has the same index-zero moduli spaces with input and output exchanged. By Definition 11.3, $V_C = (U_{\bar{C}})^\vee$. Transposing Theorem 11.11 for the reflected presentation and applying Proposition 11.10 at both ends,

$$N_p I^\#(W_3) = (N_p^{-1})^\vee I^\#(\overline{W}_3)^\vee = (U_{\bar{C}})^\vee (N_{C \circ p}^{-1})^\vee = V_C N_{C \circ p}. \quad \square$$

Remark 11.13. A one-handle followed by a three-handle does not cancel: the composite is $(M \times [0, 1]) \# (S^1 \times S^3)$. Nor do two-handle data determine the one-handle map: if a and b are maps with $b \circ a = 0$ and $b \circ f = 1$, then $b \circ (f + a \circ g) = 1$ for every g . The one-handle square therefore needs its own degeneration.

12. PRESENTATION INDEPENDENCE AND FRAME CHANGES

12.1. The presentation graph. Let (M, b) be an object. Let $\mathcal{P}(M, b)$ be the graph whose vertices are the admissible presentations of Definition 3.3 and whose edges are:

- (P1) changes of the meridian disk systems in a fixed embedded splitting (Definition 3.3 records no disk systems, so these edges are loops at a vertex; they are listed to match the edges of [20]);
- (P2) ambient isotopies of the splitting fixed near $b(B^3)$;
- (P3) stabilization and destabilization away from $b(B^3)$;
- (P4) changes of the metric data, of the primary data and of the regularizing data (Definition 3.3(v)), including the almost complex structure on the strip end, of a fixed splitting.

Proposition 12.1. *$\mathcal{P}(M, b)$ is connected.*

Proof. Remove the interior of $b(B^3)$ and put an annular suture around the equator ∂D of its boundary sphere. The result is a balanced sutured manifold $M(b)$ whose positive

and negative regions are disks. Removing D from an embedded Heegaard surface adapted to b gives a sutured Heegaard diagram with one boundary circle, and conversely. By [20, Definitions 2.34–2.35, Proposition 2.36], any two vertices of the graph of embedded sutured diagrams of $M(b)$ are joined by a path of edges given by compression-equivalent changes of attaching sets, stabilizations, and diffeomorphisms isotopic to the identity in $M(b)$ relative to the suture [20, Definition 2.34]. The first are (P1). The last are diffeomorphisms isotopic to the identity relative to the suture, which need not preserve the standard collar of the boundary sphere; since the image surface meets the boundary sphere transversely in the same circle, the collar neighbourhood theorem gives an ambient isotopy, fixed on a smaller collar, that makes it standard on the collar again, and the composite is an edge (P2). The stabilizations of [20, Definition 2.18] need not be local: the punctured torus T replacing the disk D is not required to lie in a small ball. Nevertheless every such edge $H_1 \rightarrow H_2$ factors in $\mathcal{P}(M, b)$ as a local stabilization (P3) in a ball Q with $Q \cap \Sigma_1 \subset \text{int } D$, followed by a change of disk system and an isotopy fixed near $b(B^3)$ (Lemma 12.2). An additional stabilization, in a small ball of the standard collar, makes the genus at least one throughout: fix a small ball Q_0 in the common standard collar, where all diagrams of the path are standard and all isotopies are the identity, and stabilize every vertex of the path in Q_0 . The stabilized vertices form a path of the same kind, and the two ends are joined to their unstabilized versions by edges (P3). Hence the torus genus is at least two throughout. Finally, the metric and primary data of a fixed splitting form a connected set of choices, joined by edges (P4). $\square$

Lemma 12.2. *Let $H_1 = (\Sigma_1, \alpha_1, \beta_1)$ and $H_2 = (\Sigma_2, \alpha_2, \beta_2)$ be embedded diagrams of $M(b)$, standard on a common collar of the boundary, such that H_2 is obtained from H_1 by a stabilization with data $D \subset \Sigma_1$, $T \subset \Sigma_2$, $a = \alpha_2 \cap T$, $b = \beta_2 \cap T$. Then there are a ball Q disjoint from the curves of H_1 and from $b(B^3)$, with $Q \cap \Sigma_1 \subset \text{int } D$, and an isotopy ψ_t of $M(b)$ with compact support in the interior, such that $\psi_1(\Sigma_2)$ is the local stabilization of Σ_1 in Q , with the same sides.*

Proof. The curves a and b bound disks D_a, D_b on the two sides of Σ_2 , meeting in one point, and the boundary of a regular neighbourhood of $D_a \cup D_b$ is a sphere S meeting Σ_2 in one circle, bounding a ball B_S that contains T . Compressing Σ_2 along D_a and exchanging disks in the complement of S , each half of which lies in an irreducible compression body, shows that Σ_2 is obtained from Σ_1 by adding a tube along an arc ℓ with endpoints in D , and b gives a disk exhibiting ℓ as parallel to an arc in D . Since the halves of B_S lie in irreducible compression bodies, the tube is unknotted, and shrinking it along this parallelism disk into a small ball Q around a subarc of D is an isotopy supported away from the curves of H_1 and from $b(B^3)$. The framing ball and the torus are never moved. $\square$

Lemma 12.3. *Let $Q \subset M \setminus b(B^3)$ be a ball, $W_1: M \rightarrow M_m$ a one-handle and $W_2: M_m \rightarrow M'$ a two-handle attached in $Q \times [0, 1]$ whose attaching circle meets the belt sphere of W_1 transversely once, with product bundles and vertical tube. For every diffeomorphism $\iota: M' \rightarrow M$ equal to the identity outside Q , $(W_2, \iota) \circ W_1 \cong 1_M$ in $\mathbf{B}$.*

Proof. Give $W_2 \circ W_1$ a Morse function with the two critical points and a gradient-like field equal to ∂_t on $(M \setminus Q) \times [0, 1]$. By the first cancellation theorem [28, Theorem 5.4] it can be replaced by one without critical points, the change supported near the stable and unstable manifolds of the critical points, which miss $(M \setminus Q') \times [0, 1]$ for a slightly larger ball Q' . Integrating the new field gives a diffeomorphism to $M \times [0, 1]$ equal to the identity on the incoming end and on $(M \setminus Q') \times [0, 1]$. The induced self-diffeomorphism of M at the outgoing end, composed with ι^{-1} , is supported in Q' , hence isotopic to the identity relative

to the boundary of Q' by Cerf's theorem [9], and the mapping cylinder of the isotopy is an isomorphism of decorated cobordisms fixed on the tube. $\square$

12.2. The edge squares. For each edge $e: p \rightarrow q$ we define an independent symplectic map $\Gamma_e: HF(M, p) \rightarrow HF(M, q)$ and prove

$$(12.1) \quad H(N_q) \circ G(q, p) = \Gamma_e \circ H(N_p),$$

where $G(q, p)$ is the gauge continuation map, i.e. $I^\#$ of the product cobordism with vertical tube and product bundle. In the notation of Section 11.3, $G(q, p) = \text{cont}_G(p, q)$; here the arguments are written in the order of composition, so that $G(r, q)G(q, p) = G(r, p)$.

Proposition 12.4. *Every edge of $\mathcal{P}(M, b)$ has an independently defined map Γ_e satisfying (12.1).*

Proof. (P1): The Lagrangians $L(C_i)$ depend on the embedded compression bodies, their metrics and primary perturbations, and not on the disk systems, so $\Gamma_e = 1$ and (12.1) is trivial.

(P4), primary data: Γ_e is the symplectic continuation of Section 11.3, and (12.1) is Proposition 11.9 inside one finite family; two finite families are bridged through their union, which is again a finite family containing both.

(P4), regularizing data: for changes of f, χ, η the map Γ_e is the identity and the square is Theorem 3.6(ii). For a change of the almost complex structure J on the strip end, Γ_e is the symplectic continuation. Insert a strip carrying the continuation from J to J' at the ordinary end of the mixed domain; by Theorem 5.1 at the ordinary neck, for a long neck the count on the glued domain is $\Gamma_e \circ N_p$, and the glued domain is a compactly supported deformation of the domain for J' , so Theorem 5.2 identifies the count with $N_{p'}$ up to chain homotopy.

(P3): Let r be the local stabilization of p in a ball Q . Let $W_1: M \rightarrow M \# (S^1 \times S^2)$ be the one-handle and W_2 the cancelling two-handle, both attached in $Q \times [0, 1]$. By Lemma 12.3, $W_2 \circ W_1 \cong 1_M$ for any identification of the outgoing end that is the identity outside Q , so $G(r, p) = I^\#(W_2)I^\#(W_1)$. The one-handle square (Theorem 11.11) ends at the enlarged presentation $C \circ p$, with compression bodies (V_0, V_1) carrying height Lagrangians (R_0, R_1) at a height t , while the two-handle square (Theorem 9.13) starts at a vertex whose second body is unperturbed. The two are joined by primary-data changes, as follows. Give $C \circ p$ the metric data that agrees on V_0 with the metric of r on its first body; the output of the cancelling two-handle has bodies (V_0, C_2) , where C_2 is the second body of r with the metric of r . Take first the heights t_* fixed after Proposition 11.10 for the splittings and metric data of $C \circ p$ and of r , then t and the two-handle height t_2 so small that all data below lie in the corresponding neighbourhoods and in the admissible neighbourhoods, and then the lower Lagrangians of p as in Theorem 11.11. Let R'_0 be a height Lagrangian obtained from R_0 by a small perturbation supported away from its critical set, transverse to $L(V_1)$ and to $L_{t_2 h_Q}(C_2)$ and missing their intersection; this is a generic condition. Put

- $(C \circ p)'$: the presentation $C \circ p$ with primary data $(R'_0, 0)$;
- r' : the presentation r with primary data $(R'_0, t_2 h_Q)$.

Both are admissible. The subordinate triple of the cancelling two-handle is $(C_0, C_1, C_2) = (V_0, V_1, C_2)$ with primary data $(R'_0, 0, t_2 h_Q)$, which satisfies the hypotheses of Section 9.1, so Theorem 9.13 holds from $(C \circ p)'$ to r' . Define

$$S_{r,p} = \text{cont}_S(r', r) \circ S_{W_2} \circ \text{cont}_S(C \circ p, (C \circ p)') \circ U_C.$$

The squares of Theorem 11.11, of Proposition 11.9 for the two changes of primary data, and of Theorem 9.13 compose to $H(N_r)G(r, p) = S_{r,p}H(N_p)$.

The reverse composite is a two-handle $W'_2: M \rightarrow M \# (S^1 \times S^2)$ followed by a three-handle W_3 , both in $Q \times [0, 1]$. Reversing the orientation of the cobordism exchanges them with a one-handle followed by a cancelling two-handle, so Lemma 12.3 applied to the reversed cobordism gives $W_3 \circ W'_2 \cong 1_M$ and $G(p, r) = I^\#(W_3)I^\#(W'_2)$. The triple of W'_2 is (V_0, C_2, V_1) with primary data $(R''_0, 0, t'_2 h'_Q)$, where h'_Q is the corresponding height function on V_1 , chosen in the same way, and

$$D_{p,r} = V_C \circ \text{cont}_S((C \circ p)'', C \circ p) \circ S_{W'_2} \circ \text{cont}_S(r, r''),$$

where r'' carries $(R''_0, 0)$ and $(C \circ p)''$ carries $(R''_0, t'_2 h'_Q)$. Theorems 9.13 and 11.12 and Proposition 11.9 give $H(N_p)G(p, r) = D_{p,r}H(N_r)$. Then $D_{p,r}S_{r,p}H(N_p) = H(N_p)G(p, r)G(r, p) = H(N_p)$, and since $H(N_p)$ is invertible, $D_{p,r}S_{r,p} = 1$; similarly $S_{r,p}D_{p,r} = 1$.

(P4), metric data: for two presentations p, q of one splitting that differ in their metric data, choose a common stabilization r and put $\Gamma_e = D_{q,r}S_{r,p}$. This is possible because the upper metric data in Theorems 9.13 and 11.11 is free within the class of Section 10.1: the flat-connection Lagrangians do not depend on the metric, so one upper choice serves both p and q . The square follows from the stabilization squares and $G(q, r)G(r, p) = G(q, p)$; independence of r follows by invertibility of N .

(P2): An ambient isotopy ϕ_t fixed near $b(B^3)$ transports all data. Transporting data conjugates every equation, operator and perturbation, so the counts agree exactly, and the mapping cylinder of the isotopy is the decorated product. The isotopy transports the primary data as well as the metrics, so the transported vertex differs from the prescribed one in both. Hence Γ_e is the map induced by the transported data followed by the metric-data change and then the primary-data change (P4), and the square follows from the previous cases. $\square$

Corollary 12.5. *For paths γ in $\mathcal{P}(M, b)$ from p to q , the composite Γ_γ depends only on p and q ; the maps $\Gamma(q, p)$ satisfy $\Gamma(r, q)\Gamma(q, p) = \Gamma(r, p)$ and $\Gamma(p, p) = 1$.*

Proof. By (12.1) and functoriality of gauge continuation, $\Gamma_\gamma H(N_p) = H(N_q)G(q, p)$, and $H(N_p)$ is invertible. $\square$

12.3. Frame changes.

Proposition 12.6. *Let $u: M \rightarrow \text{SO}(3)$ equal 1 near $b(B^3)$, and let p, q be admissible presentations. The restriction $u|_\Sigma$, extended by 1 over the torus summand, is an automorphism $\hat{u}$ of the odd surface bundle that extends over both compression bodies; its lifting class $c(\hat{u})$ therefore lies in $A_0 \cap A_1$, and the induced symplectomorphism $\tau_{c(\hat{u})}$ of M_h carries the Lagrangians of p to those of the transported presentation u_*p . Let u_S be the induced pushforward chain isomorphism and $S_{[u]} = \Gamma(q, u_*p) \circ u_S$. Then $H(N_q) \circ I^\#([u]) = S_{[u]} \circ H(N_p)$.*

Proof. On chains $N_{u_*p} \circ u_G = u_S \circ N_p$, where u_G is the gauge pushforward: applying the bundle automorphism u to all data transports the equations of the mixed domain, the compression bodies, the torus marking and the tube (on which $u = 1$), and preserves the lifting group, whose support is disjoint from that of u . The transported presentation u_*p is admissible, with the transported mixed map. The gauge map of the frame change is $I^\#([u]) = G(q, u_*p) \circ u_G$. Hence, by Corollary 12.5, $H(N_q)I^\#([u]) = H(N_q)G(q, u_*p)u_G = \Gamma(q, u_*p)H(N_{u_*p})u_G = \Gamma(q, u_*p)u_S H(N_p)$. $\square$

A lifting class in $A_0 + A_1$ would not suffice: if $a \in A_0 \setminus A_1$, then τ_a changes the sign of the lifted holonomy of a meridian of H_1 and moves $L(C_1)$ off itself. The condition $\zeta \in A_0 + A_1$ on the filling-frame class ζ is the triviality condition for the bundle on $H_0 \cup H_1$ and is used in the construction of the global frames of Section 4.4.

13. PROOF OF THEOREM A

13.1. A descent lemma. Let $V(M)$ denote the set of admissible presentations of an object M (Section 3). Consider the category $\mathbf{B}^V$ whose objects are pairs (M, p) with $p \in V(M)$ and whose morphisms $(M, p) \rightarrow (M', q)$ are the morphisms $M \rightarrow M'$ of $\mathbf{B}$. The projection $\mathbf{B}^V \rightarrow \mathbf{B}$ is an equivalence. For each (M, p) the gauge complex $C_G(M, p)$ computes $I^\#(M)$: the gauge continuation maps provide canonical isomorphisms $H(C_G(M, p)) \cong I^\#(M)$ compatible with the maps $I^\#(W, E, \nu)$ (Proposition 2.4). We therefore regard $I^\#$ as a functor on $\mathbf{B}^V$ with $I^\#(M, p) = H(C_G(M, p))$, and write $N_{(M, p)} = H(N_p): I^\#(M, p) \rightarrow HF(M, p)$.

An *elementary arrow* $e: (M, p) \rightarrow (M', q)$ is a triple consisting of an underlying morphism $[e] \in \mathbf{B}(M, M')$, source and target presentations, and a linear map $S_e: HF(M, p) \rightarrow HF(M', q)$ defined by holomorphic curve counts. The arrows used are listed in Section 13.2.

Lemma 13.1. *Let $\mathcal{A}$ be a set of elementary arrows such that:*

- (a) *each $N_{(M, p)}$ is an isomorphism;*
- (b) *every $e \in \mathcal{A}$ satisfies $N_{(M', q)} \circ I^\#([e]) = S_e \circ N_{(M, p)}$;*
- (c) *for all (M, p) , (M', q) and every $W \in \mathbf{B}(M, M')$ there is a word $w = (e_1, \dots, e_m)$ of elements of $\mathcal{A}$ from (M, p) to (M', q) with $[e_m] \circ \dots \circ [e_1] = W$.*

For a word w put $S_w = S_{e_m} \circ \dots \circ S_{e_1}$. Then

- (i) $N_{(M', q)} \circ I^\#(W) = S_w \circ N_{(M, p)}$ *for every word w representing W ;*
- (ii) S_w *depends only on W , p and q ; write it $S(W)_{qp}$;*
- (iii) $W \mapsto S(W)_{qp}$ *is a functor $\mathbf{B}^V \rightarrow \text{Vect}_{\mathbb{F}_2}$, and N is a natural isomorphism $I^\# \Rightarrow S$;*
- (iv) *the maps $\Gamma(q, p) = S(1_M)_{qp}$ satisfy $\Gamma(r, q)\Gamma(q, p) = \Gamma(r, p)$ and $\Gamma(p, p) = 1$. With $HF^\#(M) = \text{colim}_p HF(M, p)$ along Γ , the functor S and the transformation N descend to a functor $HF^\#$ on $\mathbf{B}$ and a natural isomorphism $I^\# \Rightarrow HF^\#$.*

Proof. (i) Induction on m . For $m = 0$ the word is empty, $W = 1_M$ and $p = q$. For $m \geq 1$ let $w' = (e_1, \dots, e_{m-1})$ end at (M'', r) . By functoriality of $I^\#$, (b) for e_m , and the induction hypothesis,

$$N_q I^\#(W) = N_q I^\#([e_m]) I^\#([w']) = S_{e_m} N_r I^\#([w']) = S_{e_m} S_{w'} N_p.$$

(ii) By (i) and (a), $S_w = N_q I^\#(W) N_p^{-1}$, which does not involve w . (iii) $S(W' \circ W) = N I^\#(W') I^\#(W) N^{-1} = S(W') S(W)$ and $S(1) = 1$; naturality is (i). (iv) is (iii) applied to identity morphisms; a system of isomorphisms satisfying the cocycle law has a colimit with isomorphisms $i_p: HF(M, p) \rightarrow HF^\#(M)$, and $N_M = i_p N_{(M, p)}$ is independent of p by (i). $\square$

Lemma 13.1 is formal. All of the content of Theorem A lies in (a), (b) and (c). In (b) the maps $N_{(M, p)}$ must be the ones fixed at the vertices: a square proved with an auxiliary mixed map at the source or target is usable only after that map has been identified, on homology, with the vertex map. This is the role of Theorem 3.6(ii) and of the data-change squares of Section 12.

13.2. Generation. The set $\mathcal{A}$ consists of the following arrows.

- (A1) *Presentation moves* $(M, p) \rightarrow (M, q)$ with $[e] = 1_M$: changes of disk system, stabilization and destabilization away from the ball, isotopies fixed near the ball, and changes of the metric and primary data (Section 12).
- (A2) *Frame changes* $[u]$ (Section 4.5), with $S_{[u]}$ the map induced by the symplectomorphism of M_h given by $u|_\Sigma$ (Proposition 12.6).

- (A3) *One-handles* $(M, p) \rightarrow (M^+, C \circ p)$, where $C \circ p$ carries height Lagrangians at some height t and $p, C \circ p$ satisfy the neighbourhood conditions of Theorem 11.11, with the compression cup.
- (A4) *Two-handle cobordisms* $(W_2, E; \phi_0^{\text{gl}}, \phi_1^{\text{gl}})$ with bundles trivial at both ends, between the presentations p_0^ϕ, p_1^ϕ of a subordinate triple with the global frames of Section 4.4, carrying the transported data of Section 9.1 with t below the threshold of Lemma 9.1(iii); these vertices are admissible presentations. The map is $S_{W_2, E}$ (Theorem 9.13).
- (A5) *Three-handles* $(M^+, C \circ p) \rightarrow (M, p)$, with the same vertex conditions as in (A3), with the compression cap (Theorem 11.12).

Proposition 13.2. *$\mathcal{A}$ satisfies condition (c) of Lemma 13.1.*

Proof. Let $(W, E, \nu): M \rightarrow M'$ and presentations p, q be given. By Propositions 4.1 and 4.2, $(W, E, \nu) = W_3 \circ (W_2, E) \circ W_1$ with product bundles on W_1 and W_3 and connected levels Z_0, Z_1 . Write W_1 and W_3 as composites of single one- and three-handles. By Lemma 4.7, $(W_2, E) = [u_1] \circ (W_2, E; \phi^{\text{gl}}) \circ [u_0]^{-1}$, a two-handle arrow between frame changes, or a single frame change if there are no two-handles. Each handle can be attached in a presentation adapted to it (Section 4.3 and the paragraph preceding it), and the two-handle cobordism in the presentation given by a subordinate triple (Proposition 4.3). By Proposition 12.1, any two presentations of the same object, and in particular consecutive presentations at each level as well as p, q at the ends, are joined by presentation moves; in particular each handle arrow can be taken between vertices satisfying the conditions of (A3)–(A5), since such vertices exist for every splitting (small heights and generic small data) and are reached by presentation moves. Concatenation gives a word of arrows in $\mathcal{A}$ from (M, p) to (M', q) whose underlying composite is (W, E, ν) , since presentation moves have underlying morphism the identity. $\square$

13.3. Proof of Theorem A. Condition (a) of Lemma 13.1 is Theorem 3.6. Condition (b) is Proposition 12.4 for (A1), Proposition 12.6 for (A2), Theorem 11.11 for (A3), at presentations in the neighbourhoods of Proposition 11.10 (arrows may be taken at any convenient vertices, since presentation moves connect them), Theorem 9.13 for (A4) and Theorem 11.12 for (A5). Condition (c) is Proposition 13.2. Lemma 13.1 gives the equation of Theorem A for every word, the independence of S_d , the functor $HF^\#$ and the natural isomorphism. The statement in the homotopy category of finite complexes follows from Lemma B.1. $\square$

Remark 13.3. No relation among symplectic words is proved directly. Independence of the decomposition follows from the squares, the connectivity of the presentation graph and the invertibility of N . The argument does not give integral statements: over $\mathbb{Z}$ the squares would be needed with signs, and Lemma B.1 fails.

13.4. The isomorphism of Daemi, Fukaya and Lipyanskiy.

Corollary 13.4. *Let p_0, p_1 be admissible presentations of M_0, M_1 with DFL-regular primary data, and let $N_{p_i}^{\text{DFL}}$ be the isomorphisms of [11, Theorem 3.29] at the data of Proposition 3.9. Then for every morphism (W, E, ν) and every admissible decomposition d ,*

$$H(N_{p_1}^{\text{DFL}} \otimes \mathbb{F}_2) \circ I^\#(W, E, \nu) = S_d(W, E, \nu) \circ H(N_{p_0}^{\text{DFL}} \otimes \mathbb{F}_2).$$

In particular, for two such choices of data at presentations p, q of one object, $H(N_q^{\text{DFL}} \otimes \mathbb{F}_2) \circ G(q, p) = \Gamma(q, p) \circ H(N_p^{\text{DFL}} \otimes \mathbb{F}_2)$ as maps from the gauge homology at p , where $G(q, p)$ is the gauge continuation and $\Gamma(q, p)$ the symplectic transitive system of Theorem A, defined by holomorphic curve counts. So within this class of data, including changes of the Heegaard

splitting, the reduction modulo two of the isomorphism of [11] on homology does not depend on the choices.

Proof. By Proposition 3.9, $N_{p_i}^{\text{DFL}} \otimes \mathbb{F}_2 = N_{p_i}$ at these data, and Theorem A applies to N_{p_i} . The second statement is the case of the identity morphism: on the gauge homology at p and q the identity of $I^\#(M)$ is the gauge continuation $G(q, p)$, and $S(1_M)_{qp} = \Gamma(q, p)$ by Lemma 13.1. $\square$

The corollary applies wherever admissible presentations with DFL-regular primary data exist. Their existence at the fixed metric data of a presentation is the transversality question of Remark 3.5. Where they exist, Theorem A shows that the vertex map at any admissible presentation of the same object is, on homology, the reduction modulo two of the isomorphism of [11] at such a presentation, composed with the maps G and Γ .

APPENDIX A. RESULTS USED FROM THE LITERATURE

We list the published results used in the proof, where they enter, and the hypotheses that have to be checked.

- (1) [11, Theorem 1, Propositions 2.21, 3.19, 3.20, Lemma 3.15, Theorem 3.29]: monotonicity with minimal Maslov number four, embeddedness of $L(C)$, the index of constant pairs, the index-energy law and triangularity of the mixed map. Used in Sections 3 and 6. Hypotheses: the bundles are odd on the matching surface; the Lagrangians are embedded, which holds for compression bodies.
- (2) [11, Theorem 4.17, Remark 4.19, Section 4.2.1, (4.20)–(4.24), Theorem 5.2, Lemma 5.23, Proposition 5.29, Section 5.4, Section 6]: local elliptic estimates for the mixed operator, theory of mixed ends, small-energy regularity, exponential decay, compactness and perturbations. Used throughout. Hypotheses: product metrics and the complex structure $*_\Sigma$ near the seams, which Convention 3.1 and Hypothesis 6.5 arrange.
- (3) [11, Section 2.3, Proposition 2.53, Section 3.1 and Figure 1, Definitions 3.8–3.9, (3.24), Proposition 3.18]: primary perturbations and the submersion property of the universal restriction map, the special quintuple and its planar region, the domains of DFL type with their weighted spaces, and the Fredholm property of the mixed operator. Used in Sections 3, 6 and 9. Hypotheses: those of (1).
- (4) [11, Section 4.1 with (4.3) and Lemma 4.4, Proposition 4.6, Section 4.2.1 with (4.16), Propositions 4.65, 4.69, 5.27 and 5.28]: the cutoff maps E_τ and a metric in which the Lagrangian is totally geodesic, the Coulomb slice (used in Section 10 in a variant), the slice operator on a cylinder quintuple, additivity of energy and index under cutoff gluing, and exponential decay on mixed ends. Used in Section 10. Hypotheses: nondegenerate or Morse–Bott ends as stated there.
- (5) [12, Theorems 1 and 3, Theorem 4.2, Lemmas 4.25 and 4.50, Theorem 4.44]: seam regularity, compactness, seam estimates and removal of seam singularities for the mixed equation. Hypotheses: fixed product collars after conformal rescaling; the disconnected odd matching surface is allowed.
- (6) [21, Sections 5.1–5.2, Definition 5.1, Propositions 5.2 and 5.4, Theorem 5.6]: the enlarged gauge groups, the quotient by ϕ , the composition law with the connecting-map condition, and excision. Used in Sections 2, 9 and 10. Hypotheses: the ϕ -non-integral condition at every end, and the connecting image ψ contained in ϕ ; both are checked where used.
- (7) [38, Sections 7.1–7.4, 7.6–7.8, Proposition 7.1]: the framed gauge transformations, the $\mathbb{Z}/4$ grading and degree formula, and $I^\#(\#^r(S^1 \times S^2))$ with its relative invariants.

- (8) [32, Definition 4.2, Proposition 4.3, Lemma 4.5]: subordinate Heegaard triples and the identification of the filled triangle with the two-handle trace; the tube is controlled by Proposition 4.3.
- (9) [20, Definitions 2.34–2.35, Proposition 2.36]: connectivity of the graph of embedded sutured diagrams; applied to the sutured manifold $M(b)$ in Proposition 12.1.
- (10) [44, Theorems 3.9, 3.13, 4.1, 4.2, Example 2.6]: quilted Floer maps, gluing, data-change isomorphisms and unit insertion. Hypotheses: monotonicity with a common constant and minimal Maslov number at least three for all Lagrangians and seams, and Hamiltonian one-forms vanishing on the boundary; the continuation of Section 11.3 is arranged in this form.
- (11) [31, Section 14.4]: continuation with moving Lagrangian boundary conditions.
- (12) [13, Propositions 3.8–3.10, Theorems 4.17 and 5.4, Section 5.3]: additivity of the index under gluing along acyclic ends, weights and kernels, neck gluing with surjectivity, and gluing at index -1 in families; the addition property of [13, Section 3.3], on which the excision argument of Lemma 6.10 is modelled.
- (13) Gluing along ordinary strip necks with transverse intersections is proved in Theorem 5.1 by the argument of [13, Section 3] with the local estimates of [12, Section 4]; monotonicity enters only through compactness. For exact Lagrangians the corresponding statement is [40, Proposition 2.22].
- (14) [27, Theorem B.4.1]: boundary regularity for J -holomorphic curves with Lagrangian boundary conditions, from $W^{1,p}$, $p > 2$.
- (15) [37, Proposition 8.6, Theorem C.1]: unique continuation for the trajectory equation.
- (16) [6], [39, Corollary 2.1.6, Theorem 2.1.7], [34], [1]: elliptic boundary value problems for first-order operators, the Gaffney–Friedrichs inequality, regularity for families of self-adjoint operators, and unique continuation.

The quilted description of the compression cup in Remark 11.5 uses in addition [44, Corollary 5.5], [45], [42] and [25, Theorem 2 and Corrigendum, Remark 1(1)]; none of these enters the proofs of Theorems A and B.

APPENDIX B. HOMOLOGICAL ALGEBRA OVER A FIELD

Lemma B.1. *Let (C, d_C) and (D, d_D) be finite-dimensional chain complexes over a field k , and $f, g: C \rightarrow D$ chain maps with $H(f) = H(g)$. Then f and g are chain homotopic. Consequently, the homology functor from the homotopy category of finite-dimensional complexes over k to graded vector spaces is fully faithful, and an equation between composites of chain maps that holds on homology holds up to chain homotopy.*

Proof. It suffices to treat $g = 0$. Over a field every finite complex splits: there are maps $i_C: H(C) \rightarrow C$, $p_C: C \rightarrow H(C)$ and $h_C: C \rightarrow C$ of degree one with $p_C i_C = 1$, $1 - i_C p_C = d_C h_C + h_C d_C$, and $d_C i_C = 0$, $p_C d_C = 0$; similarly for D . Put

$$H = h_D f + i_D p_D f h_C.$$

Using $d_D i_D = 0$ and $d_D f = f d_C$,

$$d_D H + H d_C = (d_D h_D + h_D d_D) f + i_D p_D f h_C d_C = (1 - i_D p_D) f + i_D p_D f (1 - i_C p_C - d_C h_C).$$

Since $p_D f d_C = p_D d_D f = 0$ and $p_D f i_C = H(f) = 0$, the last term equals $i_D p_D f$, and $d_D H + H d_C = f$. $\square$

The splitting maps depend on choices, so the homotopy H is not canonical, and no higher coherence is asserted. Over $\mathbb{Z}$ the lemma fails, which is one reason Theorem A is stated over $\mathbb{F}_2$.

REFERENCES

- [1] N. Aronszajn, *A unique continuation theorem for solutions of elliptic partial differential equations or inequalities of second order*, J. Math. Pures Appl. (9) **36** (1957), 235–249.
- [2] M. F. Atiyah, *New invariants of 3- and 4-dimensional manifolds*, in: The Mathematical Heritage of Hermann Weyl, Proc. Sympos. Pure Math. **48**, Amer. Math. Soc., Providence, RI, 1988, 285–299.
- [3] M. F. Atiyah and R. Bott, *The Yang–Mills equations over Riemann surfaces*, Philos. Trans. Roy. Soc. London Ser. A **308** (1983), 523–615.
- [4] M. F. Atiyah, V. K. Patodi and I. M. Singer, *Spectral asymmetry and Riemannian geometry. I*, Math. Proc. Cambridge Philos. Soc. **77** (1975), 43–69.
- [5] P. J. Braam and S. K. Donaldson, *Floer’s work on instanton homology, knots and surgery*, in: The Floer Memorial Volume, Progr. Math. **133**, Birkhäuser, Basel, 1995, 195–256.
- [6] C. Bär and W. Ballmann, *Guide to elliptic boundary value problems for Dirac-type operators*, in: Arbeitstagung Bonn 2013, Progr. Math. **319**, Birkhäuser, 2016, 43–80; arXiv:1307.3021.
- [7] N. Bottman and K. Wehrheim, *Gromov compactness for squiggly strip shrinking in pseudoholomorphic quilts*, Selecta Math. (N.S.) **24** (2018), 3381–3443.
- [8] G. Cazassus, *Symplectic instanton homology: naturality, and maps from cobordisms*, Quantum Topol. **10** (2019), 677–722.
- [9] J. Cerf, *Sur les difféomorphismes de la sphère de dimension trois ($\Gamma_4 = 0$)*, Lecture Notes in Math. **53**, Springer, 1968.
- [10] A. Daemi and K. Fukaya, *Atiyah–Floer conjecture: a formulation, a strategy of proof and generalizations*, in: Modern Geometry: A Celebration of the Work of Simon Donaldson, Proc. Sympos. Pure Math. **99**, Amer. Math. Soc., Providence, RI, 2018, 23–57.
- [11] A. Daemi, K. Fukaya and M. Lipyanskiy, *Lagrangians, $SO(3)$ -instantons and the Atiyah–Floer conjecture*, arXiv:2109.07032.
- [12] A. Daemi, K. Fukaya and M. Lipyanskiy, *Lagrangians, $SO(3)$ -instantons and mixed equation*, Geom. Funct. Anal. **34** (2024), 659–732; arXiv:2109.07038 (numbering as in the arXiv version).
- [13] S. K. Donaldson, *Floer Homology Groups in Yang–Mills Theory*, Cambridge Tracts in Math. **147**, Cambridge Univ. Press, 2002.
- [14] K. Fukaya, *Anti-self-dual equation on 4-manifolds with degenerate metric*, Geom. Funct. Anal. **8** (1998), 466–528.
- [15] S. Dostoglou and D. A. Salamon, *Self-dual instantons and holomorphic curves*, Ann. of Math. (2) **139** (1994), 581–640.
- [16] A. Floer, *An instanton-invariant for 3-manifolds*, Comm. Math. Phys. **118** (1988), 215–240.
- [17] R. E. Gompf and A. I. Stipsicz, *4-Manifolds and Kirby Calculus*, Grad. Stud. Math. **20**, Amer. Math. Soc., 1999.
- [18] H. T. Horton, *A symplectic instanton homology via traceless character varieties*, arXiv:1611.09927.
- [19] H. T. Horton, *Traceless character varieties, the link surgeries spectral sequence, and Khovanov homology*, arXiv:1702.02259.
- [20] A. Juhász, D. P. Thurston and I. Zemke, *Naturality and mapping class groups in Heegaard Floer homology*, Mem. Amer. Math. Soc. **273** (2021), no. 1338; arXiv:1210.4996.
- [21] P. B. Kronheimer and T. S. Mrowka, *Khovanov homology is an unknot-detector*, Publ. Math. Inst. Hautes Études Sci. **113** (2011), 97–208.
- [22] P. B. Kronheimer and T. S. Mrowka, *Knots, sutures, and excision*, J. Differential Geom. **84** (2010), 301–364.
- [23] F. Laudenbach and V. Poénaru, *A note on 4-dimensional handlebodies*, Bull. Soc. Math. France **100** (1972), 337–344.
- [24] M. Lipyanskiy, *Gromov–Uhlenbeck compactness*, arXiv:1409.1129.
- [25] Y. Lekili and M. Lipyanskiy, *Geometric composition in quilted Floer theory*, Adv. Math. **236** (2013), 1–23; Corrigendum, Adv. Math. **308** (2017), 1340–1345; arXiv:1003.4493v9.
- [26] C. Manolescu and C. Woodward, *Floer homology on the extended moduli space*, in: Perspectives in Analysis, Geometry, and Topology, Progr. Math. **296**, Birkhäuser/Springer, New York, 2012, 283–329.

- [27] D. McDuff and D. Salamon, *J-Holomorphic Curves and Symplectic Topology*, 2nd ed., Amer. Math. Soc. Colloq. Publ. **52**, Amer. Math. Soc., 2012.
- [28] J. Milnor, *Lectures on the h-Cobordism Theorem*, Princeton Univ. Press, 1965.
- [29] P. E. Newstead, *Characteristic classes of stable bundles of rank 2 over an algebraic curve*, Trans. Amer. Math. Soc. **169** (1972), 337–345.
- [30] Y.-G. Oh, *Floer cohomology of Lagrangian intersections and pseudo-holomorphic disks. I*, Comm. Pure Appl. Math. **46** (1993), 949–993.
- [31] Y.-G. Oh, *Symplectic Topology and Floer Homology*, Vol. 2, New Math. Monogr. **29**, Cambridge Univ. Press, 2015.
- [32] P. Ozsváth and Z. Szabó, *Holomorphic triangles and invariants for smooth four-manifolds*, Adv. Math. **202** (2006), 326–400.
- [33] J. Robbin and D. Salamon, *Asymptotic behaviour of holomorphic strips*, Ann. Inst. H. Poincaré Anal. Non Linéaire **18** (2001), 573–612.
- [34] J. Robbin and D. Salamon, *The spectral flow and the Maslov index*, Bull. London Math. Soc. **27** (1995), 1–33.
- [35] D. Salamon, *Lagrangian intersections, 3-manifolds with boundary, and the Atiyah–Floer conjecture*, in: Proceedings of the International Congress of Mathematicians (Zürich, 1994), Vol. 1, Birkhäuser, Basel, 1995, 526–536.
- [36] D. Salamon and K. Wehrheim, *An open-closed isomorphism for instanton Floer homology*, preprint, October 2025, <https://math.berkeley.edu/~katrin/papers/floer-two.pdf>.
- [37] D. Salamon and K. Wehrheim, *Instanton Floer homology with Lagrangian boundary conditions*, Geom. Topol. **12** (2008), 747–918.
- [38] C. W. Scaduto, *Instantons and odd Khovanov homology*, J. Topol. **8** (2015), 744–810.
- [39] G. Schwarz, *Hodge Decomposition—A Method for Solving Boundary Value Problems*, Lecture Notes in Math. **1607**, Springer, 1995.
- [40] P. Seidel, *A long exact sequence for symplectic Floer cohomology*, Topology **42** (2003), 1003–1063.
- [41] K. Wehrheim, *Lagrangian boundary conditions for anti-self-dual instantons and the Atiyah–Floer conjecture*, J. Symplectic Geom. **3** (2005), 703–747.
- [42] K. Wehrheim and C. T. Woodward, *Floer cohomology and geometric composition of Lagrangian correspondences*, Adv. Math. **230** (2012), 177–228.
- [43] K. Wehrheim and C. T. Woodward, *Floer field theory for coprime rank and degree*, Indiana Univ. Math. J. **69** (2020), 2035–2088; arXiv:1601.04924.
- [44] K. Wehrheim and C. T. Woodward, *Pseudoholomorphic quilts*, J. Symplectic Geom. **13** (2015), 849–904.
- [45] K. Wehrheim and C. T. Woodward, *Quilted Floer cohomology*, Geom. Topol. **14** (2010), 833–902.
- [46] G. Xu, *A compactness theorem for $SO(3)$ anti-self-dual equation with translation symmetry*, Adv. Math. **408** (2022), Paper No. 108576.